

itelescope_premium_the_easier_way

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1 iTelescope premium image set processing, the easier way

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This notebook illustrates how to process a premium image set from iTelescope using the AstroPhotography package. The python environment used corresponds to a miniconda environment `ap-env.yml`.

The example dataset used is the iTelescope Plan-20 premium dataset of the M101 supernova [SN 2023 ixf](#). Note that these files are not provided with this package. However the notebook should work with your own files if you specify a valid fits-file containing directory at the prompt below.

The processing mirrors that of the `itelescope_premium_the_hard_way.piynb` notebook, except using the `ApProcess` class. Although all of the processing stages allow many user-controlled options the default values will be used in this example.

The processing stages consists of:

1. Creating an `ApProcess` instance to perform the processing. Using the same instance for multiple processing stages on the same set of files is recommended, and simplifies identifying which images to process.
2. (To be added later) Optional bad pixel, bad column and bad row removal in the cases where the input images still have such artifacts. This is common in the iTelescope Premium image sets, where their calibration does not remove all artifacts.
3. Finding stars in the image to allow later astrometric solution finding and image quality checking.
4. Astrometric solutions using Astrometry.net
5. Resampling and stacking of multiple images onto a common footprint to create a final image in each available band..
6. Three color image composite creation using images from mutliple bands.

1.1 Notebook environment setup

```
1]: import os
import pathlib
import sys
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import time
import math
import subprocess
import shlex
import shutil
```

```
from ccdproc import ImageFileCollection
from astropy.table import Table
from astropy import wcs
import astropy
from astropy.io import fits

import AstroPhotography as ap
```

```
2]: def print_module_version(mod):
    """
    Convenience function to pretty-print Python module mod version.

    Arguments:
        mod {module} -- Imported Python module

    Returns:
        str -- module name and version
    """
    print(f"Using module {mod.__name__:30s} version: {mod.__version__}")
    return
```

```
3]: # Get Version information
print(f'Python version: {sys.version}')
print_module_version(np)
print_module_version(matplotlib)
print_module_version(astropy)
```

```
print_module_version(ap)
```

```
Python version: 3.10.13 | packaged by conda-forge | (main, Dec 23 2023,
15:36:39) [GCC 12.3.0]
Using module numpy                version: 1.26.3
Using module matplotlib            version: 3.8.2
Using module astropy               version: 6.0.0
Using module AstroPhotography      version: 0.6.0
```

```
4]: # Enable inline plotting for graphics
# %matplotlib inline
# Set default figure size to be larger
# this may only work in matplotlib 2.0+!
from IPython.core.interactiveshell import InteractiveShell
matplotlib.rcParams['figure.figsize'] = [10.0, 6.0]
# Enable multiple outputs from jupyter cells
InteractiveShell.ast_node_interactivity = "all"
```

2 Basic Pipeline Processing

2.1 Create an ApProcess instance

For convenience we'll also change into the directory containing the files we intend to process.

```
5]: default_dir = '/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline'
print(f'Default directory for input files: {default_dir}')

print('Enter the path to the directory containing the premium image set (or return for the default): ')
wdir = input('Image set path:').strip() or default_dir
try:
    os.chdir(wdir.strip())
    print('Switched directory to ' + os.getcwd())
except:
    print(f'Error, os.chdir threw an exception changing to {wdir}')
    print('Check that the path you supplied is a valid filesystem path.')
    raise
```

```
Default directory for input files:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
Enter the path to the directory containing the premium image set (or return for
the default):

Image set path:

Switched directory to
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
```

```
6]: # Choose the logging level.
loglevel = 'DEBUG'
processor = ap.ApProcess(loglevel)
```

2.2 Star Detection and Astrometry

At the end of this stage we should have valid astrometric solutions, in the form of valid WCS coordinate headers, added to copies of the input calibrated images (to avoid modifying the input calibrated images). These files, referred to here as *navigated images*, should work with your favorite FITS viewer to show Right Ascensions and Declinations.

2.2.1 Basic Parameters

```
7]: # At the most basic, only data_dir needs to be specified
data_dir = r'.'

# If the data_dir contains fits files that we do not want processed, then
# you need to specify either the inclusive and/or exclusive file patterns.
#
# Make sure to check that your pattern works on the command line using `ls`
default_include_pattern = "Calibrated-*?.fits*"
default_exclude_pattern = None

quality_summary_file = 'quality_summary_M101SN.csv'
# Need to remove existing quality summary table
pathlib.Path(quality_summary_file).unlink(True)

print(f'Default include file pattern for input files: {default_include_pattern}')
print(f'Default exclude file pattern for input files: {default_exclude_pattern}')
```

```
msg = 'Enter new include input file pattern (or return to accept default pattern)'
include_pattern = input(msg).strip() or default_include_pattern
print(f'Using "{include_pattern}" as the input include file pattern.')

msg = 'Enter new exclude input file pattern (or return to accept default pattern)'
exclude_pattern = input(msg).strip() or default_exclude_pattern
print(f'Using "{exclude_pattern}" as the input exclude file pattern.')
```

Default include file pattern for input files: Calibrated-*-.fits*
 Default exclude file pattern for input files: None

Enter new include input file pattern (or return to accept default pattern)
 Using "Calibrated-*-.fits*" as the input include file pattern.
 Enter new exclude input file pattern (or return to accept default pattern)
 Using "None" as the input exclude file pattern.

2.2.2 Optional Parameters

```
8]: # optional parameters
    # To be documented at a later time.
```

2.3 Run star finding

2.3.1 Input and Output Files

Before we run `find_stars` for the first time let us check which files it will use as inputs to `find_stars`. The processing stage is, unsurprisingly, `'inputs'`.

```
9]: # Show files names that would be generated...
    ap_filestate = 'conceptual'

    # Note we need to specify that the input_suffix is 'fits.gz' not for the inputs, which
    # are picked up by the include_pattern, but for the output file types.

    process_stage = 'input'
    no_name_and_dir = False
    with_name_and_dir = True
    input_files = processor.get_file_names(process_stage, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.
    ↪gz', no_name_and_dir)
    for file in input_files:
        print(f' {file}')
```

```
2024-11-26 21:10:12,084 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:12,084 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:12,085 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*-.fits*"
2024-11-26 21:10:12,085 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"
2024-11-26 21:10:13,655 | ApProcess | DEBUG | For ap_filetype input the relative
directory path is ./
2024-11-26 21:10:13,655 | ApProcess | DEBUG | Note that output files will be in
subdirectory ./
```

```
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
```

We can also see what the other output file names will be. **Note** that when processing it is a convention to place the output files in subdirectories. These subdirectories could be the same for a different set of input files, e.g. different input directories representing different observing nights on a specific target but having all the navigated images for that target written to a single output directory.

There are various ways to access the filename and subdirectory information, either separately or together.

You do not need to run these functions before calling `ApProcess.navigate_images`, they're performed here for purely educational reasons, and can be disabled by setting `show_files` to `False`.

```

0]: # Full set of ftypes
##ftypes = ['srclist', 'regfile', 'plotfile', 'qualfile', 'fwhmplot', 'navfile']

# Shorter subset of the files than must be generated for WCS generated and resampling/stacking to work
ftypes = ['srclist', 'navfile']

show_files = True
if show_files:
    for ftype in ftypes:
        print(80*'-')
        # Get directory info, relative path or absolute path
        rel_filedir = processor.get_directories(ftype, data_dir, False)
        abs_filedir = processor.get_directories(ftype, data_dir, True)
        print(f'File type {ftype} output directory names:')
        print(f' Relative path={rel_filedir}\n Absolute path={abs_filedir}')
        print(80*'-')

        # File names, without directory path
        print(f'File type {ftype} output file names (no directory info):')
        ofiles = processor.get_file_names(ftype, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.gz',
        ↪no_name_and_dir)
        print(f' There are {len(ofiles)} files in the file list.')
        for file in ofiles:
            print(f' {file}')
        print(80*'-')

        # File names **with** directory path
        print(f'File type {ftype} output file names (including directory):')
        ofiles = processor.get_file_names(ftype, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.gz',
        ↪with_name_and_dir)
        print(f' There are {len(ofiles)} files in the file list.')
        for file in ofiles:
            print(f' {file}')

        # Files that this instance has already processed (without directory)
        print(f'File type {ftype} output file names (no directory info), files that have been processed by this instance:')
        ofiles = processor.get_file_names(ftype, data_dir, 'processed', include_pattern, exclude_pattern, None, None, '.fits.gz',
        ↪no_name_and_dir)
        print(f' There are {len(ofiles)} files in the file list.')
        for file in ofiles:
            print(f' {file}')
        print(80*'-')

        # File that exist on disk, even if they haven't been processed by this instance
        print(f'File type {ftype} output file names (no directory info), files that exist on disk:')
        ofiles = processor.get_file_names(ftype, data_dir, 'existing', include_pattern, exclude_pattern, None, None, '.fits.gz',
        ↪no_name_and_dir)
        print(f' There are {len(ofiles)} files in the file list.')
        for file in ofiles:
            print(f' {file}')

    else:
        print(f'File name display disabled. To enable, set show_files=True')

```

```

2024-11-26 21:10:13,660 | ApProcess | DEBUG | For ap_filetype srclist the
relative directory path is ./SourceLists/
2024-11-26 21:10:13,661 | ApProcess | DEBUG | For ap_filetype srclist the
relative directory path is ./SourceLists/
2024-11-26 21:10:13,661 | ApProcess | DEBUG | For ap_filetype srclist the
absolute directory path is
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-
Pipeline/SourceLists
2024-11-26 21:10:13,661 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:13,661 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:13,662 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:13,662 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"

```

```

=====
File type srclist output directory names:
Relative path=./SourceLists/
Absolute path=/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-
Pipeline/SourceLists

```

File type srclist output file names (no directory info):

```
2024-11-26 21:10:15,148 | ApProcess | DEBUG | For ap_filetype srclist the
relative directory path is ./SourceLists/
2024-11-26 21:10:15,149 | ApProcess | DEBUG | Note that output files will be in
subdirectory ./SourceLists/
2024-11-26 21:10:15,149 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:15,149 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:15,150 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:15,150 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"
```

There are 12 files in the file list.

```
srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
srclist-M101_Supernova_2023ixf-180s-Lum-2.fits
srclist-M101_Supernova_2023ixf-180s-Lum-3.fits
srclist-M101_Supernova_2023ixf-300s-Blue-1.fits
srclist-M101_Supernova_2023ixf-300s-Blue-2.fits
srclist-M101_Supernova_2023ixf-300s-Blue-3.fits
srclist-M101_Supernova_2023ixf-300s-Green-1.fits
srclist-M101_Supernova_2023ixf-300s-Green-2.fits
srclist-M101_Supernova_2023ixf-300s-Green-3.fits
srclist-M101_Supernova_2023ixf-300s-Red-1.fits
srclist-M101_Supernova_2023ixf-300s-Red-2.fits
srclist-M101_Supernova_2023ixf-300s-Red-3.fits
```

File type srclist output file names (including directory):

```
2024-11-26 21:10:16,638 | ApProcess | DEBUG | For ap_filetype srclist the
relative directory path is ./SourceLists/
2024-11-26 21:10:16,639 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:16,639 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:16,639 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:16,639 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"
```

There are 12 files in the file list.

```
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-2.fits
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-1.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-1.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-2.fits
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits
```

File type srclist output file names (no directory info), files that have been processed by this instance:

There are 0 files in the file list.

File type srclist output file names (no directory info), files that exist on disk:

```
2024-11-26 21:10:18,146 | ApProcess | DEBUG | For ap_filetype srclist the
relative directory path is ./SourceLists/
2024-11-26 21:10:18,147 | ApProcess | DEBUG | Note that output files will be in
subdirectory ./SourceLists/
2024-11-26 21:10:18,147 | ApProcess | DEBUG | For ap_filetype navfile the
relative directory path is ./NavigatedImages/
2024-11-26 21:10:18,148 | ApProcess | DEBUG | For ap_filetype navfile the
relative directory path is ./NavigatedImages/
2024-11-26 21:10:18,148 | ApProcess | DEBUG | For ap_filetype navfile the
absolute directory path is
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-
Pipeline/NavigatedImages
2024-11-26 21:10:18,148 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:18,148 | ApProcess | DEBUG | Attempting to find stars in FITS
```

```

files within directory=.
2024-11-26 21:10:18,149 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:18,149 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"

There are 12 files in the file list.
srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
srclist-M101_Supernova_2023ixf-180s-Lum-2.fits
srclist-M101_Supernova_2023ixf-180s-Lum-3.fits
srclist-M101_Supernova_2023ixf-300s-Blue-1.fits
srclist-M101_Supernova_2023ixf-300s-Blue-2.fits
srclist-M101_Supernova_2023ixf-300s-Blue-3.fits
srclist-M101_Supernova_2023ixf-300s-Green-1.fits
srclist-M101_Supernova_2023ixf-300s-Green-2.fits
srclist-M101_Supernova_2023ixf-300s-Green-3.fits
srclist-M101_Supernova_2023ixf-300s-Red-1.fits
srclist-M101_Supernova_2023ixf-300s-Red-2.fits
srclist-M101_Supernova_2023ixf-300s-Red-3.fits
=====
File type navfile output directory names:
Relative path=./NavigatedImages/
Absolute path=/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-
Pipeline/NavigatedImages
-----
File type navfile output file names (no directory info):
2024-11-26 21:10:19,631 | ApProcess | DEBUG | For ap_filetype navfile the
relative directory path is ./NavigatedImages/
2024-11-26 21:10:19,631 | ApProcess | DEBUG | Note that output files will be in
subdirectory ./NavigatedImages/
2024-11-26 21:10:19,632 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:19,632 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:19,632 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:19,633 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"

There are 12 files in the file list.
navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
navigated-M101_Supernova_2023ixf-180s-Lum-2.fits
navigated-M101_Supernova_2023ixf-180s-Lum-3.fits
navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
navigated-M101_Supernova_2023ixf-300s-Blue-2.fits
navigated-M101_Supernova_2023ixf-300s-Blue-3.fits
navigated-M101_Supernova_2023ixf-300s-Green-1.fits
navigated-M101_Supernova_2023ixf-300s-Green-2.fits
navigated-M101_Supernova_2023ixf-300s-Green-3.fits
navigated-M101_Supernova_2023ixf-300s-Red-1.fits
navigated-M101_Supernova_2023ixf-300s-Red-2.fits
navigated-M101_Supernova_2023ixf-300s-Red-3.fits
-----
File type navfile output file names (including directory):
2024-11-26 21:10:21,118 | ApProcess | DEBUG | For ap_filetype navfile the
relative directory path is ./NavigatedImages/
2024-11-26 21:10:21,119 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:21,119 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:21,119 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:21,119 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"

There are 12 files in the file list.
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits

```

```

./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits
File type navfile output file names (no directory info), files that have been
processed by this instance:
  There are 0 files in the file list.
-----
File type navfile output file names (no directory info), files that exist on
disk:

2024-11-26 21:10:22,602 | ApProcess | DEBUG | For ap_filetype navfile the
relative directory path is ./NavigatedImages/
2024-11-26 21:10:22,602 | ApProcess | DEBUG | Note that output files will be in
subdirectory ./NavigatedImages/

```

```

  There are 12 files in the file list.
  navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
  navigated-M101_Supernova_2023ixf-180s-Lum-2.fits
  navigated-M101_Supernova_2023ixf-180s-Lum-3.fits
  navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
  navigated-M101_Supernova_2023ixf-300s-Blue-2.fits
  navigated-M101_Supernova_2023ixf-300s-Blue-3.fits
  navigated-M101_Supernova_2023ixf-300s-Green-1.fits
  navigated-M101_Supernova_2023ixf-300s-Green-2.fits
  navigated-M101_Supernova_2023ixf-300s-Green-3.fits
  navigated-M101_Supernova_2023ixf-300s-Red-1.fits
  navigated-M101_Supernova_2023ixf-300s-Red-2.fits
  navigated-M101_Supernova_2023ixf-300s-Red-3.fits

```

This matches what we expect. The order of the files is unimportant at this stage.

2.3.2 Navigating the Data

Now run `navigate_images`. There are a large number of optional parameters that we will just accept the recommended default values for.

```

1]: clean_run = True
   if clean_run:
       clean_srclists = True
       clean_navfiles = True
   else:
       clean_srclists = False
       clean_navfiles = False

   # For initial debugging it is best to make sure exceptions terminate the run.
   # And all output is sent to the screen
   initial_debugging = False
   if initial_debugging:
       stop_on_error = True
       quiet = False
       init_filelist = ['Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz',
                        'Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz']
       status = processor.navigate_images(data_dir, include_pattern, exclude_pattern, init_filelist, None, '.fits.gz',
                                         quality_summary_file, clean_srclists, clean_navfiles, stop_on_error=stop_on_error, quiet=quiet)

   else:
       # Once the first few files work, the better default is
       stop_on_error = False
       quiet = True

       status = processor.navigate_images(data_dir, include_pattern, exclude_pattern, None, None, '.fits.gz',
                                         quality_summary_file, clean_srclists, clean_navfiles, stop_on_error=stop_on_error, quiet=quiet)

```

```

2024-11-26 21:10:22,606 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:10:22,606 | ApProcess | INFO | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:10:22,607 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:10:22,607 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"
2024-11-26 21:10:24,093 | ApProcess | INFO | There are 12 input files matching
the parameters given.
2024-11-26 21:10:24,094 | ApProcess | DEBUG | Input file collection:
                                file                                naxis1 naxis2
imagetyp object filter exposure
-----
  Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz  3056   3056
--      --      Lum      --

```

```

    Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz 3056 3056
--    -- Lum --
    Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz 3056 3056
--    -- Lum --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz 3056 3056
--    -- Blue --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz 3056 3056
--    -- Blue --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz 3056 3056
--    -- Blue --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz 3056 3056
--    -- Green --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz 3056 3056
--    -- Green --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz 3056 3056
--    -- Green --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz 3056 3056
--    -- Red --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz 3056 3056
--    -- Red --
    Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz 3056 3056
--    -- Red --
2024-11-26 21:10:24,361 | ApProcess | DEBUG |
-----
2024-11-26 21:10:24,361 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:10:24,362 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:24,362 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:24,362 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:24,362 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:24,362 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:10:24,362 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:10:24,363 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-1.reg
plotfile=./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-1.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-1.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:10:24,363 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits because
clean_star_detection=True
2024-11-26 21:10:24,363 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-1.reg because
clean_star_detection=True
2024-11-26 21:10:24,363 | ApProcess | DEBUG | Removing existing
./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-1.png because
clean_star_detection=True
2024-11-26 21:10:24,364 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png because
clean_star_detection=True
2024-11-26 21:10:24,364 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:10:24,364 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:10:24,628 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:10:24,707 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67450.95, median=491.96
2024-11-26 21:10:25,527 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=492.291, median=491.025, stddev=28.186
2024-11-26 21:10:28,087 | ApFindStars | DEBUG | Source-masked image stats:
mean=489.139, median=488.849, stddev=25.656
2024-11-26 21:10:28,088 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:10:28,088 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:10:28,270 | ApFindStars | DEBUG | Retaining 17 possibly saturated

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stars in source searching and photometry.
2024-11-26 21:10:28,271 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:10:28,511 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 265 sources.
2024-11-26 21:10:28,511 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:10:28,511 | ApFindStars | DEBUG | There are 20 possibly saturated
stars in the output source list.
2024-11-26 21:10:28,512 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:10:28,512 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:10:28,556 | ApFindStars | DEBUG | Image exposure time [seconds]:
180.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:10:28,557 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 27429.19 ADU/s.
2024-11-26 21:10:28,558 | ApFindStars | DEBUG | Median source (idx=132)
generates 108.27 ADU/s.
2024-11-26 21:10:28,558 | ApFindStars | DEBUG | Faintest source (idx=264)
generates -85.00 ADU/s.
2024-11-26 21:10:28,558 | ApFindStars | DEBUG | Selecting 200 sources out of 265
to write to the output source list.
2024-11-26 21:10:28,558 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:10:28,559 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:10:29,537 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-180s-Lum-1.png
2024-11-26 21:10:29,540 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:10:29,540 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:10:29,540 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:10:29,540 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:10:29,541 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 180
2024-11-26 21:10:29,541 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 265
2024-11-26 21:10:29,542 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:10:29,542 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:10:29,542 | ApMeasureStars | DEBUG | Constructed a kdtree with 265
data points.
2024-11-26 21:10:29,545 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 13 stars from consideration.
2024-11-26 21:10:29,545 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 167 stars in 3056 row x 3056 column image.
2024-11-26 21:10:29,546 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:10:29,547 | ApMeasureStars | DEBUG | There are 166 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:10:29,547 | ApMeasureStars | DEBUG | In region CN found 55
candidates for fitting:
2024-11-26 21:10:29,548 | ApMeasureStars | DEBUG | In region TL found 25
candidates for fitting:
2024-11-26 21:10:29,549 | ApMeasureStars | DEBUG | In region TR found 29
candidates for fitting:
2024-11-26 21:10:29,550 | ApMeasureStars | DEBUG | In region BR found 31
candidates for fitting:
2024-11-26 21:10:29,552 | ApMeasureStars | DEBUG | In region BL found 26
candidates for fitting:
2024-11-26 21:10:29,553 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
min=417.57, max=42571.63
2024-11-26 21:10:29,554 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:10:29,555 | ApMeasureStars | DEBUG | Fitting star #0 (id=161) at
fixed xpos=8.66, ypos=9.07
2024-11-26 21:10:29,565 | ApMeasureStars | DEBUG | Fitting star #1 (id=122) at

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fixed xpos=8.86, ypos=9.08
2024-11-26 21:10:29,573 | ApMeasureStars | DEBUG | Fitting star #2 (id=109) at
fixed xpos=8.95, ypos=8.87
2024-11-26 21:10:29,585 | ApMeasureStars | DEBUG | Fitting star #3 (id=186) at
fixed xpos=8.85, ypos=8.62
2024-11-26 21:10:29,596 | ApMeasureStars | DEBUG | Fitting star #4 (id=176) at
fixed xpos=8.84, ypos=8.93
2024-11-26 21:10:29,604 | ApMeasureStars | DEBUG | Fitting star #5 (id=263) at
fixed xpos=9.29, ypos=9.19
2024-11-26 21:10:29,616 | ApMeasureStars | DEBUG | Fitting star #6 (id=239) at
fixed xpos=8.55, ypos=9.42
2024-11-26 21:10:29,624 | ApMeasureStars | DEBUG | Fitting star #7 (id=216) at
fixed xpos=8.91, ypos=8.90
2024-11-26 21:10:29,635 | ApMeasureStars | DEBUG | Fitting star #8 (id=250) at
fixed xpos=9.31, ypos=9.28
2024-11-26 21:10:29,648 | ApMeasureStars | DEBUG | Fitting star #9 (id=246) at
fixed xpos=8.70, ypos=8.73
2024-11-26 21:10:29,658 | ApMeasureStars | DEBUG | Fitting star #10 (id=229) at
fixed xpos=8.91, ypos=8.92
2024-11-26 21:10:29,667 | ApMeasureStars | DEBUG | Fitting star #11 (id=227) at
fixed xpos=9.05, ypos=9.49
2024-11-26 21:10:29,675 | ApMeasureStars | DEBUG | Fitting star #12 (id=163) at
fixed xpos=8.92, ypos=8.56
2024-11-26 21:10:29,684 | ApMeasureStars | DEBUG | Fitting star #13 (id=243) at
fixed xpos=8.99, ypos=9.44
2024-11-26 21:10:29,692 | ApMeasureStars | DEBUG | Fitting star #14 (id=207) at
fixed xpos=8.83, ypos=9.32
2024-11-26 21:10:29,701 | ApMeasureStars | DEBUG | Fitting star #15 (id=49) at
fixed xpos=9.45, ypos=8.67
2024-11-26 21:10:29,709 | ApMeasureStars | DEBUG | Fitting star #16 (id=43) at
fixed xpos=9.49, ypos=8.89
2024-11-26 21:10:29,718 | ApMeasureStars | DEBUG | Fitting star #17 (id=53) at
fixed xpos=9.15, ypos=9.35
2024-11-26 21:10:29,727 | ApMeasureStars | DEBUG | Fitting star #18 (id=35) at
fixed xpos=9.09, ypos=8.99
2024-11-26 21:10:29,737 | ApMeasureStars | DEBUG | Fitting star #19 (id=127) at
fixed xpos=9.07, ypos=9.40
2024-11-26 21:10:29,746 | ApMeasureStars | DEBUG | Fitting star #20 (id=36) at
fixed xpos=9.34, ypos=9.25
2024-11-26 21:10:29,757 | ApMeasureStars | DEBUG | Fitting star #21 (id=50) at
fixed xpos=9.33, ypos=9.21
2024-11-26 21:10:29,766 | ApMeasureStars | DEBUG | Fitting star #22 (id=48) at
fixed xpos=8.88, ypos=9.01
2024-11-26 21:10:29,776 | ApMeasureStars | DEBUG | Fitting star #23 (id=22) at
fixed xpos=9.35, ypos=9.20
2024-11-26 21:10:29,787 | ApMeasureStars | DEBUG | Fitting star #24 (id=58) at
fixed xpos=9.36, ypos=8.57
2024-11-26 21:10:29,800 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:29,801 | ApMeasureStars | DEBUG | Measured FWHM=4.070 +/- 0.156
pixels in direction x
2024-11-26 21:10:29,802 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:29,802 | ApMeasureStars | DEBUG | Measured FWHM=4.341 +/- 0.101
pixels in direction y
2024-11-26 21:10:29,802 | ApMeasureStars | INFO | Source average axis ratio =
1.066 +/- 0.048, overall sources are circular.
2024-11-26 21:10:29,802 | ApMeasureStars | DEBUG | Fitting completed in 0.248 s
(average 9.9 ms/star)
2024-11-26 21:10:29,802 | ApMeasureStars | DEBUG | Skipping plotting of star
cutouts and best fits.
2024-11-26 21:10:29,803 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:10:29,803 | ApMeasureStars | DEBUG | Measured FWHM=4.312 +/- 0.343
pixels in direction both
2024-11-26 21:10:29,804 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:29,804 | ApMeasureStars | DEBUG | Measured FWHM=4.070 +/- 0.156
pixels in direction x
2024-11-26 21:10:29,805 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:29,805 | ApMeasureStars | DEBUG | Measured FWHM=4.341 +/- 0.101
pixels in direction y
2024-11-26 21:10:29,805 | ApFindStars | INFO | Median FWHM (over x and y) is
4.31 +/- 0.34 pixels using 49 data points
2024-11-26 21:10:29,805 | ApProcess | DEBUG | Initial star detection and

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fitting: FWHM=4.312 +/- 0.343 pixels using 49 stars.
2024-11-26 21:10:29,805 | ApProcess | DEBUG | Updating source searching using
initial FWHM=4.312 +/- 0.343 pixels using 49 stars.
2024-11-26 21:10:29,805 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=4.31 pixels, threshold=7.0 BG sigma.
2024-11-26 21:10:30,099 | ApFindStars | INFO | Initial source list using
FWHM=4.312 pixels, threshold=7.00 x BG stddev, found 306 sources.
2024-11-26 21:10:30,100 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:10:30,100 | ApFindStars | DEBUG | There are 13 possibly saturated
stars in the output source list.
2024-11-26 21:10:30,100 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:10:30,100 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:10:30,151 | ApFindStars | DEBUG | Image exposure time [seconds]:
180.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:10:30,153 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 23173.82 ADU/s.
2024-11-26 21:10:30,153 | ApFindStars | DEBUG | Median source (idx=153)
generates 74.44 ADU/s.
2024-11-26 21:10:30,153 | ApFindStars | DEBUG | Faintest source (idx=305)
generates -43.01 ADU/s.
2024-11-26 21:10:30,153 | ApFindStars | DEBUG | Selecting 200 sources out of 306
to write to the output source list.
2024-11-26 21:10:30,154 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:10:30,154 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:10:30,157 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:10:30,157 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:10:30,157 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:10:30,157 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:10:30,158 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 187
2024-11-26 21:10:30,158 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 306
2024-11-26 21:10:30,158 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:10:30,158 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:10:30,159 | ApMeasureStars | DEBUG | Constructed a kdtree with 306
data points.
2024-11-26 21:10:30,162 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 23 stars from consideration.
2024-11-26 21:10:30,162 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 164 stars in 3056 row x 3056 column image.
2024-11-26 21:10:30,162 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:10:30,163 | ApMeasureStars | DEBUG | There are 163 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:10:30,163 | ApMeasureStars | DEBUG | In region CN found 54
candidates for fitting:
2024-11-26 21:10:30,164 | ApMeasureStars | DEBUG | In region TL found 24
candidates for fitting:
2024-11-26 21:10:30,165 | ApMeasureStars | DEBUG | In region TR found 30
candidates for fitting:
2024-11-26 21:10:30,166 | ApMeasureStars | DEBUG | In region BR found 29
candidates for fitting:
2024-11-26 21:10:30,167 | ApMeasureStars | DEBUG | In region BL found 26
candidates for fitting:
2024-11-26 21:10:30,169 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
min=417.57, max=42571.63
2024-11-26 21:10:30,170 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:10:30,171 | ApMeasureStars | DEBUG | Fitting star #0 (id=179) at
fixed xpos=8.67, ypos=9.06
2024-11-26 21:10:30,180 | ApMeasureStars | DEBUG | Fitting star #1 (id=134) at
fixed xpos=8.86, ypos=9.07

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2024-11-26 21:10:30,190 | ApMeasureStars | DEBUG | Fitting star #2 (id=122) at fixed xpos=8.96, ypos=8.88
 2024-11-26 21:10:30,202 | ApMeasureStars | DEBUG | Fitting star #3 (id=209) at fixed xpos=8.86, ypos=8.63
 2024-11-26 21:10:30,214 | ApMeasureStars | DEBUG | Fitting star #4 (id=197) at fixed xpos=8.84, ypos=8.93
 2024-11-26 21:10:30,223 | ApMeasureStars | DEBUG | Fitting star #5 (id=303) at fixed xpos=9.27, ypos=9.18
 2024-11-26 21:10:30,235 | ApMeasureStars | DEBUG | Fitting star #6 (id=271) at fixed xpos=8.57, ypos=9.40
 2024-11-26 21:10:30,243 | ApMeasureStars | DEBUG | Fitting star #7 (id=239) at fixed xpos=8.90, ypos=8.90
 2024-11-26 21:10:30,255 | ApMeasureStars | DEBUG | Fitting star #8 (id=285) at fixed xpos=9.28, ypos=9.26
 2024-11-26 21:10:30,268 | ApMeasureStars | DEBUG | Fitting star #9 (id=281) at fixed xpos=8.71, ypos=8.73
 2024-11-26 21:10:30,279 | ApMeasureStars | DEBUG | Fitting star #10 (id=258) at fixed xpos=8.91, ypos=8.92
 2024-11-26 21:10:30,287 | ApMeasureStars | DEBUG | Fitting star #11 (id=255) at fixed xpos=9.04, ypos=8.52
 2024-11-26 21:10:30,296 | ApMeasureStars | DEBUG | Fitting star #12 (id=181) at fixed xpos=8.92, ypos=8.57
 2024-11-26 21:10:30,305 | ApMeasureStars | DEBUG | Fitting star #13 (id=277) at fixed xpos=8.98, ypos=9.42
 2024-11-26 21:10:30,313 | ApMeasureStars | DEBUG | Fitting star #14 (id=230) at fixed xpos=8.84, ypos=9.30
 2024-11-26 21:10:30,322 | ApMeasureStars | DEBUG | Fitting star #15 (id=55) at fixed xpos=9.42, ypos=8.68
 2024-11-26 21:10:30,331 | ApMeasureStars | DEBUG | Fitting star #16 (id=46) at fixed xpos=8.52, ypos=8.89
 2024-11-26 21:10:30,342 | ApMeasureStars | DEBUG | Fitting star #17 (id=59) at fixed xpos=9.13, ypos=9.33
 2024-11-26 21:10:30,352 | ApMeasureStars | DEBUG | Fitting star #18 (id=36) at fixed xpos=9.08, ypos=8.98
 2024-11-26 21:10:30,362 | ApMeasureStars | DEBUG | Fitting star #19 (id=139) at fixed xpos=9.12, ypos=9.48
 2024-11-26 21:10:30,371 | ApMeasureStars | DEBUG | Fitting star #20 (id=37) at fixed xpos=9.32, ypos=9.23
 2024-11-26 21:10:30,381 | ApMeasureStars | DEBUG | Fitting star #21 (id=56) at fixed xpos=9.31, ypos=9.20
 2024-11-26 21:10:30,391 | ApMeasureStars | DEBUG | Fitting star #22 (id=54) at fixed xpos=8.88, ypos=9.00
 2024-11-26 21:10:30,401 | ApMeasureStars | DEBUG | Fitting star #23 (id=21) at fixed xpos=9.33, ypos=9.18
 2024-11-26 21:10:30,412 | ApMeasureStars | DEBUG | Fitting star #24 (id=64) at fixed xpos=9.34, ypos=8.57
 2024-11-26 21:10:30,425 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:30,426 | ApMeasureStars | DEBUG | Measured FWHM=4.127 +/- 0.251 pixels in direction x
 2024-11-26 21:10:30,426 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:30,427 | ApMeasureStars | DEBUG | Measured FWHM=4.333 +/- 0.087 pixels in direction y
 2024-11-26 21:10:30,427 | ApMeasureStars | INFO | Source average axis ratio = 1.050 +/- 0.067, overall sources are circular.
 2024-11-26 21:10:30,427 | ApMeasureStars | DEBUG | Fitting completed in 0.257 s (average 10.3 ms/star)
 2024-11-26 21:10:30,517 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 49 FWHM measurements (50 OK fits before clipping).
 2024-11-26 21:10:30,517 | ApMeasureStars | DEBUG | Measured FWHM=4.296 +/- 0.266 pixels in direction both
 2024-11-26 21:10:30,518 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:30,518 | ApMeasureStars | DEBUG | Measured FWHM=4.127 +/- 0.251 pixels in direction x
 2024-11-26 21:10:30,519 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:30,519 | ApMeasureStars | DEBUG | Measured FWHM=4.333 +/- 0.087 pixels in direction y
 2024-11-26 21:10:31,790 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png
 2024-11-26 21:10:31,791 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 49 FWHM measurements (50 OK fits before clipping).
 2024-11-26 21:10:31,792 | ApMeasureStars | DEBUG | Measured FWHM=4.296 +/- 0.266 pixels in direction both

2024-11-26 21:10:31,792 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:10:31,792 | ApMeasureStars | DEBUG | Measured FWHM=4.127 +/- 0.251 pixels in direction x

2024-11-26 21:10:31,793 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:10:31,793 | ApMeasureStars | DEBUG | Measured FWHM=4.333 +/- 0.087 pixels in direction y

2024-11-26 21:10:31,794 | ApFindStars | INFO | Median FWHM (over x and y) is 4.30 +/- 0.27 pixels using 49 data points

2024-11-26 21:10:31,794 | ApProcess | DEBUG | Final star detection and fitting: FWHM=4.296 +/- 0.266 pixels using 49 stars.

2024-11-26 21:10:32,760 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to
./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-1.png

2024-11-26 21:10:32,760 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:10:32,760 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (306, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Lum', 'Filter used')}

2024-11-26 21:10:32,761 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:10:32,769 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:10:32,769 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:10:32,769 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:10:32,770 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:10:32,770 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:10:32,770 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:10:32,770 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:10:32,770 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:10:32,770 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:10:32,771 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:10:32,771 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:10:32,771 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:10:32,771 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:10:32,773 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-1.yaml

2024-11-26 21:10:32,776 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-1.reg

2024-11-26 21:10:32,776 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:10:32,776 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (306, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Lum', 'Filter used')}

2024-11-26 21:10:32,776 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-

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OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:10:32,777 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:10:32,777 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:10:32,778 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:10:32,801 | ApProcess | DEBUG | Confirming output srclist
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits was created.
2024-11-26 21:10:32,802 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits as
clean_star_detection=True
2024-11-26 21:10:32,807 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:10:32,808 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:10:33,077 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:10:33,080 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:10:33,080 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:10:33,080 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:10:33,080 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:10:33,081 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:10:33,081 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:10:44,232 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:10:44,358 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:10:44,358 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:10:44,359 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:10:44,359 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:10:44,359 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:10:44,359 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:10:44,360 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.754804653, 'RA of reference point')
2024-11-26 21:10:44,360 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4174059435, 'DEC of reference point')
2024-11-26 21:10:44,360 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:10:44,360 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:10:44,361 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:10:44,361 | AstroPhotography.core.ApAstrometry | DEBUG | Setting

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CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:10:44,361 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.64681630389e-06, 'Transformation matrix')
2024-11-26 21:10:44,361 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175223532872, 'no comment')
2024-11-26 21:10:44,362 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175223532872, 'no comment')
2024-11-26 21:10:44,362 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.64681630389e-06, 'no comment')
2024-11-26 21:10:44,362 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:10:44,362 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:10:44,526 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:10:44,527 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-1.fits with
WCS soluton.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:10:44,625 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:10:44,906 | ApProcess | DEBUG |
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2024-11-26 21:10:44,906 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:10:44,907 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:44,907 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:44,907 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:44,907 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:10:44,907 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:10:44,908 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:10:44,908 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
src1ist=./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-2.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-2.reg
plotfile=./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-2.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-2.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-2.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits
2024-11-26 21:10:44,908 | ApProcess | DEBUG | Removing existing
./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-2.fits because
clean_star_detection=True
2024-11-26 21:10:44,909 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-2.reg because
clean_star_detection=True
2024-11-26 21:10:44,909 | ApProcess | DEBUG | Removing existing
./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-2.png because
clean_star_detection=True
2024-11-26 21:10:44,909 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-2.png because
clean_star_detection=True
2024-11-26 21:10:44,909 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-2.fits
2024-11-26 21:10:44,910 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:10:45,188 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:10:45,280 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67324.91, median=538.31
2024-11-26 21:10:46,086 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=538.669, median=537.623, stddev=28.694
2024-11-26 21:10:48,700 | ApFindStars | DEBUG | Source-masked image stats:
mean=535.657, median=535.461, stddev=26.379
2024-11-26 21:10:48,700 | ApFindStars | DEBUG | Checking for possibly saturated

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stars or regions.

2024-11-26 21:10:48,700 | ApFindStars | DEBUG | Looking for possibly saturated regions above 52428 ADU separated by {boxsize} pixels.

2024-11-26 21:10:48,878 | ApFindStars | DEBUG | Retaining 14 possibly saturated stars in source searching and photometry.

2024-11-26 21:10:48,878 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=3.00 pixels, threshold=7.0 BG sigma.

2024-11-26 21:10:49,120 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 258 sources.

2024-11-26 21:10:49,120 | ApFindStars | DEBUG | Sources with pixels exceeding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:10:49,121 | ApFindStars | DEBUG | There are 19 possibly saturated stars in the output source list.

2024-11-26 21:10:49,121 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:10:49,121 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:10:49,164 | ApFindStars | DEBUG | Image exposure time [seconds]: 180.00

2024-11-26 21:10:49,165 | ApFindStars | DEBUG | Brightest source (idx=0) generates 24904.05 ADU/s.

2024-11-26 21:10:49,166 | ApFindStars | DEBUG | Median source (idx=129) generates 85.62 ADU/s.

2024-11-26 21:10:49,166 | ApFindStars | DEBUG | Faintest source (idx=257) generates 1.38 ADU/s.

2024-11-26 21:10:49,166 | ApFindStars | DEBUG | Selecting 200 sources out of 258 to write to the output source list.

2024-11-26 21:10:49,166 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:10:49,166 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:10:50,063 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-180s-Lum-2.png

2024-11-26 21:10:50,066 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:10:50,066 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:10:50,066 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:10:50,066 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:10:50,067 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 181

2024-11-26 21:10:50,067 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 258

2024-11-26 21:10:50,067 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:10:50,067 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:10:50,068 | ApMeasureStars | DEBUG | Constructed a kdtree with 258 data points.

2024-11-26 21:10:50,071 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 11 stars from consideration.

2024-11-26 21:10:50,071 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 170 stars in 3056 row x 3056 column image.

2024-11-26 21:10:50,071 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:10:50,072 | ApMeasureStars | DEBUG | There are 169 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:10:50,072 | ApMeasureStars | DEBUG | In region CN found 53 candidates for fitting:

2024-11-26 21:10:50,073 | ApMeasureStars | DEBUG | In region TL found 25 candidates for fitting:

2024-11-26 21:10:50,075 | ApMeasureStars | DEBUG | In region TR found 31 candidates for fitting:

2024-11-26 21:10:50,076 | ApMeasureStars | DEBUG | In region BR found 34 candidates for fitting:

2024-11-26 21:10:50,077 | ApMeasureStars | DEBUG | In region BL found 26 candidates for fitting:

2024-11-26 21:10:50,079 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=465.15, max=51500.74

2024-11-26 21:10:50,081 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:10:50,081 | ApMeasureStars | DEBUG | Fitting star #0 (id=157) at fixed xpos=8.97, ypos=8.68

2024-11-26 21:10:50,089 | ApMeasureStars | DEBUG | Fitting star #1 (id=122) at fixed xpos=9.09, ypos=8.66
 2024-11-26 21:10:50,097 | ApMeasureStars | DEBUG | Fitting star #2 (id=110) at fixed xpos=9.20, ypos=8.54
 2024-11-26 21:10:50,108 | ApMeasureStars | DEBUG | Fitting star #3 (id=178) at fixed xpos=9.19, ypos=9.25
 2024-11-26 21:10:50,116 | ApMeasureStars | DEBUG | Fitting star #4 (id=171) at fixed xpos=9.19, ypos=8.66
 2024-11-26 21:10:50,125 | ApMeasureStars | DEBUG | Fitting star #5 (id=255) at fixed xpos=8.63, ypos=8.96
 2024-11-26 21:10:50,133 | ApMeasureStars | DEBUG | Fitting star #6 (id=228) at fixed xpos=8.99, ypos=9.08
 2024-11-26 21:10:50,141 | ApMeasureStars | DEBUG | Fitting star #7 (id=206) at fixed xpos=9.17, ypos=8.68
 2024-11-26 21:10:50,149 | ApMeasureStars | DEBUG | Fitting star #8 (id=240) at fixed xpos=8.56, ypos=9.00
 2024-11-26 21:10:50,158 | ApMeasureStars | DEBUG | Fitting star #9 (id=194) at fixed xpos=9.09, ypos=8.55
 2024-11-26 21:10:50,166 | ApMeasureStars | DEBUG | Fitting star #10 (id=252) at fixed xpos=8.66, ypos=9.23
 2024-11-26 21:10:50,180 | ApMeasureStars | DEBUG | Fitting star #11 (id=218) at fixed xpos=9.35, ypos=9.37
 2024-11-26 21:10:50,189 | ApMeasureStars | DEBUG | Fitting star #12 (id=216) at fixed xpos=9.44, ypos=9.11
 2024-11-26 21:10:50,197 | ApMeasureStars | DEBUG | Fitting star #13 (id=159) at fixed xpos=9.25, ypos=9.11
 2024-11-26 21:10:50,205 | ApMeasureStars | DEBUG | Fitting star #14 (id=232) at fixed xpos=9.35, ypos=9.07
 2024-11-26 21:10:50,213 | ApMeasureStars | DEBUG | Fitting star #15 (id=9) at fixed xpos=8.71, ypos=8.54
 2024-11-26 21:10:50,224 | ApMeasureStars | DEBUG | Fitting star #16 (id=57) at fixed xpos=8.51, ypos=9.28
 2024-11-26 21:10:50,232 | ApMeasureStars | DEBUG | Fitting star #17 (id=48) at fixed xpos=8.63, ypos=9.47
 2024-11-26 21:10:50,242 | ApMeasureStars | DEBUG | Fitting star #18 (id=60) at fixed xpos=9.27, ypos=8.90
 2024-11-26 21:10:50,250 | ApMeasureStars | DEBUG | Fitting star #19 (id=37) at fixed xpos=9.21, ypos=8.55
 2024-11-26 21:10:50,258 | ApMeasureStars | DEBUG | Fitting star #20 (id=38) at fixed xpos=9.38, ypos=8.93
 2024-11-26 21:10:50,267 | ApMeasureStars | DEBUG | Fitting star #21 (id=58) at fixed xpos=9.41, ypos=8.89
 2024-11-26 21:10:50,277 | ApMeasureStars | DEBUG | Fitting star #22 (id=56) at fixed xpos=8.88, ypos=8.77
 2024-11-26 21:10:50,285 | ApMeasureStars | DEBUG | Fitting star #23 (id=22) at fixed xpos=9.32, ypos=8.90
 2024-11-26 21:10:50,293 | ApMeasureStars | DEBUG | Fitting star #24 (id=65) at fixed xpos=9.35, ypos=9.36
 2024-11-26 21:10:50,303 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:50,303 | ApMeasureStars | DEBUG | Measured FWHM=4.273 +/- 0.126 pixels in direction x
 2024-11-26 21:10:50,304 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:50,304 | ApMeasureStars | DEBUG | Measured FWHM=4.053 +/- 0.105 pixels in direction y
 2024-11-26 21:10:50,304 | ApMeasureStars | INFO | Source average axis ratio = 1.054 +/- 0.041, overall sources are circular.
 2024-11-26 21:10:50,304 | ApMeasureStars | DEBUG | Fitting completed in 0.224 s (average 8.9 ms/star)
 2024-11-26 21:10:50,305 | ApMeasureStars | DEBUG | Skipping plotting of star cutouts and best fits.
 2024-11-26 21:10:50,305 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).
 2024-11-26 21:10:50,306 | ApMeasureStars | DEBUG | Measured FWHM=4.147 +/- 0.176 pixels in direction both
 2024-11-26 21:10:50,306 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:50,307 | ApMeasureStars | DEBUG | Measured FWHM=4.273 +/- 0.126 pixels in direction x
 2024-11-26 21:10:50,307 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:10:50,308 | ApMeasureStars | DEBUG | Measured FWHM=4.053 +/- 0.105 pixels in direction y
 2024-11-26 21:10:50,308 | ApFindStars | INFO | Median FWHM (over x and y) is 4.15 +/- 0.18 pixels using 50 data points

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2024-11-26 21:10:50,308 | ApProcess | DEBUG | Initial star detection and
fitting: FWHM=4.147 +/- 0.176 pixels using 50 stars.
2024-11-26 21:10:50,308 | ApProcess | DEBUG | Updating source searching using
initial FWHM=4.147 +/- 0.176 pixels using 50 stars.
2024-11-26 21:10:50,309 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=4.15 pixels, threshold=7.0 BG sigma.
2024-11-26 21:10:50,600 | ApFindStars | INFO | Initial source list using
FWHM=4.147 pixels, threshold=7.00 x BG stddev, found 287 sources.
2024-11-26 21:10:50,601 | ApFindStars | DEBUG | Sources with pixels exceeding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:10:50,601 | ApFindStars | DEBUG | There are 13 possibly saturated
stars in the output source list.
2024-11-26 21:10:50,601 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:10:50,602 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:10:50,650 | ApFindStars | DEBUG | Image exposure time [seconds]:
180.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:10:50,652 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 24566.85 ADU/s.
2024-11-26 21:10:50,652 | ApFindStars | DEBUG | Median source (idx=143)
generates 64.00 ADU/s.
2024-11-26 21:10:50,653 | ApFindStars | DEBUG | Faintest source (idx=286)
generates -91.23 ADU/s.
2024-11-26 21:10:50,654 | ApFindStars | DEBUG | Selecting 200 sources out of 287
to write to the output source list.
2024-11-26 21:10:50,654 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:10:50,654 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:10:50,656 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:10:50,657 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:10:50,657 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:10:50,657 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:10:50,658 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 187
2024-11-26 21:10:50,658 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 287
2024-11-26 21:10:50,658 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:10:50,658 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:10:50,659 | ApMeasureStars | DEBUG | Constructed a kdtree with 287
data points.
2024-11-26 21:10:50,662 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 18 stars from consideration.
2024-11-26 21:10:50,662 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 169 stars in 3056 row x 3056 column image.
2024-11-26 21:10:50,662 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:10:50,663 | ApMeasureStars | DEBUG | There are 168 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:10:50,663 | ApMeasureStars | DEBUG | In region CN found 54
candidates for fitting:
2024-11-26 21:10:50,664 | ApMeasureStars | DEBUG | In region TL found 25
candidates for fitting:
2024-11-26 21:10:50,665 | ApMeasureStars | DEBUG | In region TR found 31
candidates for fitting:
2024-11-26 21:10:50,667 | ApMeasureStars | DEBUG | In region BR found 32
candidates for fitting:
2024-11-26 21:10:50,668 | ApMeasureStars | DEBUG | In region BL found 26
candidates for fitting:
2024-11-26 21:10:50,670 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
min=465.15, max=51500.74
2024-11-26 21:10:50,671 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:10:50,672 | ApMeasureStars | DEBUG | Fitting star #0 (id=177) at
fixed xpos=8.96, ypos=8.69
2024-11-26 21:10:50,681 | ApMeasureStars | DEBUG | Fitting star #1 (id=137) at

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fixed xpos=9.08, ypos=8.67
2024-11-26 21:10:50,690 | ApMeasureStars | DEBUG | Fitting star #2 (id=126) at
fixed xpos=9.19, ypos=8.56
2024-11-26 21:10:50,700 | ApMeasureStars | DEBUG | Fitting star #3 (id=201) at
fixed xpos=9.17, ypos=9.24
2024-11-26 21:10:50,709 | ApMeasureStars | DEBUG | Fitting star #4 (id=191) at
fixed xpos=9.17, ypos=8.66
2024-11-26 21:10:50,718 | ApMeasureStars | DEBUG | Fitting star #5 (id=282) at
fixed xpos=8.64, ypos=8.96
2024-11-26 21:10:50,726 | ApMeasureStars | DEBUG | Fitting star #6 (id=255) at
fixed xpos=8.98, ypos=9.07
2024-11-26 21:10:50,734 | ApMeasureStars | DEBUG | Fitting star #7 (id=228) at
fixed xpos=9.15, ypos=8.68
2024-11-26 21:10:50,743 | ApMeasureStars | DEBUG | Fitting star #8 (id=268) at
fixed xpos=8.57, ypos=9.00
2024-11-26 21:10:50,752 | ApMeasureStars | DEBUG | Fitting star #9 (id=218) at
fixed xpos=9.07, ypos=8.52
2024-11-26 21:10:50,760 | ApMeasureStars | DEBUG | Fitting star #10 (id=279) at
fixed xpos=8.66, ypos=9.21
2024-11-26 21:10:50,774 | ApMeasureStars | DEBUG | Fitting star #11 (id=245) at
fixed xpos=9.33, ypos=9.35
2024-11-26 21:10:50,784 | ApMeasureStars | DEBUG | Fitting star #12 (id=243) at
fixed xpos=9.46, ypos=9.10
2024-11-26 21:10:50,792 | ApMeasureStars | DEBUG | Fitting star #13 (id=179) at
fixed xpos=9.23, ypos=9.10
2024-11-26 21:10:50,801 | ApMeasureStars | DEBUG | Fitting star #14 (id=259) at
fixed xpos=9.37, ypos=9.06
2024-11-26 21:10:50,809 | ApMeasureStars | DEBUG | Fitting star #15 (id=10) at
fixed xpos=8.72, ypos=8.56
2024-11-26 21:10:50,819 | ApMeasureStars | DEBUG | Fitting star #16 (id=65) at
fixed xpos=8.53, ypos=9.26
2024-11-26 21:10:50,828 | ApMeasureStars | DEBUG | Fitting star #17 (id=55) at
fixed xpos=8.64, ypos=9.45
2024-11-26 21:10:50,836 | ApMeasureStars | DEBUG | Fitting star #18 (id=68) at
fixed xpos=9.24, ypos=8.90
2024-11-26 21:10:50,845 | ApMeasureStars | DEBUG | Fitting star #19 (id=42) at
fixed xpos=9.19, ypos=8.57
2024-11-26 21:10:50,853 | ApMeasureStars | DEBUG | Fitting star #20 (id=43) at
fixed xpos=9.35, ypos=8.93
2024-11-26 21:10:50,862 | ApMeasureStars | DEBUG | Fitting star #21 (id=66) at
fixed xpos=9.38, ypos=8.89
2024-11-26 21:10:50,872 | ApMeasureStars | DEBUG | Fitting star #22 (id=64) at
fixed xpos=8.88, ypos=8.78
2024-11-26 21:10:50,880 | ApMeasureStars | DEBUG | Fitting star #23 (id=26) at
fixed xpos=9.30, ypos=8.91
2024-11-26 21:10:50,890 | ApMeasureStars | DEBUG | Fitting star #24 (id=75) at
fixed xpos=9.44, ypos=9.38
2024-11-26 21:10:50,900 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:50,900 | ApMeasureStars | DEBUG | Measured FWHM=4.266 +/- 0.119
pixels in direction x
2024-11-26 21:10:50,901 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:50,901 | ApMeasureStars | DEBUG | Measured FWHM=4.050 +/- 0.113
pixels in direction y
2024-11-26 21:10:50,901 | ApMeasureStars | INFO | Source average axis ratio =
1.053 +/- 0.042, overall sources are circular.
2024-11-26 21:10:50,902 | ApMeasureStars | DEBUG | Fitting completed in 0.230 s
(average 9.2 ms/star)
2024-11-26 21:10:50,989 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:10:50,990 | ApMeasureStars | DEBUG | Measured FWHM=4.140 +/- 0.168
pixels in direction both
2024-11-26 21:10:50,990 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:50,991 | ApMeasureStars | DEBUG | Measured FWHM=4.266 +/- 0.119
pixels in direction x
2024-11-26 21:10:50,992 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:50,992 | ApMeasureStars | DEBUG | Measured FWHM=4.050 +/- 0.113
pixels in direction y
2024-11-26 21:10:52,211 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-2.png
2024-11-26 21:10:52,212 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:10:52,212 | ApMeasureStars | DEBUG | Measured FWHM=4.140 +/- 0.168

pixels in direction both
2024-11-26 21:10:52,212 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:52,213 | ApMeasureStars | DEBUG | Measured FWHM=4.266 +/- 0.119 pixels in direction x
2024-11-26 21:10:52,213 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:10:52,214 | ApMeasureStars | DEBUG | Measured FWHM=4.050 +/- 0.113 pixels in direction y
2024-11-26 21:10:52,214 | ApFindStars | INFO | Median FWHM (over x and y) is 4.14 +/- 0.17 pixels using 50 data points
2024-11-26 21:10:52,214 | ApProcess | DEBUG | Final star detection and fitting: FWHM=4.140 +/- 0.168 pixels using 50 stars.
2024-11-26 21:10:53,105 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-180s-Lum-2.png
2024-11-26 21:10:53,105 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:10:53,106 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (287, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Lum', 'Filter used')}

2024-11-26 21:10:53,106 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:10:53,107 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:10:53,107 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.
2024-11-26 21:10:53,107 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.
2024-11-26 21:10:53,108 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.
2024-11-26 21:10:53,109 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.
2024-11-26 21:10:53,109 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.
2024-11-26 21:10:53,109 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.
2024-11-26 21:10:53,111 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-2.yaml
2024-11-26 21:10:53,113 | ApFindStars | DEBUG | Wrote ds9-format region file to
./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-2.reg
2024-11-26 21:10:53,114 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:10:53,114 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (287, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Lum', 'Filter used')}

2024-11-26 21:10:53,114 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',

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'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:10:53,115 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:10:53,115 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:10:53,115 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-M101_Supernova_2023ixf-180s-Lum-2.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:10:53,137 | ApProcess | DEBUG | Confirming output srcList
./SourceLists/srcList-M101_Supernova_2023ixf-180s-Lum-2.fits was created.
2024-11-26 21:10:53,138 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits as
clean_star_detection=True
2024-11-26 21:10:53,143 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits
2024-11-26 21:10:53,143 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:10:53,409 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:10:53,412 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srcList-M101_Supernova_2023ixf-180s-Lum-2.fits
2024-11-26 21:10:53,412 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:10:53,412 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:10:53,412 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:10:53,412 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:10:53,413 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srcList-M101_Supernova_2023ixf-180s-Lum-2.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:11:02,754 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:11:02,889 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:11:02,889 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:11:02,890 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:11:02,890 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:11:02,890 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:11:02,891 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:11:02,891 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.754169037, 'RA of reference point')
2024-11-26 21:11:02,891 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4168776265, 'DEC of reference point')
2024-11-26 21:11:02,891 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:11:02,892 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:11:02,892 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')

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2024-11-26 21:11:02,893 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:11:02,893 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.54804938463e-06, 'Transformation matrix')
2024-11-26 21:11:02,893 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175249345652, 'no comment')
2024-11-26 21:11:02,893 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175249345652, 'no comment')
2024-11-26 21:11:02,894 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.54804938463e-06, 'no comment')
2024-11-26 21:11:02,894 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:11:02,894 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:11:03,047 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits
2024-11-26 21:11:03,048 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-2.fits with
WCS soluton.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:11:03,068 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:11:03,347 | ApProcess | DEBUG |

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2024-11-26 21:11:03,347 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
2024-11-26 21:11:03,348 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:03,348 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:03,348 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:03,348 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:03,348 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:11:03,349 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:11:03,349 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
src1ist=./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-3.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-3.reg
plotfile=./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-3.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-3.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-3.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits
2024-11-26 21:11:03,349 | ApProcess | DEBUG | Removing existing
./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-3.fits because
clean_star_detection=True
2024-11-26 21:11:03,350 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-3.reg because
clean_star_detection=True
2024-11-26 21:11:03,350 | ApProcess | DEBUG | Removing existing
./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-3.png because
clean_star_detection=True
2024-11-26 21:11:03,351 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-3.png because
clean_star_detection=True
2024-11-26 21:11:03,351 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/src1ist-M101_Supernova_2023ixf-180s-Lum-3.fits
2024-11-26 21:11:03,351 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
2024-11-26 21:11:03,642 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:11:03,736 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67324.91, median=575.97
2024-11-26 21:11:04,563 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=576.314, median=575.510, stddev=28.616
2024-11-26 21:11:07,268 | ApFindStars | DEBUG | Source-masked image stats:
mean=573.739, median=573.592, stddev=26.749

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2024-11-26 21:11:07,269 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:11:07,269 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:11:07,446 | ApFindStars | DEBUG | Retaining 13 possibly saturated
stars in source searching and photometry.
2024-11-26 21:11:07,447 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:11:07,687 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 207 sources.
2024-11-26 21:11:07,688 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:11:07,688 | ApFindStars | DEBUG | There are 10 possibly saturated
stars in the output source list.
2024-11-26 21:11:07,688 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:07,689 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:07,723 | ApFindStars | DEBUG | Image exposure time [seconds]:
180.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:11:07,724 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 24758.77 ADU/s.
2024-11-26 21:11:07,724 | ApFindStars | DEBUG | Median source (idx=103)
generates 83.24 ADU/s.
2024-11-26 21:11:07,724 | ApFindStars | DEBUG | Faintest source (idx=206)
generates -93.55 ADU/s.
2024-11-26 21:11:07,724 | ApFindStars | DEBUG | Selecting 200 sources out of 207
to write to the output source list.
2024-11-26 21:11:07,725 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:07,725 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:08,644 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-180s-Lum-3.png
2024-11-26 21:11:08,646 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:11:08,647 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:11:08,647 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:11:08,647 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:11:08,648 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 190
2024-11-26 21:11:08,648 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 207
2024-11-26 21:11:08,648 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:11:08,648 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:11:08,649 | ApMeasureStars | DEBUG | Constructed a kdtree with 207
data points.
2024-11-26 21:11:08,652 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 10 stars from consideration.
2024-11-26 21:11:08,652 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 180 stars in 3056 row x 3056 column image.
2024-11-26 21:11:08,652 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:11:08,653 | ApMeasureStars | DEBUG | There are 179 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:11:08,653 | ApMeasureStars | DEBUG | In region CN found 49
candidates for fitting:
2024-11-26 21:11:08,654 | ApMeasureStars | DEBUG | In region TL found 30
candidates for fitting:
2024-11-26 21:11:08,655 | ApMeasureStars | DEBUG | In region TR found 33
candidates for fitting:
2024-11-26 21:11:08,657 | ApMeasureStars | DEBUG | In region BR found 37
candidates for fitting:
2024-11-26 21:11:08,658 | ApMeasureStars | DEBUG | In region BL found 30
candidates for fitting:
2024-11-26 21:11:08,659 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
min=492.25, max=52642.48

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2024-11-26 21:11:08,661 | ApMeasureStars | INFO | Starting to fit
 Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:11:08,661 | ApMeasureStars | DEBUG | Fitting star #0 (id=129) at
 fixed xpos=9.33, ypos=9.18

2024-11-26 21:11:08,670 | ApMeasureStars | DEBUG | Fitting star #1 (id=100) at
 fixed xpos=9.47, ypos=9.19

2024-11-26 21:11:08,679 | ApMeasureStars | DEBUG | Fitting star #2 (id=90) at
 fixed xpos=8.63, ypos=9.15

2024-11-26 21:11:08,688 | ApMeasureStars | DEBUG | Fitting star #3 (id=149) at
 fixed xpos=8.68, ypos=8.78

2024-11-26 21:11:08,697 | ApMeasureStars | DEBUG | Fitting star #4 (id=141) at
 fixed xpos=8.67, ypos=9.32

2024-11-26 21:11:08,710 | ApMeasureStars | DEBUG | Fitting star #5 (id=205) at
 fixed xpos=9.24, ypos=8.68

2024-11-26 21:11:08,720 | ApMeasureStars | DEBUG | Fitting star #6 (id=185) at
 fixed xpos=9.45, ypos=8.64

2024-11-26 21:11:08,730 | ApMeasureStars | DEBUG | Fitting star #7 (id=169) at
 fixed xpos=8.72, ypos=9.37

2024-11-26 21:11:08,741 | ApMeasureStars | DEBUG | Fitting star #8 (id=195) at
 fixed xpos=9.25, ypos=8.68

2024-11-26 21:11:08,753 | ApMeasureStars | DEBUG | Fitting star #9 (id=160) at
 fixed xpos=8.60, ypos=9.19

2024-11-26 21:11:08,764 | ApMeasureStars | DEBUG | Fitting star #10 (id=194) at
 fixed xpos=8.96, ypos=9.15

2024-11-26 21:11:08,776 | ApMeasureStars | DEBUG | Fitting star #11 (id=202) at
 fixed xpos=9.27, ypos=8.65

2024-11-26 21:11:08,784 | ApMeasureStars | DEBUG | Fitting star #12 (id=178) at
 fixed xpos=8.86, ypos=8.77

2024-11-26 21:11:08,793 | ApMeasureStars | DEBUG | Fitting star #13 (id=176) at
 fixed xpos=9.01, ypos=9.50

2024-11-26 21:11:08,804 | ApMeasureStars | DEBUG | Fitting star #14 (id=131) at
 fixed xpos=8.64, ypos=9.45

2024-11-26 21:11:08,813 | ApMeasureStars | DEBUG | Fitting star #15 (id=10) at
 fixed xpos=8.95, ypos=9.02

2024-11-26 21:11:08,823 | ApMeasureStars | DEBUG | Fitting star #16 (id=47) at
 fixed xpos=8.87, ypos=8.77

2024-11-26 21:11:08,832 | ApMeasureStars | DEBUG | Fitting star #17 (id=39) at
 fixed xpos=8.94, ypos=8.88

2024-11-26 21:11:08,840 | ApMeasureStars | DEBUG | Fitting star #18 (id=49) at
 fixed xpos=8.56, ypos=9.42

2024-11-26 21:11:08,851 | ApMeasureStars | DEBUG | Fitting star #19 (id=32) at
 fixed xpos=9.46, ypos=9.09

2024-11-26 21:11:08,860 | ApMeasureStars | DEBUG | Fitting star #20 (id=50) at
 fixed xpos=9.44, ypos=9.48

2024-11-26 21:11:08,871 | ApMeasureStars | DEBUG | Fitting star #21 (id=33) at
 fixed xpos=8.71, ypos=9.48

2024-11-26 21:11:08,880 | ApMeasureStars | DEBUG | Fitting star #22 (id=48) at
 fixed xpos=8.74, ypos=8.53

2024-11-26 21:11:08,895 | ApMeasureStars | DEBUG | Fitting star #23 (id=45) at
 fixed xpos=9.26, ypos=8.52

2024-11-26 21:11:08,904 | ApMeasureStars | DEBUG | Fitting star #24 (id=23) at
 fixed xpos=8.64, ypos=9.49

2024-11-26 21:11:08,914 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:08,915 | ApMeasureStars | DEBUG | Measured FWHM=4.223 +/- 0.190
 pixels in direction x

2024-11-26 21:11:08,915 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:08,916 | ApMeasureStars | DEBUG | Measured FWHM=4.158 +/- 0.141
 pixels in direction y

2024-11-26 21:11:08,916 | ApMeasureStars | INFO | Source average axis ratio =
 1.015 +/- 0.057, overall sources are circular.

2024-11-26 21:11:08,916 | ApMeasureStars | DEBUG | Fitting completed in 0.255 s
 (average 10.2 ms/star)

2024-11-26 21:11:08,916 | ApMeasureStars | DEBUG | Skipping plotting of star
 cutouts and best fits.

2024-11-26 21:11:08,917 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:08,917 | ApMeasureStars | DEBUG | Measured FWHM=4.210 +/- 0.162
 pixels in direction both

2024-11-26 21:11:08,918 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:08,918 | ApMeasureStars | DEBUG | Measured FWHM=4.223 +/- 0.190
 pixels in direction x

2024-11-26 21:11:08,919 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:08,919 | ApMeasureStars | DEBUG | Measured FWHM=4.158 +/- 0.141 pixels in direction y

2024-11-26 21:11:08,920 | ApFindStars | INFO | Median FWHM (over x and y) is 4.21 +/- 0.16 pixels using 50 data points

2024-11-26 21:11:08,920 | ApProcess | DEBUG | Initial star detection and fitting: FWHM=4.210 +/- 0.162 pixels using 50 stars.

2024-11-26 21:11:08,920 | ApProcess | DEBUG | Updating source searching using initial FWHM=4.210 +/- 0.162 pixels using 50 stars.

2024-11-26 21:11:08,920 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=4.21 pixels, threshold=7.0 BG sigma.

2024-11-26 21:11:09,216 | ApFindStars | INFO | Initial source list using FWHM=4.210 pixels, threshold=7.00 x BG stddev, found 240 sources.

2024-11-26 21:11:09,217 | ApFindStars | DEBUG | Sources with pixels exceding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:11:09,217 | ApFindStars | DEBUG | There are 8 possibly saturated stars in the output source list.

2024-11-26 21:11:09,217 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:11:09,217 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:11:09,257 | ApFindStars | DEBUG | Image exposure time [seconds]: 180.00

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10
 phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:11:09,258 | ApFindStars | DEBUG | Brightest source (idx=0) generates 23844.97 ADU/s.

2024-11-26 21:11:09,258 | ApFindStars | DEBUG | Median source (idx=120) generates 65.07 ADU/s.

2024-11-26 21:11:09,258 | ApFindStars | DEBUG | Faintest source (idx=239) generates -89.53 ADU/s.

2024-11-26 21:11:09,259 | ApFindStars | DEBUG | Selecting 200 sources out of 240 to write to the output source list.

2024-11-26 21:11:09,259 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:11:09,259 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:11:09,261 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:11:09,261 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:11:09,261 | ApMeasureStars | DEBUG | Count based weighting factors will used in fitting?: True

2024-11-26 21:11:09,262 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:11:09,262 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 192

2024-11-26 21:11:09,263 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 240

2024-11-26 21:11:09,263 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:11:09,263 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:11:09,263 | ApMeasureStars | DEBUG | Constructed a kdtree with 240 data points.

2024-11-26 21:11:09,266 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 16 stars from consideration.

2024-11-26 21:11:09,267 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 176 stars in 3056 row x 3056 column image.

2024-11-26 21:11:09,267 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:11:09,267 | ApMeasureStars | DEBUG | There are 175 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:11:09,268 | ApMeasureStars | DEBUG | In region CN found 51 candidates for fitting:

2024-11-26 21:11:09,269 | ApMeasureStars | DEBUG | In region TL found 28 candidates for fitting:

2024-11-26 21:11:09,270 | ApMeasureStars | DEBUG | In region TR found 31 candidates for fitting:

2024-11-26 21:11:09,271 | ApMeasureStars | DEBUG | In region BR found 36 candidates for fitting:

2024-11-26 21:11:09,273 | ApMeasureStars | DEBUG | In region BL found 29 candidates for fitting:

2024-11-26 21:11:09,274 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=492.25, max=52642.48

2024-11-26 21:11:09,276 | ApMeasureStars | INFO | Starting to fit

Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:11:09,276 | ApMeasureStars | DEBUG | Fitting star #0 (id=149) at fixed xpos=9.41, ypos=9.16

2024-11-26 21:11:09,285 | ApMeasureStars | DEBUG | Fitting star #1 (id=115) at fixed xpos=9.49, ypos=9.18

2024-11-26 21:11:09,293 | ApMeasureStars | DEBUG | Fitting star #2 (id=106) at fixed xpos=8.64, ypos=9.14

2024-11-26 21:11:09,302 | ApMeasureStars | DEBUG | Fitting star #3 (id=171) at fixed xpos=8.69, ypos=8.78

2024-11-26 21:11:09,310 | ApMeasureStars | DEBUG | Fitting star #4 (id=162) at fixed xpos=8.68, ypos=9.30

2024-11-26 21:11:09,319 | ApMeasureStars | DEBUG | Fitting star #5 (id=236) at fixed xpos=9.22, ypos=8.69

2024-11-26 21:11:09,329 | ApMeasureStars | DEBUG | Fitting star #6 (id=212) at fixed xpos=9.47, ypos=8.65

2024-11-26 21:11:09,340 | ApMeasureStars | DEBUG | Fitting star #7 (id=193) at fixed xpos=8.73, ypos=9.36

2024-11-26 21:11:09,350 | ApMeasureStars | DEBUG | Fitting star #8 (id=223) at fixed xpos=9.23, ypos=8.69

2024-11-26 21:11:09,361 | ApMeasureStars | DEBUG | Fitting star #9 (id=183) at fixed xpos=8.61, ypos=9.18

2024-11-26 21:11:09,374 | ApMeasureStars | DEBUG | Fitting star #10 (id=233) at fixed xpos=9.24, ypos=8.66

2024-11-26 21:11:09,382 | ApMeasureStars | DEBUG | Fitting star #11 (id=205) at fixed xpos=8.85, ypos=8.78

2024-11-26 21:11:09,390 | ApMeasureStars | DEBUG | Fitting star #12 (id=203) at fixed xpos=9.00, ypos=8.51

2024-11-26 21:11:09,399 | ApMeasureStars | DEBUG | Fitting star #13 (id=151) at fixed xpos=8.65, ypos=9.47

2024-11-26 21:11:09,410 | ApMeasureStars | DEBUG | Fitting star #14 (id=217) at fixed xpos=8.93, ypos=9.50

2024-11-26 21:11:09,421 | ApMeasureStars | DEBUG | Fitting star #15 (id=12) at fixed xpos=8.95, ypos=9.02

2024-11-26 21:11:09,431 | ApMeasureStars | DEBUG | Fitting star #16 (id=54) at fixed xpos=8.87, ypos=8.78

2024-11-26 21:11:09,440 | ApMeasureStars | DEBUG | Fitting star #17 (id=47) at fixed xpos=8.94, ypos=8.88

2024-11-26 21:11:09,449 | ApMeasureStars | DEBUG | Fitting star #18 (id=57) at fixed xpos=8.58, ypos=9.40

2024-11-26 21:11:09,458 | ApMeasureStars | DEBUG | Fitting star #19 (id=38) at fixed xpos=9.48, ypos=9.08

2024-11-26 21:11:09,471 | ApMeasureStars | DEBUG | Fitting star #20 (id=59) at fixed xpos=9.46, ypos=8.50

2024-11-26 21:11:09,479 | ApMeasureStars | DEBUG | Fitting star #21 (id=39) at fixed xpos=8.72, ypos=8.50

2024-11-26 21:11:09,488 | ApMeasureStars | DEBUG | Fitting star #22 (id=55) at fixed xpos=8.75, ypos=8.50

2024-11-26 21:11:09,500 | ApMeasureStars | DEBUG | Fitting star #23 (id=52) at fixed xpos=9.24, ypos=9.49

2024-11-26 21:11:09,508 | ApMeasureStars | DEBUG | Fitting star #24 (id=26) at fixed xpos=8.65, ypos=8.51

2024-11-26 21:11:09,518 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:09,518 | ApMeasureStars | DEBUG | Measured FWHM=4.219 +/- 0.157 pixels in direction x

2024-11-26 21:11:09,518 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:09,519 | ApMeasureStars | DEBUG | Measured FWHM=4.151 +/- 0.102 pixels in direction y

2024-11-26 21:11:09,519 | ApMeasureStars | INFO | Source average axis ratio = 1.017 +/- 0.045, overall sources are circular.

2024-11-26 21:11:09,519 | ApMeasureStars | DEBUG | Fitting completed in 0.244 s (average 9.7 ms/star)

2024-11-26 21:11:09,680 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:09,680 | ApMeasureStars | DEBUG | Measured FWHM=4.181 +/- 0.102 pixels in direction both

2024-11-26 21:11:09,681 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:09,681 | ApMeasureStars | DEBUG | Measured FWHM=4.219 +/- 0.157 pixels in direction x

2024-11-26 21:11:09,682 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:09,682 | ApMeasureStars | DEBUG | Measured FWHM=4.151 +/- 0.102 pixels in direction y

2024-11-26 21:11:10,903 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs

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to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-3.png
2024-11-26 21:11:10,904 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 48 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:11:10,904 | ApMeasureStars | DEBUG | Measured FWHM=4.181 +/- 0.102
pixels in direction both
2024-11-26 21:11:10,905 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:11:10,905 | ApMeasureStars | DEBUG | Measured FWHM=4.219 +/- 0.157
pixels in direction x
2024-11-26 21:11:10,906 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:11:10,906 | ApMeasureStars | DEBUG | Measured FWHM=4.151 +/- 0.102
pixels in direction y
2024-11-26 21:11:10,907 | ApFindStars | INFO | Median FWHM (over x and y) is
4.18 +/- 0.10 pixels using 48 data points
2024-11-26 21:11:10,907 | ApProcess | DEBUG | Final star detection and fitting:
FWHM=4.181 +/- 0.102 pixels using 48 stars.
2024-11-26 21:11:11,815 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-180s-Lum-3.png
2024-11-26 21:11:11,815 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:11:11,816 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz',
'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns
in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'),
'AP_NDET': (240, 'Number of sources detected in the image.'), 'AP_NPHOT': (200,
'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used
in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above
background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA':
('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target
declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'),
'FILTER': ('Lum', 'Filter used')}}
2024-11-26 21:11:11,816 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:11:11,817 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:11:11,817 | ApFindStars | WARNING | Keyword OBJECT not found, not
written to quality report.
2024-11-26 21:11:11,817 | ApFindStars | WARNING | Keyword TELESCOP not found,
not written to quality report.
2024-11-26 21:11:11,818 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.
2024-11-26 21:11:11,818 | ApFindStars | WARNING | Keyword EXPOSURE not found,
not written to quality report.
2024-11-26 21:11:11,818 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.
2024-11-26 21:11:11,818 | ApFindStars | WARNING | Keyword EGAIN not found, not
written to quality report.
2024-11-26 21:11:11,819 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.
2024-11-26 21:11:11,819 | ApFindStars | WARNING | Keyword APRX_XWD not found,
not written to quality report.
2024-11-26 21:11:11,819 | ApFindStars | WARNING | Keyword APRX_YHG not found,
not written to quality report.
2024-11-26 21:11:11,819 | ApFindStars | WARNING | Keyword APRX_XPS not found,
not written to quality report.
2024-11-26 21:11:11,819 | ApFindStars | WARNING | Keyword APRX_YPS not found,
not written to quality report.
2024-11-26 21:11:11,819 | ApFindStars | WARNING | Skipping some quality
reporting that requires an estimated platescale.
2024-11-26 21:11:11,821 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-3.yaml
2024-11-26 21:11:11,824 | ApFindStars | DEBUG | Wrote ds9-format region file to
./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-3.reg
2024-11-26 21:11:11,824 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:11:11,824 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz',
'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns
in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'),
'AP_NDET': (240, 'Number of sources detected in the image.'), 'AP_NPHOT': (200,
'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used
in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above
background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA':
('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target

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declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'),
'FILTER': ('Lum', 'Filter used'))}
2024-11-26 21:11:11,824 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:11:11,825 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:11:11,825 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:11:11,826 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:11:11,848 | ApProcess | DEBUG | Confirming output srclist
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits was created.
2024-11-26 21:11:11,849 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits as
clean_star_detection=True
2024-11-26 21:11:11,854 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits
2024-11-26 21:11:11,854 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
2024-11-26 21:11:12,140 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:11:12,143 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits
2024-11-26 21:11:12,143 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:11:12,143 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:11:12,144 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:11:12,144 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:11:12,144 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits with 200 positions
(attempt #1)
Solving...
2024-11-26 21:11:24,903 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:11:25,056 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:11:25,056 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:11:25,056 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:11:25,057 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:11:25,057 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:11:25,057 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:11:25,057 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.754467188, 'RA of reference point')
2024-11-26 21:11:25,058 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4166124875, 'DEC of reference point')
2024-11-26 21:11:25,058 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')

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2024-11-26 21:11:25,058 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:11:25,058 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:11:25,058 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:11:25,059 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.56790438677e-06, 'Transformation matrix')
2024-11-26 21:11:25,059 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175212897014, 'no comment')
2024-11-26 21:11:25,059 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175212897014, 'no comment')
2024-11-26 21:11:25,059 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.56790438677e-06, 'no comment')
2024-11-26 21:11:25,060 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:11:25,060 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:11:25,228 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits
2024-11-26 21:11:25,228 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srcList-M101_Supernova_2023ixf-180s-Lum-3.fits with
WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:11:25,246 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:11:25,515 | ApProcess | DEBUG |
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2024-11-26 21:11:25,515 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
2024-11-26 21:11:25,516 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:25,516 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:25,516 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:25,516 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:25,517 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:11:25,517 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:11:25,517 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
srcList=./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-1.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-1.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-1.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-1.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:11:25,517 | ApProcess | DEBUG | Removing existing
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits because
clean_star_detection=True
2024-11-26 21:11:25,518 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-1.reg because
clean_star_detection=True
2024-11-26 21:11:25,518 | ApProcess | DEBUG | Removing existing
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-1.png because
clean_star_detection=True
2024-11-26 21:11:25,519 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-1.png because
clean_star_detection=True
2024-11-26 21:11:25,519 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:11:25,519 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
2024-11-26 21:11:25,780 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:11:25,854 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67682.41, median=185.73

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2024-11-26 21:11:26,659 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=185.886, median=185.564, stddev=22.513
2024-11-26 21:11:29,364 | ApFindStars | DEBUG | Source-masked image stats:
mean=184.820, median=184.711, stddev=21.853
2024-11-26 21:11:29,365 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:11:29,365 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:11:29,542 | ApFindStars | DEBUG | Retaining 12 possibly saturated
stars in source searching and photometry.
2024-11-26 21:11:29,543 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:11:29,772 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 239 sources.
2024-11-26 21:11:29,772 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:11:29,772 | ApFindStars | DEBUG | There are 14 possibly saturated
stars in the output source list.
2024-11-26 21:11:29,773 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:29,773 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:29,814 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:11:29,815 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 15038.29 ADU/s.
2024-11-26 21:11:29,815 | ApFindStars | DEBUG | Median source (idx=119)
generates 27.65 ADU/s.
2024-11-26 21:11:29,815 | ApFindStars | DEBUG | Faintest source (idx=238)
generates -10.51 ADU/s.
2024-11-26 21:11:29,815 | ApFindStars | DEBUG | Selecting 200 sources out of 239
to write to the output source list.
2024-11-26 21:11:29,816 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:29,816 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:30,722 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-1.png
2024-11-26 21:11:30,724 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:11:30,725 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:11:30,725 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:11:30,725 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:11:30,726 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 186
2024-11-26 21:11:30,726 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 239
2024-11-26 21:11:30,726 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:11:30,726 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:11:30,727 | ApMeasureStars | DEBUG | Constructed a kdtree with 239
data points.
2024-11-26 21:11:30,729 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 22 stars from consideration.
2024-11-26 21:11:30,730 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 164 stars in 3056 row x 3056 column image.
2024-11-26 21:11:30,730 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:11:30,730 | ApMeasureStars | DEBUG | There are 162 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:11:30,731 | ApMeasureStars | DEBUG | In region CN found 49
candidates for fitting:
2024-11-26 21:11:30,731 | ApMeasureStars | DEBUG | In region TL found 24
candidates for fitting:
2024-11-26 21:11:30,733 | ApMeasureStars | DEBUG | In region TR found 35
candidates for fitting:
2024-11-26 21:11:30,735 | ApMeasureStars | DEBUG | In region BR found 33
candidates for fitting:

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2024-11-26 21:11:30,736 | ApMeasureStars | DEBUG | In region BL found 21 candidates for fitting:

2024-11-26 21:11:30,737 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=120.15, max=51271.86

2024-11-26 21:11:30,739 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:11:30,739 | ApMeasureStars | DEBUG | Fitting star #0 (id=132) at fixed xpos=8.72, ypos=8.58

2024-11-26 21:11:30,748 | ApMeasureStars | DEBUG | Fitting star #1 (id=99) at fixed xpos=8.69, ypos=8.67

2024-11-26 21:11:30,756 | ApMeasureStars | DEBUG | Fitting star #2 (id=86) at fixed xpos=8.82, ypos=8.84

2024-11-26 21:11:30,763 | ApMeasureStars | DEBUG | Fitting star #3 (id=157) at fixed xpos=9.09, ypos=9.12

2024-11-26 21:11:30,771 | ApMeasureStars | DEBUG | Fitting star #4 (id=146) at fixed xpos=9.09, ypos=8.91

2024-11-26 21:11:30,783 | ApMeasureStars | DEBUG | Fitting star #5 (id=176) at fixed xpos=9.33, ypos=8.62

2024-11-26 21:11:30,794 | ApMeasureStars | DEBUG | Fitting star #6 (id=234) at fixed xpos=9.14, ypos=9.37

2024-11-26 21:11:30,805 | ApMeasureStars | DEBUG | Fitting star #7 (id=197) at fixed xpos=9.14, ypos=8.89

2024-11-26 21:11:30,818 | ApMeasureStars | DEBUG | Fitting star #8 (id=213) at fixed xpos=8.99, ypos=9.34

2024-11-26 21:11:30,827 | ApMeasureStars | DEBUG | Fitting star #9 (id=178) at fixed xpos=9.31, ypos=9.08

2024-11-26 21:11:30,837 | ApMeasureStars | DEBUG | Fitting star #10 (id=210) at fixed xpos=8.55, ypos=9.17

2024-11-26 21:11:30,846 | ApMeasureStars | DEBUG | Fitting star #11 (id=231) at fixed xpos=9.00, ypos=8.73

2024-11-26 21:11:30,859 | ApMeasureStars | DEBUG | Fitting star #12 (id=184) at fixed xpos=8.51, ypos=8.58

2024-11-26 21:11:30,867 | ApMeasureStars | DEBUG | Fitting star #13 (id=187) at fixed xpos=9.46, ypos=8.73

2024-11-26 21:11:30,877 | ApMeasureStars | DEBUG | Fitting star #14 (id=203) at fixed xpos=8.54, ypos=8.67

2024-11-26 21:11:30,886 | ApMeasureStars | DEBUG | Fitting star #15 (id=9) at fixed xpos=8.78, ypos=8.93

2024-11-26 21:11:30,897 | ApMeasureStars | DEBUG | Fitting star #16 (id=48) at fixed xpos=8.88, ypos=8.64

2024-11-26 21:11:30,905 | ApMeasureStars | DEBUG | Fitting star #17 (id=45) at fixed xpos=8.91, ypos=8.50

2024-11-26 21:11:30,913 | ApMeasureStars | DEBUG | Fitting star #18 (id=50) at fixed xpos=8.52, ypos=9.14

2024-11-26 21:11:30,925 | ApMeasureStars | DEBUG | Fitting star #19 (id=35) at fixed xpos=9.39, ypos=8.98

2024-11-26 21:11:30,933 | ApMeasureStars | DEBUG | Fitting star #20 (id=21) at fixed xpos=8.76, ypos=9.08

2024-11-26 21:11:30,944 | ApMeasureStars | DEBUG | Fitting star #21 (id=55) at fixed xpos=9.49, ypos=8.51

2024-11-26 21:11:30,953 | ApMeasureStars | DEBUG | Fitting star #22 (id=53) at fixed xpos=9.44, ypos=8.84

2024-11-26 21:11:30,965 | ApMeasureStars | DEBUG | Fitting star #23 (id=37) at fixed xpos=8.69, ypos=8.58

2024-11-26 21:11:30,976 | ApMeasureStars | DEBUG | Fitting star #24 (id=49) at fixed xpos=8.80, ypos=9.49

2024-11-26 21:11:30,984 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:30,985 | ApMeasureStars | DEBUG | Measured FWHM=3.937 +/- 0.289 pixels in direction x

2024-11-26 21:11:30,985 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:30,985 | ApMeasureStars | DEBUG | Measured FWHM=3.842 +/- 0.219 pixels in direction y

2024-11-26 21:11:30,986 | ApMeasureStars | INFO | Source average axis ratio = 1.025 +/- 0.095, overall sources are circular.

2024-11-26 21:11:30,986 | ApMeasureStars | DEBUG | Fitting completed in 0.247 s (average 9.9 ms/star)

2024-11-26 21:11:30,986 | ApMeasureStars | DEBUG | Skipping plotting of star cutouts and best fits.

2024-11-26 21:11:30,987 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:30,987 | ApMeasureStars | DEBUG | Measured FWHM=3.878 +/- 0.231 pixels in direction both

2024-11-26 21:11:30,988 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:30,988 | ApMeasureStars | DEBUG | Measured FWHM=3.937 +/- 0.289 pixels in direction x

2024-11-26 21:11:30,989 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:30,989 | ApMeasureStars | DEBUG | Measured FWHM=3.842 +/- 0.219 pixels in direction y

2024-11-26 21:11:30,989 | ApFindStars | INFO | Median FWHM (over x and y) is 3.88 +/- 0.23 pixels using 48 data points

2024-11-26 21:11:30,989 | ApProcess | DEBUG | Initial star detection and fitting: FWHM=3.878 +/- 0.231 pixels using 48 stars.

2024-11-26 21:11:30,989 | ApProcess | DEBUG | Updating source searching using initial FWHM=3.878 +/- 0.231 pixels using 48 stars.

2024-11-26 21:11:30,989 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=3.88 pixels, threshold=7.0 BG sigma.

2024-11-26 21:11:31,286 | ApFindStars | INFO | Initial source list using FWHM=3.878 pixels, threshold=7.00 x BG stddev, found 243 sources.

2024-11-26 21:11:31,286 | ApFindStars | DEBUG | Sources with pixels exceeding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:11:31,287 | ApFindStars | DEBUG | There are 11 possibly saturated stars in the output source list.

2024-11-26 21:11:31,287 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:11:31,287 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:11:31,328 | ApFindStars | DEBUG | Image exposure time [seconds]: 300.00

2024-11-26 21:11:31,329 | ApFindStars | DEBUG | Brightest source (idx=0) generates 15869.78 ADU/s.

2024-11-26 21:11:31,329 | ApFindStars | DEBUG | Median source (idx=121) generates 26.10 ADU/s.

2024-11-26 21:11:31,330 | ApFindStars | DEBUG | Faintest source (idx=242) generates 0.95 ADU/s.

2024-11-26 21:11:31,330 | ApFindStars | DEBUG | Selecting 200 sources out of 243 to write to the output source list.

2024-11-26 21:11:31,330 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:11:31,330 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:11:31,333 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:11:31,333 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:11:31,333 | ApMeasureStars | DEBUG | Count based weighting factors will used in fitting?: True

2024-11-26 21:11:31,333 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:11:31,334 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 189

2024-11-26 21:11:31,334 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 243

2024-11-26 21:11:31,334 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:11:31,334 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:11:31,335 | ApMeasureStars | DEBUG | Constructed a kdtree with 243 data points.

2024-11-26 21:11:31,338 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 16 stars from consideration.

2024-11-26 21:11:31,339 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 173 stars in 3056 row x 3056 column image.

2024-11-26 21:11:31,339 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:11:31,340 | ApMeasureStars | DEBUG | There are 171 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:11:31,340 | ApMeasureStars | DEBUG | In region CN found 53 candidates for fitting:

2024-11-26 21:11:31,341 | ApMeasureStars | DEBUG | In region TL found 26 candidates for fitting:

2024-11-26 21:11:31,342 | ApMeasureStars | DEBUG | In region TR found 38 candidates for fitting:

2024-11-26 21:11:31,343 | ApMeasureStars | DEBUG | In region BR found 30 candidates for fitting:

2024-11-26 21:11:31,345 | ApMeasureStars | DEBUG | In region BL found 24 candidates for fitting:

2024-11-26 21:11:31,347 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=120.15, max=51271.86

2024-11-26 21:11:31,348 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:11:31,349 | ApMeasureStars | DEBUG | Fitting star #0 (id=136) at
fixed xpos=8.74, ypos=8.59

2024-11-26 21:11:31,358 | ApMeasureStars | DEBUG | Fitting star #1 (id=102) at
fixed xpos=8.70, ypos=8.68

2024-11-26 21:11:31,366 | ApMeasureStars | DEBUG | Fitting star #2 (id=89) at
fixed xpos=8.82, ypos=8.84

2024-11-26 21:11:31,373 | ApMeasureStars | DEBUG | Fitting star #3 (id=164) at
fixed xpos=9.09, ypos=9.11

2024-11-26 21:11:31,382 | ApMeasureStars | DEBUG | Fitting star #4 (id=152) at
fixed xpos=9.09, ypos=8.90

2024-11-26 21:11:31,394 | ApMeasureStars | DEBUG | Fitting star #5 (id=183) at
fixed xpos=9.32, ypos=8.62

2024-11-26 21:11:31,406 | ApMeasureStars | DEBUG | Fitting star #6 (id=239) at
fixed xpos=9.13, ypos=9.35

2024-11-26 21:11:31,418 | ApMeasureStars | DEBUG | Fitting star #7 (id=205) at
fixed xpos=9.14, ypos=8.89

2024-11-26 21:11:31,431 | ApMeasureStars | DEBUG | Fitting star #8 (id=219) at
fixed xpos=8.98, ypos=9.33

2024-11-26 21:11:31,440 | ApMeasureStars | DEBUG | Fitting star #9 (id=185) at
fixed xpos=9.39, ypos=9.09

2024-11-26 21:11:31,452 | ApMeasureStars | DEBUG | Fitting star #10 (id=216) at
fixed xpos=8.57, ypos=9.16

2024-11-26 21:11:31,461 | ApMeasureStars | DEBUG | Fitting star #11 (id=236) at
fixed xpos=9.00, ypos=8.74

2024-11-26 21:11:31,473 | ApMeasureStars | DEBUG | Fitting star #12 (id=192) at
fixed xpos=8.53, ypos=8.59

2024-11-26 21:11:31,483 | ApMeasureStars | DEBUG | Fitting star #13 (id=195) at
fixed xpos=9.43, ypos=8.74

2024-11-26 21:11:31,493 | ApMeasureStars | DEBUG | Fitting star #14 (id=210) at
fixed xpos=8.55, ypos=8.68

2024-11-26 21:11:31,503 | ApMeasureStars | DEBUG | Fitting star #15 (id=10) at
fixed xpos=8.79, ypos=8.91

2024-11-26 21:11:31,515 | ApMeasureStars | DEBUG | Fitting star #16 (id=44) at
fixed xpos=8.88, ypos=8.63

2024-11-26 21:11:31,522 | ApMeasureStars | DEBUG | Fitting star #17 (id=40) at
fixed xpos=8.92, ypos=8.50

2024-11-26 21:11:31,530 | ApMeasureStars | DEBUG | Fitting star #18 (id=46) at
fixed xpos=8.54, ypos=9.12

2024-11-26 21:11:31,542 | ApMeasureStars | DEBUG | Fitting star #19 (id=32) at
fixed xpos=9.47, ypos=8.92

2024-11-26 21:11:31,550 | ApMeasureStars | DEBUG | Fitting star #20 (id=21) at
fixed xpos=8.87, ypos=8.98

2024-11-26 21:11:31,559 | ApMeasureStars | DEBUG | Fitting star #21 (id=54) at
fixed xpos=8.50, ypos=9.41

2024-11-26 21:11:31,568 | ApMeasureStars | DEBUG | Fitting star #22 (id=49) at
fixed xpos=8.56, ypos=8.78

2024-11-26 21:11:31,577 | ApMeasureStars | DEBUG | Fitting star #23 (id=33) at
fixed xpos=8.69, ypos=8.58

2024-11-26 21:11:31,588 | ApMeasureStars | DEBUG | Fitting star #24 (id=45) at
fixed xpos=8.80, ypos=9.49

2024-11-26 21:11:31,597 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:31,597 | ApMeasureStars | DEBUG | Measured FWHM=4.019 +/- 0.337
pixels in direction x

2024-11-26 21:11:31,598 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:31,599 | ApMeasureStars | DEBUG | Measured FWHM=3.864 +/- 0.222
pixels in direction y

2024-11-26 21:11:31,599 | ApMeasureStars | INFO | Source average axis ratio =
1.040 +/- 0.106, overall sources are circular.

2024-11-26 21:11:31,599 | ApMeasureStars | DEBUG | Fitting completed in 0.251 s
(average 10.0 ms/star)

2024-11-26 21:11:31,686 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:31,686 | ApMeasureStars | DEBUG | Measured FWHM=3.884 +/- 0.247
pixels in direction both

2024-11-26 21:11:31,687 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:31,687 | ApMeasureStars | DEBUG | Measured FWHM=4.019 +/- 0.337
pixels in direction x

2024-11-26 21:11:31,688 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:31,688 | ApMeasureStars | DEBUG | Measured FWHM=3.864 +/- 0.222
pixels in direction y

2024-11-26 21:11:33,011 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwplot-M101_Supernova_2023ixf-300s-Blue-1.png

2024-11-26 21:11:33,012 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:33,012 | ApMeasureStars | DEBUG | Measured FWHM=3.884 +/- 0.247 pixels in direction both

2024-11-26 21:11:33,013 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:33,013 | ApMeasureStars | DEBUG | Measured FWHM=4.019 +/- 0.337 pixels in direction x

2024-11-26 21:11:33,014 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:33,014 | ApMeasureStars | DEBUG | Measured FWHM=3.864 +/- 0.222 pixels in direction y

2024-11-26 21:11:33,015 | ApFindStars | INFO | Median FWHM (over x and y) is 3.88 +/- 0.25 pixels using 48 data points

2024-11-26 21:11:33,015 | ApProcess | DEBUG | Final star detection and fitting: FWHM=3.884 +/- 0.247 pixels using 48 stars.

2024-11-26 21:11:33,922 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to ./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-1.png

2024-11-26 21:11:33,923 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:11:33,923 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (243, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Blue', 'Filter used')}

2024-11-26 21:11:33,923 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:11:33,924 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:11:33,925 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:11:33,925 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:11:33,925 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:11:33,925 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:11:33,925 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:11:33,926 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:11:33,926 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:11:33,926 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:11:33,926 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:11:33,926 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:11:33,927 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:11:33,927 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:11:33,928 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-1.yaml

2024-11-26 21:11:33,931 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-1.reg

2024-11-26 21:11:33,931 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:11:33,932 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (243, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),

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'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Blue',
'Filter used')}}
2024-11-26 21:11:33,932 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:11:33,933 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:11:33,933 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:11:33,933 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:11:33,955 | ApProcess | DEBUG | Confirming output srcList
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits was created.
2024-11-26 21:11:33,956 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits as
clean_star_detection=True
2024-11-26 21:11:33,961 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:11:33,961 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
2024-11-26 21:11:34,223 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:11:34,225 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:11:34,226 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:11:34,226 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:11:34,226 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:11:34,226 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:11:34,227 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-1.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:11:43,404 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:11:43,545 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:11:43,545 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA--TAN', 'TAN (gnomic) projection')
2024-11-26 21:11:43,546 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:11:43,546 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:11:43,547 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:11:43,547 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:11:43,547 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.753517387, 'RA of reference point')
2024-11-26 21:11:43,547 | AstroPhotography.core.ApAstrometry | DEBUG | Setting

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CRVAL2 to (54.4158146239, 'DEC of reference point')
2024-11-26 21:11:43,548 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:11:43,548 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:11:43,549 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:11:43,549 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:11:43,549 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.53011145683e-06, 'Transformation matrix')
2024-11-26 21:11:43,549 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175193469064, 'no comment')
2024-11-26 21:11:43,550 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175193469064, 'no comment')
2024-11-26 21:11:43,550 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.53011145683e-06, 'no comment')
2024-11-26 21:11:43,550 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:11:43,550 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:11:43,724 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:11:43,725 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-1.fits with
WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:11:43,744 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:11:44,040 | ApProcess | DEBUG |
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2024-11-26 21:11:44,040 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
2024-11-26 21:11:44,041 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:44,041 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:44,041 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:44,041 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:11:44,041 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:11:44,041 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:11:44,042 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-2.reg
plotfile=./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-2.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-2.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-2.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits
2024-11-26 21:11:44,042 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits because
clean_star_detection=True
2024-11-26 21:11:44,042 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-2.reg because
clean_star_detection=True
2024-11-26 21:11:44,042 | ApProcess | DEBUG | Removing existing
./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-2.png because
clean_star_detection=True
2024-11-26 21:11:44,043 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-2.png because
clean_star_detection=True
2024-11-26 21:11:44,043 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits
2024-11-26 21:11:44,043 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
2024-11-26 21:11:44,307 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056

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columns, 3056 rows
2024-11-26 21:11:44,401 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67913.47, median=180.82
2024-11-26 21:11:45,287 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=180.957, median=180.669, stddev=22.395
2024-11-26 21:11:48,388 | ApFindStars | DEBUG | Source-masked image stats:
mean=179.979, median=179.874, stddev=21.801
2024-11-26 21:11:48,388 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:11:48,389 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:11:48,565 | ApFindStars | DEBUG | Retaining 13 possibly saturated
stars in source searching and photometry.
2024-11-26 21:11:48,565 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:11:48,797 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 256 sources.
2024-11-26 21:11:48,798 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:11:48,798 | ApFindStars | DEBUG | There are 11 possibly saturated
stars in the output source list.
2024-11-26 21:11:48,798 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:48,798 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:48,841 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:11:48,842 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14230.73 ADU/s.
2024-11-26 21:11:48,842 | ApFindStars | DEBUG | Median source (idx=128)
generates 22.02 ADU/s.
2024-11-26 21:11:48,842 | ApFindStars | DEBUG | Faintest source (idx=255)
generates -8.68 ADU/s.
2024-11-26 21:11:48,842 | ApFindStars | DEBUG | Selecting 200 sources out of 256
to write to the output source list.
2024-11-26 21:11:48,843 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:48,843 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:49,753 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-2.png
2024-11-26 21:11:49,754 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:11:49,754 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:11:49,755 | ApMeasureStars | DEBUG | Count based weighting factors
will be used in fitting?: True
2024-11-26 21:11:49,755 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:11:49,755 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 189
2024-11-26 21:11:49,756 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 256
2024-11-26 21:11:49,756 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:11:49,756 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:11:49,756 | ApMeasureStars | DEBUG | Constructed a kdtree with 256
data points.
2024-11-26 21:11:49,759 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 22 stars from consideration.
2024-11-26 21:11:49,759 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 167 stars in 3056 row x 3056 column image.
2024-11-26 21:11:49,759 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:11:49,760 | ApMeasureStars | DEBUG | There are 165 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:11:49,760 | ApMeasureStars | DEBUG | In region CN found 53
candidates for fitting:
2024-11-26 21:11:49,761 | ApMeasureStars | DEBUG | In region TL found 26
candidates for fitting:
2024-11-26 21:11:49,762 | ApMeasureStars | DEBUG | In region TR found 30

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candidates for fitting:
 2024-11-26 21:11:49,764 | ApMeasureStars | DEBUG | In region BR found 32
 candidates for fitting:
 2024-11-26 21:11:49,765 | ApMeasureStars | DEBUG | In region BL found 24
 candidates for fitting:
 2024-11-26 21:11:49,767 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
 min=0.00, max=51982.92
 2024-11-26 21:11:49,768 | ApMeasureStars | INFO | Starting to fit
 Const2D+Gaussian2D model to 25 star cutouts.
 2024-11-26 21:11:49,769 | ApMeasureStars | DEBUG | Fitting star #0 (id=144) at
 fixed xpos=8.63, ypos=8.71
 2024-11-26 21:11:49,779 | ApMeasureStars | DEBUG | Fitting star #1 (id=108) at
 fixed xpos=8.58, ypos=8.87
 2024-11-26 21:11:49,790 | ApMeasureStars | DEBUG | Fitting star #2 (id=100) at
 fixed xpos=8.76, ypos=8.97
 2024-11-26 21:11:49,799 | ApMeasureStars | DEBUG | Fitting star #3 (id=159) at
 fixed xpos=9.09, ypos=8.96
 2024-11-26 21:11:49,811 | ApMeasureStars | DEBUG | Fitting star #4 (id=170) at
 fixed xpos=9.05, ypos=9.16
 2024-11-26 21:11:49,820 | ApMeasureStars | DEBUG | Fitting star #5 (id=250) at
 fixed xpos=9.36, ypos=8.94
 2024-11-26 21:11:49,828 | ApMeasureStars | DEBUG | Fitting star #6 (id=218) at
 fixed xpos=9.19, ypos=8.79
 2024-11-26 21:11:49,840 | ApMeasureStars | DEBUG | Fitting star #7 (id=233) at
 fixed xpos=9.12, ypos=9.05
 2024-11-26 21:11:49,848 | ApMeasureStars | DEBUG | Fitting star #8 (id=193) at
 fixed xpos=8.52, ypos=8.97
 2024-11-26 21:11:49,857 | ApMeasureStars | DEBUG | Fitting star #9 (id=247) at
 fixed xpos=8.99, ypos=8.71
 2024-11-26 21:11:49,867 | ApMeasureStars | DEBUG | Fitting star #10 (id=230) at
 fixed xpos=9.49, ypos=9.11
 2024-11-26 21:11:49,876 | ApMeasureStars | DEBUG | Fitting star #11 (id=245) at
 fixed xpos=9.01, ypos=9.49
 2024-11-26 21:11:49,890 | ApMeasureStars | DEBUG | Fitting star #12 (id=207) at
 fixed xpos=9.39, ypos=8.63
 2024-11-26 21:11:49,899 | ApMeasureStars | DEBUG | Fitting star #13 (id=210) at
 fixed xpos=9.30, ypos=8.69
 2024-11-26 21:11:49,908 | ApMeasureStars | DEBUG | Fitting star #14 (id=223) at
 fixed xpos=8.56, ypos=8.61
 2024-11-26 21:11:49,920 | ApMeasureStars | DEBUG | Fitting star #15 (id=8) at
 fixed xpos=8.53, ypos=9.17
 2024-11-26 21:11:49,929 | ApMeasureStars | DEBUG | Fitting star #16 (id=55) at
 fixed xpos=8.70, ypos=8.79
 2024-11-26 21:11:49,938 | ApMeasureStars | DEBUG | Fitting star #17 (id=48) at
 fixed xpos=8.68, ypos=8.74
 2024-11-26 21:11:49,948 | ApMeasureStars | DEBUG | Fitting star #18 (id=57) at
 fixed xpos=9.47, ypos=9.31
 2024-11-26 21:11:49,957 | ApMeasureStars | DEBUG | Fitting star #19 (id=33) at
 fixed xpos=9.30, ypos=9.13
 2024-11-26 21:11:49,966 | ApMeasureStars | DEBUG | Fitting star #20 (id=64) at
 fixed xpos=9.47, ypos=8.62
 2024-11-26 21:11:49,975 | ApMeasureStars | DEBUG | Fitting star #21 (id=59) at
 fixed xpos=8.63, ypos=8.91
 2024-11-26 21:11:49,984 | ApMeasureStars | DEBUG | Fitting star #22 (id=37) at
 fixed xpos=8.55, ypos=8.82
 2024-11-26 21:11:49,993 | ApMeasureStars | DEBUG | Fitting star #23 (id=56) at
 fixed xpos=8.69, ypos=8.73
 2024-11-26 21:11:50,002 | ApMeasureStars | DEBUG | Fitting star #24 (id=54) at
 fixed xpos=9.29, ypos=8.82
 2024-11-26 21:11:50,010 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=x) using 25 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:11:50,010 | ApMeasureStars | DEBUG | Measured FWHM=3.503 +/- 0.119
 pixels in direction x
 2024-11-26 21:11:50,011 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=y) using 25 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:11:50,011 | ApMeasureStars | DEBUG | Measured FWHM=3.284 +/- 0.186
 pixels in direction y
 2024-11-26 21:11:50,011 | ApMeasureStars | INFO | Source average axis ratio =
 1.066 +/- 0.071, overall sources are circular.
 2024-11-26 21:11:50,011 | ApMeasureStars | DEBUG | Fitting completed in 0.243 s
 (average 9.7 ms/star)
 2024-11-26 21:11:50,011 | ApMeasureStars | DEBUG | Skipping plotting of star
 cutouts and best fits.
 2024-11-26 21:11:50,012 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=both) using 50 FWHM measurements (50 OK fits before clipping).
 2024-11-26 21:11:50,012 | ApMeasureStars | DEBUG | Measured FWHM=3.467 +/- 0.262

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pixels in direction both
2024-11-26 21:11:50,013 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:11:50,013 | ApMeasureStars | DEBUG | Measured FWHM=3.503 +/- 0.119
pixels in direction x
2024-11-26 21:11:50,013 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:11:50,014 | ApMeasureStars | DEBUG | Measured FWHM=3.284 +/- 0.186
pixels in direction y
2024-11-26 21:11:50,014 | ApFindStars | INFO | Median FWHM (over x and y) is
3.47 +/- 0.26 pixels using 50 data points
2024-11-26 21:11:50,014 | ApProcess | DEBUG | Initial star detection and
fitting: FWHM=3.467 +/- 0.262 pixels using 50 stars.
2024-11-26 21:11:50,014 | ApProcess | DEBUG | Updating source searching using
initial FWHM=3.467 +/- 0.262 pixels using 50 stars.
2024-11-26 21:11:50,014 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.47 pixels, threshold=7.0 BG sigma.
2024-11-26 21:11:50,243 | ApFindStars | INFO | Initial source list using
FWHM=3.467 pixels, threshold=7.00 x BG stddev, found 278 sources.
2024-11-26 21:11:50,244 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:11:50,244 | ApFindStars | DEBUG | There are 10 possibly saturated
stars in the output source list.
2024-11-26 21:11:50,244 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:50,245 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:50,289 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:11:50,291 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 11116.60 ADU/s.
2024-11-26 21:11:50,291 | ApFindStars | DEBUG | Median source (idx=139)
generates 19.82 ADU/s.
2024-11-26 21:11:50,291 | ApFindStars | DEBUG | Faintest source (idx=277)
generates -8.75 ADU/s.
2024-11-26 21:11:50,291 | ApFindStars | DEBUG | Selecting 200 sources out of 278
to write to the output source list.
2024-11-26 21:11:50,291 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:11:50,291 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:11:50,293 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:11:50,294 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:11:50,294 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:11:50,294 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:11:50,294 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 190
2024-11-26 21:11:50,294 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 278
2024-11-26 21:11:50,295 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:11:50,295 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:11:50,295 | ApMeasureStars | DEBUG | Constructed a kdtree with 278
data points.
2024-11-26 21:11:50,298 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 23 stars from consideration.
2024-11-26 21:11:50,298 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 167 stars in 3056 row x 3056 column image.
2024-11-26 21:11:50,299 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:11:50,299 | ApMeasureStars | DEBUG | There are 165 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:11:50,300 | ApMeasureStars | DEBUG | In region CN found 53
candidates for fitting:
2024-11-26 21:11:50,300 | ApMeasureStars | DEBUG | In region TL found 26
candidates for fitting:
2024-11-26 21:11:50,301 | ApMeasureStars | DEBUG | In region TR found 30
candidates for fitting:

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2024-11-26 21:11:50,303 | ApMeasureStars | DEBUG | In region BR found 32 candidates for fitting:

2024-11-26 21:11:50,304 | ApMeasureStars | DEBUG | In region BL found 24 candidates for fitting:

2024-11-26 21:11:50,305 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=51982.92

2024-11-26 21:11:50,306 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:11:50,307 | ApMeasureStars | DEBUG | Fitting star #0 (id=156) at fixed xpos=8.63, ypos=8.72

2024-11-26 21:11:50,317 | ApMeasureStars | DEBUG | Fitting star #1 (id=117) at fixed xpos=8.59, ypos=8.88

2024-11-26 21:11:50,329 | ApMeasureStars | DEBUG | Fitting star #2 (id=110) at fixed xpos=8.77, ypos=8.97

2024-11-26 21:11:50,338 | ApMeasureStars | DEBUG | Fitting star #3 (id=172) at fixed xpos=9.08, ypos=8.96

2024-11-26 21:11:50,350 | ApMeasureStars | DEBUG | Fitting star #4 (id=182) at fixed xpos=9.05, ypos=9.16

2024-11-26 21:11:50,358 | ApMeasureStars | DEBUG | Fitting star #5 (id=270) at fixed xpos=9.35, ypos=8.94

2024-11-26 21:11:50,367 | ApMeasureStars | DEBUG | Fitting star #6 (id=235) at fixed xpos=9.18, ypos=8.79

2024-11-26 21:11:50,379 | ApMeasureStars | DEBUG | Fitting star #7 (id=250) at fixed xpos=9.11, ypos=9.05

2024-11-26 21:11:50,387 | ApMeasureStars | DEBUG | Fitting star #8 (id=207) at fixed xpos=8.50, ypos=8.97

2024-11-26 21:11:50,396 | ApMeasureStars | DEBUG | Fitting star #9 (id=266) at fixed xpos=8.99, ypos=8.72

2024-11-26 21:11:50,405 | ApMeasureStars | DEBUG | Fitting star #10 (id=247) at fixed xpos=8.50, ypos=9.11

2024-11-26 21:11:50,422 | ApMeasureStars | DEBUG | Fitting star #11 (id=264) at fixed xpos=9.01, ypos=9.50

2024-11-26 21:11:50,437 | ApMeasureStars | DEBUG | Fitting star #12 (id=222) at fixed xpos=9.41, ypos=8.63

2024-11-26 21:11:50,445 | ApMeasureStars | DEBUG | Fitting star #13 (id=225) at fixed xpos=9.29, ypos=8.69

2024-11-26 21:11:50,454 | ApMeasureStars | DEBUG | Fitting star #14 (id=240) at fixed xpos=8.57, ypos=8.62

2024-11-26 21:11:50,467 | ApMeasureStars | DEBUG | Fitting star #15 (id=10) at fixed xpos=8.54, ypos=9.16

2024-11-26 21:11:50,476 | ApMeasureStars | DEBUG | Fitting star #16 (id=61) at fixed xpos=8.71, ypos=8.79

2024-11-26 21:11:50,485 | ApMeasureStars | DEBUG | Fitting star #17 (id=54) at fixed xpos=8.69, ypos=8.75

2024-11-26 21:11:50,495 | ApMeasureStars | DEBUG | Fitting star #18 (id=63) at fixed xpos=9.45, ypos=9.30

2024-11-26 21:11:50,505 | ApMeasureStars | DEBUG | Fitting star #19 (id=38) at fixed xpos=9.28, ypos=9.13

2024-11-26 21:11:50,515 | ApMeasureStars | DEBUG | Fitting star #20 (id=71) at fixed xpos=9.46, ypos=8.63

2024-11-26 21:11:50,524 | ApMeasureStars | DEBUG | Fitting star #21 (id=66) at fixed xpos=8.63, ypos=8.91

2024-11-26 21:11:50,533 | ApMeasureStars | DEBUG | Fitting star #22 (id=43) at fixed xpos=8.56, ypos=8.83

2024-11-26 21:11:50,542 | ApMeasureStars | DEBUG | Fitting star #23 (id=62) at fixed xpos=8.70, ypos=8.73

2024-11-26 21:11:50,551 | ApMeasureStars | DEBUG | Fitting star #24 (id=60) at fixed xpos=9.29, ypos=8.82

2024-11-26 21:11:50,560 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:50,560 | ApMeasureStars | DEBUG | Measured FWHM=3.500 +/- 0.114 pixels in direction x

2024-11-26 21:11:50,561 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:50,561 | ApMeasureStars | DEBUG | Measured FWHM=3.258 +/- 0.126 pixels in direction y

2024-11-26 21:11:50,562 | ApMeasureStars | INFO | Source average axis ratio = 1.074 +/- 0.054, overall sources are circular.

2024-11-26 21:11:50,562 | ApMeasureStars | DEBUG | Fitting completed in 0.255 s (average 10.2 ms/star)

2024-11-26 21:11:50,648 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:50,648 | ApMeasureStars | DEBUG | Measured FWHM=3.465 +/- 0.284 pixels in direction both

2024-11-26 21:11:50,649 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:50,650 | ApMeasureStars | DEBUG | Measured FWHM=3.500 +/- 0.114 pixels in direction x

2024-11-26 21:11:50,651 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:50,652 | ApMeasureStars | DEBUG | Measured FWHM=3.258 +/- 0.126 pixels in direction y

2024-11-26 21:11:51,865 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-2.png

2024-11-26 21:11:51,866 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:11:51,866 | ApMeasureStars | DEBUG | Measured FWHM=3.465 +/- 0.284 pixels in direction both

2024-11-26 21:11:51,867 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:51,867 | ApMeasureStars | DEBUG | Measured FWHM=3.500 +/- 0.114 pixels in direction x

2024-11-26 21:11:51,867 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:11:51,868 | ApMeasureStars | DEBUG | Measured FWHM=3.258 +/- 0.126 pixels in direction y

2024-11-26 21:11:51,868 | ApFindStars | INFO | Median FWHM (over x and y) is 3.47 +/- 0.28 pixels using 50 data points

2024-11-26 21:11:51,868 | ApProcess | DEBUG | Final star detection and fitting: FWHM=3.465 +/- 0.284 pixels using 50 stars.

2024-11-26 21:11:52,772 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to

./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-2.png

2024-11-26 21:11:52,773 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:11:52,773 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (278, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Blue', 'Filter used')}

2024-11-26 21:11:52,774 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:11:52,775 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:11:52,775 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:11:52,775 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:11:52,775 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:11:52,775 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:11:52,775 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:11:52,776 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:11:52,779 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-2.yaml

2024-11-26 21:11:52,782 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-2.reg

2024-11-26 21:11:52,782 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:11:52,782 | ApFindStars | DEBUG | FITS keywords found in image:

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{'IMG_FILE': ('Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz', 'Name of image file
searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'),
'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (278, 'Number of
sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final
photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),
'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Blue',
'Filter used')}}
2024-11-26 21:11:52,782 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:11:52,783 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:11:52,783 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:11:52,784 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-2.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:11:52,806 | ApProcess | DEBUG | Confirming output srcList
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-2.fits was created.
2024-11-26 21:11:52,807 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits as
clean_star_detection=True
2024-11-26 21:11:52,812 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits
2024-11-26 21:11:52,812 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
2024-11-26 21:11:53,075 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:11:53,078 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-2.fits
2024-11-26 21:11:53,078 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:11:53,078 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:11:53,078 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:11:53,079 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:11:53,079 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Blue-2.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:12:02,457 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:12:02,597 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:12:02,597 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:12:02,598 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:12:02,598 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:12:02,598 | AstroPhotography.core.ApAstrometry | DEBUG | Setting

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LONPOLE to (180.0, 'no comment')
2024-11-26 21:12:02,598 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:12:02,599 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.765829971, 'RA of reference point')
2024-11-26 21:12:02,599 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4177920309, 'DEC of reference point')
2024-11-26 21:12:02,599 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:12:02,600 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:12:02,600 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:12:02,600 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:12:02,600 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.4531970053e-06, 'Transformation matrix')
2024-11-26 21:12:02,600 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175223048501, 'no comment')
2024-11-26 21:12:02,601 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175223048501, 'no comment')
2024-11-26 21:12:02,601 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.4531970053e-06, 'no comment')
2024-11-26 21:12:02,601 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:12:02,601 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:12:02,774 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits
2024-11-26 21:12:02,774 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits with
WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:12:02,794 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:12:03,085 | ApProcess | DEBUG |
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2024-11-26 21:12:03,086 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
2024-11-26 21:12:03,086 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:03,086 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:03,086 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:03,086 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:03,086 | ApProcess | DEBUG | Subdirectory ./Metadata/ already
exists.
2024-11-26 21:12:03,087 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:12:03,087 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-3.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-3.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-3.png
qualfile=./Metadata/qual-M101_Supernova_2023ixf-300s-Blue-3.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits
2024-11-26 21:12:03,087 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits because
clean_star_detection=True
2024-11-26 21:12:03,087 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-3.reg because
clean_star_detection=True
2024-11-26 21:12:03,088 | ApProcess | DEBUG | Removing existing
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-3.png because
clean_star_detection=True
2024-11-26 21:12:03,088 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-3.png because

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clean_star_detection=True
2024-11-26 21:12:03,089 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits
2024-11-26 21:12:03,089 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
2024-11-26 21:12:03,351 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:12:03,436 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=68250.33, median=184.30
2024-11-26 21:12:04,239 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=184.423, median=184.190, stddev=22.288
2024-11-26 21:12:07,202 | ApFindStars | DEBUG | Source-masked image stats:
mean=183.636, median=183.536, stddev=21.837
2024-11-26 21:12:07,202 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:12:07,203 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:12:07,381 | ApFindStars | DEBUG | Retaining 11 possibly saturated
stars in source searching and photometry.
2024-11-26 21:12:07,382 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:12:07,617 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 244 sources.
2024-11-26 21:12:07,617 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:12:07,618 | ApFindStars | DEBUG | There are 8 possibly saturated
stars in the output source list.
2024-11-26 21:12:07,618 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:07,618 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:07,660 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:12:07,661 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 13743.08 ADU/s.
2024-11-26 21:12:07,661 | ApFindStars | DEBUG | Median source (idx=122)
generates 17.83 ADU/s.
2024-11-26 21:12:07,662 | ApFindStars | DEBUG | Faintest source (idx=243)
generates -0.60 ADU/s.
2024-11-26 21:12:07,662 | ApFindStars | DEBUG | Selecting 200 sources out of 244
to write to the output source list.
2024-11-26 21:12:07,662 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:07,662 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:08,582 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-3.png
2024-11-26 21:12:08,583 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:12:08,584 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:12:08,584 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:12:08,584 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:12:08,585 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 192
2024-11-26 21:12:08,585 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 244
2024-11-26 21:12:08,586 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:12:08,586 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:12:08,586 | ApMeasureStars | DEBUG | Constructed a kdtree with 244
data points.
2024-11-26 21:12:08,589 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 17 stars from consideration.
2024-11-26 21:12:08,590 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 175 stars in 3056 row x 3056 column image.
2024-11-26 21:12:08,590 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:12:08,590 | ApMeasureStars | DEBUG | There are 173 stars more than

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10 pixels away from the edge of the detector.

2024-11-26 21:12:08,591 | ApMeasureStars | DEBUG | In region CN found 53 candidates for fitting:

2024-11-26 21:12:08,592 | ApMeasureStars | DEBUG | In region TL found 28 candidates for fitting:

2024-11-26 21:12:08,593 | ApMeasureStars | DEBUG | In region TR found 29 candidates for fitting:

2024-11-26 21:12:08,595 | ApMeasureStars | DEBUG | In region BR found 33 candidates for fitting:

2024-11-26 21:12:08,596 | ApMeasureStars | DEBUG | In region BL found 30 candidates for fitting:

2024-11-26 21:12:08,598 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=51606.07

2024-11-26 21:12:08,599 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:12:08,600 | ApMeasureStars | DEBUG | Fitting star #0 (id=142) at fixed xpos=9.08, ypos=8.76

2024-11-26 21:12:08,613 | ApMeasureStars | DEBUG | Fitting star #1 (id=111) at fixed xpos=9.10, ypos=8.90

2024-11-26 21:12:08,626 | ApMeasureStars | DEBUG | Fitting star #2 (id=102) at fixed xpos=9.27, ypos=8.96

2024-11-26 21:12:08,639 | ApMeasureStars | DEBUG | Fitting star #3 (id=177) at fixed xpos=8.91, ypos=9.10

2024-11-26 21:12:08,651 | ApMeasureStars | DEBUG | Fitting star #4 (id=163) at fixed xpos=9.43, ypos=9.14

2024-11-26 21:12:08,663 | ApMeasureStars | DEBUG | Fitting star #5 (id=241) at fixed xpos=8.75, ypos=9.46

2024-11-26 21:12:08,672 | ApMeasureStars | DEBUG | Fitting star #6 (id=210) at fixed xpos=8.59, ypos=8.64

2024-11-26 21:12:08,683 | ApMeasureStars | DEBUG | Fitting star #7 (id=227) at fixed xpos=9.46, ypos=8.73

2024-11-26 21:12:08,692 | ApMeasureStars | DEBUG | Fitting star #8 (id=187) at fixed xpos=8.90, ypos=8.66

2024-11-26 21:12:08,704 | ApMeasureStars | DEBUG | Fitting star #9 (id=240) at fixed xpos=9.36, ypos=8.51

2024-11-26 21:12:08,714 | ApMeasureStars | DEBUG | Fitting star #10 (id=188) at fixed xpos=8.92, ypos=9.06

2024-11-26 21:12:08,727 | ApMeasureStars | DEBUG | Fitting star #11 (id=224) at fixed xpos=8.91, ypos=9.07

2024-11-26 21:12:08,742 | ApMeasureStars | DEBUG | Fitting star #12 (id=238) at fixed xpos=9.42, ypos=9.46

2024-11-26 21:12:08,751 | ApMeasureStars | DEBUG | Fitting star #13 (id=196) at fixed xpos=8.86, ypos=8.62

2024-11-26 21:12:08,763 | ApMeasureStars | DEBUG | Fitting star #14 (id=199) at fixed xpos=8.62, ypos=8.78

2024-11-26 21:12:08,777 | ApMeasureStars | DEBUG | Fitting star #15 (id=8) at fixed xpos=9.09, ypos=9.31

2024-11-26 21:12:08,788 | ApMeasureStars | DEBUG | Fitting star #16 (id=47) at fixed xpos=9.25, ypos=8.84

2024-11-26 21:12:08,801 | ApMeasureStars | DEBUG | Fitting star #17 (id=37) at fixed xpos=9.19, ypos=8.93

2024-11-26 21:12:08,816 | ApMeasureStars | DEBUG | Fitting star #18 (id=50) at fixed xpos=8.92, ypos=9.50

2024-11-26 21:12:08,825 | ApMeasureStars | DEBUG | Fitting star #19 (id=31) at fixed xpos=8.79, ypos=9.23

2024-11-26 21:12:08,839 | ApMeasureStars | DEBUG | Fitting star #20 (id=18) at fixed xpos=9.34, ypos=9.04

2024-11-26 21:12:08,851 | ApMeasureStars | DEBUG | Fitting star #21 (id=58) at fixed xpos=8.96, ypos=9.50

2024-11-26 21:12:08,860 | ApMeasureStars | DEBUG | Fitting star #22 (id=51) at fixed xpos=9.14, ypos=8.68

2024-11-26 21:12:08,870 | ApMeasureStars | DEBUG | Fitting star #23 (id=32) at fixed xpos=9.14, ypos=8.69

2024-11-26 21:12:08,879 | ApMeasureStars | DEBUG | Fitting star #24 (id=48) at fixed xpos=9.24, ypos=8.59

2024-11-26 21:12:08,891 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:08,892 | ApMeasureStars | DEBUG | Measured FWHM=3.676 +/- 0.314 pixels in direction x

2024-11-26 21:12:08,892 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:08,893 | ApMeasureStars | DEBUG | Measured FWHM=3.563 +/- 0.274 pixels in direction y

2024-11-26 21:12:08,893 | ApMeasureStars | INFO | Source average axis ratio = 1.032 +/- 0.119, overall sources are circular.

2024-11-26 21:12:08,893 | ApMeasureStars | DEBUG | Fitting completed in 0.294 s

(average 11.7 ms/star)

2024-11-26 21:12:08,893 | ApMeasureStars | DEBUG | Skipping plotting of star cutouts and best fits.

2024-11-26 21:12:08,894 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:12:08,894 | ApMeasureStars | DEBUG | Measured FWHM=3.652 +/- 0.272 pixels in direction both

2024-11-26 21:12:08,895 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:08,895 | ApMeasureStars | DEBUG | Measured FWHM=3.676 +/- 0.314 pixels in direction x

2024-11-26 21:12:08,896 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:08,896 | ApMeasureStars | DEBUG | Measured FWHM=3.563 +/- 0.274 pixels in direction y

2024-11-26 21:12:08,897 | ApFindStars | INFO | Median FWHM (over x and y) is 3.65 +/- 0.27 pixels using 48 data points

2024-11-26 21:12:08,897 | ApProcess | DEBUG | Initial star detection and fitting: FWHM=3.652 +/- 0.272 pixels using 48 stars.

2024-11-26 21:12:08,897 | ApProcess | DEBUG | Updating source searching using initial FWHM=3.652 +/- 0.272 pixels using 48 stars.

2024-11-26 21:12:08,897 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=3.65 pixels, threshold=7.0 BG sigma.

2024-11-26 21:12:09,196 | ApFindStars | INFO | Initial source list using FWHM=3.652 pixels, threshold=7.00 x BG stddev, found 266 sources.

2024-11-26 21:12:09,197 | ApFindStars | DEBUG | Sources with pixels exceeding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:12:09,197 | ApFindStars | DEBUG | There are 9 possibly saturated stars in the output source list.

2024-11-26 21:12:09,197 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:12:09,197 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:12:09,245 | ApFindStars | DEBUG | Image exposure time [seconds]: 300.00

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10
 phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:12:09,246 | ApFindStars | DEBUG | Brightest source (idx=0) generates 13977.91 ADU/s.

2024-11-26 21:12:09,246 | ApFindStars | DEBUG | Median source (idx=133) generates 17.01 ADU/s.

2024-11-26 21:12:09,246 | ApFindStars | DEBUG | Faintest source (idx=265) generates -0.58 ADU/s.

2024-11-26 21:12:09,246 | ApFindStars | DEBUG | Selecting 200 sources out of 266 to write to the output source list.

2024-11-26 21:12:09,246 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:12:09,247 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:12:09,249 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:12:09,249 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:12:09,249 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:12:09,249 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:12:09,250 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 191

2024-11-26 21:12:09,250 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 266

2024-11-26 21:12:09,251 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:12:09,251 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:12:09,251 | ApMeasureStars | DEBUG | Constructed a kdtree with 266 data points.

2024-11-26 21:12:09,254 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 22 stars from consideration.

2024-11-26 21:12:09,254 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 169 stars in 3056 row x 3056 column image.

2024-11-26 21:12:09,255 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:12:09,255 | ApMeasureStars | DEBUG | There are 167 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:12:09,256 | ApMeasureStars | DEBUG | In region CN found 56
 candidates for fitting:
 2024-11-26 21:12:09,256 | ApMeasureStars | DEBUG | In region TL found 27
 candidates for fitting:
 2024-11-26 21:12:09,257 | ApMeasureStars | DEBUG | In region TR found 30
 candidates for fitting:
 2024-11-26 21:12:09,259 | ApMeasureStars | DEBUG | In region BR found 29
 candidates for fitting:
 2024-11-26 21:12:09,260 | ApMeasureStars | DEBUG | In region BL found 25
 candidates for fitting:
 2024-11-26 21:12:09,261 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
 min=0.00, max=51606.07
 2024-11-26 21:12:09,263 | ApMeasureStars | INFO | Starting to fit
 Const2D+Gaussian2D model to 25 star cutouts.
 2024-11-26 21:12:09,264 | ApMeasureStars | DEBUG | Fitting star #0 (id=159) at
 fixed xpos=9.07, ypos=8.77
 2024-11-26 21:12:09,276 | ApMeasureStars | DEBUG | Fitting star #1 (id=121) at
 fixed xpos=9.09, ypos=8.90
 2024-11-26 21:12:09,289 | ApMeasureStars | DEBUG | Fitting star #2 (id=109) at
 fixed xpos=9.25, ypos=8.96
 2024-11-26 21:12:09,303 | ApMeasureStars | DEBUG | Fitting star #3 (id=200) at
 fixed xpos=8.91, ypos=9.10
 2024-11-26 21:12:09,313 | ApMeasureStars | DEBUG | Fitting star #4 (id=183) at
 fixed xpos=9.45, ypos=9.13
 2024-11-26 21:12:09,326 | ApMeasureStars | DEBUG | Fitting star #5 (id=262) at
 fixed xpos=8.76, ypos=9.47
 2024-11-26 21:12:09,336 | ApMeasureStars | DEBUG | Fitting star #6 (id=236) at
 fixed xpos=8.60, ypos=8.65
 2024-11-26 21:12:09,347 | ApMeasureStars | DEBUG | Fitting star #7 (id=249) at
 fixed xpos=9.48, ypos=8.73
 2024-11-26 21:12:09,356 | ApMeasureStars | DEBUG | Fitting star #8 (id=211) at
 fixed xpos=8.91, ypos=8.67
 2024-11-26 21:12:09,368 | ApMeasureStars | DEBUG | Fitting star #9 (id=261) at
 fixed xpos=9.47, ypos=9.49
 2024-11-26 21:12:09,381 | ApMeasureStars | DEBUG | Fitting star #10 (id=212) at
 fixed xpos=8.91, ypos=9.05
 2024-11-26 21:12:09,394 | ApMeasureStars | DEBUG | Fitting star #11 (id=246) at
 fixed xpos=8.91, ypos=9.06
 2024-11-26 21:12:09,409 | ApMeasureStars | DEBUG | Fitting star #12 (id=259) at
 fixed xpos=9.39, ypos=9.45
 2024-11-26 21:12:09,417 | ApMeasureStars | DEBUG | Fitting star #13 (id=221) at
 fixed xpos=8.86, ypos=8.63
 2024-11-26 21:12:09,431 | ApMeasureStars | DEBUG | Fitting star #14 (id=224) at
 fixed xpos=8.63, ypos=8.78
 2024-11-26 21:12:09,444 | ApMeasureStars | DEBUG | Fitting star #15 (id=10) at
 fixed xpos=9.08, ypos=9.30
 2024-11-26 21:12:09,453 | ApMeasureStars | DEBUG | Fitting star #16 (id=50) at
 fixed xpos=9.24, ypos=8.85
 2024-11-26 21:12:09,465 | ApMeasureStars | DEBUG | Fitting star #17 (id=41) at
 fixed xpos=9.18, ypos=8.93
 2024-11-26 21:12:09,480 | ApMeasureStars | DEBUG | Fitting star #18 (id=52) at
 fixed xpos=8.92, ypos=9.48
 2024-11-26 21:12:09,490 | ApMeasureStars | DEBUG | Fitting star #19 (id=34) at
 fixed xpos=8.80, ypos=9.23
 2024-11-26 21:12:09,504 | ApMeasureStars | DEBUG | Fitting star #20 (id=21) at
 fixed xpos=9.33, ypos=9.04
 2024-11-26 21:12:09,518 | ApMeasureStars | DEBUG | Fitting star #21 (id=58) at
 fixed xpos=8.96, ypos=8.51
 2024-11-26 21:12:09,527 | ApMeasureStars | DEBUG | Fitting star #22 (id=53) at
 fixed xpos=9.13, ypos=8.68
 2024-11-26 21:12:09,537 | ApMeasureStars | DEBUG | Fitting star #23 (id=35) at
 fixed xpos=9.13, ypos=8.70
 2024-11-26 21:12:09,550 | ApMeasureStars | DEBUG | Fitting star #24 (id=51) at
 fixed xpos=9.23, ypos=8.60
 2024-11-26 21:12:09,559 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=x) using 24 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:12:09,559 | ApMeasureStars | DEBUG | Measured FWHM=3.656 +/- 0.220
 pixels in direction x
 2024-11-26 21:12:09,560 | ApMeasureStars | DEBUG | Estimating median FWHM
 (direction=y) using 24 FWHM measurements (25 OK fits before clipping).
 2024-11-26 21:12:09,560 | ApMeasureStars | DEBUG | Measured FWHM=3.548 +/- 0.338
 pixels in direction y
 2024-11-26 21:12:09,560 | ApMeasureStars | INFO | Source average axis ratio =
 1.031 +/- 0.116, overall sources are circular.
 2024-11-26 21:12:09,560 | ApMeasureStars | DEBUG | Fitting completed in 0.297 s
 (average 11.9 ms/star)

2024-11-26 21:12:09,648 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:12:09,648 | ApMeasureStars | DEBUG | Measured FWHM=3.626 +/- 0.287 pixels in direction both

2024-11-26 21:12:09,649 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:09,649 | ApMeasureStars | DEBUG | Measured FWHM=3.656 +/- 0.220 pixels in direction x

2024-11-26 21:12:09,650 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:09,650 | ApMeasureStars | DEBUG | Measured FWHM=3.548 +/- 0.338 pixels in direction y

2024-11-26 21:12:10,896 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-3.png

2024-11-26 21:12:10,897 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:12:10,897 | ApMeasureStars | DEBUG | Measured FWHM=3.626 +/- 0.287 pixels in direction both

2024-11-26 21:12:10,898 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:10,898 | ApMeasureStars | DEBUG | Measured FWHM=3.656 +/- 0.220 pixels in direction x

2024-11-26 21:12:10,899 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:10,899 | ApMeasureStars | DEBUG | Measured FWHM=3.548 +/- 0.338 pixels in direction y

2024-11-26 21:12:10,899 | ApFindStars | INFO | Median FWHM (over x and y) is 3.63 +/- 0.29 pixels using 48 data points

2024-11-26 21:12:10,899 | ApProcess | DEBUG | Final star detection and fitting: FWHM=3.626 +/- 0.287 pixels using 48 stars.

2024-11-26 21:12:11,844 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to ./SourceLists/imshow-M101_Supernova_2023ixf-300s-Blue-3.png

2024-11-26 21:12:11,844 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:12:11,845 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (266, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Blue', 'Filter used')}

2024-11-26 21:12:11,845 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:12:11,846 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:12:11,846 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:12:11,846 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:12:11,846 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:12:11,847 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:12:11,847 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:12:11,847 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:12:11,847 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:12:11,847 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:12:11,847 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:12:11,848 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:12:11,848 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:12:11,848 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

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2024-11-26 21:12:11,849 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-3.yaml
2024-11-26 21:12:11,852 | ApFindStars | DEBUG | Wrote ds9-format region file to
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-3.reg
2024-11-26 21:12:11,853 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:12:11,853 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz', 'Name of image file
searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'),
'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (266, 'Number of
sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final
photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),
'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Blue',
'Filter used')}}
2024-11-26 21:12:11,853 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:12:11,854 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:12:11,854 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:12:11,855 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:12:11,878 | ApProcess | DEBUG | Confirming output srclist
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits was created.
2024-11-26 21:12:11,878 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits as
clean_star_detection=True
2024-11-26 21:12:11,883 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits
2024-11-26 21:12:11,883 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
2024-11-26 21:12:12,149 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:12:12,152 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits
2024-11-26 21:12:12,152 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:12:12,152 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:12:12,152 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:12:12,153 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:12:12,153 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:12:23,557 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:12:23,682 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:12:23,683 | AstroPhotography.core.ApAstrometry | DEBUG | Setting

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CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:12:23,683 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:12:23,683 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:12:23,683 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:12:23,684 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:12:23,684 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.768525715, 'RA of reference point')
2024-11-26 21:12:23,684 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4176364229, 'DEC of reference point')
2024-11-26 21:12:23,684 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:12:23,684 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:12:23,685 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:12:23,685 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:12:23,685 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.49475579249e-06, 'Transformation matrix')
2024-11-26 21:12:23,685 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175217638519, 'no comment')
2024-11-26 21:12:23,686 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175217638519, 'no comment')
2024-11-26 21:12:23,686 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.49475579249e-06, 'no comment')
2024-11-26 21:12:23,686 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:12:23,686 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:12:23,841 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits
2024-11-26 21:12:23,842 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/src1ist-M101_Supernova_2023ixf-300s-Blue-3.fits with
WCS soluton.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:12:23,859 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:12:24,116 | ApProcess | DEBUG |
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2024-11-26 21:12:24,117 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
2024-11-26 21:12:24,117 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:24,117 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:24,118 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:24,118 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:24,118 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:12:24,118 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
src1ist=./SourceLists/src1ist-M101_Supernova_2023ixf-300s-Green-1.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-1.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-1.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-1.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Green-1.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits
2024-11-26 21:12:24,118 | ApProcess | DEBUG | Removing existing
./SourceLists/src1ist-M101_Supernova_2023ixf-300s-Green-1.fits because
clean_star_detection=True
2024-11-26 21:12:24,119 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-1.reg because

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clean_star_detection=True
2024-11-26 21:12:24,119 | ApProcess | DEBUG | Removing existing
./SourceLists/implot-M101_Supernova_2023ixf-300s-Green-1.png because
clean_star_detection=True
2024-11-26 21:12:24,119 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-1.png because
clean_star_detection=True
2024-11-26 21:12:24,120 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Green-1.fits
2024-11-26 21:12:24,120 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
2024-11-26 21:12:24,375 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:12:24,443 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67357.68, median=234.65
2024-11-26 21:12:25,275 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=234.820, median=234.439, stddev=23.422
2024-11-26 21:12:28,583 | ApFindStars | DEBUG | Source-masked image stats:
mean=233.577, median=233.455, stddev=22.624
2024-11-26 21:12:28,583 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:12:28,584 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:12:28,766 | ApFindStars | DEBUG | Retaining 14 possibly saturated
stars in source searching and photometry.
2024-11-26 21:12:28,767 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:12:29,006 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 251 sources.
2024-11-26 21:12:29,007 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:12:29,007 | ApFindStars | DEBUG | There are 11 possibly saturated
stars in the output source list.
2024-11-26 21:12:29,007 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:29,008 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:29,047 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:12:29,048 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 15539.66 ADU/s.
2024-11-26 21:12:29,048 | ApFindStars | DEBUG | Median source (idx=125)
generates 30.15 ADU/s.
2024-11-26 21:12:29,048 | ApFindStars | DEBUG | Faintest source (idx=250)
generates -6.25 ADU/s.
2024-11-26 21:12:29,049 | ApFindStars | DEBUG | Selecting 200 sources out of 251
to write to the output source list.
2024-11-26 21:12:29,049 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:29,049 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:29,952 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/implot-M101_Supernova_2023ixf-300s-Green-1.png
2024-11-26 21:12:29,953 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:12:29,954 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:12:29,954 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:12:29,954 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:12:29,955 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 189
2024-11-26 21:12:29,955 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 251
2024-11-26 21:12:29,955 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:12:29,955 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:12:29,956 | ApMeasureStars | DEBUG | Constructed a kdtree with 251
data points.
2024-11-26 21:12:29,959 | ApMeasureStars | DEBUG | Nearest neighbor filtering

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removed 12 stars from consideration.

2024-11-26 21:12:29,959 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 177 stars in 3056 row x 3056 column image.

2024-11-26 21:12:29,959 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:12:29,960 | ApMeasureStars | DEBUG | There are 176 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:12:29,960 | ApMeasureStars | DEBUG | In region CN found 53 candidates for fitting:

2024-11-26 21:12:29,961 | ApMeasureStars | DEBUG | In region TL found 29 candidates for fitting:

2024-11-26 21:12:29,962 | ApMeasureStars | DEBUG | In region TR found 35 candidates for fitting:

2024-11-26 21:12:29,963 | ApMeasureStars | DEBUG | In region BR found 32 candidates for fitting:

2024-11-26 21:12:29,965 | ApMeasureStars | DEBUG | In region BL found 27 candidates for fitting:

2024-11-26 21:12:29,967 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=166.69, max=52511.56

2024-11-26 21:12:29,969 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:12:29,969 | ApMeasureStars | DEBUG | Fitting star #0 (id=133) at fixed xpos=9.03, ypos=8.71

2024-11-26 21:12:29,978 | ApMeasureStars | DEBUG | Fitting star #1 (id=103) at fixed xpos=9.00, ypos=8.80

2024-11-26 21:12:29,987 | ApMeasureStars | DEBUG | Fitting star #2 (id=94) at fixed xpos=9.06, ypos=8.94

2024-11-26 21:12:29,995 | ApMeasureStars | DEBUG | Fitting star #3 (id=165) at fixed xpos=9.37, ypos=9.27

2024-11-26 21:12:30,003 | ApMeasureStars | DEBUG | Fitting star #4 (id=151) at fixed xpos=9.38, ypos=9.06

2024-11-26 21:12:30,010 | ApMeasureStars | DEBUG | Fitting star #5 (id=249) at fixed xpos=9.41, ypos=9.48

2024-11-26 21:12:30,021 | ApMeasureStars | DEBUG | Fitting star #6 (id=217) at fixed xpos=9.46, ypos=9.03

2024-11-26 21:12:30,029 | ApMeasureStars | DEBUG | Fitting star #7 (id=192) at fixed xpos=8.55, ypos=9.25

2024-11-26 21:12:30,038 | ApMeasureStars | DEBUG | Fitting star #8 (id=231) at fixed xpos=9.20, ypos=9.46

2024-11-26 21:12:30,050 | ApMeasureStars | DEBUG | Fitting star #9 (id=181) at fixed xpos=9.45, ypos=9.08

2024-11-26 21:12:30,064 | ApMeasureStars | DEBUG | Fitting star #10 (id=229) at fixed xpos=8.84, ypos=9.36

2024-11-26 21:12:30,073 | ApMeasureStars | DEBUG | Fitting star #11 (id=245) at fixed xpos=9.30, ypos=8.86

2024-11-26 21:12:30,083 | ApMeasureStars | DEBUG | Fitting star #12 (id=208) at fixed xpos=8.68, ypos=8.87

2024-11-26 21:12:30,092 | ApMeasureStars | DEBUG | Fitting star #13 (id=205) at fixed xpos=8.82, ypos=8.76

2024-11-26 21:12:30,101 | ApMeasureStars | DEBUG | Fitting star #14 (id=222) at fixed xpos=8.87, ypos=8.85

2024-11-26 21:12:30,110 | ApMeasureStars | DEBUG | Fitting star #15 (id=5) at fixed xpos=8.96, ypos=8.97

2024-11-26 21:12:30,121 | ApMeasureStars | DEBUG | Fitting star #16 (id=44) at fixed xpos=9.12, ypos=8.63

2024-11-26 21:12:30,130 | ApMeasureStars | DEBUG | Fitting star #17 (id=36) at fixed xpos=9.14, ypos=9.50

2024-11-26 21:12:30,141 | ApMeasureStars | DEBUG | Fitting star #18 (id=47) at fixed xpos=8.80, ypos=9.18

2024-11-26 21:12:30,149 | ApMeasureStars | DEBUG | Fitting star #19 (id=28) at fixed xpos=8.64, ypos=9.00

2024-11-26 21:12:30,157 | ApMeasureStars | DEBUG | Fitting star #20 (id=53) at fixed xpos=8.75, ypos=8.53

2024-11-26 21:12:30,169 | ApMeasureStars | DEBUG | Fitting star #21 (id=49) at fixed xpos=8.83, ypos=8.88

2024-11-26 21:12:30,180 | ApMeasureStars | DEBUG | Fitting star #22 (id=29) at fixed xpos=8.93, ypos=8.64

2024-11-26 21:12:30,188 | ApMeasureStars | DEBUG | Fitting star #23 (id=45) at fixed xpos=9.06, ypos=8.58

2024-11-26 21:12:30,195 | ApMeasureStars | DEBUG | Fitting star #24 (id=43) at fixed xpos=8.51, ypos=8.77

2024-11-26 21:12:30,206 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:30,207 | ApMeasureStars | DEBUG | Measured FWHM=3.819 +/- 0.345 pixels in direction x

2024-11-26 21:12:30,208 | ApMeasureStars | DEBUG | Estimating median FWHM

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(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:30,208 | ApMeasureStars | DEBUG | Measured FWHM=3.471 +/- 0.166
pixels in direction y
2024-11-26 21:12:30,208 | ApMeasureStars | INFO | Source average axis ratio =
1.100 +/- 0.112, overall sources are circular.
2024-11-26 21:12:30,209 | ApMeasureStars | DEBUG | Fitting completed in 0.240 s
(average 9.6 ms/star)
2024-11-26 21:12:30,209 | ApMeasureStars | DEBUG | Skipping plotting of star
cutouts and best fits.
2024-11-26 21:12:30,209 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:12:30,210 | ApMeasureStars | DEBUG | Measured FWHM=3.684 +/- 0.354
pixels in direction both
2024-11-26 21:12:30,210 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:30,211 | ApMeasureStars | DEBUG | Measured FWHM=3.819 +/- 0.345
pixels in direction x
2024-11-26 21:12:30,211 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:30,211 | ApMeasureStars | DEBUG | Measured FWHM=3.471 +/- 0.166
pixels in direction y
2024-11-26 21:12:30,212 | ApFindStars | INFO | Median FWHM (over x and y) is
3.68 +/- 0.35 pixels using 50 data points
2024-11-26 21:12:30,212 | ApProcess | DEBUG | Initial star detection and
fitting: FWHM=3.684 +/- 0.354 pixels using 50 stars.
2024-11-26 21:12:30,212 | ApProcess | DEBUG | Updating source searching using
initial FWHM=3.684 +/- 0.354 pixels using 50 stars.
2024-11-26 21:12:30,212 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.68 pixels, threshold=7.0 BG sigma.
2024-11-26 21:12:30,509 | ApFindStars | INFO | Initial source list using
FWHM=3.684 pixels, threshold=7.00 x BG stddev, found 286 sources.
2024-11-26 21:12:30,510 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:12:30,510 | ApFindStars | DEBUG | There are 8 possibly saturated
stars in the output source list.
2024-11-26 21:12:30,510 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:30,511 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:30,556 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:12:30,557 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 15190.42 ADU/s.
2024-11-26 21:12:30,557 | ApFindStars | DEBUG | Median source (idx=143)
generates 22.93 ADU/s.
2024-11-26 21:12:30,557 | ApFindStars | DEBUG | Faintest source (idx=285)
generates -7.06 ADU/s.
2024-11-26 21:12:30,557 | ApFindStars | DEBUG | Selecting 200 sources out of 286
to write to the output source list.
2024-11-26 21:12:30,558 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:30,558 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:30,560 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:12:30,560 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:12:30,561 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:12:30,561 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:12:30,561 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 192
2024-11-26 21:12:30,562 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 286
2024-11-26 21:12:30,562 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:12:30,562 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:12:30,562 | ApMeasureStars | DEBUG | Constructed a kdtree with 286
data points.
2024-11-26 21:12:30,565 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 21 stars from consideration.

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2024-11-26 21:12:30,565 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 171 stars in 3056 row x 3056 column image.

2024-11-26 21:12:30,566 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:12:30,566 | ApMeasureStars | DEBUG | There are 170 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:12:30,567 | ApMeasureStars | DEBUG | In region CN found 54 candidates for fitting:

2024-11-26 21:12:30,567 | ApMeasureStars | DEBUG | In region TL found 28 candidates for fitting:

2024-11-26 21:12:30,569 | ApMeasureStars | DEBUG | In region TR found 31 candidates for fitting:

2024-11-26 21:12:30,570 | ApMeasureStars | DEBUG | In region BR found 31 candidates for fitting:

2024-11-26 21:12:30,571 | ApMeasureStars | DEBUG | In region BL found 26 candidates for fitting:

2024-11-26 21:12:30,573 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=166.69, max=52158.44

2024-11-26 21:12:30,574 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:12:30,574 | ApMeasureStars | DEBUG | Fitting star #0 (id=161) at fixed xpos=9.02, ypos=8.72

2024-11-26 21:12:30,584 | ApMeasureStars | DEBUG | Fitting star #1 (id=118) at fixed xpos=9.00, ypos=8.81

2024-11-26 21:12:30,593 | ApMeasureStars | DEBUG | Fitting star #2 (id=106) at fixed xpos=9.06, ypos=8.94

2024-11-26 21:12:30,601 | ApMeasureStars | DEBUG | Fitting star #3 (id=196) at fixed xpos=9.36, ypos=9.26

2024-11-26 21:12:30,609 | ApMeasureStars | DEBUG | Fitting star #4 (id=179) at fixed xpos=9.37, ypos=9.06

2024-11-26 21:12:30,616 | ApMeasureStars | DEBUG | Fitting star #5 (id=285) at fixed xpos=9.40, ypos=9.46

2024-11-26 21:12:30,627 | ApMeasureStars | DEBUG | Fitting star #6 (id=255) at fixed xpos=9.45, ypos=9.03

2024-11-26 21:12:30,635 | ApMeasureStars | DEBUG | Fitting star #7 (id=227) at fixed xpos=8.56, ypos=9.24

2024-11-26 21:12:30,644 | ApMeasureStars | DEBUG | Fitting star #8 (id=269) at fixed xpos=9.20, ypos=9.44

2024-11-26 21:12:30,655 | ApMeasureStars | DEBUG | Fitting star #9 (id=213) at fixed xpos=9.46, ypos=9.08

2024-11-26 21:12:30,668 | ApMeasureStars | DEBUG | Fitting star #10 (id=281) at fixed xpos=9.29, ypos=8.86

2024-11-26 21:12:30,677 | ApMeasureStars | DEBUG | Fitting star #11 (id=244) at fixed xpos=8.70, ypos=8.88

2024-11-26 21:12:30,686 | ApMeasureStars | DEBUG | Fitting star #12 (id=241) at fixed xpos=8.82, ypos=8.77

2024-11-26 21:12:30,695 | ApMeasureStars | DEBUG | Fitting star #13 (id=260) at fixed xpos=8.88, ypos=8.85

2024-11-26 21:12:30,704 | ApMeasureStars | DEBUG | Fitting star #14 (id=163) at fixed xpos=9.30, ypos=8.89

2024-11-26 21:12:30,713 | ApMeasureStars | DEBUG | Fitting star #15 (id=9) at fixed xpos=8.96, ypos=8.97

2024-11-26 21:12:30,724 | ApMeasureStars | DEBUG | Fitting star #16 (id=48) at fixed xpos=9.11, ypos=8.64

2024-11-26 21:12:30,733 | ApMeasureStars | DEBUG | Fitting star #17 (id=42) at fixed xpos=9.14, ypos=8.51

2024-11-26 21:12:30,742 | ApMeasureStars | DEBUG | Fitting star #18 (id=51) at fixed xpos=8.81, ypos=9.17

2024-11-26 21:12:30,751 | ApMeasureStars | DEBUG | Fitting star #19 (id=34) at fixed xpos=8.65, ypos=9.00

2024-11-26 21:12:30,759 | ApMeasureStars | DEBUG | Fitting star #20 (id=57) at fixed xpos=8.75, ypos=8.54

2024-11-26 21:12:30,771 | ApMeasureStars | DEBUG | Fitting star #21 (id=53) at fixed xpos=8.83, ypos=8.88

2024-11-26 21:12:30,784 | ApMeasureStars | DEBUG | Fitting star #22 (id=35) at fixed xpos=8.94, ypos=8.65

2024-11-26 21:12:30,791 | ApMeasureStars | DEBUG | Fitting star #23 (id=49) at fixed xpos=9.06, ypos=8.59

2024-11-26 21:12:30,799 | ApMeasureStars | DEBUG | Fitting star #24 (id=47) at fixed xpos=8.51, ypos=8.77

2024-11-26 21:12:30,810 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:30,811 | ApMeasureStars | DEBUG | Measured FWHM=3.813 +/- 0.348 pixels in direction x

2024-11-26 21:12:30,812 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

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2024-11-26 21:12:30,812 | ApMeasureStars | DEBUG | Measured FWHM=3.472 +/- 0.195
pixels in direction y
2024-11-26 21:12:30,812 | ApMeasureStars | INFO | Source average axis ratio =
1.098 +/- 0.118, overall sources are circular.
2024-11-26 21:12:30,812 | ApMeasureStars | DEBUG | Fitting completed in 0.238 s
(average 9.5 ms/star)
2024-11-26 21:12:30,900 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:12:30,900 | ApMeasureStars | DEBUG | Measured FWHM=3.671 +/- 0.351
pixels in direction both
2024-11-26 21:12:30,901 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:30,901 | ApMeasureStars | DEBUG | Measured FWHM=3.813 +/- 0.348
pixels in direction x
2024-11-26 21:12:30,902 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:30,902 | ApMeasureStars | DEBUG | Measured FWHM=3.472 +/- 0.195
pixels in direction y
2024-11-26 21:12:32,130 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-1.png
2024-11-26 21:12:32,131 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:12:32,131 | ApMeasureStars | DEBUG | Measured FWHM=3.671 +/- 0.351
pixels in direction both
2024-11-26 21:12:32,131 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:32,132 | ApMeasureStars | DEBUG | Measured FWHM=3.813 +/- 0.348
pixels in direction x
2024-11-26 21:12:32,132 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:12:32,133 | ApMeasureStars | DEBUG | Measured FWHM=3.472 +/- 0.195
pixels in direction y
2024-11-26 21:12:32,133 | ApFindStars | INFO | Median FWHM (over x and y) is
3.67 +/- 0.35 pixels using 49 data points
2024-11-26 21:12:32,133 | ApProcess | DEBUG | Final star detection and fitting:
FWHM=3.671 +/- 0.351 pixels using 49 stars.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:323:
RuntimeWarning: More than 20 figures have been opened. Figures created through
the pyplot interface (`matplotlib.pyplot.figure`) are retained until explicitly
closed and may consume too much memory. (To control this warning, see the
rcParam `figure.max_open_warning`). Consider using `matplotlib.pyplot.close()`.
    fig, ax = plt.subplots()
2024-11-26 21:12:33,053 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-1.png
2024-11-26 21:12:33,053 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:12:33,054 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz', 'Name of image file
searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'),
'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (286, 'Number of
sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final
photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),
'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Green',
'Filter used')}}
2024-11-26 21:12:33,054 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:12:33,055 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:12:33,055 | ApFindStars | WARNING | Keyword OBJECT not found, not
written to quality report.
2024-11-26 21:12:33,055 | ApFindStars | WARNING | Keyword TELESCOP not found,
not written to quality report.
2024-11-26 21:12:33,056 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.
2024-11-26 21:12:33,056 | ApFindStars | WARNING | Keyword EXPOSURE not found,
not written to quality report.
2024-11-26 21:12:33,056 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.
2024-11-26 21:12:33,056 | ApFindStars | WARNING | Keyword EGAIN not found, not
written to quality report.

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2024-11-26 21:12:33,057 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:12:33,057 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:12:33,057 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:12:33,057 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:12:33,057 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:12:33,057 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:12:33,059 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Green-1.yaml

2024-11-26 21:12:33,062 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-1.reg

2024-11-26 21:12:33,062 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:12:33,063 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (286, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Green', 'Filter used')}

2024-11-26 21:12:33,063 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:12:33,064 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:12:33,064 | ApFindStars | DEBUG | Converting python 0-based coordinates to FITS 1-based pixel coordinates for XY table.

2024-11-26 21:12:33,064 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-1.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:12:33,088 | ApProcess | DEBUG | Confirming output srclist ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-1.fits was created.

2024-11-26 21:12:33,088 | ApProcess | DEBUG | Removing existing ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits as clean_star_detection=True

2024-11-26 21:12:33,093 | ApProcess | DEBUG | Performing astrometry, output navigated image: ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits

2024-11-26 21:12:33,093 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz

2024-11-26 21:12:33,339 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D BITPIX=-32 image with 3056 columns, 3056 rows

2024-11-26 21:12:33,342 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-1.fits

2024-11-26 21:12:33,342 | AstroPhotography.core.ApAstrometry | DEBUG | The source list was generated from this input image. Proceeding.

2024-11-26 21:12:33,342 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated center_ra=210.802, center_dec=54.349, radius=8.000 deg.

2024-11-26 21:12:33,342 | AstroPhotography.core.ApAstrometry | WARNING | Could not generate scale hints.

2024-11-26 21:12:33,343 | AstroPhotography.core.ApAstrometry | DEBUG | Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':

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54.34944444444444, 'radius': 8}
2024-11-26 21:12:33,343 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Green-1.fits with 200
positions (attempt #1)

Solving...

2024-11-26 21:12:39,090 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:12:39,210 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:12:39,211 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:12:39,211 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:12:39,211 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:12:39,212 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:12:39,212 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:12:39,212 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.752042779, 'RA of reference point')
2024-11-26 21:12:39,212 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4164198315, 'DEC of reference point')
2024-11-26 21:12:39,213 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:12:39,213 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:12:39,213 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:12:39,213 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:12:39,214 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.54696541574e-06, 'Transformation matrix')
2024-11-26 21:12:39,214 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175236287304, 'no comment')
2024-11-26 21:12:39,214 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175236287304, 'no comment')
2024-11-26 21:12:39,214 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.54696541574e-06, 'no comment')
2024-11-26 21:12:39,215 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:12:39,215 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:12:39,354 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits
2024-11-26 21:12:39,355 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Green-1.fits
with WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:12:39,372 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:12:39,623 | ApProcess | DEBUG |

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2024-11-26 21:12:39,624 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
2024-11-26 21:12:39,624 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:39,624 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:39,624 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:39,625 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:12:39,625 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:12:39,625 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:12:39,625 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz,

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input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-2.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-2.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-2.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Green-2.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits
2024-11-26 21:12:39,625 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits because
clean_star_detection=True
2024-11-26 21:12:39,626 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-2.reg because
clean_star_detection=True
2024-11-26 21:12:39,626 | ApProcess | DEBUG | Removing existing
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-2.png because
clean_star_detection=True
2024-11-26 21:12:39,627 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-2.png because
clean_star_detection=True
2024-11-26 21:12:39,627 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits
2024-11-26 21:12:39,628 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
2024-11-26 21:12:39,892 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:12:39,971 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67348.50, median=177.25
2024-11-26 21:12:40,814 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=177.320, median=177.173, stddev=21.884
2024-11-26 21:12:43,887 | ApFindStars | DEBUG | Source-masked image stats:
mean=176.877, median=176.804, stddev=21.651
2024-11-26 21:12:43,888 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:12:43,888 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:12:44,067 | ApFindStars | DEBUG | Retaining 11 possibly saturated
stars in source searching and photometry.
2024-11-26 21:12:44,068 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:12:44,301 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 209 sources.
2024-11-26 21:12:44,302 | ApFindStars | DEBUG | Sources with pixels exceeding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:12:44,302 | ApFindStars | DEBUG | There are 8 possibly saturated
stars in the output source list.
2024-11-26 21:12:44,302 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:44,303 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:44,337 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
2024-11-26 21:12:44,339 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14715.41 ADU/s.
2024-11-26 21:12:44,339 | ApFindStars | DEBUG | Median source (idx=104)
generates 19.76 ADU/s.
2024-11-26 21:12:44,339 | ApFindStars | DEBUG | Faintest source (idx=208)
generates 1.11 ADU/s.
2024-11-26 21:12:44,339 | ApFindStars | DEBUG | Selecting 200 sources out of 209
to write to the output source list.
2024-11-26 21:12:44,339 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:12:44,340 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:12:45,250 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-2.png
2024-11-26 21:12:45,251 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:12:45,252 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:12:45,252 | ApMeasureStars | DEBUG | Count based weighting factors
will be used in fitting?: True
2024-11-26 21:12:45,252 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True
2024-11-26 21:12:45,253 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 192

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2024-11-26 21:12:45,254 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 209

2024-11-26 21:12:45,254 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:12:45,254 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:12:45,254 | ApMeasureStars | DEBUG | Constructed a kdtree with 209 data points.

2024-11-26 21:12:45,257 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 4 stars from consideration.

2024-11-26 21:12:45,258 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 188 stars in 3056 row x 3056 column image.

2024-11-26 21:12:45,258 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:12:45,258 | ApMeasureStars | DEBUG | There are 185 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:12:45,259 | ApMeasureStars | DEBUG | In region CN found 48 candidates for fitting:

2024-11-26 21:12:45,260 | ApMeasureStars | DEBUG | In region TL found 26 candidates for fitting:

2024-11-26 21:12:45,261 | ApMeasureStars | DEBUG | In region TR found 40 candidates for fitting:

2024-11-26 21:12:45,263 | ApMeasureStars | DEBUG | In region BR found 38 candidates for fitting:

2024-11-26 21:12:45,264 | ApMeasureStars | DEBUG | In region BL found 33 candidates for fitting:

2024-11-26 21:12:45,266 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=46847.90

2024-11-26 21:12:45,267 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:12:45,268 | ApMeasureStars | DEBUG | Fitting star #0 (id=131) at fixed xpos=9.19, ypos=9.39

2024-11-26 21:12:45,278 | ApMeasureStars | DEBUG | Fitting star #1 (id=102) at fixed xpos=9.19, ypos=8.55

2024-11-26 21:12:45,286 | ApMeasureStars | DEBUG | Fitting star #2 (id=91) at fixed xpos=9.34, ypos=8.66

2024-11-26 21:12:45,295 | ApMeasureStars | DEBUG | Fitting star #3 (id=151) at fixed xpos=8.55, ypos=8.86

2024-11-26 21:12:45,309 | ApMeasureStars | DEBUG | Fitting star #4 (id=143) at fixed xpos=8.61, ypos=8.68

2024-11-26 21:12:45,320 | ApMeasureStars | DEBUG | Fitting star #5 (id=167) at fixed xpos=8.82, ypos=9.43

2024-11-26 21:12:45,329 | ApMeasureStars | DEBUG | Fitting star #6 (id=207) at fixed xpos=8.69, ypos=8.92

2024-11-26 21:12:45,338 | ApMeasureStars | DEBUG | Fitting star #7 (id=188) at fixed xpos=8.71, ypos=8.57

2024-11-26 21:12:45,350 | ApMeasureStars | DEBUG | Fitting star #8 (id=168) at fixed xpos=8.88, ypos=8.83

2024-11-26 21:12:45,361 | ApMeasureStars | DEBUG | Fitting star #9 (id=197) at fixed xpos=8.50, ypos=8.93

2024-11-26 21:12:45,371 | ApMeasureStars | DEBUG | Fitting star #10 (id=169) at fixed xpos=9.23, ypos=9.48

2024-11-26 21:12:45,380 | ApMeasureStars | DEBUG | Fitting star #11 (id=196) at fixed xpos=9.11, ypos=8.84

2024-11-26 21:12:45,393 | ApMeasureStars | DEBUG | Fitting star #12 (id=203) at fixed xpos=8.53, ypos=9.37

2024-11-26 21:12:45,402 | ApMeasureStars | DEBUG | Fitting star #13 (id=178) at fixed xpos=8.94, ypos=9.46

2024-11-26 21:12:45,411 | ApMeasureStars | DEBUG | Fitting star #14 (id=176) at fixed xpos=9.09, ypos=9.34

2024-11-26 21:12:45,425 | ApMeasureStars | DEBUG | Fitting star #15 (id=4) at fixed xpos=9.15, ypos=8.72

2024-11-26 21:12:45,433 | ApMeasureStars | DEBUG | Fitting star #16 (id=45) at fixed xpos=9.33, ypos=9.44

2024-11-26 21:12:45,442 | ApMeasureStars | DEBUG | Fitting star #17 (id=41) at fixed xpos=9.35, ypos=9.33

2024-11-26 21:12:45,450 | ApMeasureStars | DEBUG | Fitting star #18 (id=47) at fixed xpos=9.02, ypos=8.93

2024-11-26 21:12:45,459 | ApMeasureStars | DEBUG | Fitting star #19 (id=32) at fixed xpos=8.89, ypos=8.77

2024-11-26 21:12:45,467 | ApMeasureStars | DEBUG | Fitting star #20 (id=18) at fixed xpos=9.35, ypos=8.85

2024-11-26 21:12:45,478 | ApMeasureStars | DEBUG | Fitting star #21 (id=53) at fixed xpos=8.99, ypos=9.24

2024-11-26 21:12:45,486 | ApMeasureStars | DEBUG | Fitting star #22 (id=50) at fixed xpos=9.04, ypos=8.66

2024-11-26 21:12:45,500 | ApMeasureStars | DEBUG | Fitting star #23 (id=33) at fixed xpos=9.15, ypos=9.44

2024-11-26 21:12:45,512 | ApMeasureStars | DEBUG | Fitting star #24 (id=46) at fixed xpos=9.26, ypos=9.38

2024-11-26 21:12:45,524 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:45,525 | ApMeasureStars | DEBUG | Measured FWHM=3.949 +/- 0.334 pixels in direction x

2024-11-26 21:12:45,525 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:45,526 | ApMeasureStars | DEBUG | Measured FWHM=3.692 +/- 0.122 pixels in direction y

2024-11-26 21:12:45,526 | ApMeasureStars | INFO | Source average axis ratio = 1.070 +/- 0.097, overall sources are circular.

2024-11-26 21:12:45,526 | ApMeasureStars | DEBUG | Fitting completed in 0.259 s (average 10.3 ms/star)

2024-11-26 21:12:45,526 | ApMeasureStars | DEBUG | Skipping plotting of star cutouts and best fits.

2024-11-26 21:12:45,527 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:12:45,527 | ApMeasureStars | DEBUG | Measured FWHM=3.773 +/- 0.248 pixels in direction both

2024-11-26 21:12:45,527 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:45,528 | ApMeasureStars | DEBUG | Measured FWHM=3.949 +/- 0.334 pixels in direction x

2024-11-26 21:12:45,528 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:45,528 | ApMeasureStars | DEBUG | Measured FWHM=3.692 +/- 0.122 pixels in direction y

2024-11-26 21:12:45,529 | ApFindStars | INFO | Median FWHM (over x and y) is 3.77 +/- 0.25 pixels using 48 data points

2024-11-26 21:12:45,529 | ApProcess | DEBUG | Initial star detection and fitting: FWHM=3.773 +/- 0.248 pixels using 48 stars.

2024-11-26 21:12:45,529 | ApProcess | DEBUG | Updating source searching using initial FWHM=3.773 +/- 0.248 pixels using 48 stars.

2024-11-26 21:12:45,529 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=3.77 pixels, threshold=7.0 BG sigma.

2024-11-26 21:12:45,822 | ApFindStars | INFO | Initial source list using FWHM=3.773 pixels, threshold=7.00 x BG stddev, found 222 sources.

2024-11-26 21:12:45,822 | ApFindStars | DEBUG | Sources with pixels exceeding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:12:45,823 | ApFindStars | DEBUG | There are 6 possibly saturated stars in the output source list.

2024-11-26 21:12:45,823 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:12:45,823 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:12:45,859 | ApFindStars | DEBUG | Image exposure time [seconds]: 300.00

2024-11-26 21:12:45,860 | ApFindStars | DEBUG | Brightest source (idx=0) generates 13735.26 ADU/s.

2024-11-26 21:12:45,860 | ApFindStars | DEBUG | Median source (idx=111) generates 18.28 ADU/s.

2024-11-26 21:12:45,861 | ApFindStars | DEBUG | Faintest source (idx=221) generates 0.35 ADU/s.

2024-11-26 21:12:45,861 | ApFindStars | DEBUG | Selecting 200 sources out of 222 to write to the output source list.

2024-11-26 21:12:45,861 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:12:45,861 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:12:45,864 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:12:45,864 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:12:45,864 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:12:45,864 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:12:45,865 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 194

2024-11-26 21:12:45,865 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 222

2024-11-26 21:12:45,866 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:12:45,866 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:12:45,866 | ApMeasureStars | DEBUG | Constructed a kdtree with 222 data points.

2024-11-26 21:12:45,869 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 6 stars from consideration.

2024-11-26 21:12:45,869 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 188 stars in 3056 row x 3056 column image.

2024-11-26 21:12:45,870 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:12:45,870 | ApMeasureStars | DEBUG | There are 186 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:12:45,871 | ApMeasureStars | DEBUG | In region CN found 54 candidates for fitting:

2024-11-26 21:12:45,872 | ApMeasureStars | DEBUG | In region TL found 28 candidates for fitting:

2024-11-26 21:12:45,873 | ApMeasureStars | DEBUG | In region TR found 36 candidates for fitting:

2024-11-26 21:12:45,875 | ApMeasureStars | DEBUG | In region BR found 40 candidates for fitting:

2024-11-26 21:12:45,876 | ApMeasureStars | DEBUG | In region BL found 28 candidates for fitting:

2024-11-26 21:12:45,877 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=46847.90

2024-11-26 21:12:45,879 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:12:45,879 | ApMeasureStars | DEBUG | Fitting star #0 (id=134) at fixed xpos=9.18, ypos=9.37

2024-11-26 21:12:45,888 | ApMeasureStars | DEBUG | Fitting star #1 (id=104) at fixed xpos=9.18, ypos=8.53

2024-11-26 21:12:45,897 | ApMeasureStars | DEBUG | Fitting star #2 (id=95) at fixed xpos=9.33, ypos=8.67

2024-11-26 21:12:45,906 | ApMeasureStars | DEBUG | Fitting star #3 (id=154) at fixed xpos=8.57, ypos=8.86

2024-11-26 21:12:45,916 | ApMeasureStars | DEBUG | Fitting star #4 (id=146) at fixed xpos=8.62, ypos=8.69

2024-11-26 21:12:45,928 | ApMeasureStars | DEBUG | Fitting star #5 (id=175) at fixed xpos=8.83, ypos=9.41

2024-11-26 21:12:45,937 | ApMeasureStars | DEBUG | Fitting star #6 (id=220) at fixed xpos=8.69, ypos=8.92

2024-11-26 21:12:45,946 | ApMeasureStars | DEBUG | Fitting star #7 (id=197) at fixed xpos=8.72, ypos=8.58

2024-11-26 21:12:45,958 | ApMeasureStars | DEBUG | Fitting star #8 (id=176) at fixed xpos=8.87, ypos=8.84

2024-11-26 21:12:45,969 | ApMeasureStars | DEBUG | Fitting star #9 (id=207) at fixed xpos=8.51, ypos=8.93

2024-11-26 21:12:45,979 | ApMeasureStars | DEBUG | Fitting star #10 (id=178) at fixed xpos=9.22, ypos=9.49

2024-11-26 21:12:45,989 | ApMeasureStars | DEBUG | Fitting star #11 (id=206) at fixed xpos=9.11, ypos=8.85

2024-11-26 21:12:46,000 | ApMeasureStars | DEBUG | Fitting star #12 (id=217) at fixed xpos=8.55, ypos=9.35

2024-11-26 21:12:46,009 | ApMeasureStars | DEBUG | Fitting star #13 (id=186) at fixed xpos=8.94, ypos=9.44

2024-11-26 21:12:46,018 | ApMeasureStars | DEBUG | Fitting star #14 (id=184) at fixed xpos=9.09, ypos=9.33

2024-11-26 21:12:46,029 | ApMeasureStars | DEBUG | Fitting star #15 (id=7) at fixed xpos=9.15, ypos=8.73

2024-11-26 21:12:46,038 | ApMeasureStars | DEBUG | Fitting star #16 (id=49) at fixed xpos=9.32, ypos=9.42

2024-11-26 21:12:46,047 | ApMeasureStars | DEBUG | Fitting star #17 (id=44) at fixed xpos=9.34, ypos=9.31

2024-11-26 21:12:46,055 | ApMeasureStars | DEBUG | Fitting star #18 (id=51) at fixed xpos=9.02, ypos=8.94

2024-11-26 21:12:46,064 | ApMeasureStars | DEBUG | Fitting star #19 (id=34) at fixed xpos=8.90, ypos=8.77

2024-11-26 21:12:46,073 | ApMeasureStars | DEBUG | Fitting star #20 (id=21) at fixed xpos=9.35, ypos=8.85

2024-11-26 21:12:46,083 | ApMeasureStars | DEBUG | Fitting star #21 (id=58) at fixed xpos=8.99, ypos=9.23

2024-11-26 21:12:46,093 | ApMeasureStars | DEBUG | Fitting star #22 (id=54) at fixed xpos=9.08, ypos=8.55

2024-11-26 21:12:46,106 | ApMeasureStars | DEBUG | Fitting star #23 (id=36) at fixed xpos=9.15, ypos=9.42

2024-11-26 21:12:46,118 | ApMeasureStars | DEBUG | Fitting star #24 (id=50) at fixed xpos=9.26, ypos=9.37

2024-11-26 21:12:46,130 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:46,131 | ApMeasureStars | DEBUG | Measured FWHM=3.903 +/- 0.278 pixels in direction x

2024-11-26 21:12:46,131 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 23 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:46,132 | ApMeasureStars | DEBUG | Measured FWHM=3.692 +/- 0.121 pixels in direction y

2024-11-26 21:12:46,132 | ApMeasureStars | INFO | Source average axis ratio = 1.057 +/- 0.083, overall sources are circular.

2024-11-26 21:12:46,132 | ApMeasureStars | DEBUG | Fitting completed in 0.253 s (average 10.1 ms/star)

2024-11-26 21:12:46,221 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:12:46,221 | ApMeasureStars | DEBUG | Measured FWHM=3.771 +/- 0.250 pixels in direction both

2024-11-26 21:12:46,222 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:46,222 | ApMeasureStars | DEBUG | Measured FWHM=3.903 +/- 0.278 pixels in direction x

2024-11-26 21:12:46,223 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 23 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:46,223 | ApMeasureStars | DEBUG | Measured FWHM=3.692 +/- 0.121 pixels in direction y

2024-11-26 21:12:47,449 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-2.png

2024-11-26 21:12:47,450 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 48 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:12:47,450 | ApMeasureStars | DEBUG | Measured FWHM=3.771 +/- 0.250 pixels in direction both

2024-11-26 21:12:47,450 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:47,451 | ApMeasureStars | DEBUG | Measured FWHM=3.903 +/- 0.278 pixels in direction x

2024-11-26 21:12:47,451 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 23 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:12:47,452 | ApMeasureStars | DEBUG | Measured FWHM=3.692 +/- 0.121 pixels in direction y

2024-11-26 21:12:47,452 | ApFindStars | INFO | Median FWHM (over x and y) is 3.77 +/- 0.25 pixels using 48 data points

2024-11-26 21:12:47,452 | ApProcess | DEBUG | Final star detection and fitting: FWHM=3.771 +/- 0.250 pixels using 48 stars.

2024-11-26 21:12:48,361 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to ./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-2.png

2024-11-26 21:12:48,362 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:12:48,362 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (222, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Green', 'Filter used')}

2024-11-26 21:12:48,362 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:12:48,363 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:12:48,363 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

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2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.
2024-11-26 21:12:48,364 | ApFindStars | WARNING | Keyword APRX_XWD not found,
not written to quality report.
2024-11-26 21:12:48,365 | ApFindStars | WARNING | Keyword APRX_YHG not found,
not written to quality report.
2024-11-26 21:12:48,365 | ApFindStars | WARNING | Keyword APRX_XPS not found,
not written to quality report.
2024-11-26 21:12:48,365 | ApFindStars | WARNING | Keyword APRX_YPS not found,
not written to quality report.
2024-11-26 21:12:48,366 | ApFindStars | WARNING | Skipping some quality
reporting that requires an estimated platescale.
2024-11-26 21:12:48,367 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-M101_Supernova_2023ixf-300s-Green-2.yaml
2024-11-26 21:12:48,370 | ApFindStars | DEBUG | Wrote ds9-format region file to
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-2.reg
2024-11-26 21:12:48,371 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:12:48,371 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz', 'Name of image file
searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'),
'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (222, 'Number of
sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final
photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'),
'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'),
'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40',
'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'),
'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Green',
'Filter used')}}
2024-11-26 21:12:48,371 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:12:48,372 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:12:48,372 | ApFindStars | DEBUG | Converting python 0-based
coordinates to FITS 1-based pixel coordinates for XY table.
2024-11-26 21:12:48,372 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:12:48,395 | ApProcess | DEBUG | Confirming output srclist
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits was created.
2024-11-26 21:12:48,396 | ApProcess | DEBUG | Removing existing
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits as
clean_star_detection=True
2024-11-26 21:12:48,402 | ApProcess | DEBUG | Performing astrometry, output
navigated image:
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits
2024-11-26 21:12:48,402 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
2024-11-26 21:12:48,654 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D
BITPIX=-32 image with 3056 columns, 3056 rows
2024-11-26 21:12:48,657 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits
2024-11-26 21:12:48,657 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:12:48,658 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:12:48,658 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:12:48,658 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':

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54.34944444444444, 'radius': 8}
2024-11-26 21:12:48,659 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits with 200
positions (attempt #1)

Solving...

2024-11-26 21:13:02,904 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:13:03,030 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:13:03,030 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:13:03,030 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:13:03,031 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:13:03,031 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:13:03,031 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:13:03,031 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.762560589, 'RA of reference point')
2024-11-26 21:13:03,032 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4158409652, 'DEC of reference point')
2024-11-26 21:13:03,032 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:13:03,032 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:13:03,033 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:13:03,033 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:13:03,034 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.52108905512e-06, 'Transformation matrix')
2024-11-26 21:13:03,034 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175016672539, 'no comment')
2024-11-26 21:13:03,034 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175016672539, 'no comment')
2024-11-26 21:13:03,034 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.52108905512e-06, 'no comment')
2024-11-26 21:13:03,035 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:13:03,035 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:13:03,185 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits
2024-11-26 21:13:03,186 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits
with WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:13:03,204 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:13:03,459 | ApProcess | DEBUG |

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2024-11-26 21:13:03,459 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:13:03,460 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:03,460 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:03,460 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:03,460 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:03,460 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:13:03,460 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:13:03,461 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz,

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input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-3.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-3.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-3.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Green-3.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits
2024-11-26 21:13:03,461 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits because
clean_star_detection=True
2024-11-26 21:13:03,461 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-3.reg because
clean_star_detection=True
2024-11-26 21:13:03,462 | ApProcess | DEBUG | Removing existing
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-3.png because
clean_star_detection=True
2024-11-26 21:13:03,462 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-3.png because
clean_star_detection=True
2024-11-26 21:13:03,462 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits
2024-11-26 21:13:03,463 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:13:03,718 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:13:03,809 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67424.05, median=224.37
2024-11-26 21:13:04,715 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=224.560, median=224.159, stddev=23.302
2024-11-26 21:13:07,780 | ApFindStars | DEBUG | Source-masked image stats:
mean=223.306, median=223.169, stddev=22.501
2024-11-26 21:13:07,781 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:13:07,781 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:13:07,959 | ApFindStars | DEBUG | Retaining 15 possibly saturated
stars in source searching and photometry.
2024-11-26 21:13:07,959 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:13:08,191 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 275 sources.
2024-11-26 21:13:08,191 | ApFindStars | DEBUG | Sources with pixels exceeding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:13:08,192 | ApFindStars | DEBUG | There are 12 possibly saturated
stars in the output source list.
2024-11-26 21:13:08,192 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:08,192 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:08,238 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
    phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:13:08,239 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 11633.61 ADU/s.
2024-11-26 21:13:08,240 | ApFindStars | DEBUG | Median source (idx=137)
generates 23.91 ADU/s.
2024-11-26 21:13:08,240 | ApFindStars | DEBUG | Faintest source (idx=274)
generates -6.99 ADU/s.
2024-11-26 21:13:08,240 | ApFindStars | DEBUG | Selecting 200 sources out of 275
to write to the output source list.
2024-11-26 21:13:08,240 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:08,240 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:09,167 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-3.png
2024-11-26 21:13:09,169 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:13:09,170 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:13:09,170 | ApMeasureStars | DEBUG | Count based weighting factors
will be used in fitting?: True
2024-11-26 21:13:09,170 | ApMeasureStars | DEBUG | Will the background level be

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fitted for?           : True
2024-11-26 21:13:09,171 | ApMeasureStars | DEBUG | Size of input trimmed source
list (filtered): 188
2024-11-26 21:13:09,171 | ApMeasureStars | DEBUG | Size of full source list used
for neighbor removal: 275
2024-11-26 21:13:09,171 | ApMeasureStars | DEBUG | Adopting a fit box width of
18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:13:09,172 | ApMeasureStars | DEBUG | Preparing to trim the input
source list of stars with neighbors within 18 pixels.
2024-11-26 21:13:09,172 | ApMeasureStars | DEBUG | Constructed a kdtree with 275
data points.
2024-11-26 21:13:09,177 | ApMeasureStars | DEBUG | Nearest neighbor filtering
removed 27 stars from consideration.
2024-11-26 21:13:09,177 | ApMeasureStars | DEBUG | Selecting candidate stars for
fitting out of 161 stars in 3056 row x 3056 column image.
2024-11-26 21:13:09,178 | ApMeasureStars | DEBUG | Center has radius 764.0
pixels centered on x=1528.0, y=1528.0
2024-11-26 21:13:09,178 | ApMeasureStars | DEBUG | There are 159 stars more than
10 pixels away from the edge of the detector.
2024-11-26 21:13:09,179 | ApMeasureStars | DEBUG | In region CN found 46
candidates for fitting:
2024-11-26 21:13:09,179 | ApMeasureStars | DEBUG | In region TL found 27
candidates for fitting:
2024-11-26 21:13:09,181 | ApMeasureStars | DEBUG | In region TR found 30
candidates for fitting:
2024-11-26 21:13:09,182 | ApMeasureStars | DEBUG | In region BR found 29
candidates for fitting:
2024-11-26 21:13:09,184 | ApMeasureStars | DEBUG | In region BL found 27
candidates for fitting:
2024-11-26 21:13:09,185 | ApMeasureStars | DEBUG | In 25x18x18 pixel array
min=0.00, max=50402.15
2024-11-26 21:13:09,186 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:13:09,187 | ApMeasureStars | DEBUG | Fitting star #0 (id=157) at
fixed xpos=8.70, ypos=8.69
2024-11-26 21:13:09,202 | ApMeasureStars | DEBUG | Fitting star #1 (id=119) at
fixed xpos=8.61, ypos=8.81
2024-11-26 21:13:09,216 | ApMeasureStars | DEBUG | Fitting star #2 (id=107) at
fixed xpos=8.81, ypos=8.87
2024-11-26 21:13:09,230 | ApMeasureStars | DEBUG | Fitting star #3 (id=183) at
fixed xpos=9.12, ypos=9.20
2024-11-26 21:13:09,243 | ApMeasureStars | DEBUG | Fitting star #4 (id=173) at
fixed xpos=9.17, ypos=8.94
2024-11-26 21:13:09,256 | ApMeasureStars | DEBUG | Fitting star #5 (id=270) at
fixed xpos=9.42, ypos=8.90
2024-11-26 21:13:09,266 | ApMeasureStars | DEBUG | Fitting star #6 (id=239) at
fixed xpos=9.33, ypos=8.77
2024-11-26 21:13:09,276 | ApMeasureStars | DEBUG | Fitting star #7 (id=211) at
fixed xpos=9.43, ypos=8.96
2024-11-26 21:13:09,285 | ApMeasureStars | DEBUG | Fitting star #8 (id=250) at
fixed xpos=9.15, ypos=9.03
2024-11-26 21:13:09,297 | ApMeasureStars | DEBUG | Fitting star #9 (id=200) at
fixed xpos=9.36, ypos=8.83
2024-11-26 21:13:09,306 | ApMeasureStars | DEBUG | Fitting star #10 (id=249) at
fixed xpos=8.66, ypos=9.09
2024-11-26 21:13:09,323 | ApMeasureStars | DEBUG | Fitting star #11 (id=264) at
fixed xpos=9.19, ypos=8.52
2024-11-26 21:13:09,332 | ApMeasureStars | DEBUG | Fitting star #12 (id=229) at
fixed xpos=9.36, ypos=8.70
2024-11-26 21:13:09,342 | ApMeasureStars | DEBUG | Fitting star #13 (id=226) at
fixed xpos=8.51, ypos=8.61
2024-11-26 21:13:09,355 | ApMeasureStars | DEBUG | Fitting star #14 (id=244) at
fixed xpos=8.76, ypos=8.62
2024-11-26 21:13:09,368 | ApMeasureStars | DEBUG | Fitting star #15 (id=8) at
fixed xpos=9.49, ypos=8.91
2024-11-26 21:13:09,377 | ApMeasureStars | DEBUG | Fitting star #16 (id=54) at
fixed xpos=8.64, ypos=8.62
2024-11-26 21:13:09,393 | ApMeasureStars | DEBUG | Fitting star #17 (id=47) at
fixed xpos=8.58, ypos=8.50
2024-11-26 21:13:09,410 | ApMeasureStars | DEBUG | Fitting star #18 (id=57) at
fixed xpos=9.39, ypos=9.17
2024-11-26 21:13:09,425 | ApMeasureStars | DEBUG | Fitting star #19 (id=38) at
fixed xpos=9.23, ypos=8.94
2024-11-26 21:13:09,439 | ApMeasureStars | DEBUG | Fitting star #20 (id=39) at
fixed xpos=9.49, ypos=9.48
2024-11-26 21:13:09,454 | ApMeasureStars | DEBUG | Fitting star #21 (id=55) at

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fixed xpos=8.68, ypos=8.55
2024-11-26 21:13:09,467 | ApMeasureStars | DEBUG | Fitting star #22 (id=53) at
fixed xpos=9.24, ypos=8.56
2024-11-26 21:13:09,476 | ApMeasureStars | DEBUG | Fitting star #23 (id=61) at
fixed xpos=8.85, ypos=9.24
2024-11-26 21:13:09,489 | ApMeasureStars | DEBUG | Fitting star #24 (id=77) at
fixed xpos=8.59, ypos=9.40
2024-11-26 21:13:09,503 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:09,504 | ApMeasureStars | DEBUG | Measured FWHM=3.531 +/- 0.414
pixels in direction x
2024-11-26 21:13:09,504 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:09,505 | ApMeasureStars | DEBUG | Measured FWHM=3.672 +/- 0.466
pixels in direction y
2024-11-26 21:13:09,505 | ApMeasureStars | INFO | Source average axis ratio =
1.040 +/- 0.180, overall sources are circular.
2024-11-26 21:13:09,505 | ApMeasureStars | DEBUG | Fitting completed in 0.318 s
(average 12.7 ms/star)
2024-11-26 21:13:09,505 | ApMeasureStars | DEBUG | Skipping plotting of star
cutouts and best fits.
2024-11-26 21:13:09,506 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:13:09,506 | ApMeasureStars | DEBUG | Measured FWHM=3.637 +/- 0.362
pixels in direction both
2024-11-26 21:13:09,507 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:09,507 | ApMeasureStars | DEBUG | Measured FWHM=3.531 +/- 0.414
pixels in direction x
2024-11-26 21:13:09,508 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:09,508 | ApMeasureStars | DEBUG | Measured FWHM=3.672 +/- 0.466
pixels in direction y
2024-11-26 21:13:09,508 | ApFindStars | INFO | Median FWHM (over x and y) is
3.64 +/- 0.36 pixels using 50 data points
2024-11-26 21:13:09,509 | ApProcess | DEBUG | Initial star detection and
fitting: FWHM=3.637 +/- 0.362 pixels using 50 stars.
2024-11-26 21:13:09,509 | ApProcess | DEBUG | Updating source searching using
initial FWHM=3.637 +/- 0.362 pixels using 50 stars.
2024-11-26 21:13:09,509 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.64 pixels, threshold=7.0 BG sigma.
2024-11-26 21:13:09,806 | ApFindStars | INFO | Initial source list using
FWHM=3.637 pixels, threshold=7.00 x BG stddev, found 299 sources.
2024-11-26 21:13:09,807 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:13:09,807 | ApFindStars | DEBUG | There are 12 possibly saturated
stars in the output source list.
2024-11-26 21:13:09,807 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:09,808 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:09,861 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:13:09,862 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14237.18 ADU/s.
2024-11-26 21:13:09,862 | ApFindStars | DEBUG | Median source (idx=149)
generates 22.15 ADU/s.
2024-11-26 21:13:09,862 | ApFindStars | DEBUG | Faintest source (idx=298)
generates -6.78 ADU/s.
2024-11-26 21:13:09,862 | ApFindStars | DEBUG | Selecting 200 sources out of 299
to write to the output source list.
2024-11-26 21:13:09,863 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:09,863 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:09,866 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:13:09,866 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:13:09,866 | ApMeasureStars | DEBUG | Count based weighting factors
will used in fitting?: True
2024-11-26 21:13:09,866 | ApMeasureStars | DEBUG | Will the background level be
fitted for? : True

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2024-11-26 21:13:09,867 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 188
2024-11-26 21:13:09,867 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 299
2024-11-26 21:13:09,868 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.
2024-11-26 21:13:09,868 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.
2024-11-26 21:13:09,868 | ApMeasureStars | DEBUG | Constructed a kdtree with 299 data points.
2024-11-26 21:13:09,871 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 27 stars from consideration.
2024-11-26 21:13:09,871 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 161 stars in 3056 row x 3056 column image.
2024-11-26 21:13:09,871 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0
2024-11-26 21:13:09,872 | ApMeasureStars | DEBUG | There are 159 stars more than 10 pixels away from the edge of the detector.
2024-11-26 21:13:09,873 | ApMeasureStars | DEBUG | In region CN found 48 candidates for fitting:
2024-11-26 21:13:09,873 | ApMeasureStars | DEBUG | In region TL found 25 candidates for fitting:
2024-11-26 21:13:09,875 | ApMeasureStars | DEBUG | In region TR found 29 candidates for fitting:
2024-11-26 21:13:09,877 | ApMeasureStars | DEBUG | In region BR found 29 candidates for fitting:
2024-11-26 21:13:09,878 | ApMeasureStars | DEBUG | In region BL found 28 candidates for fitting:
2024-11-26 21:13:09,880 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=50402.15
2024-11-26 21:13:09,881 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:13:09,882 | ApMeasureStars | DEBUG | Fitting star #0 (id=179) at fixed xpos=8.70, ypos=8.68
2024-11-26 21:13:09,896 | ApMeasureStars | DEBUG | Fitting star #1 (id=136) at fixed xpos=8.62, ypos=8.81
2024-11-26 21:13:09,910 | ApMeasureStars | DEBUG | Fitting star #2 (id=120) at fixed xpos=8.82, ypos=8.86
2024-11-26 21:13:09,924 | ApMeasureStars | DEBUG | Fitting star #3 (id=205) at fixed xpos=9.11, ypos=9.18
2024-11-26 21:13:09,939 | ApMeasureStars | DEBUG | Fitting star #4 (id=194) at fixed xpos=9.16, ypos=8.93
2024-11-26 21:13:09,953 | ApMeasureStars | DEBUG | Fitting star #5 (id=291) at fixed xpos=9.40, ypos=8.89
2024-11-26 21:13:09,967 | ApMeasureStars | DEBUG | Fitting star #6 (id=260) at fixed xpos=9.31, ypos=8.76
2024-11-26 21:13:09,976 | ApMeasureStars | DEBUG | Fitting star #7 (id=229) at fixed xpos=9.46, ypos=8.94
2024-11-26 21:13:09,985 | ApMeasureStars | DEBUG | Fitting star #8 (id=273) at fixed xpos=9.14, ypos=9.01
2024-11-26 21:13:09,998 | ApMeasureStars | DEBUG | Fitting star #9 (id=219) at fixed xpos=9.34, ypos=8.82
2024-11-26 21:13:10,008 | ApMeasureStars | DEBUG | Fitting star #10 (id=272) at fixed xpos=8.67, ypos=9.07
2024-11-26 21:13:10,024 | ApMeasureStars | DEBUG | Fitting star #11 (id=285) at fixed xpos=9.17, ypos=8.52
2024-11-26 21:13:10,033 | ApMeasureStars | DEBUG | Fitting star #12 (id=247) at fixed xpos=9.33, ypos=8.69
2024-11-26 21:13:10,043 | ApMeasureStars | DEBUG | Fitting star #13 (id=244) at fixed xpos=8.53, ypos=8.60
2024-11-26 21:13:10,057 | ApMeasureStars | DEBUG | Fitting star #14 (id=267) at fixed xpos=8.76, ypos=8.61
2024-11-26 21:13:10,070 | ApMeasureStars | DEBUG | Fitting star #15 (id=7) at fixed xpos=9.46, ypos=8.91
2024-11-26 21:13:10,082 | ApMeasureStars | DEBUG | Fitting star #16 (id=57) at fixed xpos=8.65, ypos=8.62
2024-11-26 21:13:10,099 | ApMeasureStars | DEBUG | Fitting star #17 (id=49) at fixed xpos=8.59, ypos=8.51
2024-11-26 21:13:10,113 | ApMeasureStars | DEBUG | Fitting star #18 (id=60) at fixed xpos=9.37, ypos=9.16
2024-11-26 21:13:10,128 | ApMeasureStars | DEBUG | Fitting star #19 (id=40) at fixed xpos=9.22, ypos=8.94
2024-11-26 21:13:10,141 | ApMeasureStars | DEBUG | Fitting star #20 (id=41) at fixed xpos=8.51, ypos=9.49
2024-11-26 21:13:10,150 | ApMeasureStars | DEBUG | Fitting star #21 (id=58) at fixed xpos=8.69, ypos=8.53

2024-11-26 21:13:10,164 | ApMeasureStars | DEBUG | Fitting star #22 (id=56) at fixed xpos=9.23, ypos=8.57

2024-11-26 21:13:10,173 | ApMeasureStars | DEBUG | Fitting star #23 (id=66) at fixed xpos=8.86, ypos=9.23

2024-11-26 21:13:10,186 | ApMeasureStars | DEBUG | Fitting star #24 (id=86) at fixed xpos=8.61, ypos=9.39

2024-11-26 21:13:10,199 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:10,200 | ApMeasureStars | DEBUG | Measured FWHM=3.628 +/- 0.286 pixels in direction x

2024-11-26 21:13:10,200 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:10,200 | ApMeasureStars | DEBUG | Measured FWHM=3.670 +/- 0.502 pixels in direction y

2024-11-26 21:13:10,201 | ApMeasureStars | INFO | Source average axis ratio = 1.012 +/- 0.160, overall sources are circular.

2024-11-26 21:13:10,201 | ApMeasureStars | DEBUG | Fitting completed in 0.320 s (average 12.8 ms/star)

2024-11-26 21:13:10,289 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:13:10,290 | ApMeasureStars | DEBUG | Measured FWHM=3.641 +/- 0.365 pixels in direction both

2024-11-26 21:13:10,290 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:10,290 | ApMeasureStars | DEBUG | Measured FWHM=3.628 +/- 0.286 pixels in direction x

2024-11-26 21:13:10,291 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:10,292 | ApMeasureStars | DEBUG | Measured FWHM=3.670 +/- 0.502 pixels in direction y

2024-11-26 21:13:11,558 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-3.png

2024-11-26 21:13:11,559 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:13:11,559 | ApMeasureStars | DEBUG | Measured FWHM=3.641 +/- 0.365 pixels in direction both

2024-11-26 21:13:11,559 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:11,560 | ApMeasureStars | DEBUG | Measured FWHM=3.628 +/- 0.286 pixels in direction x

2024-11-26 21:13:11,560 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:11,561 | ApMeasureStars | DEBUG | Measured FWHM=3.670 +/- 0.502 pixels in direction y

2024-11-26 21:13:11,561 | ApFindStars | INFO | Median FWHM (over x and y) is 3.64 +/- 0.37 pixels using 50 data points

2024-11-26 21:13:11,561 | ApProcess | DEBUG | Final star detection and fitting: FWHM=3.641 +/- 0.365 pixels using 50 stars.

2024-11-26 21:13:12,488 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to ./SourceLists/imshow-M101_Supernova_2023ixf-300s-Green-3.png

2024-11-26 21:13:12,488 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:13:12,488 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (299, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Green', 'Filter used')}

2024-11-26 21:13:12,489 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:13:12,489 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:13:12,490 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:13:12,490 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:13:12,490 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:13:12,491 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:13:12,491 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:13:12,491 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:13:12,491 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:13:12,491 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:13:12,491 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:13:12,492 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:13:12,492 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:13:12,492 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:13:12,494 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Green-3.yaml

2024-11-26 21:13:12,497 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-3.reg

2024-11-26 21:13:12,498 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:13:12,498 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (299, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Green', 'Filter used')}

2024-11-26 21:13:12,498 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:13:12,499 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:13:12,499 | ApFindStars | DEBUG | Converting python 0-based coordinates to FITS 1-based pixel coordinates for XY table.

2024-11-26 21:13:12,500 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astrophy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astrophy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astrophy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:13:12,522 | ApProcess | DEBUG | Confirming output srclist ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits was created.

2024-11-26 21:13:12,522 | ApProcess | DEBUG | Removing existing ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits as clean_star_detection=True

2024-11-26 21:13:12,528 | ApProcess | DEBUG | Performing astrometry, output navigated image: ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits

2024-11-26 21:13:12,528 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz

2024-11-26 21:13:12,773 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D BITPIX=-32 image with 3056 columns, 3056 rows

2024-11-26 21:13:12,775 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits

2024-11-26 21:13:12,775 | AstroPhotography.core.ApAstrometry | DEBUG | The source list was generated from this input image. Proceeding.

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2024-11-26 21:13:12,776 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:13:12,776 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:13:12,776 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:13:12,777 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits with 200
positions (attempt #1)

Solving...

2024-11-26 21:13:22,586 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:13:22,724 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:13:22,724 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:13:22,724 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:13:22,725 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:13:22,725 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:13:22,725 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:13:22,726 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.765500189, 'RA of reference point')
2024-11-26 21:13:22,726 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4173494558, 'DEC of reference point')
2024-11-26 21:13:22,726 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:13:22,726 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:13:22,726 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:13:22,726 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:13:22,727 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.51760502153e-06, 'Transformation matrix')
2024-11-26 21:13:22,727 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175254279407, 'no comment')
2024-11-26 21:13:22,727 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175254279407, 'no comment')
2024-11-26 21:13:22,727 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.51760502153e-06, 'no comment')
2024-11-26 21:13:22,728 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:13:22,728 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:13:22,884 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits
2024-11-26 21:13:22,884 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits
with WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astropy:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:13:22,903 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:13:23,190 | ApProcess | DEBUG |

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2024-11-26 21:13:23,190 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
2024-11-26 21:13:23,190 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:23,191 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:23,191 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:23,191 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.

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2024-11-26 21:13:23,191 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.
2024-11-26 21:13:23,191 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.
2024-11-26 21:13:23,191 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-1.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-1.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-1.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Red-1.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits
2024-11-26 21:13:23,192 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits because
clean_star_detection=True
2024-11-26 21:13:23,192 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-1.reg because
clean_star_detection=True
2024-11-26 21:13:23,192 | ApProcess | DEBUG | Removing existing
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-1.png because
clean_star_detection=True
2024-11-26 21:13:23,193 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-1.png because
clean_star_detection=True
2024-11-26 21:13:23,193 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits
2024-11-26 21:13:23,193 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
2024-11-26 21:13:23,456 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows
2024-11-26 21:13:23,550 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67116.45, median=189.07
2024-11-26 21:13:24,392 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=189.221, median=188.805, stddev=22.865
2024-11-26 21:13:27,301 | ApFindStars | DEBUG | Source-masked image stats:
mean=187.867, median=187.760, stddev=21.954
2024-11-26 21:13:27,302 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.
2024-11-26 21:13:27,302 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.
2024-11-26 21:13:27,484 | ApFindStars | DEBUG | Retaining 13 possibly saturated
stars in source searching and photometry.
2024-11-26 21:13:27,484 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.
2024-11-26 21:13:27,724 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 297 sources.
2024-11-26 21:13:27,724 | ApFindStars | DEBUG | Sources with pixels exceeding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:13:27,725 | ApFindStars | DEBUG | There are 13 possibly saturated
stars in the output source list.
2024-11-26 21:13:27,725 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:27,725 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:27,774 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:13:27,775 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14432.07 ADU/s.
2024-11-26 21:13:27,776 | ApFindStars | DEBUG | Median source (idx=148)
generates 28.19 ADU/s.
2024-11-26 21:13:27,776 | ApFindStars | DEBUG | Faintest source (idx=296)
generates -2.96 ADU/s.
2024-11-26 21:13:27,776 | ApFindStars | DEBUG | Selecting 200 sources out of 297
to write to the output source list.
2024-11-26 21:13:27,776 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:27,777 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:28,705 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-1.png
2024-11-26 21:13:28,707 | ApMeasureStars | INFO | Measuring the source extent

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(Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:13:28,707 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:13:28,707 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:13:28,707 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:13:28,708 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 187

2024-11-26 21:13:28,708 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 297

2024-11-26 21:13:28,708 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:13:28,708 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:13:28,709 | ApMeasureStars | DEBUG | Constructed a kdtree with 297 data points.

2024-11-26 21:13:28,711 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 18 stars from consideration.

2024-11-26 21:13:28,712 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 169 stars in 3056 row x 3056 column image.

2024-11-26 21:13:28,712 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:13:28,712 | ApMeasureStars | DEBUG | There are 168 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:13:28,713 | ApMeasureStars | DEBUG | In region CN found 52 candidates for fitting:

2024-11-26 21:13:28,713 | ApMeasureStars | DEBUG | In region TL found 22 candidates for fitting:

2024-11-26 21:13:28,714 | ApMeasureStars | DEBUG | In region TR found 34 candidates for fitting:

2024-11-26 21:13:28,716 | ApMeasureStars | DEBUG | In region BR found 35 candidates for fitting:

2024-11-26 21:13:28,717 | ApMeasureStars | DEBUG | In region BL found 25 candidates for fitting:

2024-11-26 21:13:28,718 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=46607.87

2024-11-26 21:13:28,720 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:13:28,721 | ApMeasureStars | DEBUG | Fitting star #0 (id=164) at fixed xpos=8.69, ypos=9.50

2024-11-26 21:13:28,731 | ApMeasureStars | DEBUG | Fitting star #1 (id=130) at fixed xpos=8.65, ypos=8.57

2024-11-26 21:13:28,742 | ApMeasureStars | DEBUG | Fitting star #2 (id=117) at fixed xpos=8.76, ypos=8.70

2024-11-26 21:13:28,752 | ApMeasureStars | DEBUG | Fitting star #3 (id=69) at fixed xpos=8.51, ypos=9.48

2024-11-26 21:13:28,765 | ApMeasureStars | DEBUG | Fitting star #4 (id=197) at fixed xpos=9.11, ypos=8.98

2024-11-26 21:13:28,777 | ApMeasureStars | DEBUG | Fitting star #5 (id=292) at fixed xpos=9.35, ypos=9.25

2024-11-26 21:13:28,789 | ApMeasureStars | DEBUG | Fitting star #6 (id=228) at fixed xpos=9.44, ypos=9.05

2024-11-26 21:13:28,799 | ApMeasureStars | DEBUG | Fitting star #7 (id=261) at fixed xpos=8.56, ypos=9.40

2024-11-26 21:13:28,811 | ApMeasureStars | DEBUG | Fitting star #8 (id=215) at fixed xpos=9.29, ypos=8.93

2024-11-26 21:13:28,821 | ApMeasureStars | DEBUG | Fitting star #9 (id=265) at fixed xpos=9.12, ypos=9.27

2024-11-26 21:13:28,832 | ApMeasureStars | DEBUG | Fitting star #10 (id=289) at fixed xpos=9.07, ypos=8.56

2024-11-26 21:13:28,840 | ApMeasureStars | DEBUG | Fitting star #11 (id=244) at fixed xpos=9.47, ypos=8.52

2024-11-26 21:13:28,850 | ApMeasureStars | DEBUG | Fitting star #12 (id=241) at fixed xpos=8.53, ypos=9.48

2024-11-26 21:13:28,859 | ApMeasureStars | DEBUG | Fitting star #13 (id=166) at fixed xpos=8.99, ypos=8.52

2024-11-26 21:13:28,868 | ApMeasureStars | DEBUG | Fitting star #14 (id=257) at fixed xpos=8.59, ypos=8.56

2024-11-26 21:13:28,881 | ApMeasureStars | DEBUG | Fitting star #15 (id=5) at fixed xpos=8.60, ypos=8.70

2024-11-26 21:13:28,891 | ApMeasureStars | DEBUG | Fitting star #16 (id=49) at fixed xpos=8.77, ypos=9.42

2024-11-26 21:13:28,902 | ApMeasureStars | DEBUG | Fitting star #17 (id=51) at fixed xpos=9.42, ypos=8.90

2024-11-26 21:13:28,911 | ApMeasureStars | DEBUG | Fitting star #18 (id=44) at

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fixed xpos=8.76, ypos=9.25
2024-11-26 21:13:28,920 | ApMeasureStars | DEBUG | Fitting star #19 (id=33) at
fixed xpos=9.37, ypos=8.73
2024-11-26 21:13:28,928 | ApMeasureStars | DEBUG | Fitting star #20 (id=21) at
fixed xpos=8.89, ypos=8.97
2024-11-26 21:13:28,938 | ApMeasureStars | DEBUG | Fitting star #21 (id=59) at
fixed xpos=9.37, ypos=9.36
2024-11-26 21:13:28,949 | ApMeasureStars | DEBUG | Fitting star #22 (id=54) at
fixed xpos=8.66, ypos=8.78
2024-11-26 21:13:28,959 | ApMeasureStars | DEBUG | Fitting star #23 (id=35) at
fixed xpos=8.57, ypos=9.49
2024-11-26 21:13:28,971 | ApMeasureStars | DEBUG | Fitting star #24 (id=50) at
fixed xpos=9.17, ypos=8.56
2024-11-26 21:13:28,986 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:28,987 | ApMeasureStars | DEBUG | Measured FWHM=4.320 +/- 0.537
pixels in direction x
2024-11-26 21:13:28,987 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:28,988 | ApMeasureStars | DEBUG | Measured FWHM=3.659 +/- 0.238
pixels in direction y
2024-11-26 21:13:28,988 | ApMeasureStars | INFO | Source average axis ratio =
1.181 +/- 0.165, overall sources are circular.
2024-11-26 21:13:28,988 | ApMeasureStars | DEBUG | Fitting completed in 0.268 s
(average 10.7 ms/star)
2024-11-26 21:13:28,989 | ApMeasureStars | DEBUG | Skipping plotting of star
cutouts and best fits.
2024-11-26 21:13:28,989 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:13:28,989 | ApMeasureStars | DEBUG | Measured FWHM=3.859 +/- 0.533
pixels in direction both
2024-11-26 21:13:28,990 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:28,990 | ApMeasureStars | DEBUG | Measured FWHM=4.320 +/- 0.537
pixels in direction x
2024-11-26 21:13:28,991 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:28,991 | ApMeasureStars | DEBUG | Measured FWHM=3.659 +/- 0.238
pixels in direction y
2024-11-26 21:13:28,991 | ApFindStars | INFO | Median FWHM (over x and y) is
3.86 +/- 0.53 pixels using 50 data points
2024-11-26 21:13:28,992 | ApProcess | DEBUG | Initial star detection and
fitting: FWHM=3.859 +/- 0.533 pixels using 50 stars.
2024-11-26 21:13:28,992 | ApProcess | DEBUG | Updating source searching using
initial FWHM=3.859 +/- 0.533 pixels using 50 stars.
2024-11-26 21:13:28,992 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.86 pixels, threshold=7.0 BG sigma.
2024-11-26 21:13:29,287 | ApFindStars | INFO | Initial source list using
FWHM=3.859 pixels, threshold=7.00 x BG stddev, found 327 sources.
2024-11-26 21:13:29,288 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:13:29,288 | ApFindStars | DEBUG | There are 10 possibly saturated
stars in the output source list.
2024-11-26 21:13:29,288 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:29,288 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:29,344 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
2024-11-26 21:13:29,345 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14602.34 ADU/s.
2024-11-26 21:13:29,345 | ApFindStars | DEBUG | Median source (idx=163)
generates 24.53 ADU/s.
2024-11-26 21:13:29,345 | ApFindStars | DEBUG | Faintest source (idx=326)
generates 0.56 ADU/s.
2024-11-26 21:13:29,346 | ApFindStars | DEBUG | Selecting 200 sources out of 327
to write to the output source list.
2024-11-26 21:13:29,346 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:29,346 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:29,348 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:13:29,348 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.
2024-11-26 21:13:29,349 | ApMeasureStars | DEBUG | Count based weighting factors

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will used in fitting?: True

2024-11-26 21:13:29,349 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:13:29,349 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 190

2024-11-26 21:13:29,350 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 327

2024-11-26 21:13:29,350 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:13:29,350 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:13:29,350 | ApMeasureStars | DEBUG | Constructed a kdtree with 327 data points.

2024-11-26 21:13:29,353 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 21 stars from consideration.

2024-11-26 21:13:29,354 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 169 stars in 3056 row x 3056 column image.

2024-11-26 21:13:29,354 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:13:29,355 | ApMeasureStars | DEBUG | There are 168 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:13:29,355 | ApMeasureStars | DEBUG | In region CN found 54 candidates for fitting:

2024-11-26 21:13:29,356 | ApMeasureStars | DEBUG | In region TL found 20 candidates for fitting:

2024-11-26 21:13:29,357 | ApMeasureStars | DEBUG | In region TR found 34 candidates for fitting:

2024-11-26 21:13:29,359 | ApMeasureStars | DEBUG | In region BR found 31 candidates for fitting:

2024-11-26 21:13:29,360 | ApMeasureStars | DEBUG | In region BL found 29 candidates for fitting:

2024-11-26 21:13:29,361 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=0.00, max=46607.87

2024-11-26 21:13:29,363 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:13:29,364 | ApMeasureStars | DEBUG | Fitting star #0 (id=178) at fixed xpos=8.70, ypos=9.47

2024-11-26 21:13:29,373 | ApMeasureStars | DEBUG | Fitting star #1 (id=136) at fixed xpos=8.66, ypos=8.55

2024-11-26 21:13:29,383 | ApMeasureStars | DEBUG | Fitting star #2 (id=123) at fixed xpos=8.76, ypos=8.72

2024-11-26 21:13:29,393 | ApMeasureStars | DEBUG | Fitting star #3 (id=76) at fixed xpos=8.51, ypos=9.46

2024-11-26 21:13:29,404 | ApMeasureStars | DEBUG | Fitting star #4 (id=217) at fixed xpos=9.11, ypos=8.98

2024-11-26 21:13:29,416 | ApMeasureStars | DEBUG | Fitting star #5 (id=320) at fixed xpos=9.33, ypos=9.24

2024-11-26 21:13:29,427 | ApMeasureStars | DEBUG | Fitting star #6 (id=250) at fixed xpos=9.41, ypos=9.05

2024-11-26 21:13:29,437 | ApMeasureStars | DEBUG | Fitting star #7 (id=288) at fixed xpos=8.56, ypos=9.38

2024-11-26 21:13:29,447 | ApMeasureStars | DEBUG | Fitting star #8 (id=238) at fixed xpos=9.28, ypos=8.94

2024-11-26 21:13:29,457 | ApMeasureStars | DEBUG | Fitting star #9 (id=292) at fixed xpos=9.11, ypos=9.26

2024-11-26 21:13:29,467 | ApMeasureStars | DEBUG | Fitting star #10 (id=316) at fixed xpos=9.07, ypos=8.58

2024-11-26 21:13:29,476 | ApMeasureStars | DEBUG | Fitting star #11 (id=268) at fixed xpos=9.44, ypos=8.54

2024-11-26 21:13:29,486 | ApMeasureStars | DEBUG | Fitting star #12 (id=263) at fixed xpos=8.55, ypos=9.46

2024-11-26 21:13:29,495 | ApMeasureStars | DEBUG | Fitting star #13 (id=180) at fixed xpos=8.99, ypos=8.55

2024-11-26 21:13:29,504 | ApMeasureStars | DEBUG | Fitting star #14 (id=284) at fixed xpos=8.61, ypos=8.58

2024-11-26 21:13:29,519 | ApMeasureStars | DEBUG | Fitting star #15 (id=9) at fixed xpos=8.60, ypos=8.69

2024-11-26 21:13:29,529 | ApMeasureStars | DEBUG | Fitting star #16 (id=52) at fixed xpos=8.77, ypos=9.39

2024-11-26 21:13:29,540 | ApMeasureStars | DEBUG | Fitting star #17 (id=55) at fixed xpos=8.51, ypos=8.87

2024-11-26 21:13:29,551 | ApMeasureStars | DEBUG | Fitting star #18 (id=46) at fixed xpos=8.77, ypos=9.23

2024-11-26 21:13:29,560 | ApMeasureStars | DEBUG | Fitting star #19 (id=35) at fixed xpos=9.37, ypos=8.73

2024-11-26 21:13:29,568 | ApMeasureStars | DEBUG | Fitting star #20 (id=22) at

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fixed xpos=8.85, ypos=8.96
2024-11-26 21:13:29,579 | ApMeasureStars | DEBUG | Fitting star #21 (id=63) at
fixed xpos=9.45, ypos=9.33
2024-11-26 21:13:29,590 | ApMeasureStars | DEBUG | Fitting star #22 (id=58) at
fixed xpos=8.63, ypos=8.72
2024-11-26 21:13:29,600 | ApMeasureStars | DEBUG | Fitting star #23 (id=37) at
fixed xpos=8.56, ypos=9.47
2024-11-26 21:13:29,614 | ApMeasureStars | DEBUG | Fitting star #24 (id=53) at
fixed xpos=9.10, ypos=9.47
2024-11-26 21:13:29,628 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:29,628 | ApMeasureStars | DEBUG | Measured FWHM=4.321 +/- 0.539
pixels in direction x
2024-11-26 21:13:29,629 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:29,629 | ApMeasureStars | DEBUG | Measured FWHM=3.653 +/- 0.228
pixels in direction y
2024-11-26 21:13:29,629 | ApMeasureStars | INFO | Source average axis ratio =
1.183 +/- 0.165, overall sources are circular.
2024-11-26 21:13:29,630 | ApMeasureStars | DEBUG | Fitting completed in 0.267 s
(average 10.7 ms/star)
2024-11-26 21:13:29,718 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:13:29,718 | ApMeasureStars | DEBUG | Measured FWHM=3.863 +/- 0.550
pixels in direction both
2024-11-26 21:13:29,719 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:29,720 | ApMeasureStars | DEBUG | Measured FWHM=4.321 +/- 0.539
pixels in direction x
2024-11-26 21:13:29,721 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:29,721 | ApMeasureStars | DEBUG | Measured FWHM=3.653 +/- 0.228
pixels in direction y
2024-11-26 21:13:31,234 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-1.png
2024-11-26 21:13:31,234 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:13:31,235 | ApMeasureStars | DEBUG | Measured FWHM=3.863 +/- 0.550
pixels in direction both
2024-11-26 21:13:31,235 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:31,235 | ApMeasureStars | DEBUG | Measured FWHM=4.321 +/- 0.539
pixels in direction x
2024-11-26 21:13:31,236 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:31,236 | ApMeasureStars | DEBUG | Measured FWHM=3.653 +/- 0.228
pixels in direction y
2024-11-26 21:13:31,237 | ApFindStars | INFO | Median FWHM (over x and y) is
3.86 +/- 0.55 pixels using 50 data points
2024-11-26 21:13:31,237 | ApProcess | DEBUG | Final star detection and fitting:
FWHM=3.863 +/- 0.550 pixels using 50 stars.
2024-11-26 21:13:32,157 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-1.png
2024-11-26 21:13:32,158 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:13:32,158 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz',
'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns
in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'),
'AP_NDET': (327, 'Number of sources detected in the image.'), 'AP_NPHOT': (200,
'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used
in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above
background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA':
('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target
declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'),
'FILTER': ('Red', 'Filter used')}]
2024-11-26 21:13:32,158 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:13:32,159 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:13:32,160 | ApFindStars | WARNING | Keyword OBJECT not found, not
written to quality report.
2024-11-26 21:13:32,160 | ApFindStars | WARNING | Keyword TELESCOP not found,
not written to quality report.

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2024-11-26 21:13:32,160 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:13:32,160 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:13:32,161 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:13:32,161 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:13:32,161 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:13:32,161 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:13:32,162 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:13:32,162 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:13:32,162 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:13:32,162 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:13:32,163 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Red-1.yaml

2024-11-26 21:13:32,167 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-1.reg

2024-11-26 21:13:32,167 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:13:32,167 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (327, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Red', 'Filter used')}

2024-11-26 21:13:32,167 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:13:32,168 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:13:32,168 | ApFindStars | DEBUG | Converting python 0-based coordinates to FITS 1-based pixel coordinates for XY table.

2024-11-26 21:13:32,169 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:13:32,192 | ApProcess | DEBUG | Confirming output srclist ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits was created.

2024-11-26 21:13:32,192 | ApProcess | DEBUG | Removing existing ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits as clean_star_detection=True

2024-11-26 21:13:32,197 | ApProcess | DEBUG | Performing astrometry, output navigated image: ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits

2024-11-26 21:13:32,197 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz

2024-11-26 21:13:32,458 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D BITPIX=-32 image with 3056 columns, 3056 rows

2024-11-26 21:13:32,460 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits

2024-11-26 21:13:32,461 | AstroPhotography.core.ApAstrometry | DEBUG | The

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source list was generated from this input image. Proceeding.
2024-11-26 21:13:32,461 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:13:32,461 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:13:32,461 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:13:32,462 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-1.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:13:44,685 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:13:44,825 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:13:44,825 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:13:44,825 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:13:44,826 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:13:44,826 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:13:44,826 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:13:44,827 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.756379633, 'RA of reference point')
2024-11-26 21:13:44,827 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4159564919, 'DEC of reference point')
2024-11-26 21:13:44,827 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:13:44,827 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:13:44,827 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:13:44,828 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:13:44,828 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.48845470175e-06, 'Transformation matrix')
2024-11-26 21:13:44,828 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175222838507, 'no comment')
2024-11-26 21:13:44,829 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175222838507, 'no comment')
2024-11-26 21:13:44,829 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.48845470175e-06, 'no comment')
2024-11-26 21:13:44,829 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:13:44,830 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:13:45,001 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits
2024-11-26 21:13:45,002 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-1.fits with
WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:13:45,020 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:13:45,318 | ApProcess | DEBUG |

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2024-11-26 21:13:45,319 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
2024-11-26 21:13:45,319 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:45,319 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:45,319 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:13:45,320 | ApProcess | DEBUG | Subdirectory ./SourceLists/

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already exists.

2024-11-26 21:13:45,320 | ApProcess | DEBUG | Subdirectory ./MetaData/ already exists.

2024-11-26 21:13:45,320 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/ already exists.

2024-11-26 21:13:45,320 | ApProcess | DEBUG | Output file names for input Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz, input_rootname=None, input_suffix=.fits.gz are srcList=./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-2.fits, regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-2.reg plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-2.png, fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-2.png qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Red-2.yaml and navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits

2024-11-26 21:13:45,320 | ApProcess | DEBUG | Removing existing ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-2.fits because clean_star_detection=True

2024-11-26 21:13:45,321 | ApProcess | DEBUG | Removing existing ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-2.reg because clean_star_detection=True

2024-11-26 21:13:45,321 | ApProcess | DEBUG | Removing existing ./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-2.png because clean_star_detection=True

2024-11-26 21:13:45,321 | ApProcess | DEBUG | Removing existing ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-2.png because clean_star_detection=True

2024-11-26 21:13:45,321 | ApProcess | DEBUG | Finding stars, output source list: ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-2.fits

2024-11-26 21:13:45,322 | ApFindStars | INFO | Loading extension 0 of FITS file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz

2024-11-26 21:13:45,581 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056 columns, 3056 rows

2024-11-26 21:13:45,667 | ApFindStars | DEBUG | Raw data statistics are min=0.00, max=66810.24, median=203.06

2024-11-26 21:13:46,446 | ApFindStars | DEBUG | Sigma clipped image stats: mean=203.200, median=202.900, stddev=22.697

2024-11-26 21:13:49,360 | ApFindStars | DEBUG | Source-masked image stats: mean=202.257, median=202.138, stddev=22.116

2024-11-26 21:13:49,361 | ApFindStars | DEBUG | Checking for possibly saturated stars or regions.

2024-11-26 21:13:49,361 | ApFindStars | DEBUG | Looking for possibly saturated regions above 52428 ADU separated by {boxsize} pixels.

2024-11-26 21:13:49,541 | ApFindStars | DEBUG | Retaining 11 possibly saturated stars in source searching and photometry.

2024-11-26 21:13:49,541 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=3.00 pixels, threshold=7.0 BG sigma.

2024-11-26 21:13:49,775 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 240 sources.

2024-11-26 21:13:49,776 | ApFindStars | DEBUG | Sources with pixels exceeding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:13:49,776 | ApFindStars | DEBUG | There are 8 possibly saturated stars in the output source list.

2024-11-26 21:13:49,776 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:13:49,777 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:13:49,815 | ApFindStars | DEBUG | Image exposure time [seconds]: 300.00

2024-11-26 21:13:49,816 | ApFindStars | DEBUG | Brightest source (idx=0) generates 11140.65 ADU/s.

2024-11-26 21:13:49,816 | ApFindStars | DEBUG | Median source (idx=120) generates 28.56 ADU/s.

2024-11-26 21:13:49,817 | ApFindStars | DEBUG | Faintest source (idx=239) generates 0.99 ADU/s.

2024-11-26 21:13:49,817 | ApFindStars | DEBUG | Selecting 200 sources out of 240 to write to the output source list.

2024-11-26 21:13:49,817 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:13:49,817 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:13:50,735 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to ./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-2.png

2024-11-26 21:13:50,737 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:13:50,737 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region

will be fitted, excluding the brightest 0 stars.

2024-11-26 21:13:50,738 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:13:50,738 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:13:50,738 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 192

2024-11-26 21:13:50,739 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 240

2024-11-26 21:13:50,739 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:13:50,739 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:13:50,739 | ApMeasureStars | DEBUG | Constructed a kdtree with 240 data points.

2024-11-26 21:13:50,742 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 9 stars from consideration.

2024-11-26 21:13:50,743 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 183 stars in 3056 row x 3056 column image.

2024-11-26 21:13:50,743 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:13:50,744 | ApMeasureStars | DEBUG | There are 183 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:13:50,745 | ApMeasureStars | DEBUG | In region CN found 50 candidates for fitting:

2024-11-26 21:13:50,745 | ApMeasureStars | DEBUG | In region TL found 32 candidates for fitting:

2024-11-26 21:13:50,747 | ApMeasureStars | DEBUG | In region TR found 34 candidates for fitting:

2024-11-26 21:13:50,748 | ApMeasureStars | DEBUG | In region BR found 42 candidates for fitting:

2024-11-26 21:13:50,749 | ApMeasureStars | DEBUG | In region BL found 25 candidates for fitting:

2024-11-26 21:13:50,751 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=128.59, max=41154.30

2024-11-26 21:13:50,752 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:13:50,753 | ApMeasureStars | DEBUG | Fitting star #0 (id=143) at fixed xpos=8.78, ypos=8.56

2024-11-26 21:13:50,762 | ApMeasureStars | DEBUG | Fitting star #1 (id=109) at fixed xpos=8.73, ypos=8.73

2024-11-26 21:13:50,775 | ApMeasureStars | DEBUG | Fitting star #2 (id=99) at fixed xpos=8.78, ypos=8.81

2024-11-26 21:13:50,785 | ApMeasureStars | DEBUG | Fitting star #3 (id=63) at fixed xpos=8.65, ypos=8.55

2024-11-26 21:13:50,796 | ApMeasureStars | DEBUG | Fitting star #4 (id=165) at fixed xpos=9.14, ypos=9.09

2024-11-26 21:13:50,804 | ApMeasureStars | DEBUG | Fitting star #5 (id=239) at fixed xpos=9.18, ypos=9.19

2024-11-26 21:13:50,812 | ApMeasureStars | DEBUG | Fitting star #6 (id=212) at fixed xpos=9.21, ypos=8.96

2024-11-26 21:13:50,820 | ApMeasureStars | DEBUG | Fitting star #7 (id=191) at fixed xpos=9.37, ypos=9.06

2024-11-26 21:13:50,828 | ApMeasureStars | DEBUG | Fitting star #8 (id=221) at fixed xpos=8.50, ypos=8.50

2024-11-26 21:13:50,837 | ApMeasureStars | DEBUG | Fitting star #9 (id=182) at fixed xpos=9.26, ypos=8.91

2024-11-26 21:13:50,845 | ApMeasureStars | DEBUG | Fitting star #10 (id=222) at fixed xpos=8.64, ypos=9.26

2024-11-26 21:13:50,858 | ApMeasureStars | DEBUG | Fitting star #11 (id=236) at fixed xpos=9.07, ypos=8.68

2024-11-26 21:13:50,867 | ApMeasureStars | DEBUG | Fitting star #12 (id=204) at fixed xpos=9.46, ypos=8.73

2024-11-26 21:13:50,876 | ApMeasureStars | DEBUG | Fitting star #13 (id=201) at fixed xpos=8.61, ypos=8.61

2024-11-26 21:13:50,887 | ApMeasureStars | DEBUG | Fitting star #14 (id=218) at fixed xpos=8.64, ypos=8.77

2024-11-26 21:13:50,899 | ApMeasureStars | DEBUG | Fitting star #15 (id=7) at fixed xpos=8.77, ypos=8.82

2024-11-26 21:13:50,912 | ApMeasureStars | DEBUG | Fitting star #16 (id=47) at fixed xpos=8.90, ypos=8.53

2024-11-26 21:13:50,923 | ApMeasureStars | DEBUG | Fitting star #17 (id=50) at fixed xpos=8.55, ypos=9.10

2024-11-26 21:13:50,936 | ApMeasureStars | DEBUG | Fitting star #18 (id=42) at fixed xpos=8.85, ypos=9.46

2024-11-26 21:13:50,949 | ApMeasureStars | DEBUG | Fitting star #19 (id=32) at

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fixed xpos=9.36, ypos=8.91
2024-11-26 21:13:50,959 | ApMeasureStars | DEBUG | Fitting star #20 (id=19) at
fixed xpos=8.87, ypos=8.92
2024-11-26 21:13:50,966 | ApMeasureStars | DEBUG | Fitting star #21 (id=56) at
fixed xpos=9.45, ypos=9.45
2024-11-26 21:13:50,974 | ApMeasureStars | DEBUG | Fitting star #22 (id=52) at
fixed xpos=8.59, ypos=8.72
2024-11-26 21:13:50,986 | ApMeasureStars | DEBUG | Fitting star #23 (id=33) at
fixed xpos=8.66, ypos=8.50
2024-11-26 21:13:50,995 | ApMeasureStars | DEBUG | Fitting star #24 (id=48) at
fixed xpos=8.80, ypos=8.50
2024-11-26 21:13:51,002 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:51,003 | ApMeasureStars | DEBUG | Measured FWHM=4.150 +/- 0.368
pixels in direction x
2024-11-26 21:13:51,003 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:51,003 | ApMeasureStars | DEBUG | Measured FWHM=3.633 +/- 0.116
pixels in direction y
2024-11-26 21:13:51,004 | ApMeasureStars | INFO | Source average axis ratio =
1.142 +/- 0.108, overall sources are circular.
2024-11-26 21:13:51,004 | ApMeasureStars | DEBUG | Fitting completed in 0.251 s
(average 10.1 ms/star)
2024-11-26 21:13:51,004 | ApMeasureStars | DEBUG | Skipping plotting of star
cutouts and best fits.
2024-11-26 21:13:51,005 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 50 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:13:51,005 | ApMeasureStars | DEBUG | Measured FWHM=3.805 +/- 0.358
pixels in direction both
2024-11-26 21:13:51,005 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:51,006 | ApMeasureStars | DEBUG | Measured FWHM=4.150 +/- 0.368
pixels in direction x
2024-11-26 21:13:51,007 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:13:51,007 | ApMeasureStars | DEBUG | Measured FWHM=3.633 +/- 0.116
pixels in direction y
2024-11-26 21:13:51,007 | ApFindStars | INFO | Median FWHM (over x and y) is
3.80 +/- 0.36 pixels using 50 data points
2024-11-26 21:13:51,007 | ApProcess | DEBUG | Initial star detection and
fitting: FWHM=3.805 +/- 0.358 pixels using 50 stars.
2024-11-26 21:13:51,007 | ApProcess | DEBUG | Updating source searching using
initial FWHM=3.805 +/- 0.358 pixels using 50 stars.
2024-11-26 21:13:51,008 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.80 pixels, threshold=7.0 BG sigma.
2024-11-26 21:13:51,313 | ApFindStars | INFO | Initial source list using
FWHM=3.805 pixels, threshold=7.00 x BG stddev, found 266 sources.
2024-11-26 21:13:51,314 | ApFindStars | DEBUG | Sources with pixels exceding
52428 ADU will be flagged as possibly saturated.
2024-11-26 21:13:51,314 | ApFindStars | DEBUG | There are 6 possibly saturated
stars in the output source list.
2024-11-26 21:13:51,314 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:51,315 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:51,360 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:13:51,361 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14407.23 ADU/s.
2024-11-26 21:13:51,361 | ApFindStars | DEBUG | Median source (idx=133)
generates 24.28 ADU/s.
2024-11-26 21:13:51,361 | ApFindStars | DEBUG | Faintest source (idx=265)
generates -0.50 ADU/s.
2024-11-26 21:13:51,362 | ApFindStars | DEBUG | Selecting 200 sources out of 266
to write to the output source list.
2024-11-26 21:13:51,362 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.
2024-11-26 21:13:51,362 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.
2024-11-26 21:13:51,364 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:13:51,364 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region
will be fitted, excluding the brightest 0 stars.

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2024-11-26 21:13:51,365 | ApMeasureStars | DEBUG | Count based weighting factors will used in fitting?: True

2024-11-26 21:13:51,365 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:13:51,365 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 194

2024-11-26 21:13:51,366 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 266

2024-11-26 21:13:51,366 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:13:51,366 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:13:51,366 | ApMeasureStars | DEBUG | Constructed a kdtree with 266 data points.

2024-11-26 21:13:51,369 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 14 stars from consideration.

2024-11-26 21:13:51,369 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 180 stars in 3056 row x 3056 column image.

2024-11-26 21:13:51,370 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:13:51,370 | ApMeasureStars | DEBUG | There are 180 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:13:51,371 | ApMeasureStars | DEBUG | In region CN found 52 candidates for fitting:

2024-11-26 21:13:51,371 | ApMeasureStars | DEBUG | In region TL found 29 candidates for fitting:

2024-11-26 21:13:51,373 | ApMeasureStars | DEBUG | In region TR found 32 candidates for fitting:

2024-11-26 21:13:51,374 | ApMeasureStars | DEBUG | In region BR found 40 candidates for fitting:

2024-11-26 21:13:51,376 | ApMeasureStars | DEBUG | In region BL found 27 candidates for fitting:

2024-11-26 21:13:51,377 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=128.59, max=41154.30

2024-11-26 21:13:51,379 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:13:51,379 | ApMeasureStars | DEBUG | Fitting star #0 (id=150) at fixed xpos=8.79, ypos=8.57

2024-11-26 21:13:51,388 | ApMeasureStars | DEBUG | Fitting star #1 (id=115) at fixed xpos=8.74, ypos=8.73

2024-11-26 21:13:51,400 | ApMeasureStars | DEBUG | Fitting star #2 (id=104) at fixed xpos=8.79, ypos=8.82

2024-11-26 21:13:51,411 | ApMeasureStars | DEBUG | Fitting star #3 (id=68) at fixed xpos=8.66, ypos=8.57

2024-11-26 21:13:51,421 | ApMeasureStars | DEBUG | Fitting star #4 (id=179) at fixed xpos=9.13, ypos=9.09

2024-11-26 21:13:51,429 | ApMeasureStars | DEBUG | Fitting star #5 (id=263) at fixed xpos=9.17, ypos=9.18

2024-11-26 21:13:51,438 | ApMeasureStars | DEBUG | Fitting star #6 (id=235) at fixed xpos=9.20, ypos=8.96

2024-11-26 21:13:51,446 | ApMeasureStars | DEBUG | Fitting star #7 (id=208) at fixed xpos=9.35, ypos=9.06

2024-11-26 21:13:51,453 | ApMeasureStars | DEBUG | Fitting star #8 (id=243) at fixed xpos=8.52, ypos=8.51

2024-11-26 21:13:51,462 | ApMeasureStars | DEBUG | Fitting star #9 (id=196) at fixed xpos=9.24, ypos=8.91

2024-11-26 21:13:51,470 | ApMeasureStars | DEBUG | Fitting star #10 (id=260) at fixed xpos=9.06, ypos=8.68

2024-11-26 21:13:51,479 | ApMeasureStars | DEBUG | Fitting star #11 (id=224) at fixed xpos=9.48, ypos=8.74

2024-11-26 21:13:51,489 | ApMeasureStars | DEBUG | Fitting star #12 (id=221) at fixed xpos=8.62, ypos=8.62

2024-11-26 21:13:51,500 | ApMeasureStars | DEBUG | Fitting star #13 (id=241) at fixed xpos=8.65, ypos=8.77

2024-11-26 21:13:51,511 | ApMeasureStars | DEBUG | Fitting star #14 (id=199) at fixed xpos=9.16, ypos=9.34

2024-11-26 21:13:51,522 | ApMeasureStars | DEBUG | Fitting star #15 (id=8) at fixed xpos=8.78, ypos=8.82

2024-11-26 21:13:51,535 | ApMeasureStars | DEBUG | Fitting star #16 (id=48) at fixed xpos=8.91, ypos=8.54

2024-11-26 21:13:51,547 | ApMeasureStars | DEBUG | Fitting star #17 (id=51) at fixed xpos=8.56, ypos=9.10

2024-11-26 21:13:51,559 | ApMeasureStars | DEBUG | Fitting star #18 (id=42) at fixed xpos=8.85, ypos=9.44

2024-11-26 21:13:51,574 | ApMeasureStars | DEBUG | Fitting star #19 (id=35) at fixed xpos=9.46, ypos=8.81

2024-11-26 21:13:51,583 | ApMeasureStars | DEBUG | Fitting star #20 (id=19) at fixed xpos=8.86, ypos=8.92

2024-11-26 21:13:51,590 | ApMeasureStars | DEBUG | Fitting star #21 (id=57) at fixed xpos=9.44, ypos=9.44

2024-11-26 21:13:51,598 | ApMeasureStars | DEBUG | Fitting star #22 (id=53) at fixed xpos=8.59, ypos=8.73

2024-11-26 21:13:51,608 | ApMeasureStars | DEBUG | Fitting star #23 (id=36) at fixed xpos=8.67, ypos=8.51

2024-11-26 21:13:51,616 | ApMeasureStars | DEBUG | Fitting star #24 (id=49) at fixed xpos=8.80, ypos=8.52

2024-11-26 21:13:51,625 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:51,625 | ApMeasureStars | DEBUG | Measured FWHM=4.144 +/- 0.463 pixels in direction x

2024-11-26 21:13:51,626 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:51,626 | ApMeasureStars | DEBUG | Measured FWHM=3.631 +/- 0.146 pixels in direction y

2024-11-26 21:13:51,627 | ApMeasureStars | INFO | Source average axis ratio = 1.141 +/- 0.135, overall sources are circular.

2024-11-26 21:13:51,627 | ApMeasureStars | DEBUG | Fitting completed in 0.248 s (average 9.9 ms/star)

2024-11-26 21:13:51,715 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:13:51,715 | ApMeasureStars | DEBUG | Measured FWHM=3.795 +/- 0.352 pixels in direction both

2024-11-26 21:13:51,716 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:51,717 | ApMeasureStars | DEBUG | Measured FWHM=4.144 +/- 0.463 pixels in direction x

2024-11-26 21:13:51,717 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:51,718 | ApMeasureStars | DEBUG | Measured FWHM=3.631 +/- 0.146 pixels in direction y

2024-11-26 21:13:52,939 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-2.png

2024-11-26 21:13:52,940 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:13:52,940 | ApMeasureStars | DEBUG | Measured FWHM=3.795 +/- 0.352 pixels in direction both

2024-11-26 21:13:52,941 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:52,941 | ApMeasureStars | DEBUG | Measured FWHM=4.144 +/- 0.463 pixels in direction x

2024-11-26 21:13:52,942 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:13:52,942 | ApMeasureStars | DEBUG | Measured FWHM=3.631 +/- 0.146 pixels in direction y

2024-11-26 21:13:52,942 | ApFindStars | INFO | Median FWHM (over x and y) is 3.80 +/- 0.35 pixels using 50 data points

2024-11-26 21:13:52,942 | ApProcess | DEBUG | Final star detection and fitting: FWHM=3.795 +/- 0.352 pixels using 50 stars.

2024-11-26 21:13:54,174 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap of image and sources to ./SourceLists/implot-M101_Supernova_2023ixf-300s-Red-2.png

2024-11-26 21:13:54,175 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:13:54,175 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (266, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTIDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Red', 'Filter used')}

2024-11-26 21:13:54,175 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:13:54,176 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:13:54,176 | ApFindStars | WARNING | Keyword OBJECT not found, not written to quality report.

2024-11-26 21:13:54,176 | ApFindStars | WARNING | Keyword TELESCOP not found,

not written to quality report.

2024-11-26 21:13:54,176 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:13:54,177 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:13:54,177 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:13:54,177 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:13:54,177 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:13:54,177 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:13:54,178 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:13:54,178 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:13:54,178 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:13:54,178 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:13:54,180 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Red-2.yaml

2024-11-26 21:13:54,183 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-2.reg

2024-11-26 21:13:54,183 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:13:54,184 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (266, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Red', 'Filter used')}

2024-11-26 21:13:54,184 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:13:54,184 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:13:54,185 | ApFindStars | DEBUG | Converting python 0-based coordinates to FITS 1-based pixel coordinates for XY table.

2024-11-26 21:13:54,185 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-2.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:13:54,208 | ApProcess | DEBUG | Confirming output srcList ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-2.fits was created.

2024-11-26 21:13:54,208 | ApProcess | DEBUG | Removing existing ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits as clean_star_detection=True

2024-11-26 21:13:54,213 | ApProcess | DEBUG | Performing astrometry, output navigated image: ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits

2024-11-26 21:13:54,214 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz

2024-11-26 21:13:54,474 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D BITPIX=-32 image with 3056 columns, 3056 rows

2024-11-26 21:13:54,476 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-2.fits

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2024-11-26 21:13:54,477 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:13:54,477 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:13:54,477 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:13:54,477 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:13:54,478 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-2.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:14:03,313 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:14:03,453 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:14:03,453 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA--TAN', 'TAN (gnomic) projection')
2024-11-26 21:14:03,453 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:14:03,454 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:14:03,454 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:14:03,454 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:14:03,454 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.758594665, 'RA of reference point')
2024-11-26 21:14:03,455 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4152949456, 'DEC of reference point')
2024-11-26 21:14:03,455 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:14:03,455 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:14:03,456 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:14:03,456 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:14:03,456 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.51102189485e-06, 'Transformation matrix')
2024-11-26 21:14:03,456 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000174989478108, 'no comment')
2024-11-26 21:14:03,456 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000174989478108, 'no comment')
2024-11-26 21:14:03,457 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.51102189485e-06, 'no comment')
2024-11-26 21:14:03,457 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:14:03,457 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:14:03,628 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits
2024-11-26 21:14:03,629 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-2.fits with
WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:14:03,647 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:14:03,943 | ApProcess | DEBUG |
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2024-11-26 21:14:03,943 | ApProcess | INFO | Processing input file Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
2024-11-26 21:14:03,944 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:14:03,944 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.
2024-11-26 21:14:03,944 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.

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2024-11-26 21:14:03,944 | ApProcess | DEBUG | Subdirectory ./SourceLists/
already exists.

2024-11-26 21:14:03,944 | ApProcess | DEBUG | Subdirectory ./MetaData/ already
exists.

2024-11-26 21:14:03,945 | ApProcess | DEBUG | Subdirectory ./NavigatedImages/
already exists.

2024-11-26 21:14:03,945 | ApProcess | DEBUG | Output file names for input
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz,
input_rootname=None, input_suffix=.fits.gz are
srclist=./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits,
regfile=./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-3.reg
plotfile=./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-3.png,
fwhmplot=./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-3.png
qualfile=./MetaData/qual-M101_Supernova_2023ixf-300s-Red-3.yaml and
navfile=./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits

2024-11-26 21:14:03,945 | ApProcess | DEBUG | Removing existing
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits because
clean_star_detection=True

2024-11-26 21:14:03,945 | ApProcess | DEBUG | Removing existing
./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-3.reg because
clean_star_detection=True

2024-11-26 21:14:03,946 | ApProcess | DEBUG | Removing existing
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-3.png because
clean_star_detection=True

2024-11-26 21:14:03,946 | ApProcess | DEBUG | Removing existing
./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-3.png because
clean_star_detection=True

2024-11-26 21:14:03,946 | ApProcess | DEBUG | Finding stars, output source list:
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits

2024-11-26 21:14:03,947 | ApFindStars | INFO | Loading extension 0 of FITS file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

2024-11-26 21:14:04,210 | ApFindStars | DEBUG | 2-D BITPIX=-32 image with 3056
columns, 3056 rows

2024-11-26 21:14:04,275 | ApFindStars | DEBUG | Raw data statistics are
min=0.00, max=67009.20, median=184.31

2024-11-26 21:14:05,052 | ApFindStars | DEBUG | Sigma clipped image stats:
mean=184.460, median=184.037, stddev=22.871

2024-11-26 21:14:07,945 | ApFindStars | DEBUG | Source-masked image stats:
mean=183.042, median=182.942, stddev=21.921

2024-11-26 21:14:07,946 | ApFindStars | DEBUG | Checking for possibly saturated
stars or regions.

2024-11-26 21:14:07,946 | ApFindStars | DEBUG | Looking for possibly saturated
regions above 52428 ADU separated by {boxsize} pixels.

2024-11-26 21:14:08,126 | ApFindStars | DEBUG | Retaining 18 possibly saturated
stars in source searching and photometry.

2024-11-26 21:14:08,126 | ApFindStars | DEBUG | Running DAOStarFinder with
FWHM=3.00 pixels, threshold=7.0 BG sigma.

2024-11-26 21:14:08,360 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 334 sources.

2024-11-26 21:14:08,360 | ApFindStars | DEBUG | Sources with pixels exceeding
52428 ADU will be flagged as possibly saturated.

2024-11-26 21:14:08,360 | ApFindStars | DEBUG | There are 18 possibly saturated
stars in the output source list.

2024-11-26 21:14:08,361 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.

2024-11-26 21:14:08,361 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:14:08,416 | ApFindStars | DEBUG | Image exposure time [seconds]:
300.00

2024-11-26 21:14:08,417 | ApFindStars | DEBUG | Brightest source (idx=0)
generates 14552.86 ADU/s.

2024-11-26 21:14:08,417 | ApFindStars | DEBUG | Median source (idx=167)
generates 24.83 ADU/s.

2024-11-26 21:14:08,417 | ApFindStars | DEBUG | Faintest source (idx=333)
generates 1.35 ADU/s.

2024-11-26 21:14:08,418 | ApFindStars | DEBUG | Selecting 200 sources out of 334
to write to the output source list.

2024-11-26 21:14:08,418 | ApFindStars | DEBUG | Radius of circular aperture
photometry is 6 pixels.

2024-11-26 21:14:08,418 | ApFindStars | DEBUG | Local BG estimation in annulus
outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:14:09,320 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-3.png

2024-11-26 21:14:09,322 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:14:09,322 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region will be fitted, excluding the brightest 0 stars.

2024-11-26 21:14:09,323 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:14:09,323 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:14:09,324 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 182

2024-11-26 21:14:09,324 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 334

2024-11-26 21:14:09,324 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:14:09,324 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:14:09,325 | ApMeasureStars | DEBUG | Constructed a kdtree with 334 data points.

2024-11-26 21:14:09,328 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 22 stars from consideration.

2024-11-26 21:14:09,328 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 160 stars in 3056 row x 3056 column image.

2024-11-26 21:14:09,328 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:14:09,329 | ApMeasureStars | DEBUG | There are 158 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:14:09,329 | ApMeasureStars | DEBUG | In region CN found 49 candidates for fitting:

2024-11-26 21:14:09,330 | ApMeasureStars | DEBUG | In region TL found 25 candidates for fitting:

2024-11-26 21:14:09,331 | ApMeasureStars | DEBUG | In region TR found 29 candidates for fitting:

2024-11-26 21:14:09,333 | ApMeasureStars | DEBUG | In region BR found 31 candidates for fitting:

2024-11-26 21:14:09,334 | ApMeasureStars | DEBUG | In region BL found 24 candidates for fitting:

2024-11-26 21:14:09,335 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=106.15, max=48580.03

2024-11-26 21:14:09,337 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:14:09,338 | ApMeasureStars | DEBUG | Fitting star #0 (id=194) at fixed xpos=8.55, ypos=8.92

2024-11-26 21:14:09,347 | ApMeasureStars | DEBUG | Fitting star #1 (id=148) at fixed xpos=9.49, ypos=9.06

2024-11-26 21:14:09,356 | ApMeasureStars | DEBUG | Fitting star #2 (id=136) at fixed xpos=8.60, ypos=9.22

2024-11-26 21:14:09,365 | ApMeasureStars | DEBUG | Fitting star #3 (id=216) at fixed xpos=9.00, ypos=9.25

2024-11-26 21:14:09,377 | ApMeasureStars | DEBUG | Fitting star #4 (id=229) at fixed xpos=9.02, ypos=9.48

2024-11-26 21:14:09,391 | ApMeasureStars | DEBUG | Fitting star #5 (id=326) at fixed xpos=9.29, ypos=9.34

2024-11-26 21:14:09,400 | ApMeasureStars | DEBUG | Fitting star #6 (id=286) at fixed xpos=9.21, ypos=9.08

2024-11-26 21:14:09,411 | ApMeasureStars | DEBUG | Fitting star #7 (id=260) at fixed xpos=9.33, ypos=9.36

2024-11-26 21:14:09,423 | ApMeasureStars | DEBUG | Fitting star #8 (id=250) at fixed xpos=9.19, ypos=9.19

2024-11-26 21:14:09,434 | ApMeasureStars | DEBUG | Fitting star #9 (id=298) at fixed xpos=9.47, ypos=8.66

2024-11-26 21:14:09,446 | ApMeasureStars | DEBUG | Fitting star #10 (id=321) at fixed xpos=9.04, ypos=8.82

2024-11-26 21:14:09,456 | ApMeasureStars | DEBUG | Fitting star #11 (id=277) at fixed xpos=9.32, ypos=8.87

2024-11-26 21:14:09,467 | ApMeasureStars | DEBUG | Fitting star #12 (id=274) at fixed xpos=9.42, ypos=8.81

2024-11-26 21:14:09,478 | ApMeasureStars | DEBUG | Fitting star #13 (id=196) at fixed xpos=8.79, ypos=9.06

2024-11-26 21:14:09,492 | ApMeasureStars | DEBUG | Fitting star #14 (id=294) at fixed xpos=8.60, ypos=8.88

2024-11-26 21:14:09,501 | ApMeasureStars | DEBUG | Fitting star #15 (id=69) at fixed xpos=9.49, ypos=8.92

2024-11-26 21:14:09,511 | ApMeasureStars | DEBUG | Fitting star #16 (id=73) at fixed xpos=9.25, ypos=8.51

2024-11-26 21:14:09,521 | ApMeasureStars | DEBUG | Fitting star #17 (id=61) at fixed xpos=9.43, ypos=8.80

2024-11-26 21:14:09,531 | ApMeasureStars | DEBUG | Fitting star #18 (id=50) at fixed xpos=9.08, ypos=9.26

2024-11-26 21:14:09,539 | ApMeasureStars | DEBUG | Fitting star #19 (id=116) at fixed xpos=8.58, ypos=8.81

2024-11-26 21:14:09,552 | ApMeasureStars | DEBUG | Fitting star #20 (id=53) at fixed xpos=9.41, ypos=8.89

2024-11-26 21:14:09,562 | ApMeasureStars | DEBUG | Fitting star #21 (id=70) at fixed xpos=9.43, ypos=8.90

2024-11-26 21:14:09,572 | ApMeasureStars | DEBUG | Fitting star #22 (id=34) at fixed xpos=9.29, ypos=9.10

2024-11-26 21:14:09,581 | ApMeasureStars | DEBUG | Fitting star #23 (id=68) at fixed xpos=9.03, ypos=9.02

2024-11-26 21:14:09,589 | ApMeasureStars | DEBUG | Fitting star #24 (id=104) at fixed xpos=9.45, ypos=8.78

2024-11-26 21:14:09,599 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:14:09,599 | ApMeasureStars | DEBUG | Measured FWHM=3.410 +/- 0.129 pixels in direction x

2024-11-26 21:14:09,600 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:14:09,601 | ApMeasureStars | DEBUG | Measured FWHM=3.234 +/- 0.207 pixels in direction y

2024-11-26 21:14:09,601 | ApMeasureStars | INFO | Source average axis ratio = 1.054 +/- 0.078, overall sources are circular.

2024-11-26 21:14:09,601 | ApMeasureStars | DEBUG | Fitting completed in 0.264 s (average 10.6 ms/star)

2024-11-26 21:14:09,601 | ApMeasureStars | DEBUG | Skipping plotting of star cutouts and best fits.

2024-11-26 21:14:09,602 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=both) using 50 FWHM measurements (50 OK fits before clipping).

2024-11-26 21:14:09,602 | ApMeasureStars | DEBUG | Measured FWHM=3.362 +/- 0.197 pixels in direction both

2024-11-26 21:14:09,603 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=x) using 25 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:14:09,603 | ApMeasureStars | DEBUG | Measured FWHM=3.410 +/- 0.129 pixels in direction x

2024-11-26 21:14:09,603 | ApMeasureStars | DEBUG | Estimating median FWHM (direction=y) using 24 FWHM measurements (25 OK fits before clipping).

2024-11-26 21:14:09,604 | ApMeasureStars | DEBUG | Measured FWHM=3.234 +/- 0.207 pixels in direction y

2024-11-26 21:14:09,604 | ApFindStars | INFO | Median FWHM (over x and y) is 3.36 +/- 0.20 pixels using 50 data points

2024-11-26 21:14:09,604 | ApProcess | DEBUG | Initial star detection and fitting: FWHM=3.362 +/- 0.197 pixels using 50 stars.

2024-11-26 21:14:09,604 | ApProcess | DEBUG | Updating source searching using initial FWHM=3.362 +/- 0.197 pixels using 50 stars.

2024-11-26 21:14:09,604 | ApFindStars | DEBUG | Running DAOStarFinder with FWHM=3.36 pixels, threshold=7.0 BG sigma.

2024-11-26 21:14:09,834 | ApFindStars | INFO | Initial source list using FWHM=3.362 pixels, threshold=7.00 x BG stddev, found 349 sources.

2024-11-26 21:14:09,835 | ApFindStars | DEBUG | Sources with pixels exceding 52428 ADU will be flagged as possibly saturated.

2024-11-26 21:14:09,835 | ApFindStars | DEBUG | There are 18 possibly saturated stars in the output source list.

2024-11-26 21:14:09,835 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:14:09,835 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:14:09,892 | ApFindStars | DEBUG | Image exposure time [seconds]: 300.00

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10

phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:14:09,893 | ApFindStars | DEBUG | Brightest source (idx=0) generates 14603.44 ADU/s.

2024-11-26 21:14:09,893 | ApFindStars | DEBUG | Median source (idx=174) generates 22.77 ADU/s.

2024-11-26 21:14:09,893 | ApFindStars | DEBUG | Faintest source (idx=348) generates -1.93 ADU/s.

2024-11-26 21:14:09,893 | ApFindStars | DEBUG | Selecting 200 sources out of 349 to write to the output source list.

2024-11-26 21:14:09,893 | ApFindStars | DEBUG | Radius of circular aperture photometry is 6 pixels.

2024-11-26 21:14:09,894 | ApFindStars | DEBUG | Local BG estimation in annulus outer radius 9 pixels, inner radius 6 pixels.

2024-11-26 21:14:09,896 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:14:09,896 | ApMeasureStars | DEBUG | Up to 5 stars per sub-region

will be fitted, excluding the brightest 0 stars.

2024-11-26 21:14:09,897 | ApMeasureStars | DEBUG | Count based weighting factors will be used in fitting?: True

2024-11-26 21:14:09,897 | ApMeasureStars | DEBUG | Will the background level be fitted for? : True

2024-11-26 21:14:09,897 | ApMeasureStars | DEBUG | Size of input trimmed source list (filtered): 182

2024-11-26 21:14:09,898 | ApMeasureStars | DEBUG | Size of full source list used for neighbor removal: 349

2024-11-26 21:14:09,898 | ApMeasureStars | DEBUG | Adopting a fit box width of 18 pixels, edge exclusion of 10 pixels.

2024-11-26 21:14:09,898 | ApMeasureStars | DEBUG | Preparing to trim the input source list of stars with neighbors within 18 pixels.

2024-11-26 21:14:09,899 | ApMeasureStars | DEBUG | Constructed a kdtree with 349 data points.

2024-11-26 21:14:09,901 | ApMeasureStars | DEBUG | Nearest neighbor filtering removed 23 stars from consideration.

2024-11-26 21:14:09,901 | ApMeasureStars | DEBUG | Selecting candidate stars for fitting out of 159 stars in 3056 row x 3056 column image.

2024-11-26 21:14:09,902 | ApMeasureStars | DEBUG | Center has radius 764.0 pixels centered on x=1528.0, y=1528.0

2024-11-26 21:14:09,903 | ApMeasureStars | DEBUG | There are 157 stars more than 10 pixels away from the edge of the detector.

2024-11-26 21:14:09,903 | ApMeasureStars | DEBUG | In region CN found 49 candidates for fitting:

2024-11-26 21:14:09,904 | ApMeasureStars | DEBUG | In region TL found 25 candidates for fitting:

2024-11-26 21:14:09,906 | ApMeasureStars | DEBUG | In region TR found 29 candidates for fitting:

2024-11-26 21:14:09,907 | ApMeasureStars | DEBUG | In region BR found 30 candidates for fitting:

2024-11-26 21:14:09,908 | ApMeasureStars | DEBUG | In region BL found 24 candidates for fitting:

2024-11-26 21:14:09,910 | ApMeasureStars | DEBUG | In 25x18x18 pixel array min=106.15, max=48580.03

2024-11-26 21:14:09,911 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:14:09,912 | ApMeasureStars | DEBUG | Fitting star #0 (id=202) at fixed xpos=8.56, ypos=8.92

2024-11-26 21:14:09,921 | ApMeasureStars | DEBUG | Fitting star #1 (id=155) at fixed xpos=8.50, ypos=9.06

2024-11-26 21:14:09,930 | ApMeasureStars | DEBUG | Fitting star #2 (id=142) at fixed xpos=8.61, ypos=9.21

2024-11-26 21:14:09,939 | ApMeasureStars | DEBUG | Fitting star #3 (id=237) at fixed xpos=9.02, ypos=9.47

2024-11-26 21:14:09,952 | ApMeasureStars | DEBUG | Fitting star #4 (id=224) at fixed xpos=9.00, ypos=9.24

2024-11-26 21:14:09,964 | ApMeasureStars | DEBUG | Fitting star #5 (id=340) at fixed xpos=9.29, ypos=9.33

2024-11-26 21:14:09,972 | ApMeasureStars | DEBUG | Fitting star #6 (id=296) at fixed xpos=9.20, ypos=9.07

2024-11-26 21:14:09,985 | ApMeasureStars | DEBUG | Fitting star #7 (id=268) at fixed xpos=9.32, ypos=9.35

2024-11-26 21:14:09,997 | ApMeasureStars | DEBUG | Fitting star #8 (id=258) at fixed xpos=9.19, ypos=9.18

2024-11-26 21:14:10,009 | ApMeasureStars | DEBUG | Fitting star #9 (id=307) at fixed xpos=9.48, ypos=8.67

2024-11-26 21:14:10,018 | ApMeasureStars | DEBUG | Fitting star #10 (id=334) at fixed xpos=9.03, ypos=8.82

2024-11-26 21:14:10,029 | ApMeasureStars | DEBUG | Fitting star #11 (id=287) at fixed xpos=9.31, ypos=8.87

2024-11-26 21:14:10,039 | ApMeasureStars | DEBUG | Fitting star #12 (id=284) at fixed xpos=9.44, ypos=8.81

2024-11-26 21:14:10,048 | ApMeasureStars | DEBUG | Fitting star #13 (id=204) at fixed xpos=8.80, ypos=9.06

2024-11-26 21:14:10,062 | ApMeasureStars | DEBUG | Fitting star #14 (id=303) at fixed xpos=8.61, ypos=8.88

2024-11-26 21:14:10,070 | ApMeasureStars | DEBUG | Fitting star #15 (id=71) at fixed xpos=8.50, ypos=8.92

2024-11-26 21:14:10,079 | ApMeasureStars | DEBUG | Fitting star #16 (id=75) at fixed xpos=9.24, ypos=9.50

2024-11-26 21:14:10,088 | ApMeasureStars | DEBUG | Fitting star #17 (id=63) at fixed xpos=9.44, ypos=8.81

2024-11-26 21:14:10,097 | ApMeasureStars | DEBUG | Fitting star #18 (id=52) at fixed xpos=9.08, ypos=9.25

2024-11-26 21:14:10,106 | ApMeasureStars | DEBUG | Fitting star #19 (id=121) at

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fixed xpos=8.59, ypos=8.81
2024-11-26 21:14:10,119 | ApMeasureStars | DEBUG | Fitting star #20 (id=55) at
fixed xpos=9.40, ypos=8.89
2024-11-26 21:14:10,129 | ApMeasureStars | DEBUG | Fitting star #21 (id=72) at
fixed xpos=9.44, ypos=8.90
2024-11-26 21:14:10,140 | ApMeasureStars | DEBUG | Fitting star #22 (id=35) at
fixed xpos=9.28, ypos=9.10
2024-11-26 21:14:10,148 | ApMeasureStars | DEBUG | Fitting star #23 (id=70) at
fixed xpos=9.03, ypos=9.02
2024-11-26 21:14:10,157 | ApMeasureStars | DEBUG | Fitting star #24 (id=109) at
fixed xpos=9.44, ypos=8.79
2024-11-26 21:14:10,167 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:14:10,167 | ApMeasureStars | DEBUG | Measured FWHM=3.378 +/- 0.143
pixels in direction x
2024-11-26 21:14:10,168 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:14:10,168 | ApMeasureStars | DEBUG | Measured FWHM=3.187 +/- 0.118
pixels in direction y
2024-11-26 21:14:10,168 | ApMeasureStars | INFO | Source average axis ratio =
1.060 +/- 0.060, overall sources are circular.
2024-11-26 21:14:10,168 | ApMeasureStars | DEBUG | Fitting completed in 0.257 s
(average 10.3 ms/star)
2024-11-26 21:14:10,257 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:14:10,257 | ApMeasureStars | DEBUG | Measured FWHM=3.325 +/- 0.219
pixels in direction both
2024-11-26 21:14:10,258 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:14:10,258 | ApMeasureStars | DEBUG | Measured FWHM=3.378 +/- 0.143
pixels in direction x
2024-11-26 21:14:10,259 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:14:10,259 | ApMeasureStars | DEBUG | Measured FWHM=3.187 +/- 0.118
pixels in direction y
2024-11-26 21:14:11,471 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-3.png
2024-11-26 21:14:11,472 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=both) using 49 FWHM measurements (50 OK fits before clipping).
2024-11-26 21:14:11,472 | ApMeasureStars | DEBUG | Measured FWHM=3.325 +/- 0.219
pixels in direction both
2024-11-26 21:14:11,473 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=x) using 25 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:14:11,473 | ApMeasureStars | DEBUG | Measured FWHM=3.378 +/- 0.143
pixels in direction x
2024-11-26 21:14:11,474 | ApMeasureStars | DEBUG | Estimating median FWHM
(direction=y) using 24 FWHM measurements (25 OK fits before clipping).
2024-11-26 21:14:11,474 | ApMeasureStars | DEBUG | Measured FWHM=3.187 +/- 0.118
pixels in direction y
2024-11-26 21:14:11,474 | ApFindStars | INFO | Median FWHM (over x and y) is
3.32 +/- 0.22 pixels using 49 data points
2024-11-26 21:14:11,474 | ApProcess | DEBUG | Final star detection and fitting:
FWHM=3.325 +/- 0.219 pixels using 49 stars.
2024-11-26 21:14:12,361 | ApFindStars | DEBUG | Plotted asinh-stretched bitmap
of image and sources to
./SourceLists/imshow-M101_Supernova_2023ixf-300s-Red-3.png
2024-11-26 21:14:12,362 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:14:12,362 | ApFindStars | DEBUG | FITS keywords found in image:
{'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz',
'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns
in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'),
'AP_NDET': (349, 'Number of sources detected in the image.'), 'AP_NPHOT': (200,
'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used
in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above
background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA':
('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target
declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'),
'FILTER': ('Red', 'Filter used')}}
2024-11-26 21:14:12,362 | ApFindStars | DEBUG | FITS keywords missing from
image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME',
'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-
OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:14:12,363 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:14:12,363 | ApFindStars | WARNING | Keyword OBJECT not found, not
written to quality report.

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2024-11-26 21:14:12,363 | ApFindStars | WARNING | Keyword TELESCOP not found, not written to quality report.

2024-11-26 21:14:12,364 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:14:12,364 | ApFindStars | WARNING | Keyword EXPOSURE not found, not written to quality report.

2024-11-26 21:14:12,364 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:14:12,364 | ApFindStars | WARNING | Keyword EGAIN not found, not written to quality report.

2024-11-26 21:14:12,364 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:14:12,365 | ApFindStars | WARNING | Keyword APRX_XWD not found, not written to quality report.

2024-11-26 21:14:12,365 | ApFindStars | WARNING | Keyword APRX_YHG not found, not written to quality report.

2024-11-26 21:14:12,365 | ApFindStars | WARNING | Keyword APRX_XPS not found, not written to quality report.

2024-11-26 21:14:12,365 | ApFindStars | WARNING | Keyword APRX_YPS not found, not written to quality report.

2024-11-26 21:14:12,365 | ApFindStars | WARNING | Skipping some quality reporting that requires an estimated platescale.

2024-11-26 21:14:12,367 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-M101_Supernova_2023ixf-300s-Red-3.yaml

2024-11-26 21:14:12,369 | ApFindStars | DEBUG | Wrote ds9-format region file to ./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-3.reg

2024-11-26 21:14:12,370 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:14:12,370 | ApFindStars | DEBUG | FITS keywords found in image: {'IMG_FILE': ('Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz', 'Name of image file searched for stars'), 'IMG_COLS': (3056, 'Number of columns in input image'), 'IMG_ROWS': (3056, 'Number of rows in input image'), 'AP_NDET': (349, 'Number of sources detected in the image.'), 'AP_NPHOT': (200, 'Number of sources final photometry.'), 'AP_NFIT': (25, 'Number of sources used in FWHM fitting.'), 'AP_NSIGM': (7.0, 'Source searching threshold (sigma above background)'), 'APTDIA': (610.0, '[mm] Diameter of telescope aperture'), 'RA': ('14 03 12.40', 'Target right ascension'), 'DEC': ('+54 20 58.0', 'Target declination'), 'FOCALLEN': (3962.0, '[mm] Stated telescope focal length'), 'FILTER': ('Red', 'Filter used')}

2024-11-26 21:14:12,370 | ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'RA-OBJ', 'DEC-OBJ', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']

2024-11-26 21:14:12,371 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:14:12,371 | ApFindStars | DEBUG | Converting python 0-based coordinates to FITS 1-based pixel coordinates for XY table.

2024-11-26 21:14:12,372 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-3.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:14:12,394 | ApProcess | DEBUG | Confirming output srcList

./SourceLists/srcList-M101_Supernova_2023ixf-300s-Red-3.fits was created.

2024-11-26 21:14:12,394 | ApProcess | DEBUG | Removing existing

./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits as clean_star_detection=True

2024-11-26 21:14:12,399 | ApProcess | DEBUG | Performing astrometry, output navigated image:

./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits

2024-11-26 21:14:12,399 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

2024-11-26 21:14:12,648 | AstroPhotography.core.ApAstrometry | DEBUG | 2-D BITPIX=-32 image with 3056 columns, 3056 rows

2024-11-26 21:14:12,651 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from

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./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits
2024-11-26 21:14:12,651 | AstroPhotography.core.ApAstrometry | DEBUG | The
source list was generated from this input image. Proceeding.
2024-11-26 21:14:12,651 | AstroPhotography.core.ApAstrometry | DEBUG | Estimated
center_ra=210.802, center_dec=54.349, radius=8.000 deg.
2024-11-26 21:14:12,652 | AstroPhotography.core.ApAstrometry | WARNING | Could
not generate scale hints.
2024-11-26 21:14:12,652 | AstroPhotography.core.ApAstrometry | DEBUG |
Astrometry.net hints dictionary: {'center_ra': 210.80166666666665, 'center_dec':
54.34944444444444, 'radius': 8}
2024-11-26 21:14:12,652 | AstroPhotography.core.ApAstrometry | DEBUG |
Submitting astrometry.net solve from source list
./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits with 200 positions
(attempt #1)

Solving...

2024-11-26 21:14:22,510 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:14:22,650 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
WCSAXES to (2, 'no comment')
2024-11-26 21:14:22,650 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE1 to ('RA---TAN', 'TAN (gnomic) projection')
2024-11-26 21:14:22,650 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CTYPE2 to ('DEC--TAN', 'TAN (gnomic) projection')
2024-11-26 21:14:22,651 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
EQUINOX to (2000.0, 'Equatorial coordinates definition (yr)')
2024-11-26 21:14:22,651 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LONPOLE to (180.0, 'no comment')
2024-11-26 21:14:22,651 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
LATPOLE to (0.0, 'no comment')
2024-11-26 21:14:22,651 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL1 to (210.76658025, 'RA of reference point')
2024-11-26 21:14:22,652 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRVAL2 to (54.4173533954, 'DEC of reference point')
2024-11-26 21:14:22,652 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX1 to (1528.5, 'X reference pixel')
2024-11-26 21:14:22,652 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CRPIX2 to (1528.5, 'Y reference pixel')
2024-11-26 21:14:22,652 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT1 to ('deg', 'X pixel scale units')
2024-11-26 21:14:22,652 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CUNIT2 to ('deg', 'Y pixel scale units')
2024-11-26 21:14:22,653 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_1 to (-4.49304627965e-06, 'Transformation matrix')
2024-11-26 21:14:22,653 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD1_2 to (0.000175218750927, 'no comment')
2024-11-26 21:14:22,653 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_1 to (-0.000175218750927, 'no comment')
2024-11-26 21:14:22,653 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
CD2_2 to (-4.49304627965e-06, 'no comment')
2024-11-26 21:14:22,654 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEW to (3056, 'Image width, in pixels.')
2024-11-26 21:14:22,654 | AstroPhotography.core.ApAstrometry | DEBUG | Setting
IMAGEH to (3056, 'Image height, in pixels.')
2024-11-26 21:14:22,818 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits
2024-11-26 21:14:22,818 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits with
WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:14:22,837 | ApProcess | DEBUG | ApAstrometry return status: 0
2024-11-26 21:14:22,838 | ApProcess | DEBUG | For ap_filetype qualfile the
relative directory path is ./MetaData/
2024-11-26 21:14:22,838 | ApQualitySummarizer | INFO | Searching for file
matching pattern "qual*.yaml" in ./MetaData/
2024-11-26 21:14:22,839 | ApQualitySummarizer | DEBUG | Found 12 matching path
and pattern.
2024-11-26 21:14:22,839 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Blue-2.yaml
2024-11-26 21:14:22,841 | ApQualitySummarizer | DEBUG | Loading

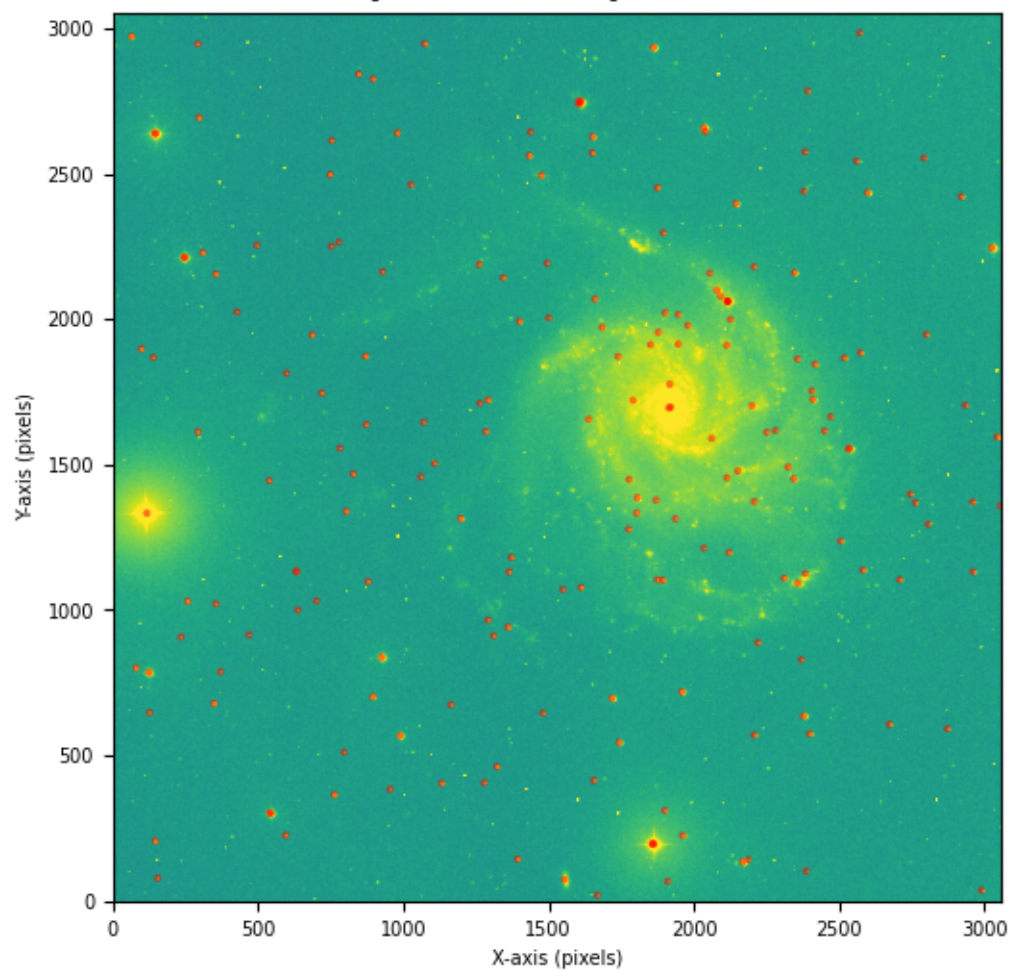
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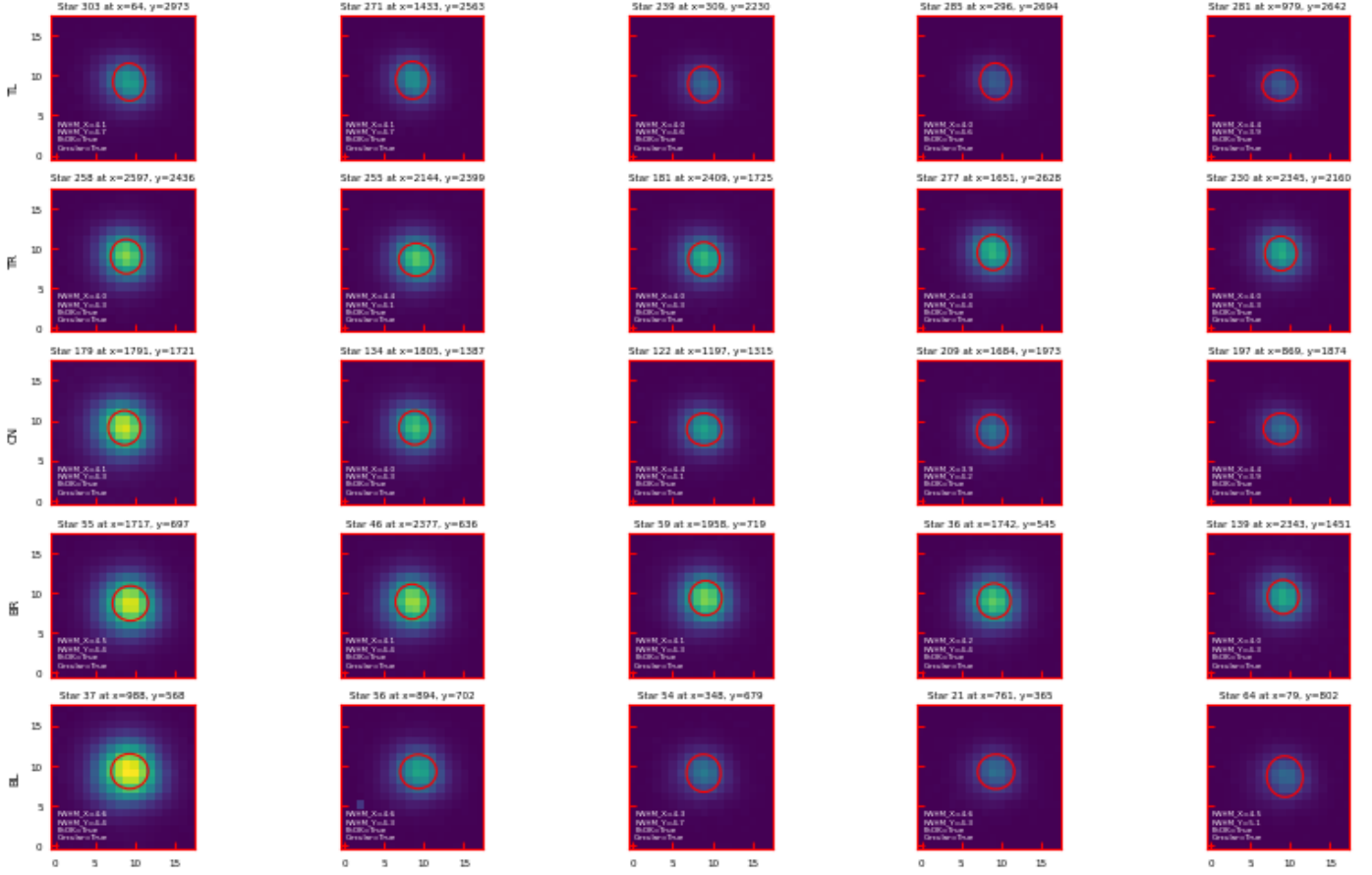
qual-M101_Supernova_2023ixf-300s-Green-1.yaml
2024-11-26 21:14:22,843 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Red-1.yaml
2024-11-26 21:14:22,845 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Red-2.yaml
2024-11-26 21:14:22,846 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-180s-Lum-2.yaml
2024-11-26 21:14:22,848 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-180s-Lum-1.yaml
2024-11-26 21:14:22,850 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Green-3.yaml
2024-11-26 21:14:22,852 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Green-2.yaml
2024-11-26 21:14:22,854 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Blue-3.yaml
2024-11-26 21:14:22,856 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Red-3.yaml
2024-11-26 21:14:22,858 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-300s-Blue-1.yaml
2024-11-26 21:14:22,859 | ApQualitySummarizer | DEBUG | Loading
qual-M101_Supernova_2023ixf-180s-Lum-3.yaml
2024-11-26 21:14:22,861 | ApQualitySummarizer | INFO | There are 4 unique
target/telescope/filter groupings: unknown_target:unknown_telescope:Blue,
unknown_target:unknown_telescope:Green, unknown_target:unknown_telescope:Red,
unknown_target:unknown_telescope:Lum
2024-11-26 21:14:22,862 | ApQualitySummarizer | DEBUG | Processing
target:telescope:filter unknown_target:unknown_telescope:Blue with 3 datasets.
2024-11-26 21:14:22,862 | ApQualitySummarizer | DEBUG | Processing
target:telescope:filter unknown_target:unknown_telescope:Green with 3 datasets.
2024-11-26 21:14:22,863 | ApQualitySummarizer | DEBUG | Processing
target:telescope:filter unknown_target:unknown_telescope:Red with 3 datasets.
2024-11-26 21:14:22,863 | ApQualitySummarizer | DEBUG | Processing
target:telescope:filter unknown_target:unknown_telescope:Lum with 3 datasets.
2024-11-26 21:14:22,866 | ApQualitySummarizer | INFO | Wrote quality summary to
quality_summary_M101SN.csv
2024-11-26 21:14:22,866 | ApProcess | INFO | Finished processing 12 files in
238.771 seconds.
2024-11-26 21:14:22,866 | ApProcess | DEBUG | Skipped 0/12 navigations and 0/12
astrometric solutions.

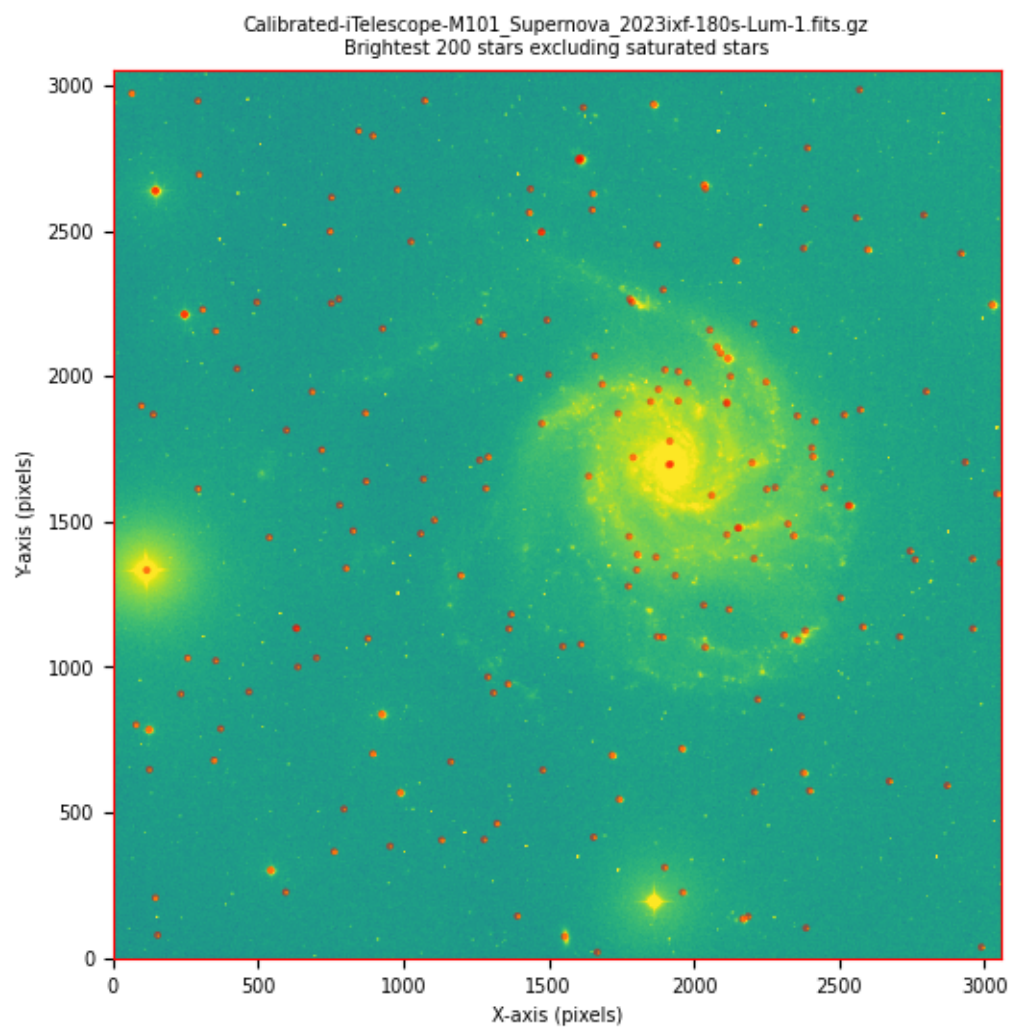
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Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
Brightest 200 stars excluding saturated stars

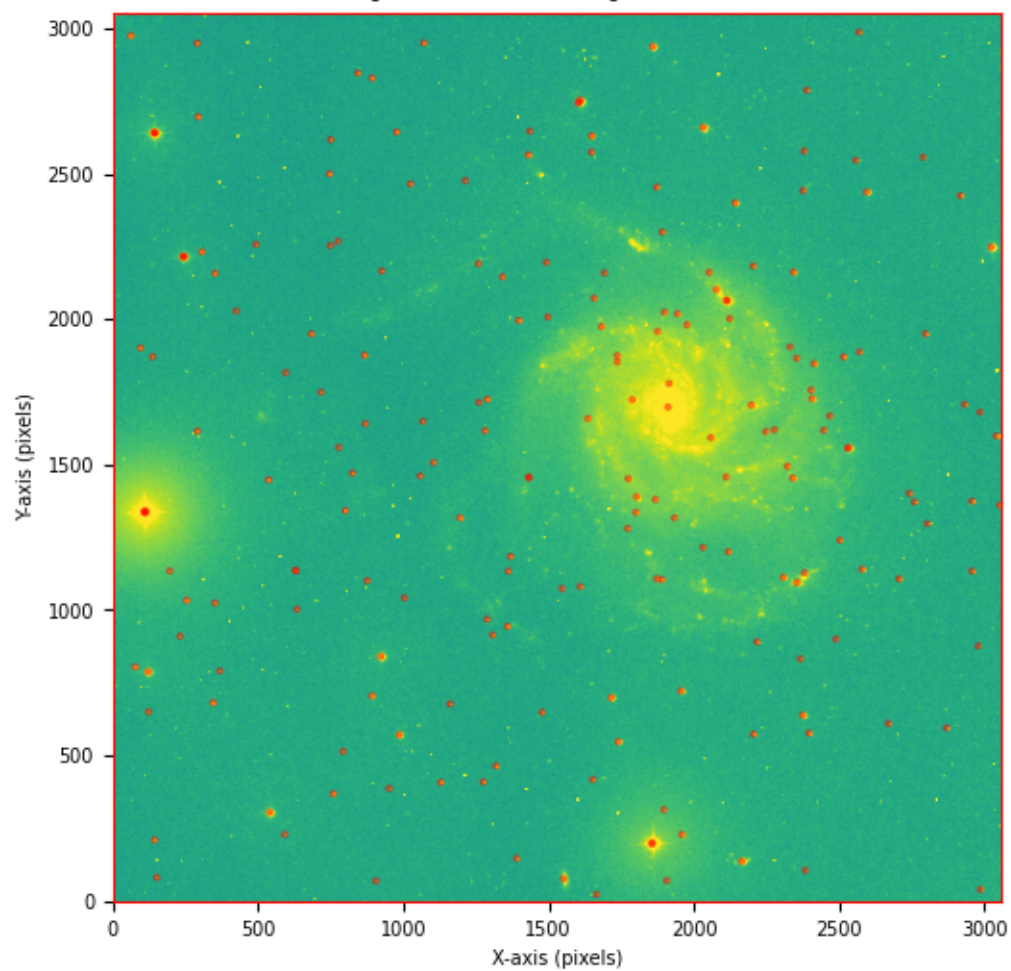


Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_2023xf-180s-Lum-1.fits.gz
 Median FWHM=4.30 +/- 0.27 (MAD stddev) pixels
 Along X FWHM=4.13 +/- 0.25 pixels, along Y FWHM=4.33 +/- 0.09 pixels

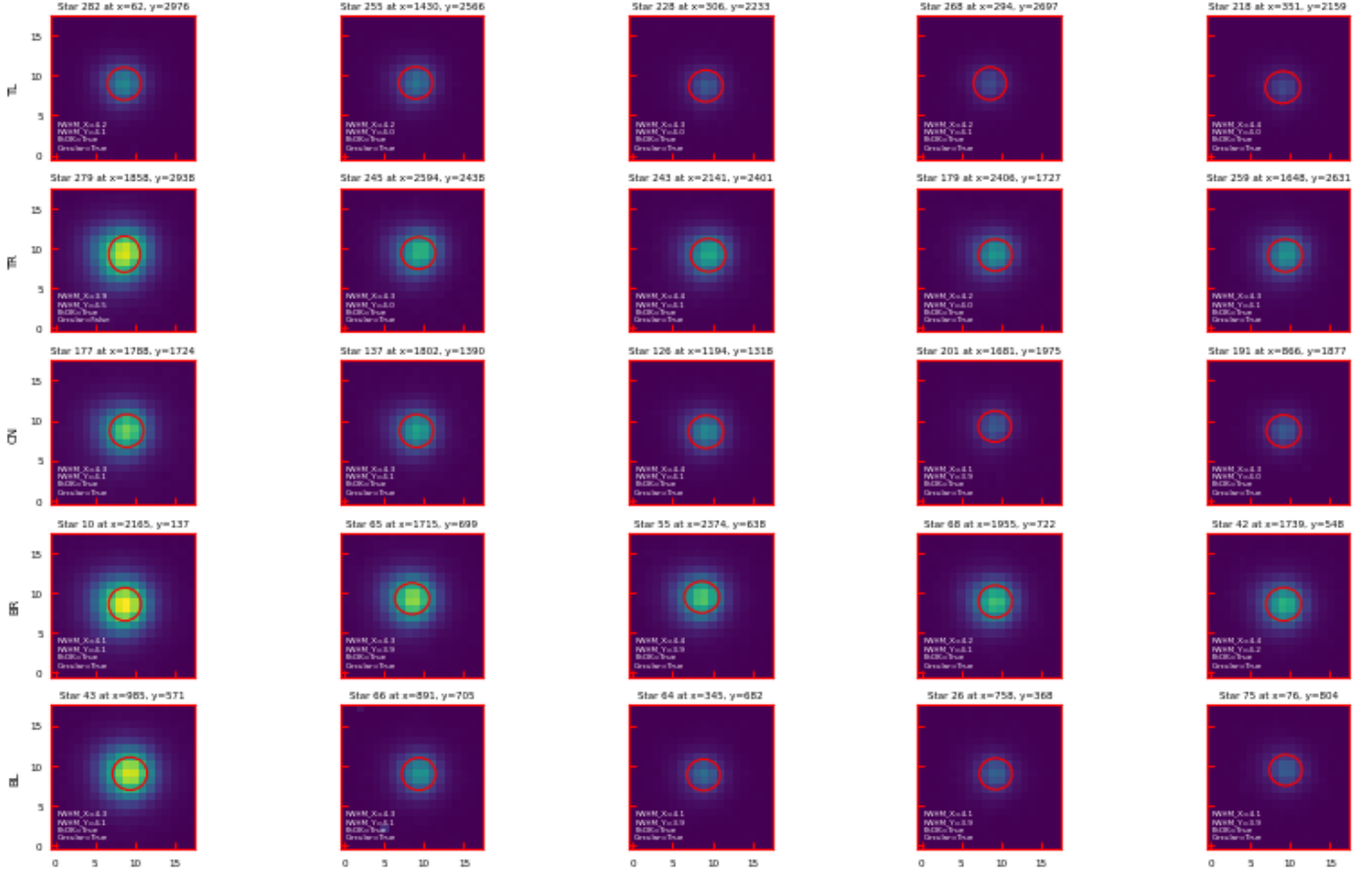




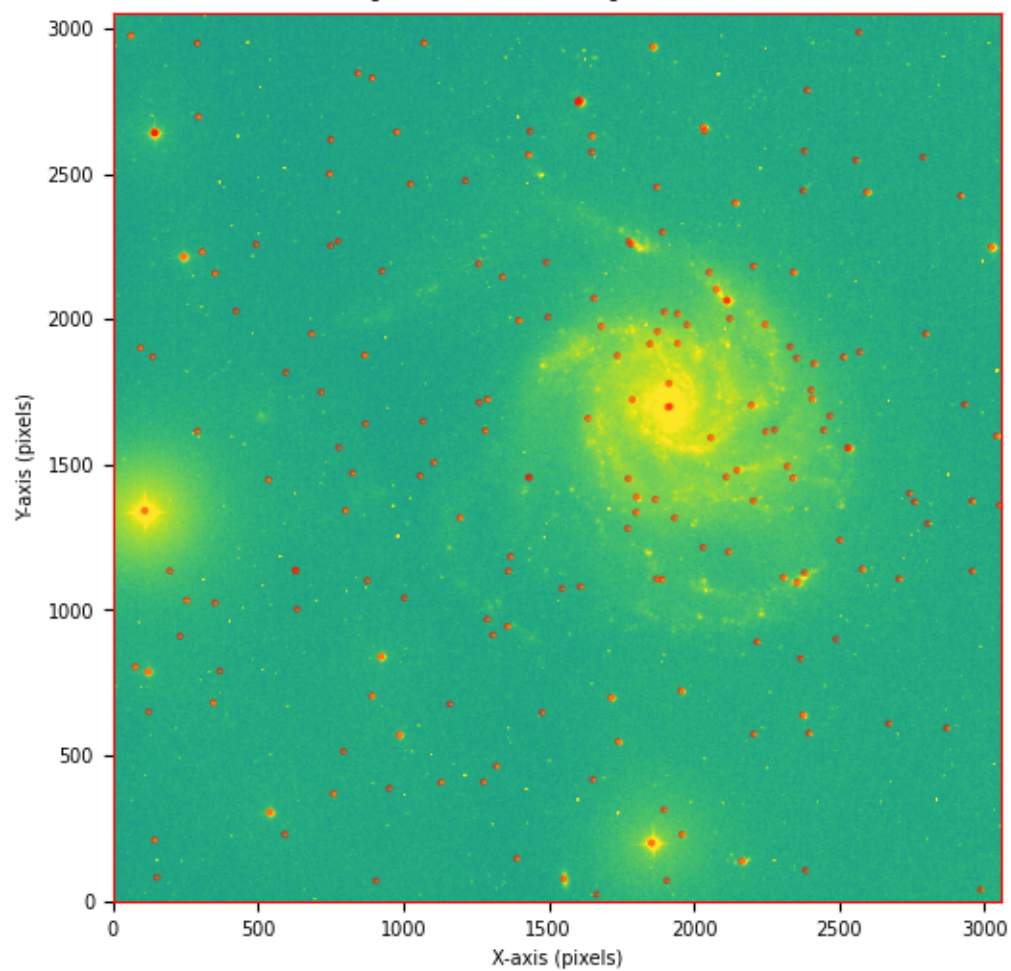
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
Brightest 200 stars excluding saturated stars

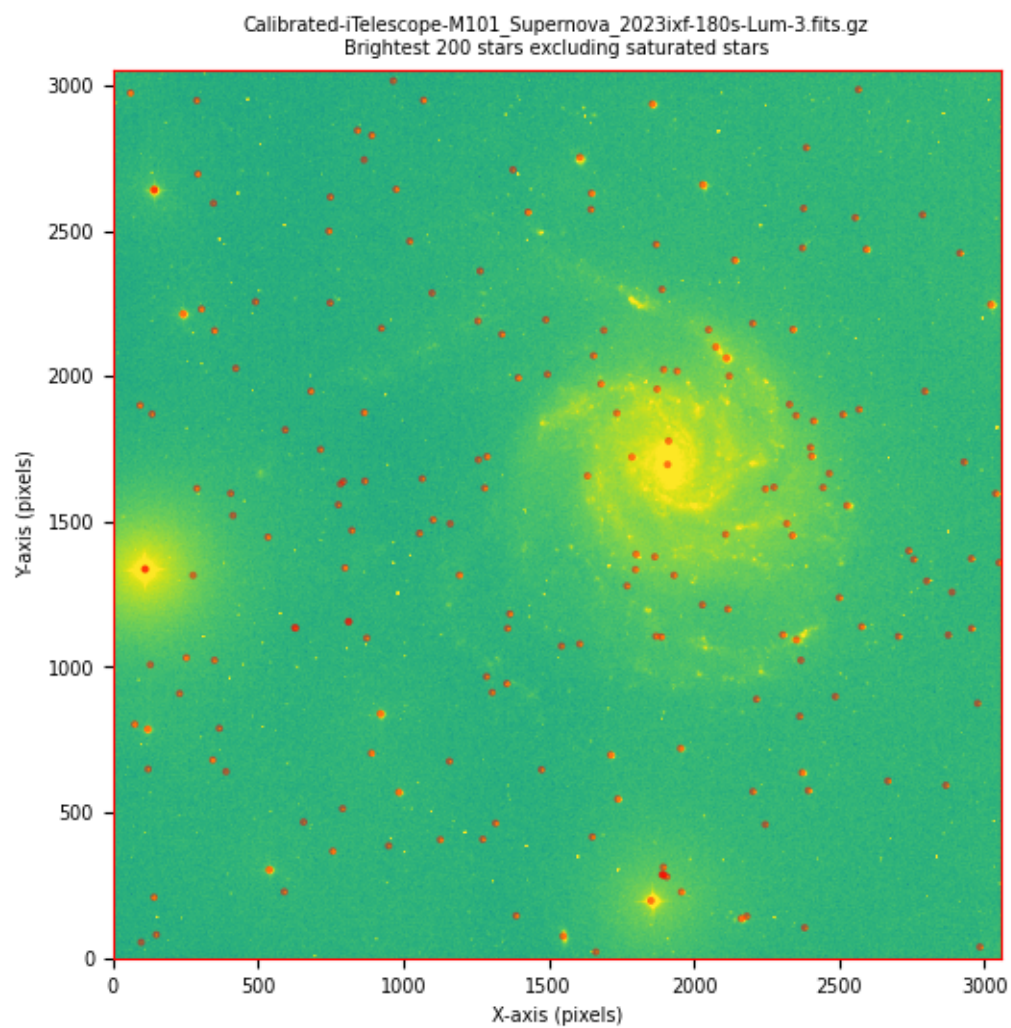


Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_2023xf-180s-Lum-2.fits.gz
 Median FWHM=4.14 +/- 0.17 (MAD stddev) pixels
 Along X FWHM=4.27 +/- 0.12 pixels, along Y FWHM=4.05 +/- 0.11 pixels

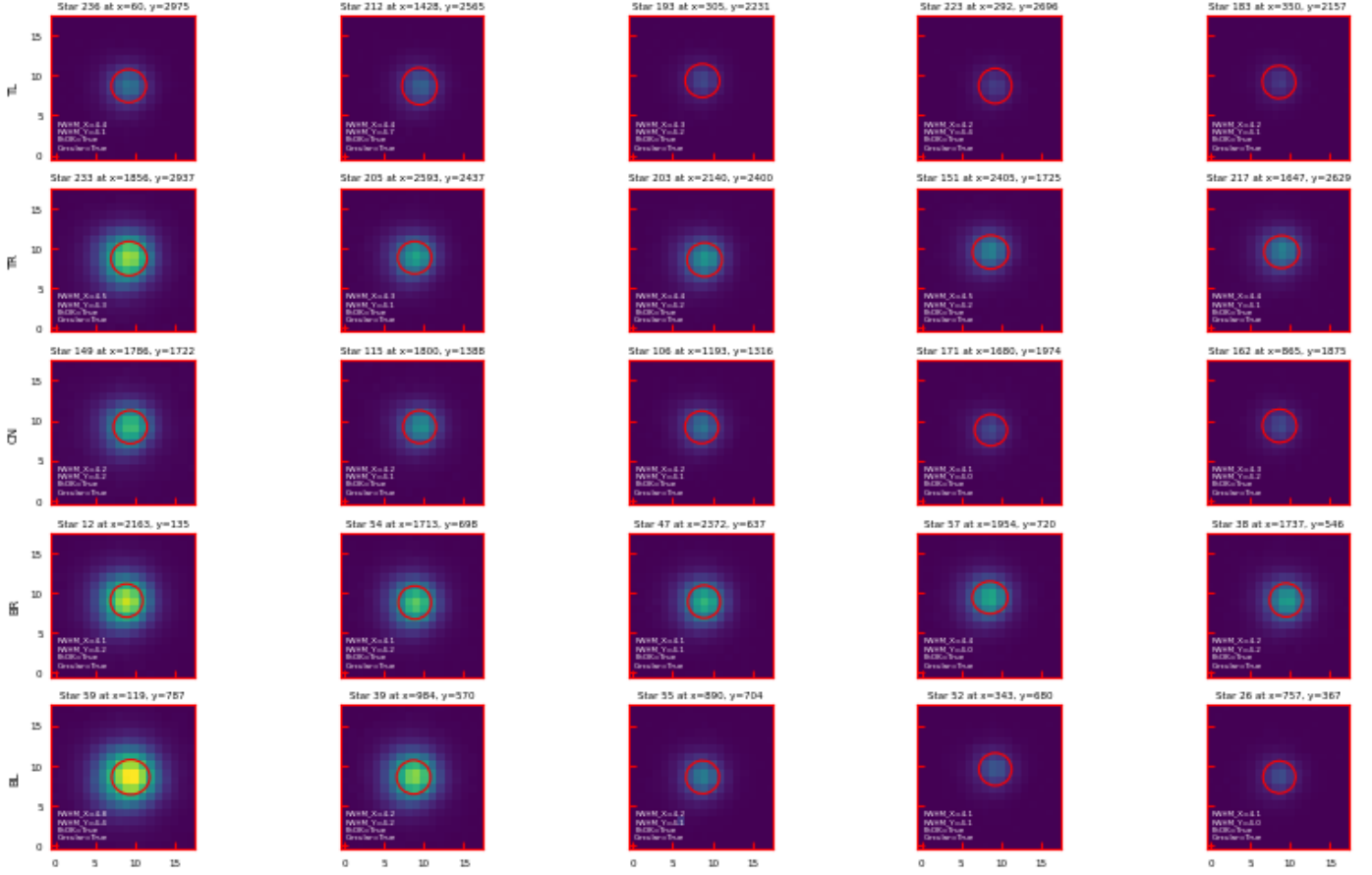


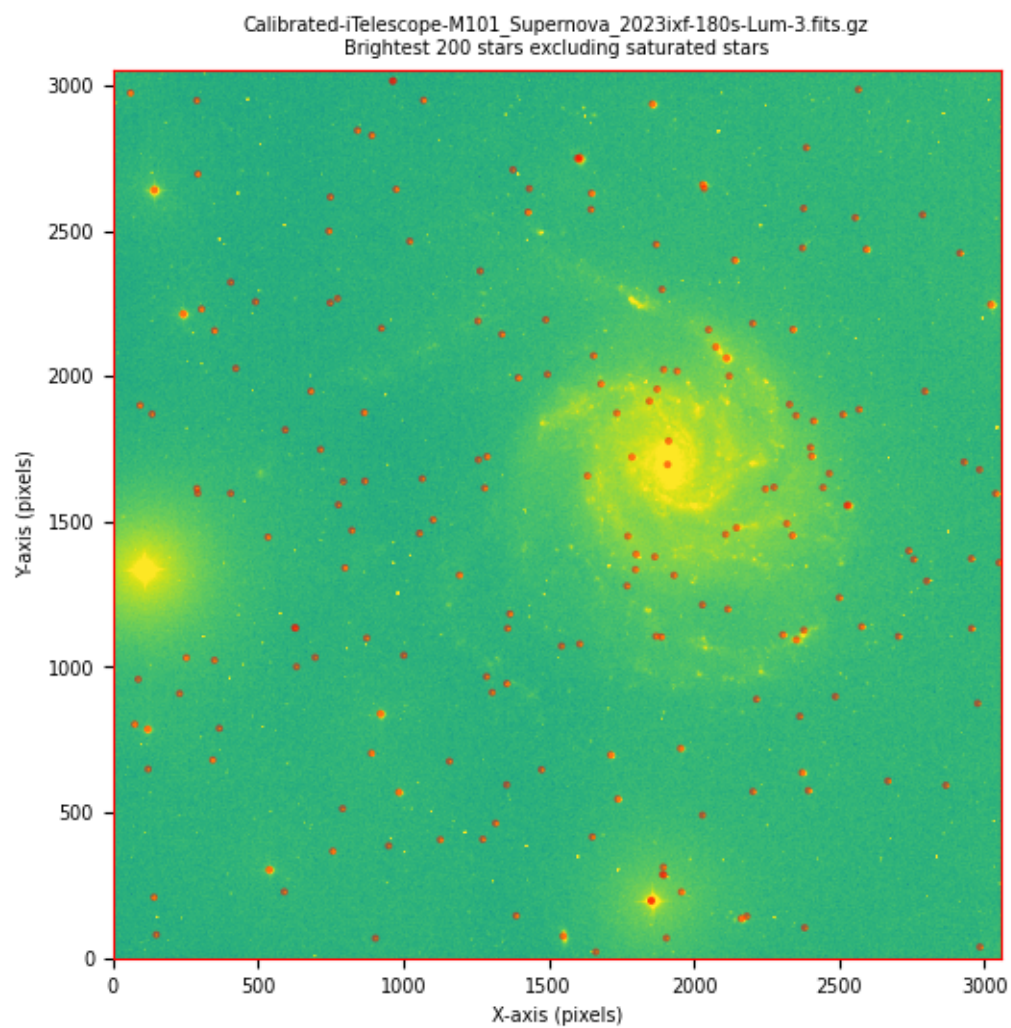
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
Brightest 200 stars excluding saturated stars



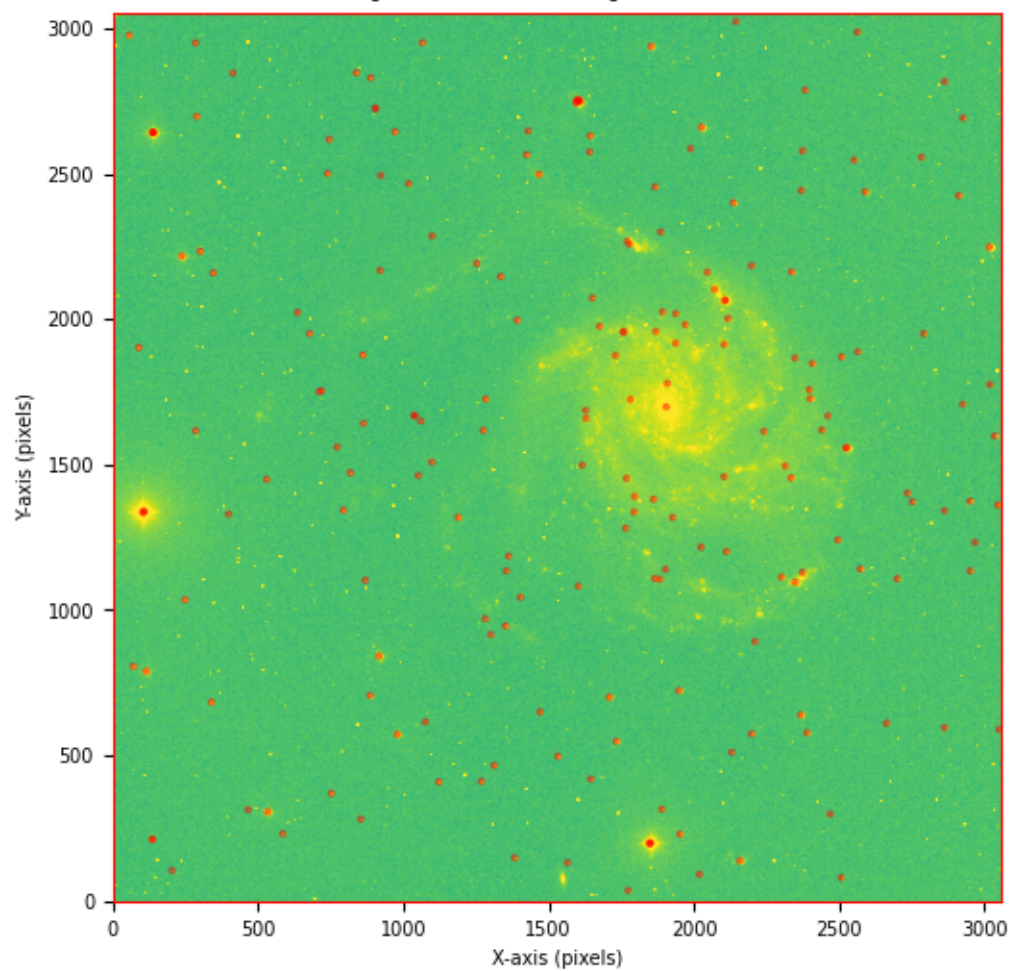


Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_2023xf-180s-Lum-3.fits.gz
 Median FWHM=4.18 +/- 0.10 (MAD stddev) pixels
 Along X FWHM=4.22 +/- 0.10 pixels, along Y FWHM=4.15 +/- 0.10 pixels

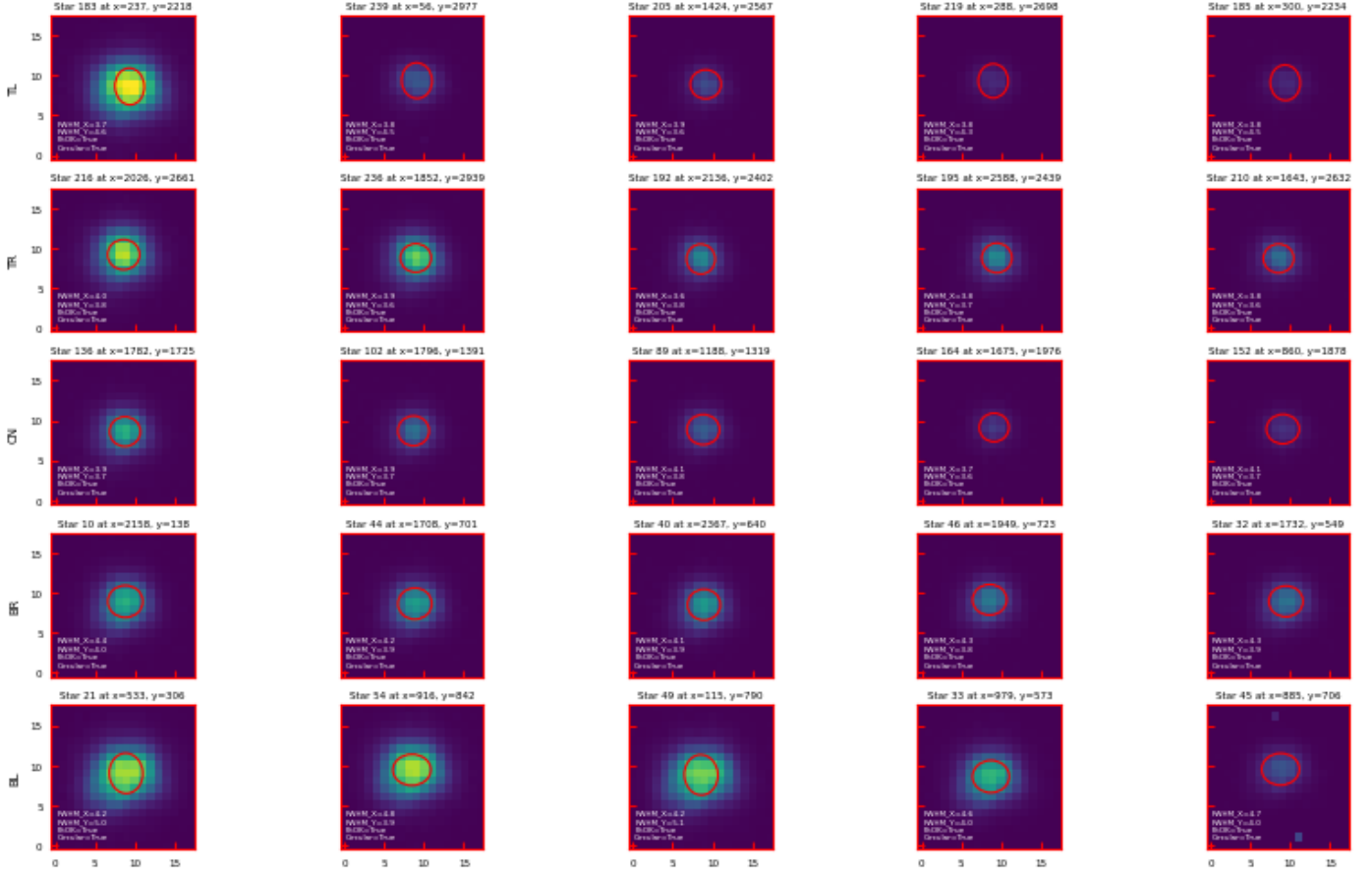


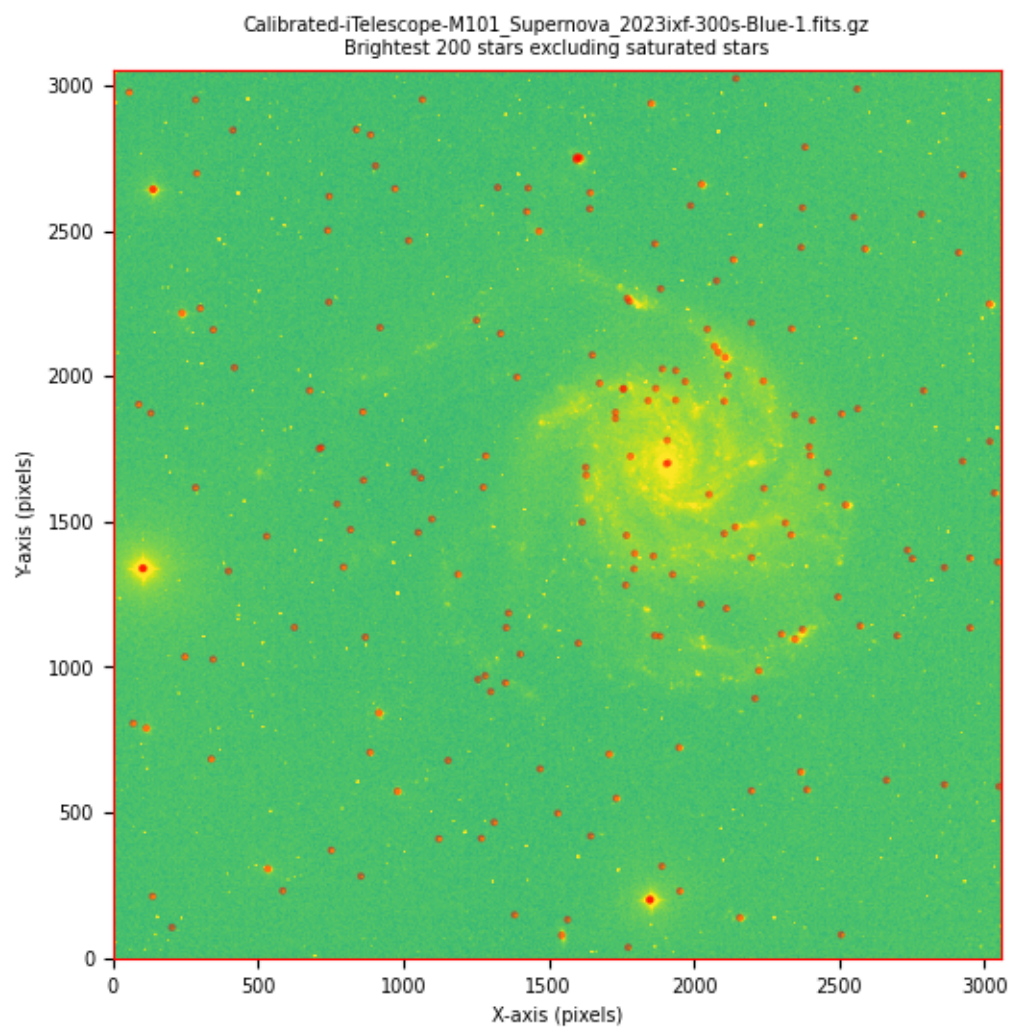


Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
Brightest 200 stars excluding saturated stars

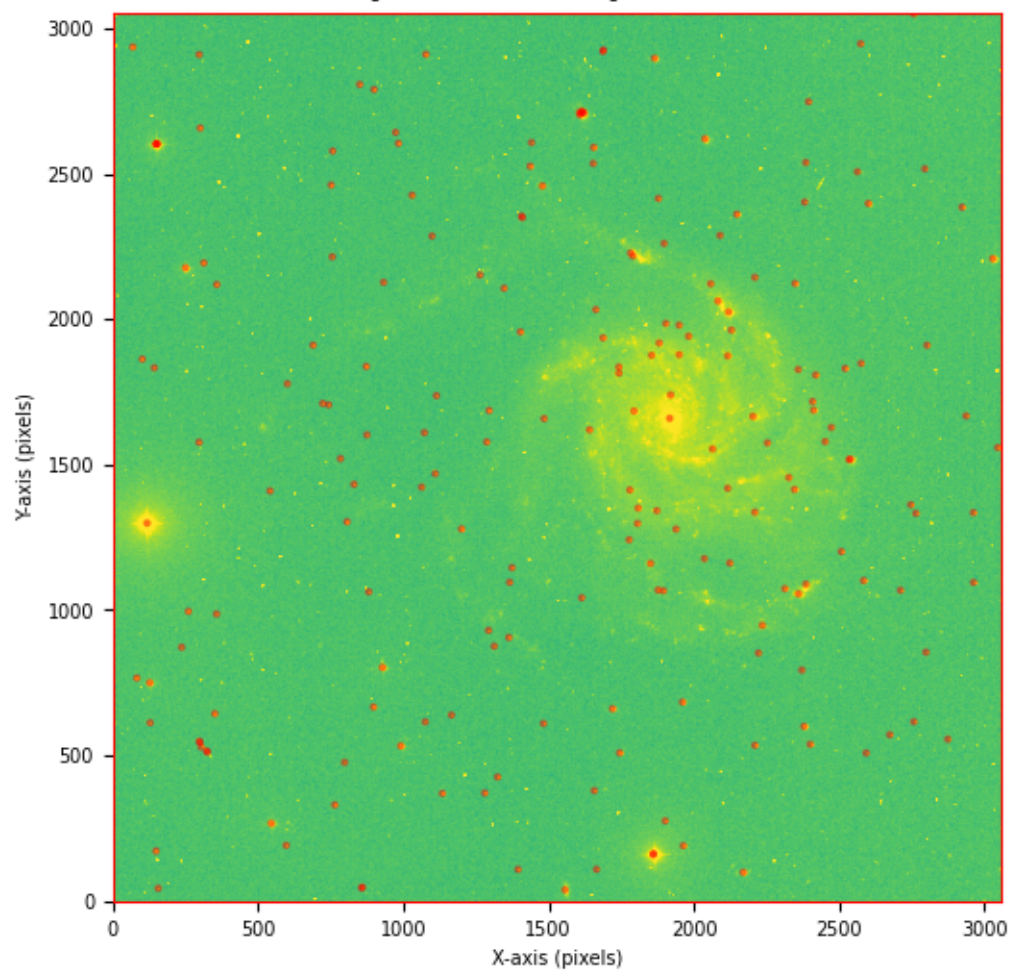


Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_2023xf-300s-Blue-1.fits.gz
 Median FWHM=3.88 +/- 0.23 (MAD stddev) pixels
 Along X FWHM=4.02 +/- 0.34 pixels, along Y FWHM=3.86 +/- 0.22 pixels

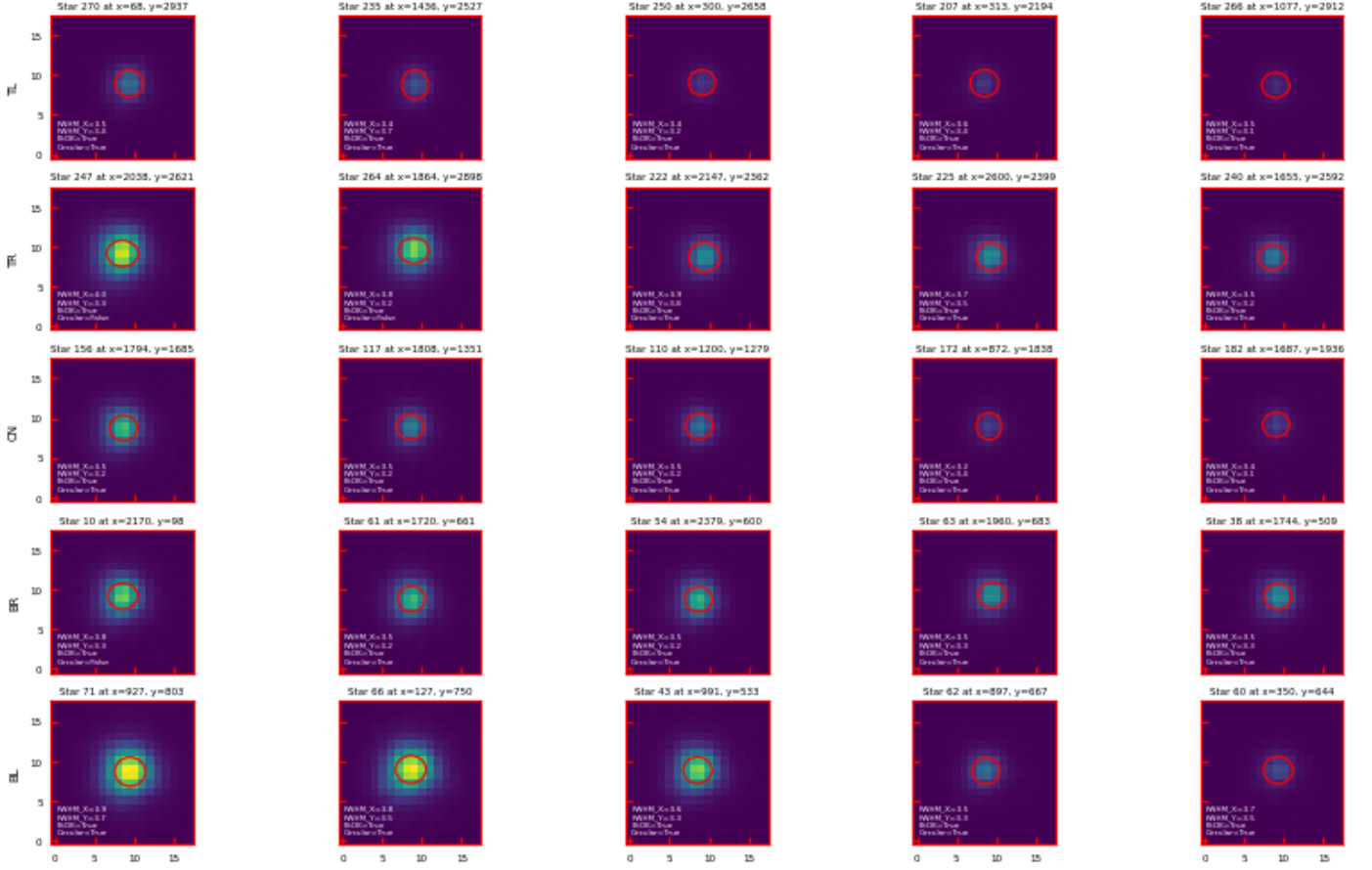


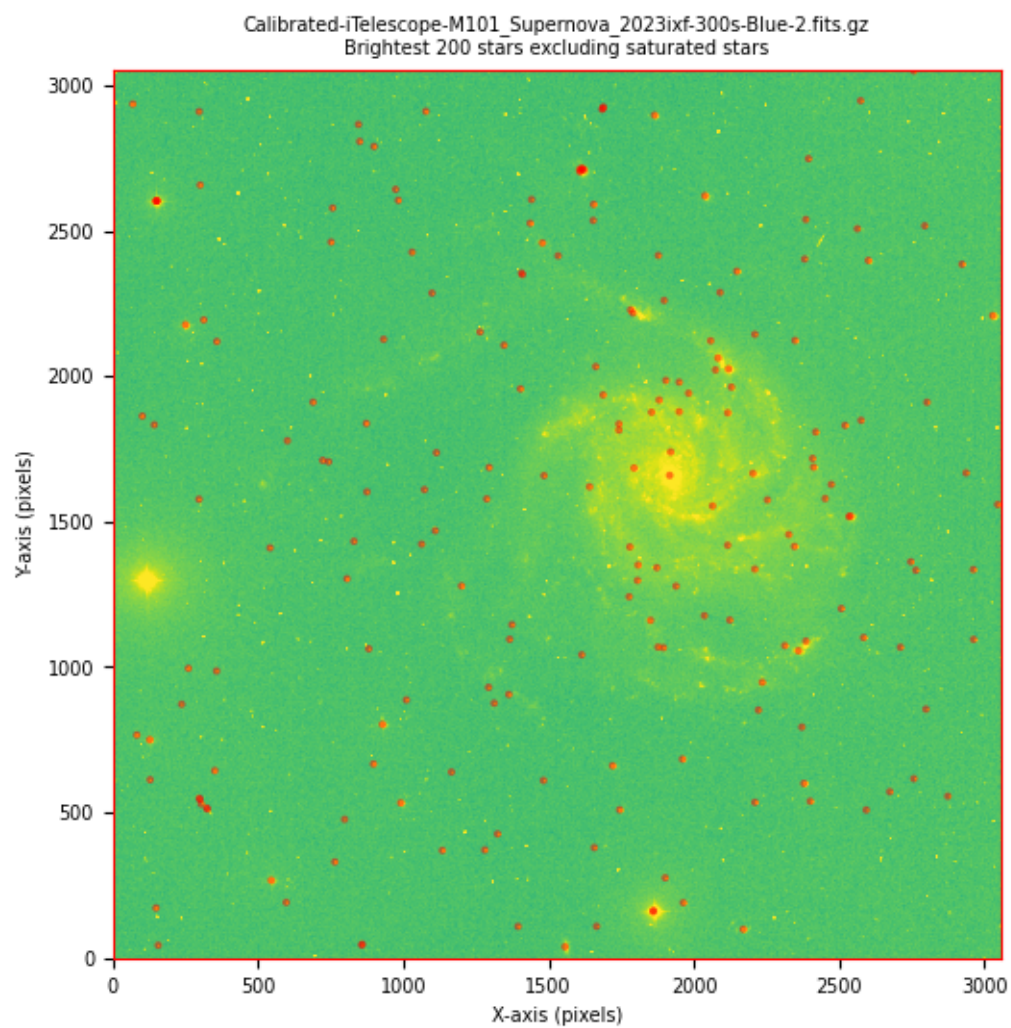


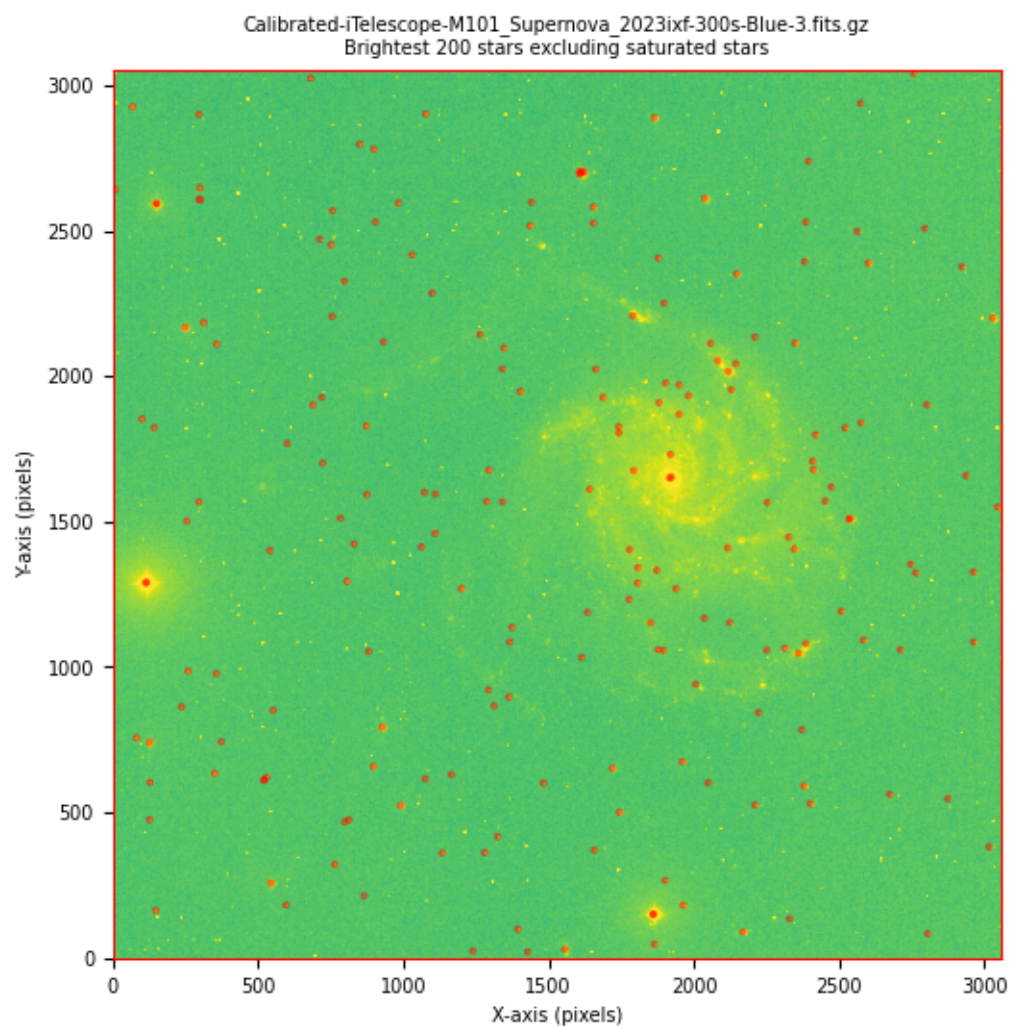
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
Brightest 200 stars excluding saturated stars



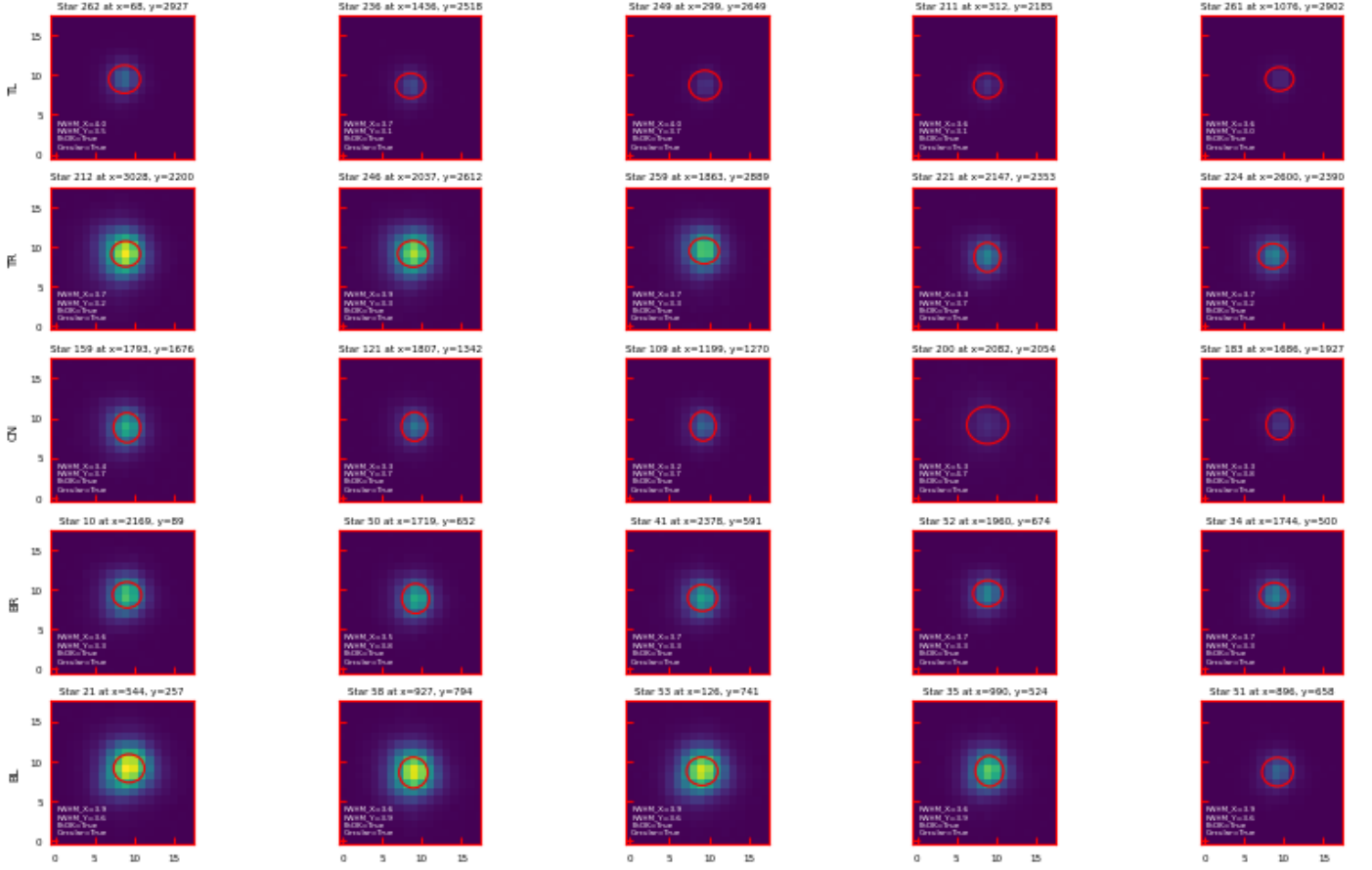
Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_2023xf-300s-Blue-2.fits.gz
 Median FWHM=3.47 +/- 0.28 (MAD stddev) pixels
 Along X FWHM=3.50 +/- 0.11 pixels, along Y FWHM=3.26 +/- 0.13 pixels

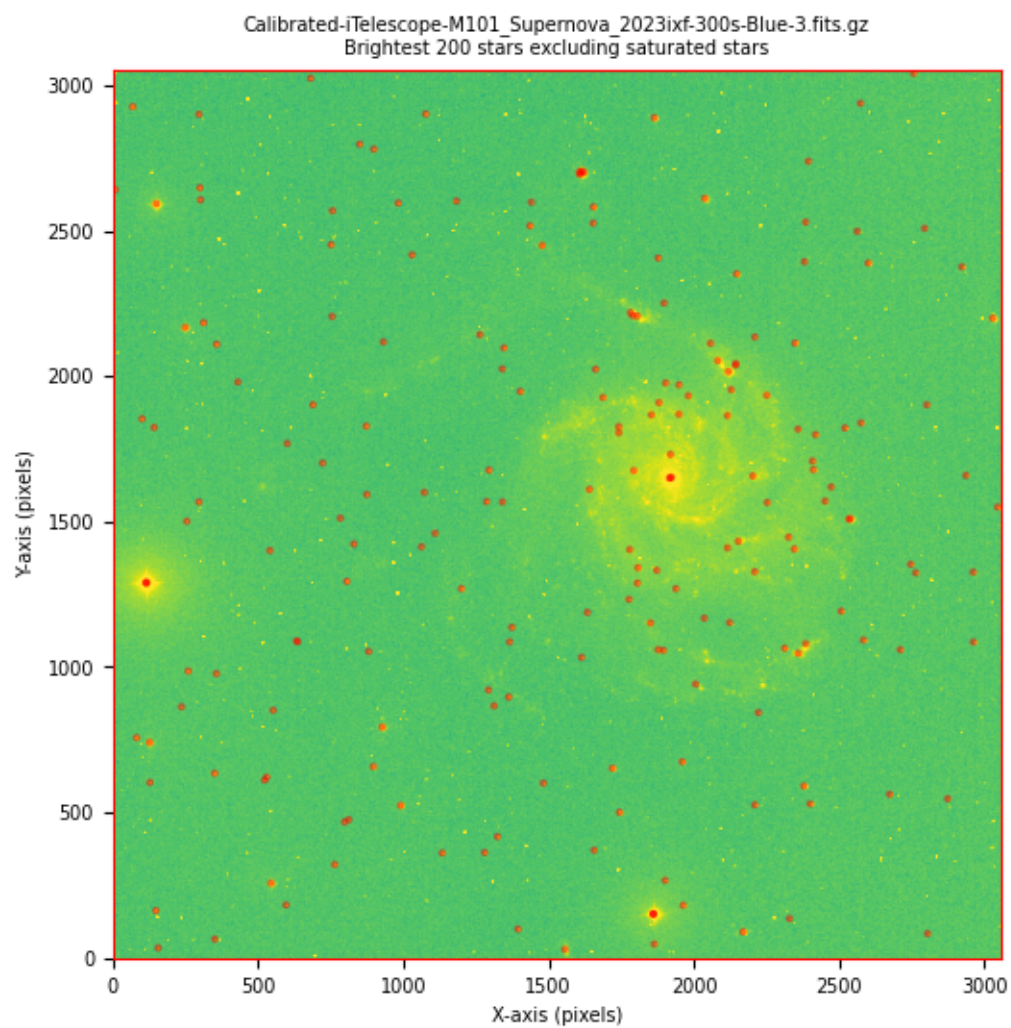




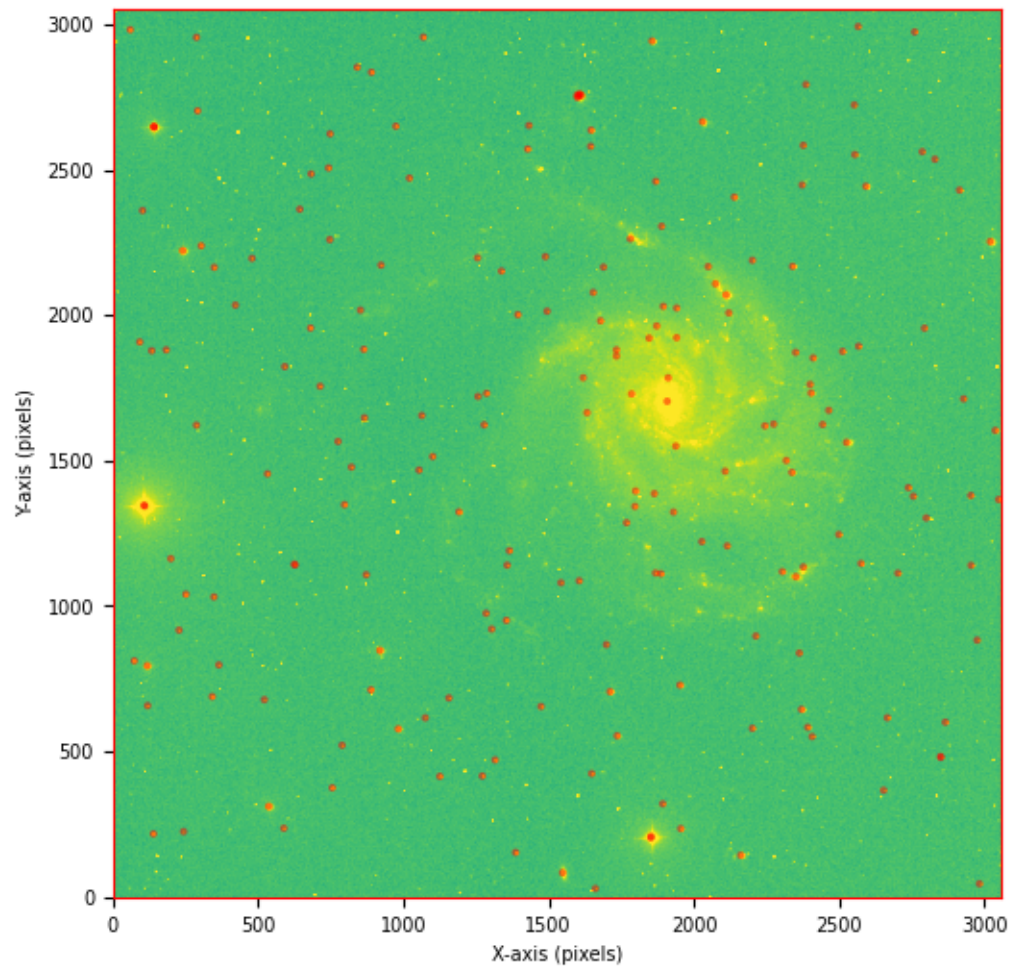


Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_2023xf-300s-Blue-3.fits.gz
 Median FWHM=3.63 +/- 0.29 (MAD stddev) pixels
 Along X FWHM=3.66 +/- 0.22 pixels, along Y FWHM=3.55 +/- 0.34 pixels

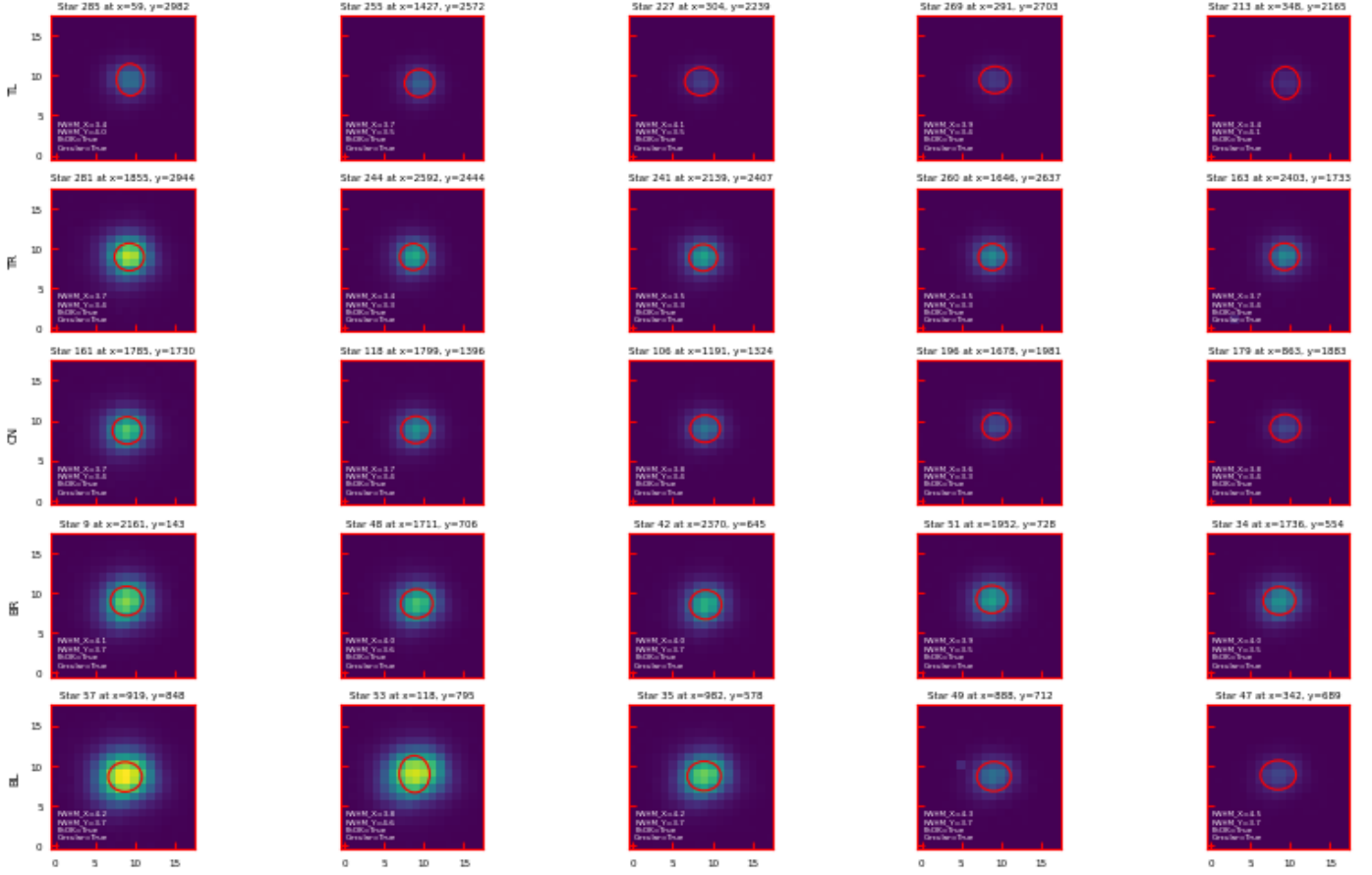




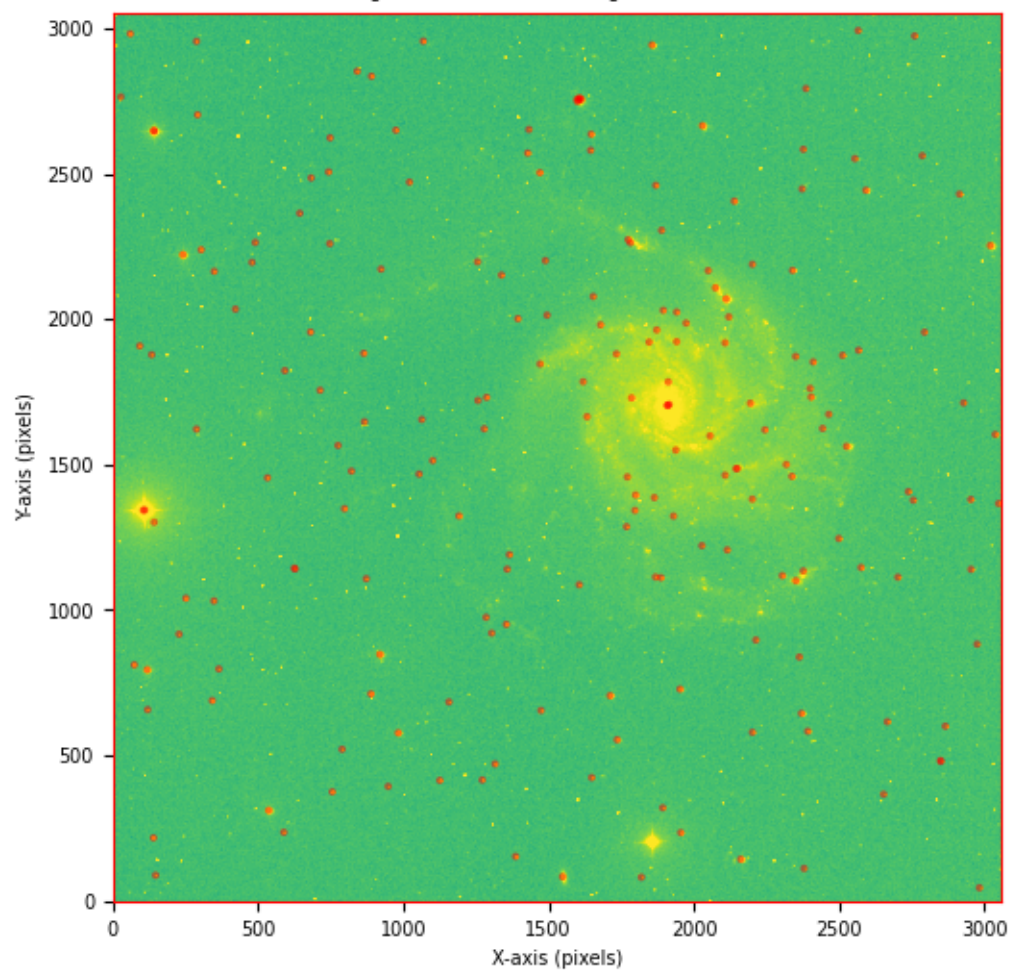
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
Brightest 200 stars excluding saturated stars



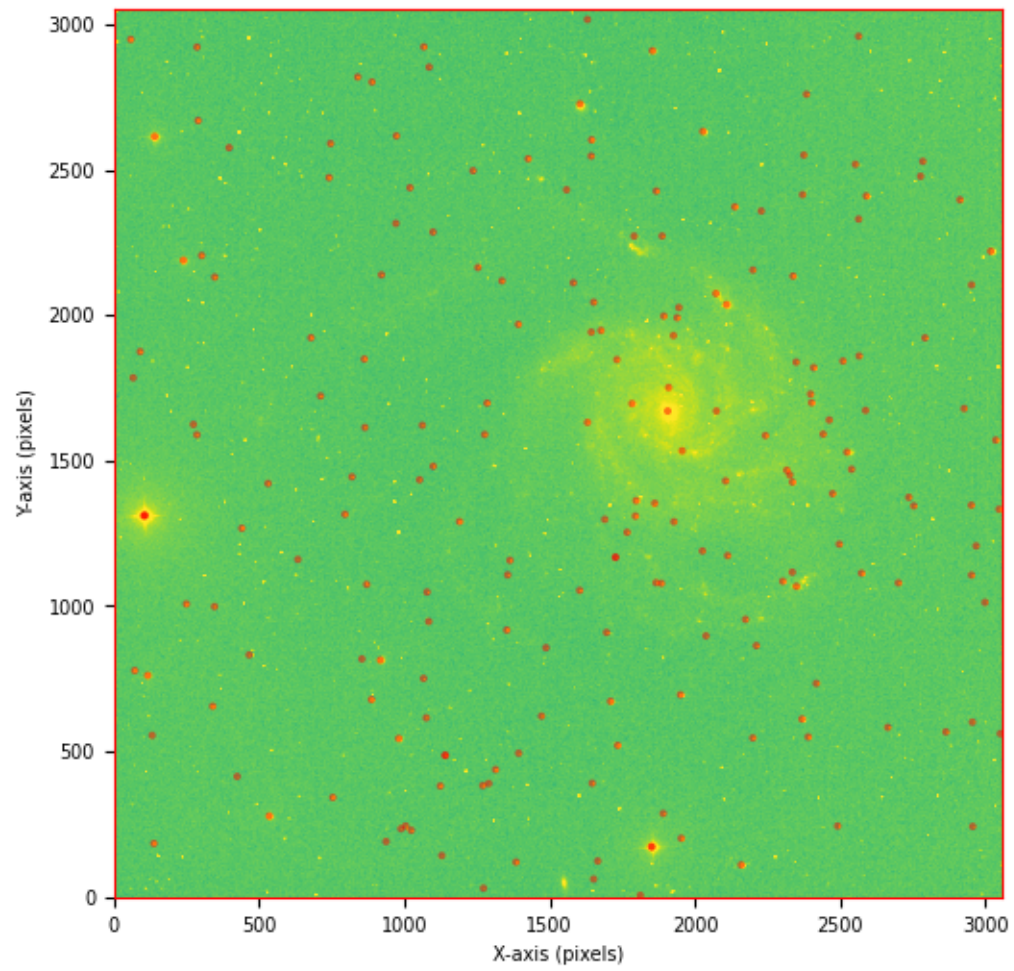
Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_20231xf-300s-Green-1.fits.gz
 Median FWHM=3.67 +/- 0.35 [MAD stddev] pixels
 Along X FWHM=3.81 +/- 0.35 pixels, along Y FWHM=3.47 +/- 0.19 pixels



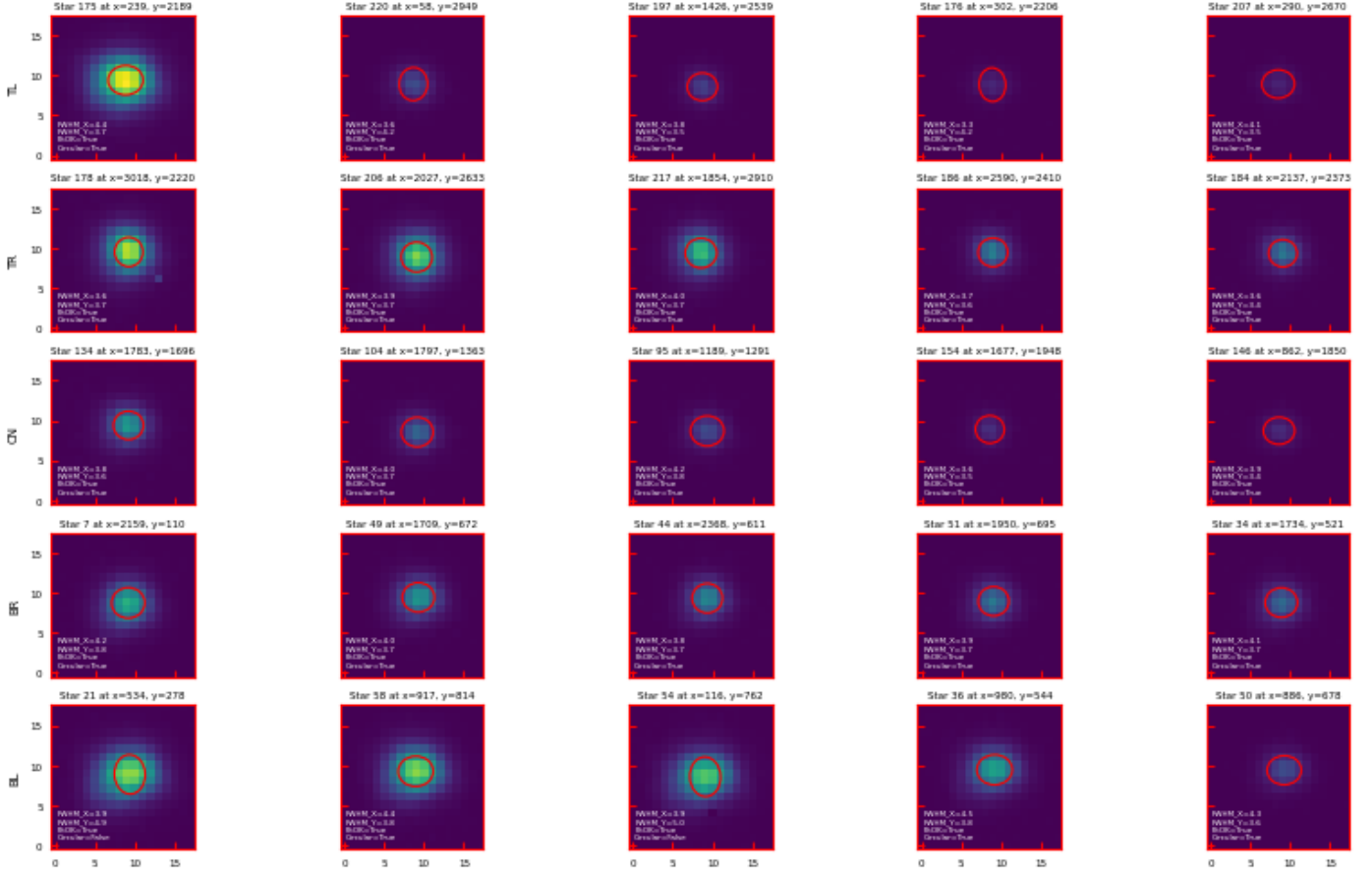
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
Brightest 200 stars excluding saturated stars



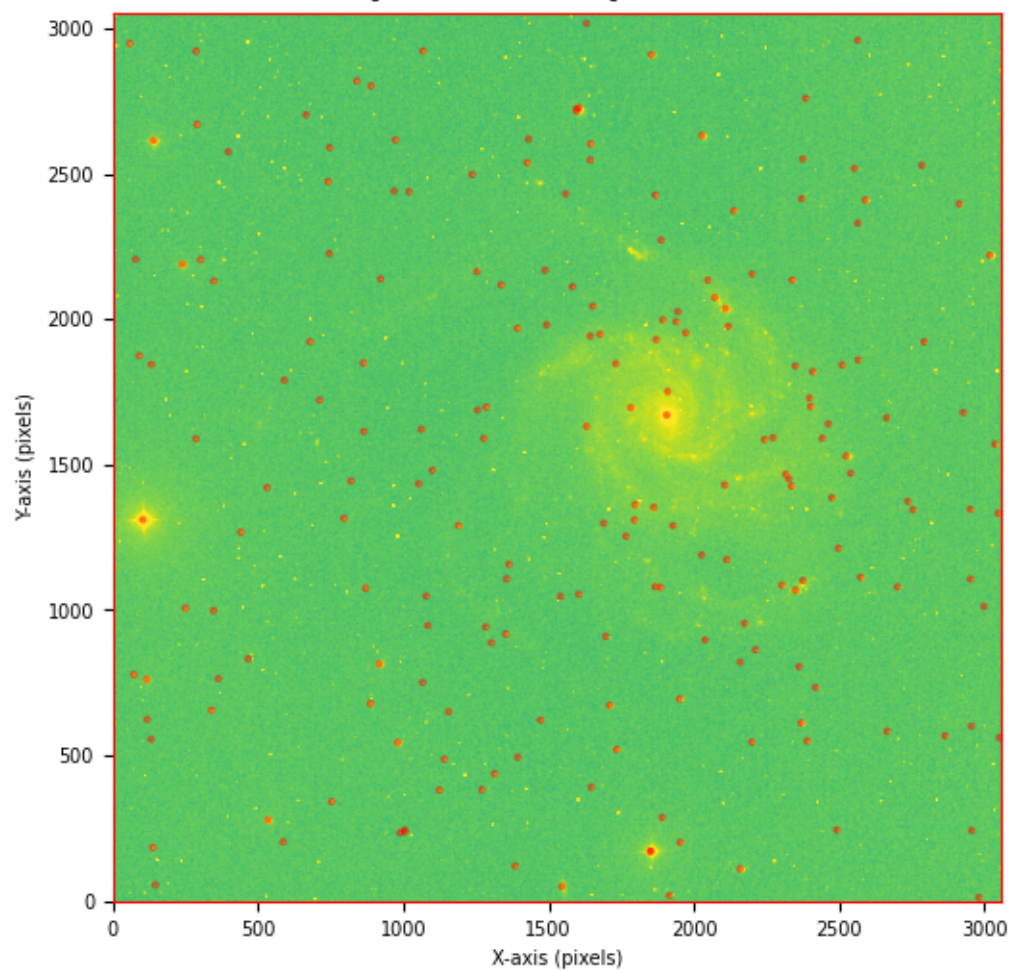
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
Brightest 200 stars excluding saturated stars



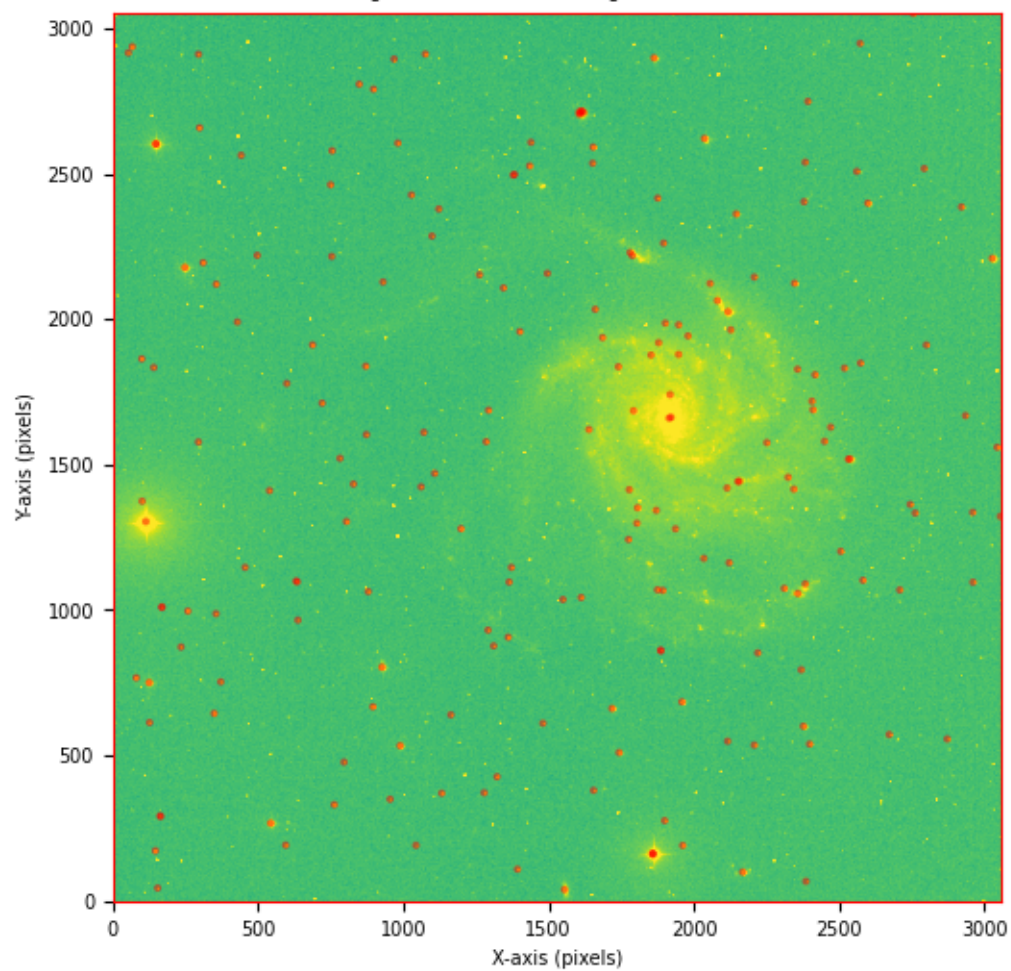
Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_20231xf-300s-Green-2.fits.gz
 Median FWHM=3.77 +/- 0.25 [MAD stddev] pixels
 Along X FWHM=3.90 +/- 0.28 pixels, along Y FWHM=3.65 +/- 0.12 pixels



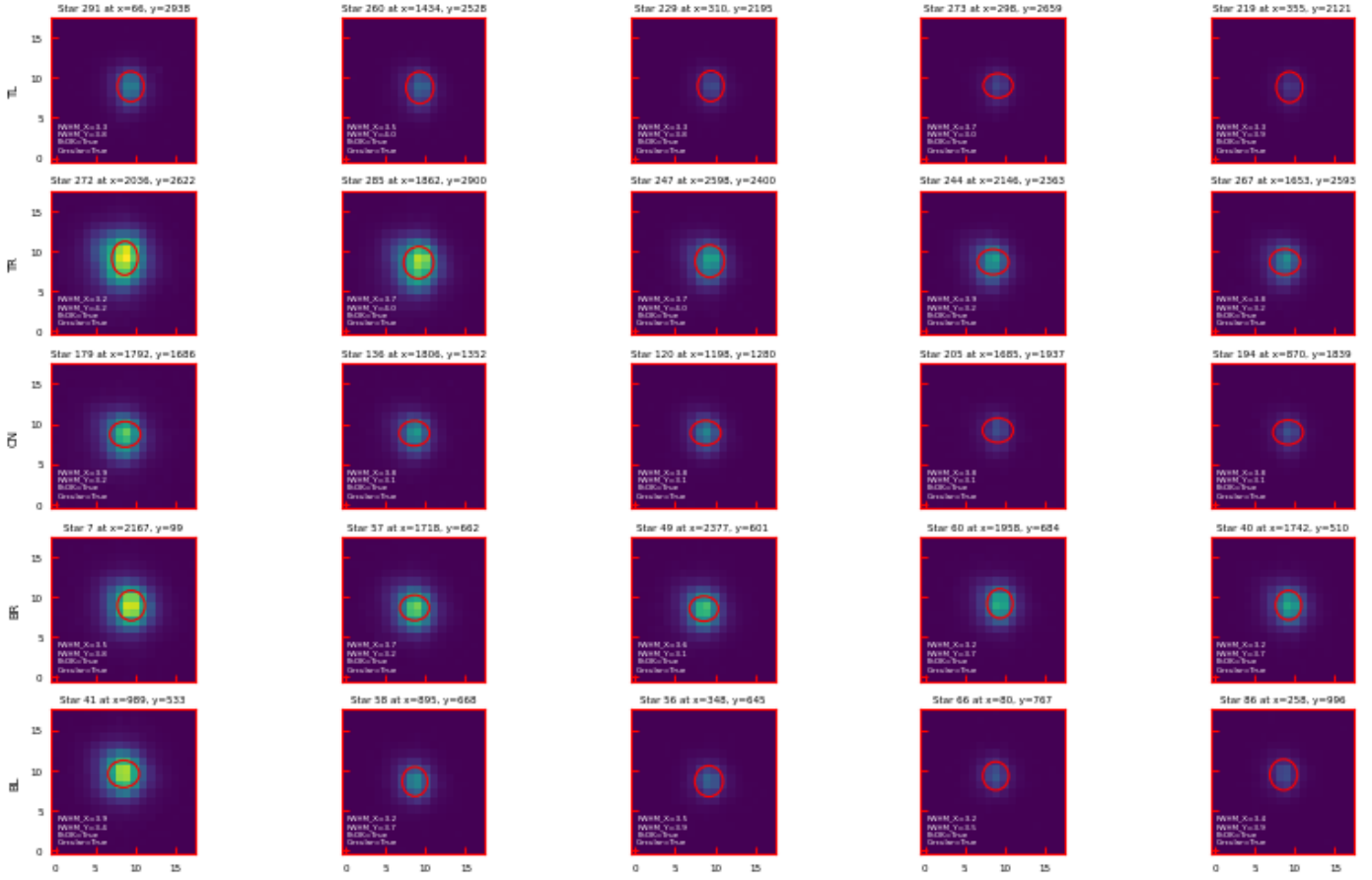
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
Brightest 200 stars excluding saturated stars



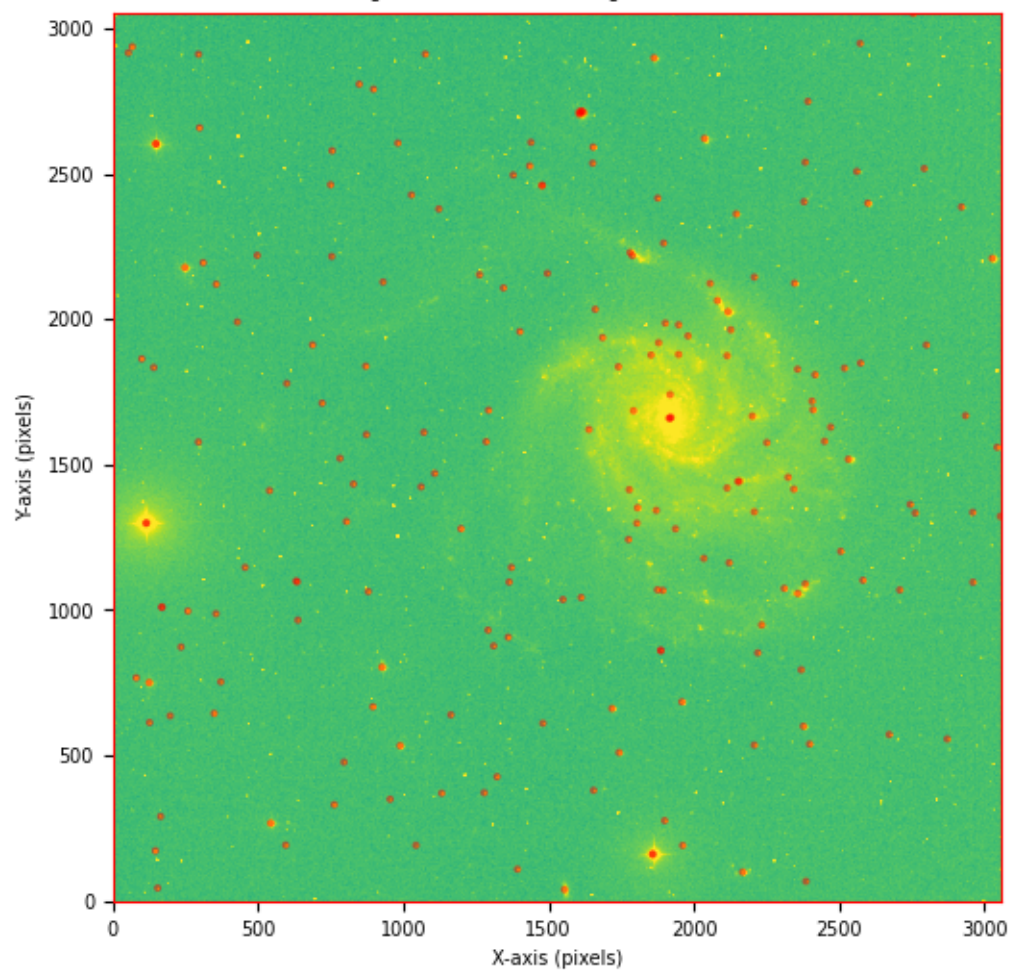
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
Brightest 200 stars excluding saturated stars



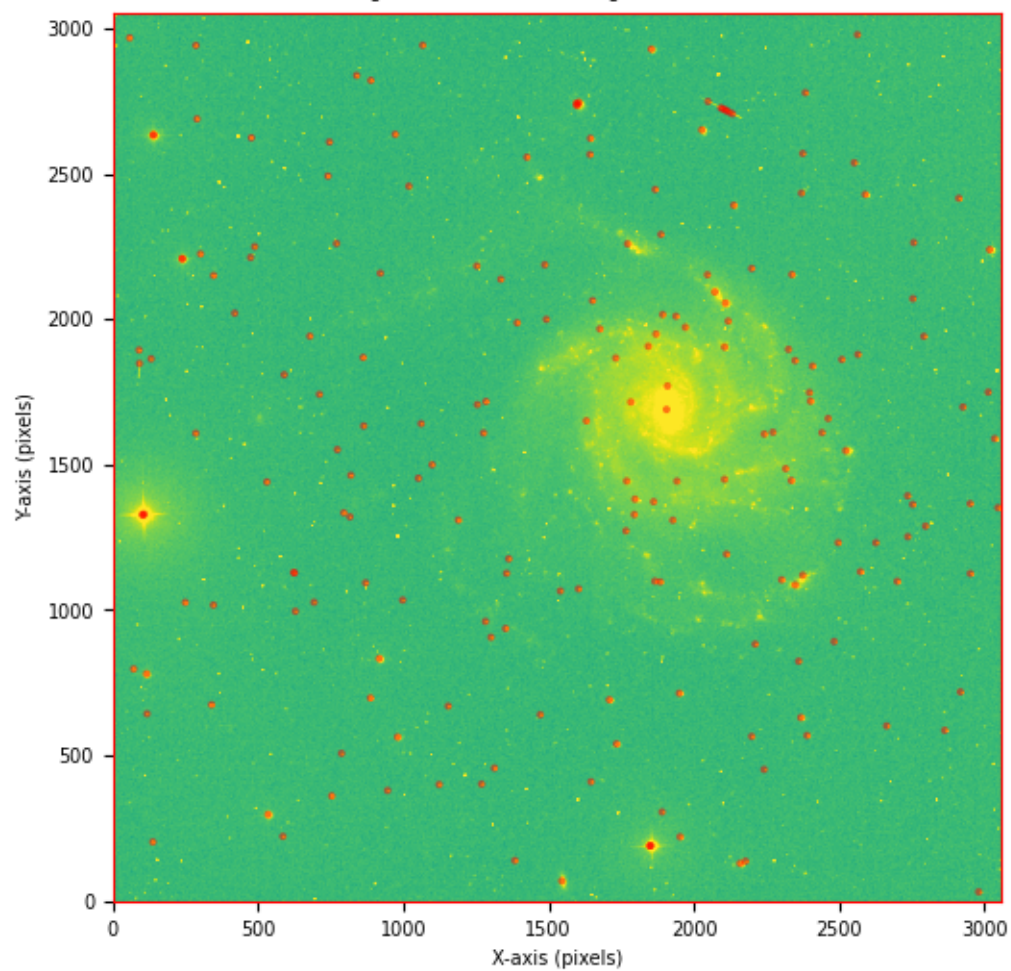
Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_20231xf-300s-Green-3.fits.gz
 Median FWHM=3.64 +/- 0.37 [MAD stddev] pixels
 Along X FWHM=3.63 +/- 0.29 pixels, along Y FWHM=3.67 +/- 0.50 pixels



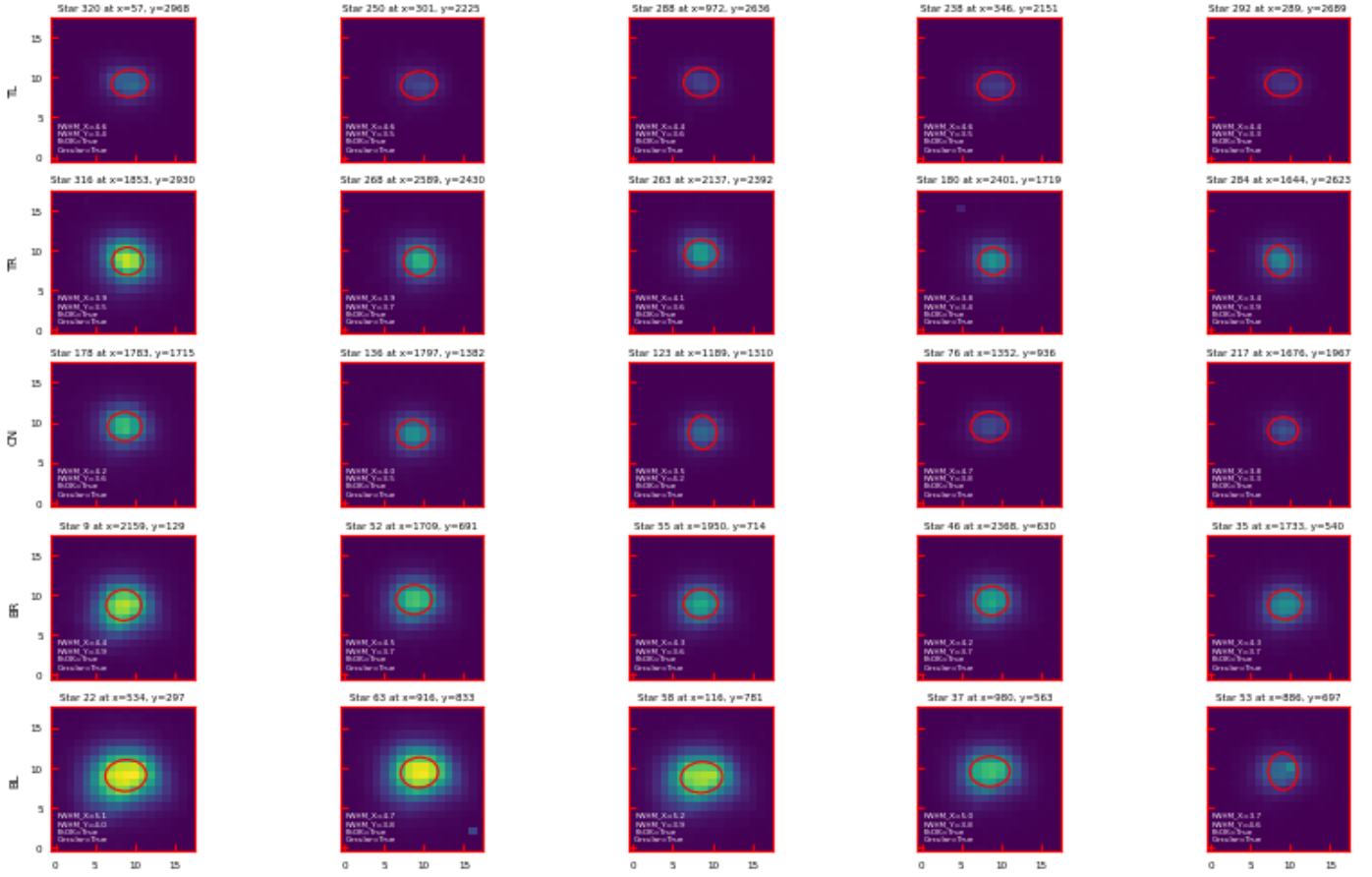
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
Brightest 200 stars excluding saturated stars



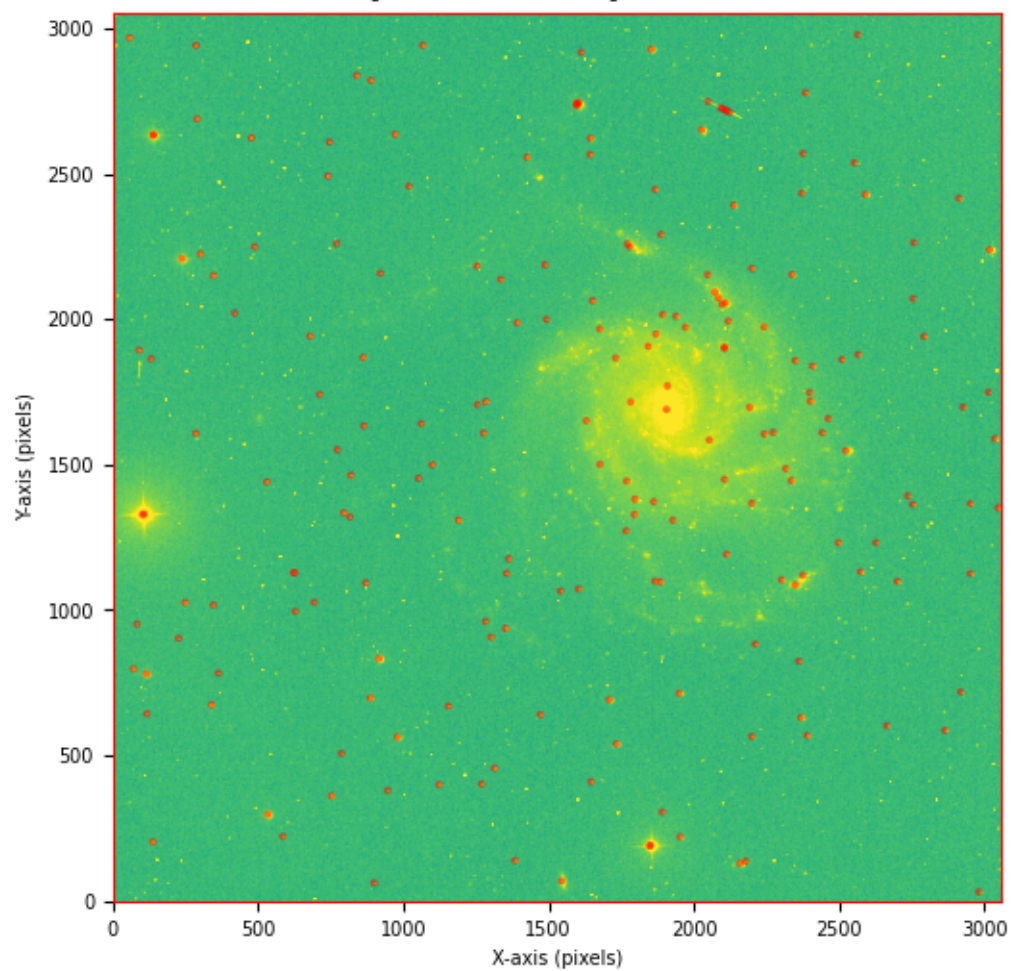
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
Brightest 200 stars excluding saturated stars



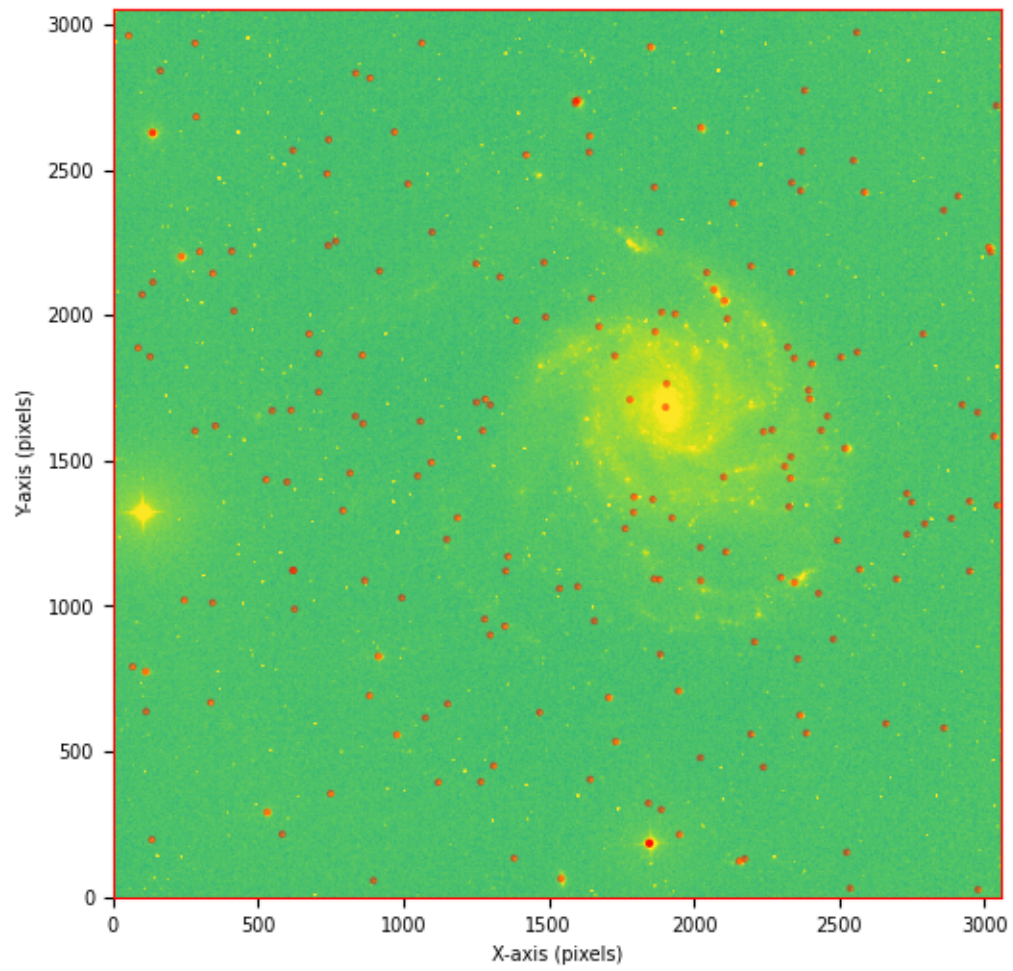
Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_20231xf-300s-Red-1.fits.gz
 Median FWHM=3.86 +/- 0.53 (RAD stddev) pixels
 Along X FWHM=4.32 +/- 0.54 pixels, along Y FWHM=3.65 +/- 0.23 pixels



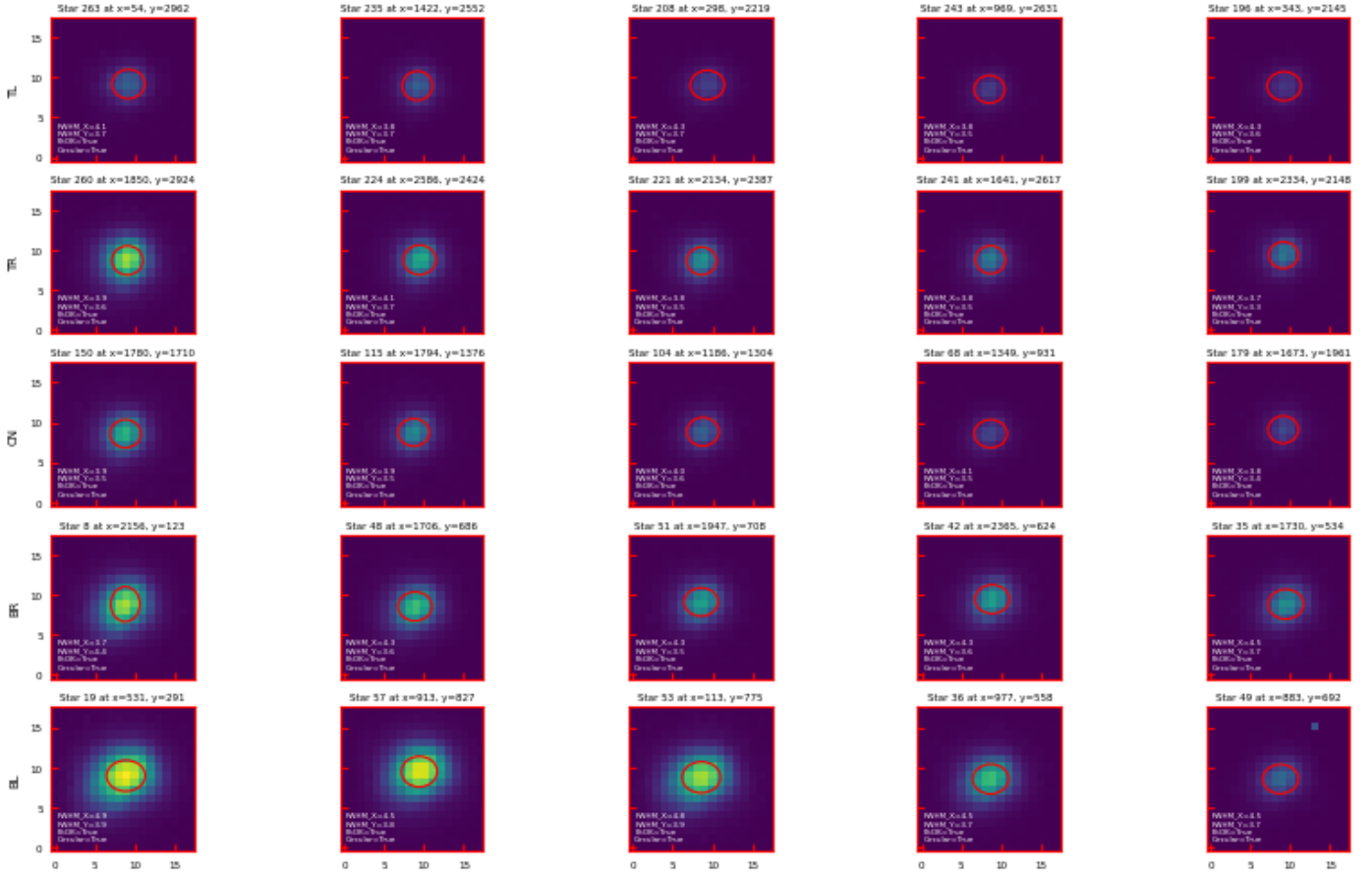
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
Brightest 200 stars excluding saturated stars



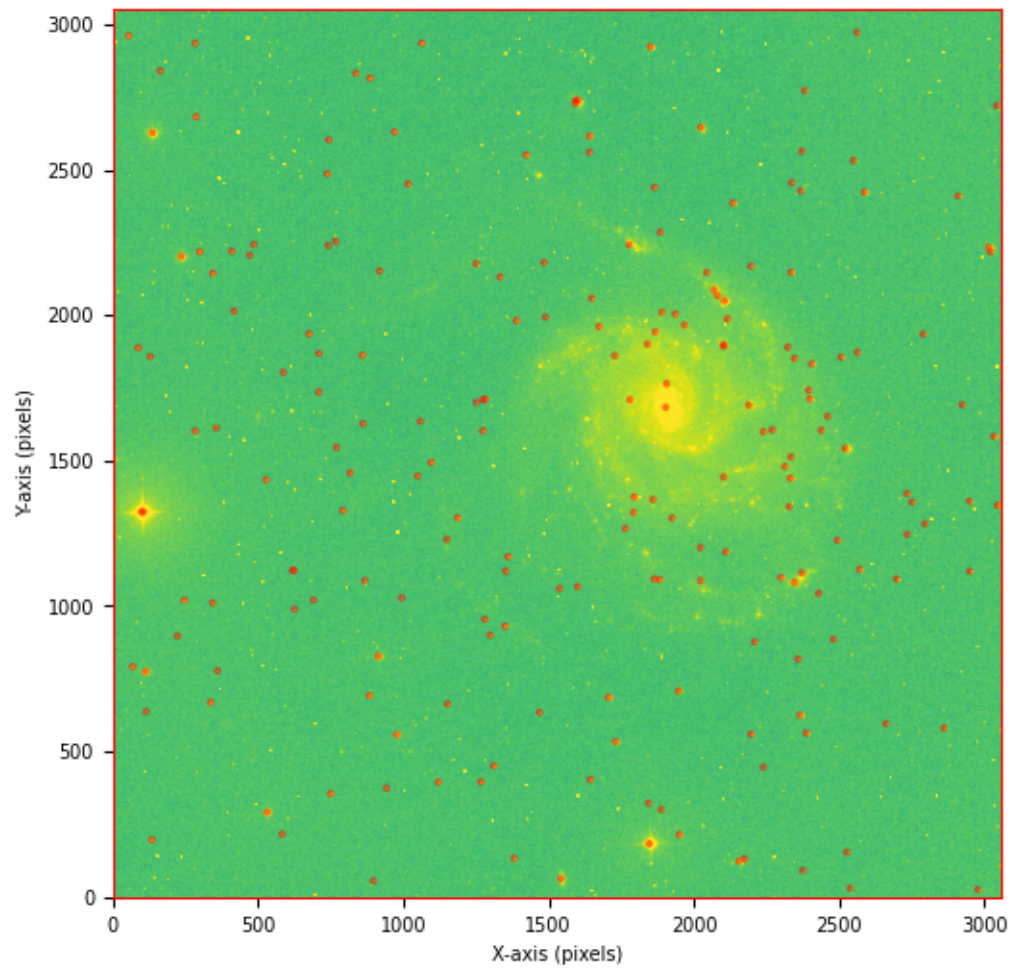
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
Brightest 200 stars excluding saturated stars

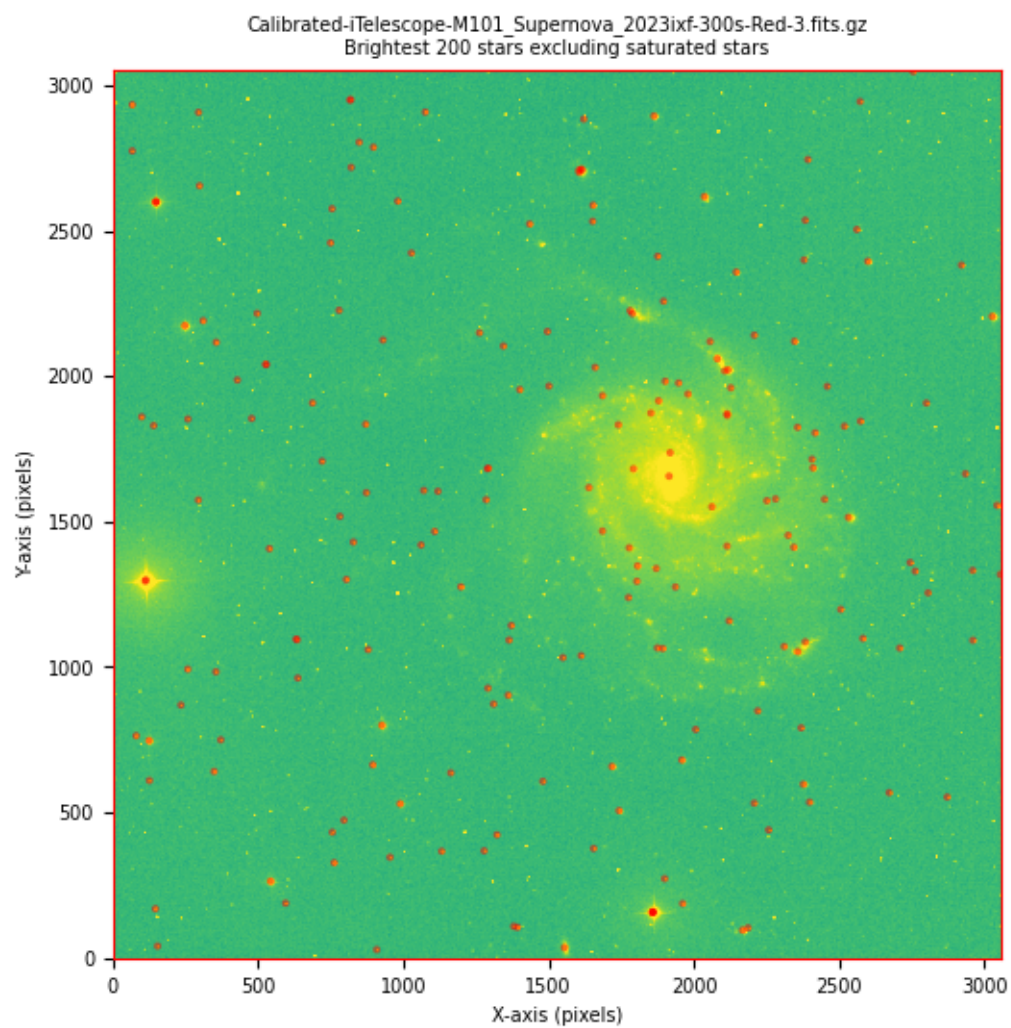


Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_20231xf-300s-Red-2.fits.gz
 Median FWHM=3.80 +/- 0.33 (RAD stddev) pixels
 Along X FWHM=4.14 +/- 0.46 pixels, along Y FWHM=3.63 +/- 0.15 pixels

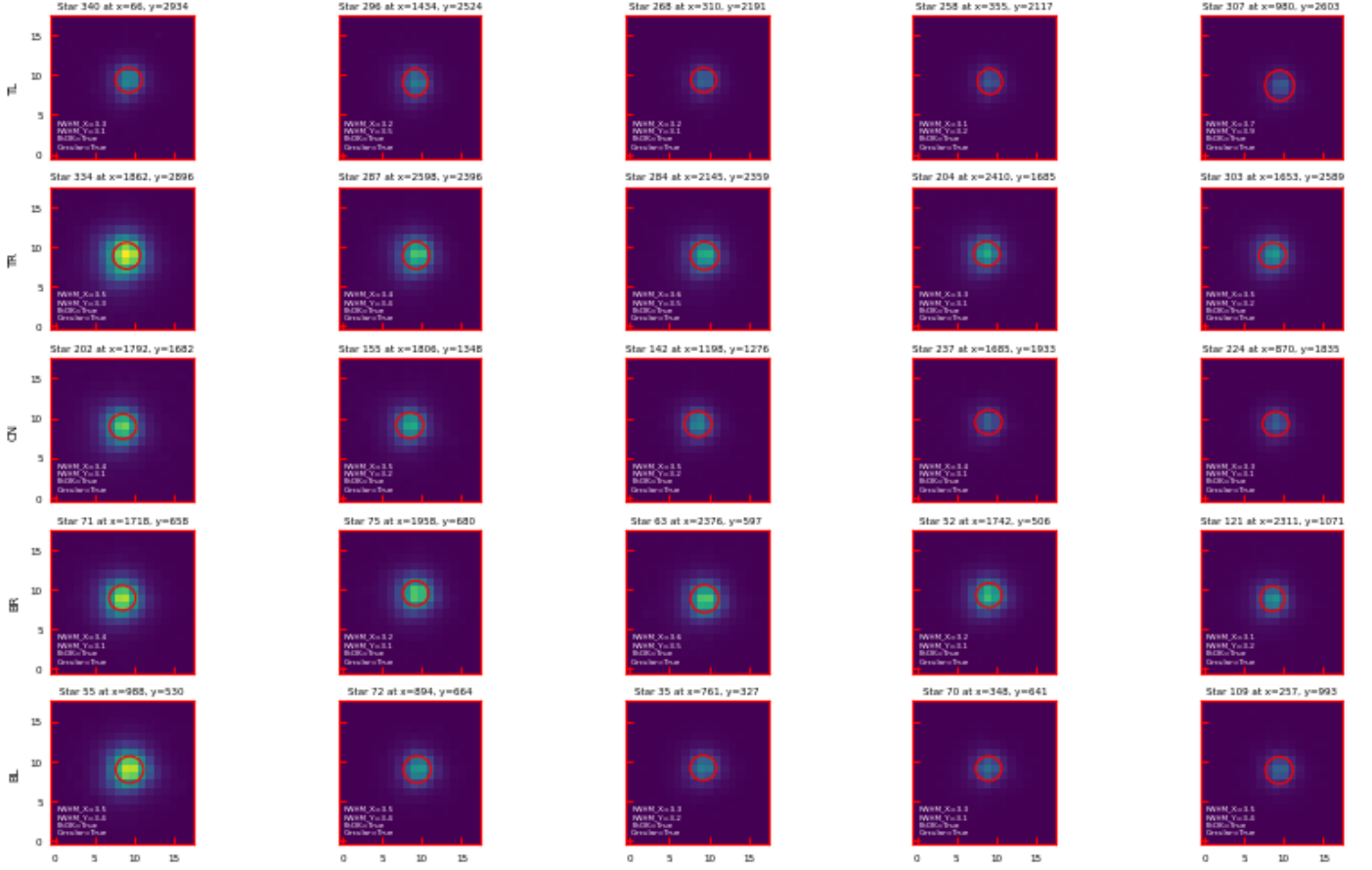


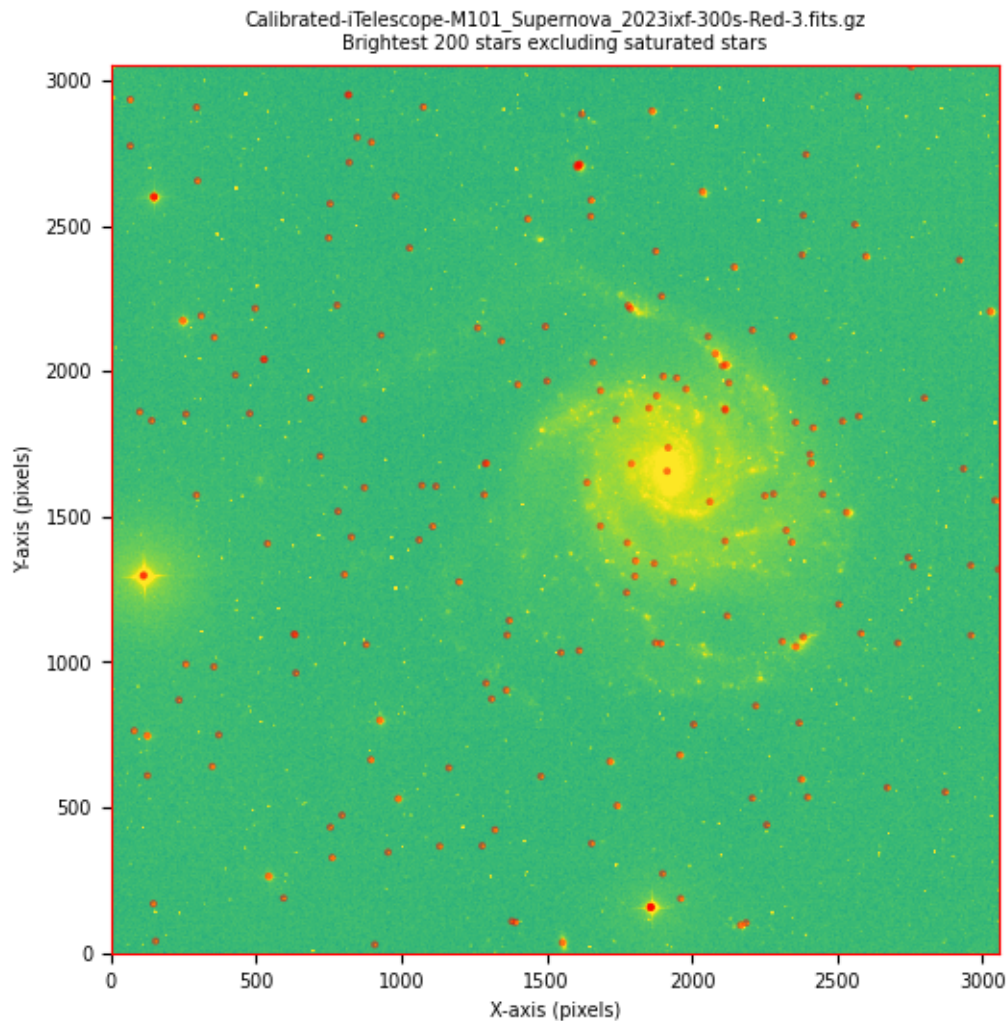
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
Brightest 200 stars excluding saturated stars





Star FWHM measurements for:
 Calibrated-Telescope-M101_Supernova_20231xf-300s-Red-3.fits.gz
 Median FWHM=3.32 +/- 0.22 (RAD stddev) pixels
 Along X FWHM=3.38 +/- 0.14 pixels, along Y FWHM=3.19 +/- 0.12 pixels





If you ran `ApProcess` with a `DEBUG` or lower log-level you will see a message about missing FITS keywords.

2024-09-20 09:00:50,425 | AstroPhotography.core.ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBSID']

These keywords are not critical (otherwise the message would be `WARNING` or `ERROR` level) but having them set can speed processing. In Section 3 of this notebook we add some of these keywords in a pre-processing stage and rerun the image navigation pipeline.

```
# show status table
status.pprint(max_width=200)
```

filename	find_stars_status	find_stars_time	astrometry_status	astrometry_time
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz	pass	8.438	pass	11.818
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz	pass	8.228	pass	9.926
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz	pass	8.498	pass	13.392
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz	pass	8.437	pass	9.783
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz	pass	8.764	pass	9.982
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz	pass	8.790	pass	11.976
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz	pass	8.968	pass	6.279
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz	pass	8.768	pass	14.803
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz	pass	9.060	pass	10.376

```
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
pass      8.999      pass      12.824
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
pass      8.887      pass      9.434
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
pass      8.448      pass      10.439
```

The processing times for each stage depend on the host processor single CPU speed (single threaded python), but will typically be longer for the astrometry stage as the time depends on the nova.astrometry.net web service. This depends on their CPU oad, but also very strongly on the number of possible locations and scales it needs to consider. `ApAstrometry` tries to provide a number of hints to `Astrometry.net`, the more of which are present the more likely a successful solution in a shorter time.

In particular, if `ApFindStars` can estimate the approximate plate scale of the images, then astrometric solutions are much faster although astrometry is always slower than the source searching. Note that raw and calibrated images from `iTelescope` often have the FITS header keywords needed for approximate plate scale estimation, but they are often stripped out of `iTelescope` Premium image sets.

```
3]: # Load and display the quality summary table (NOTE this is a very wide table)
qual_summ_table = None
if pathlib.Path(quality_summary_file).exists():
    qual_summ_table = Table.read(quality_summary_file, format="ascii.csv")
    qual_summ_table.columns # Dict of table columns (access by column name, index, or slice)
    qual_summ_table.colnames # List of column names
    qual_summ_table.meta # Dict of meta-data
    len(qual_summ_table) # Number of table rows

    # Show some data from the quality summary data.
    qual_summ_table['file', 'ncols', 'nrows', 'filter', 'median', 'stddev',
        'num_detected', 'fwhm_val_pix', 'fwhm_err_pix', 'fwhm_val_arcs', 'fwhm_err_arcs', 'num_data_pts'].pprint_all()
else:
    print(f'Quality summary file {quality_summary_file} does not exist. Skipping...')
```

```
3]: <TableColumns names=('targ:tel:filter','file','ncols','nrows','filter','median',
'stddev','num_detected','num_with_photometry','search_nsigma','adups_brightest',
'adups_median','adups_faintest','num_saturated_in_image','num_saturated_in_photo
metry','num_fit','circular_psf','fwhm_val_pix','fwhm_err_pix','fwhm_val_arcs','f
whm_err_arcs','num_data_pts')>
```

```
3]: ['targ:tel:filter',
'file',
'ncols',
'nrows',
'filter',
'median',
'stddev',
'num_detected',
'num_with_photometry',
'search_nsigma',
'adups_brightest',
'adups_median',
'adups_faintest',
'num_saturated_in_image',
'num_saturated_in_photometry',
'num_fit',
'circular_psf',
'fwhm_val_pix',
'fwhm_err_pix',
'fwhm_val_arcs',
'fwhm_err_arcs',
'num_data_pts']
```

3]: OrderedDict()

3]: 12

file			ncols		nrows	
filter	median	stddev	num_detected	fwhm_val_pix	fwhm_err_pix	fwhm_val_arcs
fwhm_err_arcs num_data_pts						

Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz			3056	3056		
Blue 179.874023	21.801167		278	3.465217	0.284413	-999.0
-999.0	50					
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz			3056	3056		
Blue 183.536133	21.837498		266	3.625976	0.287341	-999.0
-999.0	48					

Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz	3056	3056
Blue 184.711258 21.853487 243 3.883582 0.247384 -999.0 48		
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz	3056	3056
Green 233.455261 22.624113 286 3.671051 0.350572 -999.0 49		
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz	3056	3056
Green 223.168976 22.500845 299 3.641141 0.365279 -999.0 50		
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz	3056	3056
Green 176.803589 21.651184 222 3.77143 0.249502 -999.0 48		
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz	3056	3056
Red 187.759628 21.953953 327 3.863207 0.549708 -999.0 50		
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz	3056	3056
Red 202.138245 22.116404 266 3.795447 0.351523 -999.0 50		
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz	3056	3056
Red 182.942108 21.920551 349 3.324884 0.218721 -999.0 49		
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz	3056	3056
Lum 535.460938 26.379311 287 4.139703 0.168107 -999.0 50		
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz	3056	3056
Lum 488.849457 25.656437 306 4.295939 0.2659 -999.0 49		
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz	3056	3056
Lum 573.592041 26.749031 240 4.180753 0.101949 -999.0 48		

The quality summary file provides a general overview of the images that have been processed. (Note it summarizes all quality YaML files in the specified directory, not just the ones generated in the latest run of the `navigate_images` function.)

- The file is grouped by Filter, even if the lexical order of the files names is different.
 - The sigma-clipped background level, with detected sources removed, `median +/- stddev` (standard deviation) is one way to check if a particular image is affected by clouds. For example the Blue filter images have very similar background levels of 180+/-22, 184+/-22, and 185+/-22 ADU. The third Green filter image, on the other hand, differs from the others in have a background level of 177+/-22 compared to 233+/-23 and 223+/-23.
 - The number of detected sources should likewise be similar, and again the third Green image differs from the first and second Green images.
- The most important statistic are the measured source full width at half maximum, a measure of the telescope resolution, seeing conditions and tracking stability. Outliers here should like be excluded from image stacking.
 - In this case only the values in pixel units, not arcseconds are valid. That is because the pixel-to-arcsecond conversion used by `ApMeasureStars` comes from plate scale FITS header keywords. The are often missing from iTelescope Premium datasets.

Looking at the WCS headers We can look at the WCS headers of the navigated files we just generated to assess what the WCS solutions are like, e.g.

```
4]: # List of navigated file processed by this instance
navigated_file_list = processor.get_file_names('navfile', data_dir, 'processed', include_pattern, exclude_pattern, None, None, '.fits.
↪gz', with_name_and_dir)
print(f'Navigated image files generated by this ApProcess instance: \n{navigated_file_list}')

navigated_file_list = processor.get_file_names('navfile', data_dir, 'existing', include_pattern, exclude_pattern, None, None, '.fits.
↪gz', with_name_and_dir)
print(f'Navigated image files on disk: \n{navigated_file_list}')

first_file = navigated_file_list[0]
w, pix_scales, pix_area, im_scales = ap.util.load_wcs_from_file(first_file)
print(f'\nSummarizing WCS information from {first_file}:\n')
ap.util.summarize_wcs(w)
```

```
2024-11-26 21:14:39,989 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:14:39,990 | ApProcess | DEBUG | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:14:39,990 | ApProcess | DEBUG | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:14:39,990 | ApProcess | DEBUG | Excluding files that match
exclude_pattern="None"
```

```
Navigated image files generated by this ApProcess instance:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
```

```
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
```

Navigated image files on disk:

```
['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
```

Summarizing WCS information from

./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits:

```
NAXIS      = 2
NAXIS1     = 3056
NAXIS2     = 3056
CTYPE1     = 'RA---TAN'
CTYPE2     = 'DEC--TAN'
CRVAL1     = 210.754804653
CRVAL2     = 54.4174059435
CRPIX1     = 1528.5
CRPIX2     = 1528.5
CD1_1      = -4.64681630389e-06
CD1_2      = 0.000175223532872
CD2_1      = -0.000175223532872
CD2_2      = -4.64681630389e-06
CRVAL1,2 RA, Dec as HMS, DMS    = 14:03:01.2 +54:25:02.7
Pixel size equivalent CDELTA1    = 0.631 arcseconds
Pixel size equivalent CDELTA2    = 0.631 arcseconds
Image X-axis angular size        = 32.140 arcminutes
Pixel Y-axis angular size        = 32.140 arcminutes
Image rotation equivalent CROTA2 = 91.519 degrees
END
```

```
4]: ("NAXIS      = 2\nNAXIS1     = 3056\nNAXIS2     = 3056\nCTYPE1     = 'RA---TAN'\nCTYPE2\n= 'DEC--TAN'\nCRVAL1     = 210.754804653\nCRVAL2     = 54.4174059435\nCRPIX1     =\n1528.5\nCRPIX2     = 1528.5\nCD1_1      = -4.64681630389e-06\nCD1_2      =\n0.000175223532872\nCD2_1      = -0.000175223532872\nCD2_2      =\n-4.64681630389e-06\nCRVAL1,2 RA, Dec as HMS, DMS    = 14:03:01.2\n+54:25:02.7\nPixel size equivalent CDELTA1    = 0.631 arcseconds\nPixel size\nequivalent CDELTA2    = 0.631 arcseconds\nImage X-axis angular size        =\n32.140 arcminutes\nPixel Y-axis angular size        = 32.140 arcminutes\nImage\nrotation equivalent CROTA2 = 91.519 degrees\nEND      ",\n0.6310264944404745,\n0.6310264944404745,\n32.140282783501505,\n32.140282783501505,\n91.51909147248065)
```

2.3.3 Checking the Navigated Images

It is recommended that you check that the WCS headers give reasonable results and are self consistent. I personally prefer to use `SAOImage ds9` for this from the command line, by loading the images, aligning them and then blinking between the images. If the astrometry is good then the same star should be in the same place on the screen in each image, and not move or jump from frame to frame.

```
> cd ./NavigatedImages/
> ls
navigated-M101_Supernova_2023ixf-180s-Lum-1.fits    navigated-M101_Supernova_2023ixf-300s-Green-1.fits
navigated-M101_Supernova_2023ixf-180s-Lum-2.fits    navigated-M101_Supernova_2023ixf-300s-Green-2.fits
navigated-M101_Supernova_2023ixf-180s-Lum-3.fits    navigated-M101_Supernova_2023ixf-300s-Green-3.fits
navigated-M101_Supernova_2023ixf-300s-Blue-1.fits    navigated-M101_Supernova_2023ixf-300s-Red-1.fits
navigated-M101_Supernova_2023ixf-300s-Blue-2.fits    navigated-M101_Supernova_2023ixf-300s-Red-2.fits
navigated-M101_Supernova_2023ixf-300s-Blue-3.fits    navigated-M101_Supernova_2023ixf-300s-Red-3.fits
> ds9 -asinh -zoom to fit -cmap viridis -scale mode 99.5 navigated-M101_Supernova_2023ixf-*fits &
```

Then: - expand the ds9 window to make it larger - click on the first image (or any image) so that it is outlined in blue - Mousing over pixels in that image should change the WCS RA and Dec values shown - Click the menu `Zoom -> Align` and the first image and the zoomed-out view in the top right should change orientation - Click `Frame -> Single Frame`, then `Zoom -> Fit` - Now to match all the images by RA and Dec, `Frame -> Match -> Frame -> WCS` - Make sure

the default blink interval is 1 second (less than 1 second is too fast) by `Frame -> Frame Parameters -> Blink Interval` - Finally, set the images to blink using `Frame -> Blink Frames`

2.4 Image Stacking (Resampling and Mosaicing)

`ApProcessor` can also resample images to a common WCS system, and mosaic multiple images sharing a common WCS together, using `Astromatic swarp` under the hood. These operations are more commonly referred to as image stacking in the amateur astronomy community.

As discussed in the `itelescope_premium_the_hard_way.ipynb` notebook there are a number of ways of doing this, but for now we only support the first method listed there: Resample all images to match the WCS of another image, for example the first image in the list of navigated images, irrespective of how uniform the platescale is or whether the image is North-up, East-left aligned.

The `ApProcess.resample_images_to_match` function provides this method. By default it processes all the navigated files in all available `FILTER` combinations that the current `ApProcess` instance is aware of, irrespective of whether the files have the same exposure time, which may not be ideal in cases where you have images with different seeing quality and or different exposure times.

If cases where you already have pre-existing navigated images not processed by your current `ApProcess` instance you will need to either use `ApProcess.set_file_names` or the

```
[5]: unique_filters = processor.get_filter_list()
    print(unique_filters)
```

```
2024-11-26 21:14:41,481 | ApProcess | DEBUG | get_filter_list: there are 12
files of type input found under .
2024-11-26 21:14:41,484 | ApProcess | DEBUG | Finding files and exposure times
for filter Blue
2024-11-26 21:14:41,484 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:14:41,997 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz has FILTER=Blue
2024-11-26 21:14:42,120 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz has FILTER=Blue
2024-11-26 21:14:42,244 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz has FILTER=Blue
2024-11-26 21:14:42,974 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:14:43,342 | ApProcess | INFO | Found the following 3 files for
filter Blue: ['Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz', 'Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz', 'Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz']
2024-11-26 21:14:43,342 | ApProcess | DEBUG | Finding files and exposure times
for filter Green
2024-11-26 21:14:43,342 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:14:44,217 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz has FILTER=Green
2024-11-26 21:14:44,340 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz has FILTER=Green
2024-11-26 21:14:44,459 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz has FILTER=Green
2024-11-26 21:14:44,825 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:14:45,185 | ApProcess | INFO | Found the following 3 files for
filter Green: ['Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz', 'Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz', 'Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz']
2024-11-26 21:14:45,185 | ApProcess | DEBUG | Finding files and exposure times
for filter Lum
2024-11-26 21:14:45,185 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:14:45,309 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz has FILTER=Lum
2024-11-26 21:14:45,440 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz has FILTER=Lum
2024-11-26 21:14:45,576 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz has FILTER=Lum
2024-11-26 21:14:46,667 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:14:47,055 | ApProcess | INFO | Found the following 3 files for
filter Lum: ['Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz',
'Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz', 'Calibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz']
2024-11-26 21:14:47,055 | ApProcess | DEBUG | Finding files and exposure times
for filter Red
2024-11-26 21:14:47,056 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:14:48,291 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz has FILTER=Red
2024-11-26 21:14:48,413 | ApProcess | DEBUG | File Calibrated-
```

```

iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz has FILTER=Red
2024-11-26 21:14:48,535 | ApProcess | DEBUG | File Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz has FILTER=Red
2024-11-26 21:14:48,535 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:14:48,899 | ApProcess | INFO | Found the following 3 files for
filter Red: ['Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz',
'Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz', 'Calibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz']
2024-11-26 21:14:48,900 | ApProcess | DEBUG | Unique filter and exposure time
data:
2024-11-26 21:14:48,900 | ApProcess | DEBUG | filter num_navfiles tot_exptime_m
min_exptime_m max_exptime_m
2024-11-26 21:14:48,900 | ApProcess | DEBUG | -----
-----
2024-11-26 21:14:48,900 | ApProcess | DEBUG | Blue 3 15.0
5.0 5.0
2024-11-26 21:14:48,901 | ApProcess | DEBUG | Green 3 15.0
5.0 5.0
2024-11-26 21:14:48,901 | ApProcess | DEBUG | Lum 3 9.0
3.0 3.0
2024-11-26 21:14:48,901 | ApProcess | DEBUG | Red 3 15.0
5.0 5.0
2024-11-26 21:14:48,901 | ApProcess | INFO | There are 4 unique filters among 12
files: ['Blue', 'Green', 'Lum', 'Red']

['Blue', 'Green', 'Lum', 'Red']

```

Note that the ImageFileCollection shows only two files, because we set `initial_debugging=True` and only processed two images.

To process any other images that already exist we can use `ApProcess.set_file_names`.

```

6] processor.set_file_names(data_dir, include_pattern, exclude_pattern, None, None, '.fits.gz')

```

```

2024-11-26 21:14:48,903 | ApProcess | INFO | Finding input and output files
starting in directory=.
2024-11-26 21:14:48,904 | ApProcess | DEBUG | Current working directory:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
2024-11-26 21:14:48,904 | ApProcess | INFO | Using files that match
include_pattern="Calibrated-*?.fits*"
2024-11-26 21:14:48,904 | ApProcess | INFO | Excluding files that match
exclude_pattern="None"
2024-11-26 21:14:50,384 | ApProcess | INFO | There are 12 input files matching
the parameters given.
2024-11-26 21:14:50,384 | ApProcess | DEBUG | Input file collection:
ImageFileCollection(location='.', keywords=['naxis1', 'naxis2', 'imagetyp',
'object', 'filter', 'exposure'], glob_include='Calibrated-*?.fits*')
2024-11-26 21:14:50,384 | ApProcess | INFO | Looking for existing files matching
name conventions...
2024-11-26 21:14:50,385 | ApProcess | DEBUG | Looking for existing srclist
files.
2024-11-26 21:14:50,385 | ApProcess | INFO | Found the following 12 srclist
files: ['./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-180s-Lum-3.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Blue-3.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-1.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-2.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Green-3.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-1.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-2.fits',
'./SourceLists/srclist-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:14:50,386 | ApProcess | DEBUG | Looking for existing regfile
files.
2024-11-26 21:14:50,386 | ApProcess | INFO | Found the following 12 regfile
files: ['./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-1.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-2.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-180s-Lum-3.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-1.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-2.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Blue-3.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-1.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-2.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Green-3.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-1.reg',
'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-2.reg',

```

```

'./SourceLists/ds9-M101_Supernova_2023ixf-300s-Red-3.reg']
2024-11-26 21:14:50,386 | ApProcess | DEBUG | Looking for existing plotfile
files.
2024-11-26 21:14:50,387 | ApProcess | INFO | Found the following 12 plotfile
files: ['./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-1.png',
'./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-2.png',
'./SourceLists/implot-M101_Supernova_2023ixf-180s-Lum-3.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-1.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-2.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Blue-3.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Green-1.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Green-2.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Green-3.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Red-1.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Red-2.png',
'./SourceLists/implot-M101_Supernova_2023ixf-300s-Red-3.png']
2024-11-26 21:14:50,387 | ApProcess | DEBUG | Looking for existing fwhmplot
files.
2024-11-26 21:14:50,388 | ApProcess | INFO | Found the following 12 fwhmplot
files: ['./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-1.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-2.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-180s-Lum-3.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-1.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-2.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Blue-3.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-1.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-2.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Green-3.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-1.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-2.png',
'./SourceLists/fwhmplot-M101_Supernova_2023ixf-300s-Red-3.png']
2024-11-26 21:14:50,388 | ApProcess | DEBUG | Looking for existing qualfile
files.
2024-11-26 21:14:50,388 | ApProcess | INFO | Found the following 12 qualfile
files: ['./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-1.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-2.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-180s-Lum-3.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-1.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-2.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Blue-3.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Green-1.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Green-2.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Green-3.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Red-1.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Red-2.yaml',
'./MetaData/qual-M101_Supernova_2023ixf-300s-Red-3.yaml']
2024-11-26 21:14:50,389 | ApProcess | DEBUG | Looking for existing navfile
files.
2024-11-26 21:14:50,389 | ApProcess | INFO | Found the following 12 navfile
files: ['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']

```

```

7]: unique_filters = processor.get_filter_list('navfile')
print(f'unique_filters = {unique_filters}')

```

```

2024-11-26 21:14:50,405 | ApProcess | DEBUG | get_filter_list: there are 12
files of type navfile found under .
2024-11-26 21:14:50,407 | ApProcess | DEBUG | Finding files and exposure times
for filter Blue
2024-11-26 21:14:50,407 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:14:50,409 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits has
FILTER=Blue
2024-11-26 21:14:50,409 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits has

```

FILTER=Blue
2024-11-26 21:14:50,410 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits has
FILTER=Blue
2024-11-26 21:14:50,412 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:14:50,413 | ApProcess | INFO | Found the following 3 files for
filter Blue:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:14:50,413 | ApProcess | DEBUG | Finding files and exposure times
for filter Green
2024-11-26 21:14:50,414 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:14:50,416 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits has
FILTER=Green
2024-11-26 21:14:50,417 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits has
FILTER=Green
2024-11-26 21:14:50,417 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits has
FILTER=Green
2024-11-26 21:14:50,419 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:14:50,420 | ApProcess | INFO | Found the following 3 files for
filter Green:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:14:50,420 | ApProcess | DEBUG | Finding files and exposure times
for filter Lum
2024-11-26 21:14:50,420 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:14:50,421 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits has
FILTER=Lum
2024-11-26 21:14:50,421 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits has
FILTER=Lum
2024-11-26 21:14:50,422 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits has
FILTER=Lum
2024-11-26 21:14:50,425 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:14:50,426 | ApProcess | INFO | Found the following 3 files for
filter Lum:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:14:50,427 | ApProcess | DEBUG | Finding files and exposure times
for filter Red
2024-11-26 21:14:50,427 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:14:50,431 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits has
FILTER=Red
2024-11-26 21:14:50,431 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits has
FILTER=Red
2024-11-26 21:14:50,432 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits has
FILTER=Red
2024-11-26 21:14:50,432 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:14:50,434 | ApProcess | INFO | Found the following 3 files for
filter Red:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:14:50,434 | ApProcess | DEBUG | Unique filter and exposure time
data:
2024-11-26 21:14:50,434 | ApProcess | DEBUG | filter num_navfiles tot_exptime_m
min_exptime_m max_exptime_m
2024-11-26 21:14:50,435 | ApProcess | DEBUG | -----

2024-11-26 21:14:50,435 | ApProcess | DEBUG | Blue 3 15.0
5.0 5.0
2024-11-26 21:14:50,435 | ApProcess | DEBUG | Green 3 15.0

```

5.0          5.0
2024-11-26 21:14:50,435 | ApProcess | DEBUG | Lum          3          9.0
3.0          3.0
2024-11-26 21:14:50,435 | ApProcess | DEBUG | Red          3          15.0
5.0          5.0
2024-11-26 21:14:50,436 | ApProcess | INFO | There are 4 unique filters among 12
files: ['Blue', 'Green', 'Lum', 'Red']

unique_filters = ['Blue', 'Green', 'Lum', 'Red']

```

Having dispensed with an example of how to find which filters are available, let's just let `resample_images_to_match` process all the images.

```

8]: target_wcs_file = './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits'
    resam_prefix   = 'M101_Supernova_2023ixf_'
    resam_suffix   = '_resamp_weighted.fits'
    resam_dir      = 'Resampled'
    resampled_summary_plot = 'resampled_summary_plot.png'
    composite_summary_plot = 'composite_summary_plot.png'
    resampled_img_dict, info_table = processor.resample_images_to_match(target_wcs_file, resam_prefix, resam_suffix,
    ↪resampled_dir=resam_dir, filter_list=None,
                                resampled_summary_plot=resampled_summary_plot,
    ↪composite_summary_plot=composite_summary_plot)

```

```

2024-11-26 21:14:50,438 | ApProcess | DEBUG | Checking if WCS target file
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits exists.
2024-11-26 21:14:50,451 | ApProcess | DEBUG | get_filter_list: there are 12
files of type navfile found under .
2024-11-26 21:14:50,453 | ApProcess | DEBUG | Finding files and exposure times
for filter Blue
2024-11-26 21:14:50,453 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:14:50,455 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits has
FILTER=Blue
2024-11-26 21:14:50,456 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits has
FILTER=Blue
2024-11-26 21:14:50,457 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits has
FILTER=Blue
2024-11-26 21:14:50,459 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:14:50,461 | ApProcess | INFO | Found the following 3 files for
filter Blue:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:14:50,461 | ApProcess | DEBUG | Finding files and exposure times
for filter Green
2024-11-26 21:14:50,461 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:14:50,464 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits has
FILTER=Green
2024-11-26 21:14:50,464 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits has
FILTER=Green
2024-11-26 21:14:50,465 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits has
FILTER=Green
2024-11-26 21:14:50,466 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:14:50,467 | ApProcess | INFO | Found the following 3 files for
filter Green:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:14:50,468 | ApProcess | DEBUG | Finding files and exposure times
for filter Lum
2024-11-26 21:14:50,468 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:14:50,468 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits has
FILTER=Lum
2024-11-26 21:14:50,469 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits has
FILTER=Lum
2024-11-26 21:14:50,470 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits has

```

FILTER=Lum
2024-11-26 21:14:50,473 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:14:50,474 | ApProcess | INFO | Found the following 3 files for
filter Lum:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:14:50,474 | ApProcess | DEBUG | Finding files and exposure times
for filter Red
2024-11-26 21:14:50,474 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:14:50,478 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits has
FILTER=Red
2024-11-26 21:14:50,478 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits has
FILTER=Red
2024-11-26 21:14:50,479 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits has
FILTER=Red
2024-11-26 21:14:50,479 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:14:50,480 | ApProcess | INFO | Found the following 3 files for
filter Red:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:14:50,481 | ApProcess | DEBUG | Unique filter and exposure time
data:
2024-11-26 21:14:50,481 | ApProcess | DEBUG | filter num_navfiles tot_exptime_m
min_exptime_m max_exptime_m
2024-11-26 21:14:50,481 | ApProcess | DEBUG | -----

2024-11-26 21:14:50,481 | ApProcess | DEBUG | Blue 3 15.0
5.0 5.0
2024-11-26 21:14:50,482 | ApProcess | DEBUG | Green 3 15.0
5.0 5.0
2024-11-26 21:14:50,482 | ApProcess | DEBUG | Lum 3 9.0
3.0 3.0
2024-11-26 21:14:50,482 | ApProcess | DEBUG | Red 3 15.0
5.0 5.0
2024-11-26 21:14:50,482 | ApProcess | INFO | There are 4 unique filters among 12
files: ['Blue', 'Green', 'Lum', 'Red']
2024-11-26 21:14:50,485 | ApProcess | INFO | Target WCS file is 32.14 x 32.14
arcmin (wxh) with 0.63 x 0.63 arcsec pixels at an angle of 91.52 deg from North.
2024-11-26 21:14:50,485 | ApProcess | DEBUG | Subdirectory Resampled already
exists.
2024-11-26 21:14:50,485 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:14:50,487 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits has
FILTER=Blue
2024-11-26 21:14:50,487 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits has
FILTER=Blue
2024-11-26 21:14:50,488 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits has
FILTER=Blue
2024-11-26 21:14:50,490 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:14:50,490 | ApProcess | INFO | Processing 3 images for filter Blue
2024-11-26 21:14:50,504 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:14:50,505 | ApProcess | DEBUG | FITS keywords found in image:
['APTDIA', 'FOCALLEN', 'FILTER']
2024-11-26 21:14:50,505 | ApProcess | DEBUG | FITS keywords missing from image:
['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-
TEMP', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:14:50,506 | ApProcess | INFO | Filter Blue with 3 images totalling
900.0 seconds.
2024-11-26 21:14:50,506 | ApProcess | DEBUG | Input files:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:14:50,507 | ApProcess | DEBUG | Input weights: ['0.0033333',
 '0.0033333', '0.0033333']
2024-11-26 21:14:55,149 | ApProcess | INFO | Success: swarp process took 4.580
seconds, return code 0

2024-11-26 21:14:55,175 | ApProcess | INFO | Modified FITS header keywords in Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
2024-11-26 21:14:55,175 | ApProcess | INFO | Searching 12 files for headers in extension=0 that have FILTER=Green
2024-11-26 21:14:55,178 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits has FILTER=Green
2024-11-26 21:14:55,179 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits has FILTER=Green
2024-11-26 21:14:55,179 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits has FILTER=Green
2024-11-26 21:14:55,180 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:14:55,181 | ApProcess | INFO | Processing 3 images for filter Green
2024-11-26 21:14:55,195 | ApProcess | INFO | Reading FITS header keywords from ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits
2024-11-26 21:14:55,195 | ApProcess | DEBUG | FITS keywords found in image: ['APTDIA', 'FOCALLEN', 'FILTER']
2024-11-26 21:14:55,196 | ApProcess | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:14:55,197 | ApProcess | INFO | Filter Green with 3 images totalling 900.0 seconds.
2024-11-26 21:14:55,197 | ApProcess | DEBUG | Input files: ['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits', './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits', './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:14:55,198 | ApProcess | DEBUG | Input weights: ['0.0033333', '0.0033333', '0.0033333']
2024-11-26 21:14:59,814 | ApProcess | INFO | Success: swarp process took 4.560 seconds, return code 0
2024-11-26 21:14:59,839 | ApProcess | INFO | Modified FITS header keywords in Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
2024-11-26 21:14:59,839 | ApProcess | INFO | Searching 12 files for headers in extension=0 that have FILTER=Lum
2024-11-26 21:14:59,840 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits has FILTER=Lum
2024-11-26 21:14:59,841 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits has FILTER=Lum
2024-11-26 21:14:59,841 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits has FILTER=Lum
2024-11-26 21:14:59,844 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:14:59,845 | ApProcess | INFO | Processing 3 images for filter Lum
2024-11-26 21:14:59,859 | ApProcess | INFO | Reading FITS header keywords from ./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:14:59,859 | ApProcess | DEBUG | FITS keywords found in image: ['APTDIA', 'FOCALLEN', 'FILTER']
2024-11-26 21:14:59,859 | ApProcess | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-TEMP', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:14:59,861 | ApProcess | INFO | Filter Lum with 3 images totalling 540.0 seconds.
2024-11-26 21:14:59,861 | ApProcess | DEBUG | Input files: ['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits', './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits', './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:14:59,861 | ApProcess | DEBUG | Input weights: ['0.0055556', '0.0055556', '0.0055556']
2024-11-26 21:15:04,524 | ApProcess | INFO | Success: swarp process took 4.608 seconds, return code 0
2024-11-26 21:15:04,549 | ApProcess | INFO | Modified FITS header keywords in Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
2024-11-26 21:15:04,549 | ApProcess | INFO | Searching 12 files for headers in extension=0 that have FILTER=Red
2024-11-26 21:15:04,553 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits has FILTER=Red
2024-11-26 21:15:04,553 | ApProcess | DEBUG | File ./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits has FILTER=Red
2024-11-26 21:15:04,554 | ApProcess | DEBUG | File

```

./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits has
FILTER=Red
2024-11-26 21:15:04,554 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:15:04,554 | ApProcess | INFO | Processing 3 images for filter Red
2024-11-26 21:15:04,568 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits
2024-11-26 21:15:04,568 | ApProcess | DEBUG | FITS keywords found in image:
['APTDIA', 'FOCALLEN', 'FILTER']
2024-11-26 21:15:04,568 | ApProcess | DEBUG | FITS keywords missing from image:
['EXPOSURE', 'DATE-OBS', 'OBJECT', 'OBJNAME', 'TELESCOP', 'INSTRUME', 'CCD-
TEMP', 'XPIXSZ', 'YPIXSZ', 'EGAIN', 'LAT-OBS', 'LONG-OBS', 'ALT-OBS', 'AIRMASS']
2024-11-26 21:15:04,570 | ApProcess | INFO | Filter Red with 3 images totalling
900.0 seconds.
2024-11-26 21:15:04,570 | ApProcess | DEBUG | Input files:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:15:04,570 | ApProcess | DEBUG | Input weights: ['0.0033333',
'0.0033333', '0.0033333']
2024-11-26 21:15:09,164 | ApProcess | INFO | Success: swarp process took 4.540
seconds, return code 0
2024-11-26 21:15:09,233 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
2024-11-26 21:15:09,233 | ApProcess | DEBUG | Finding files and exposure times
for filter Blue
2024-11-26 21:15:09,233 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:15:09,235 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits has
FILTER=Blue
2024-11-26 21:15:09,236 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits has
FILTER=Blue
2024-11-26 21:15:09,236 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits has
FILTER=Blue
2024-11-26 21:15:09,239 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:15:09,240 | ApProcess | INFO | Found the following 3 files for
filter Blue:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:15:09,240 | ApProcess | DEBUG | Finding files and exposure times
for filter Green
2024-11-26 21:15:09,240 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:15:09,243 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits has
FILTER=Green
2024-11-26 21:15:09,243 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits has
FILTER=Green
2024-11-26 21:15:09,244 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits has
FILTER=Green
2024-11-26 21:15:09,245 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:15:09,247 | ApProcess | INFO | Found the following 3 files for
filter Green:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:15:09,247 | ApProcess | DEBUG | Finding files and exposure times
for filter Lum
2024-11-26 21:15:09,247 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:15:09,248 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits has
FILTER=Lum
2024-11-26 21:15:09,248 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits has
FILTER=Lum
2024-11-26 21:15:09,249 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits has
FILTER=Lum
2024-11-26 21:15:09,252 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:15:09,253 | ApProcess | INFO | Found the following 3 files for

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filter Lum:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:15:09,253 | ApProcess | DEBUG | Finding files and exposure times
for filter Red
2024-11-26 21:15:09,254 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:15:09,257 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits has
FILTER=Red
2024-11-26 21:15:09,258 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits has
FILTER=Red
2024-11-26 21:15:09,258 | ApProcess | DEBUG | File
./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits has
FILTER=Red
2024-11-26 21:15:09,258 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:15:09,260 | ApProcess | INFO | Found the following 3 files for
filter Red:
['./NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-1.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-2.fits',
 './NavigatedImages/navigated-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:15:09,260 | ApProcess | INFO | Finished. Generated 4 resampled
images in 18.822 seconds: {'Blue':
'Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits', 'Green':
'Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits', 'Lum':
'Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits', 'Red':
'Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits'}
2024-11-26 21:15:09,261 | ApProcess | DEBUG | Setting swap_axis True because
crot=91.52 degrees.
2024-11-26 21:15:09,261 | ApProcess | DEBUG | Generating a summary plot of 4
using WCS info, tick spacing 10 arcmin, and swap_radec_axis=True.

The are 4 FITS images in the input imlist
  With 4 files the plot layout is 2 rows x 2 columns.
Attempting to plot Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
  Input image data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
  Applying asinh stretch between 0.5249 and 0.8013
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Attempting to plot Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
  Input image data 0.5th percentile = 0.6154, 99.5th percentile = 0.9078
  Applying asinh stretch between 0.6154 and 0.9078
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Attempting to plot Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
  Input image data 0.5th percentile = 2.7678, 99.5th percentile = 3.7362
  Applying asinh stretch between 2.7678 and 3.7362
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Attempting to plot Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
  Input image data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944
  Applying asinh stretch between 0.5399 and 0.8944
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)

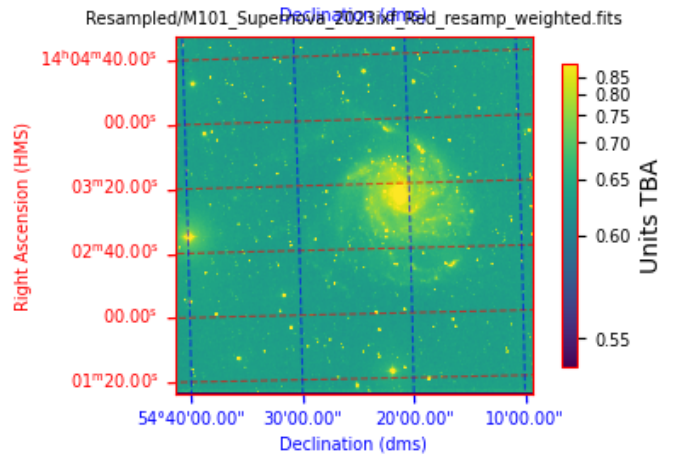
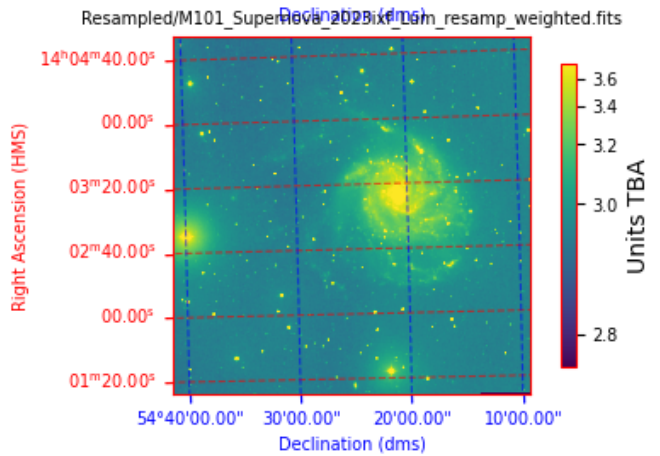
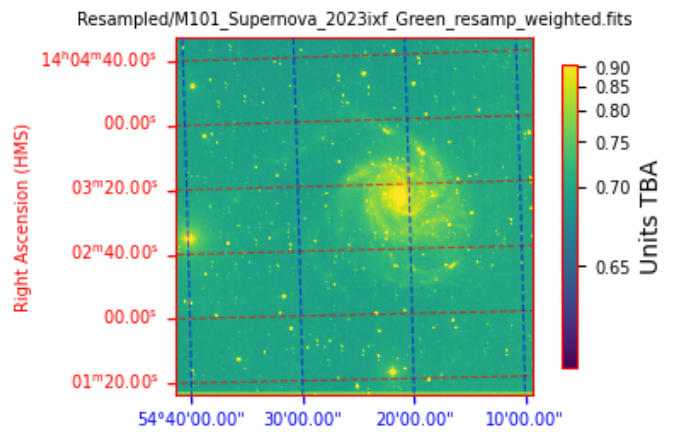
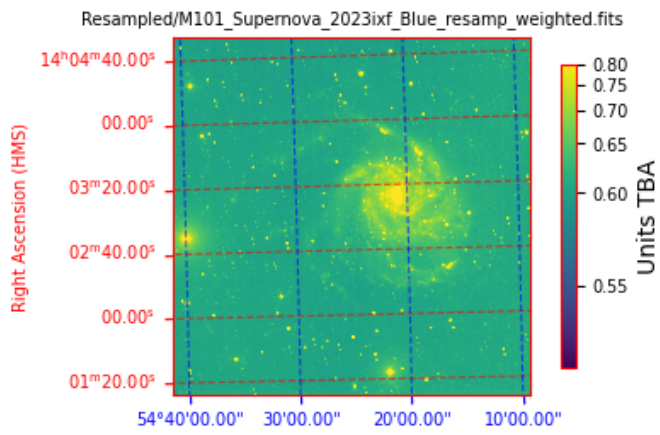
2024-11-26 21:15:11,491 | ApProcess | INFO | Resampled image summary plot
generated: resampled_summary_plot.png
2024-11-26 21:15:11,492 | ApProcess | DEBUG | Generating a summary color
composite plot using WCS info, tick spacing 10 arcmin, and swap_radec_axis=True.

Wrote plot to resampled_summary_plot.png
There are enough files to generate a RGB threecolor composite.
NB: insufficient files to generate a SHO threecolor composite.
Input red channel data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944
Using an image minimum of 0.5398805156350136 for all channels.
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)

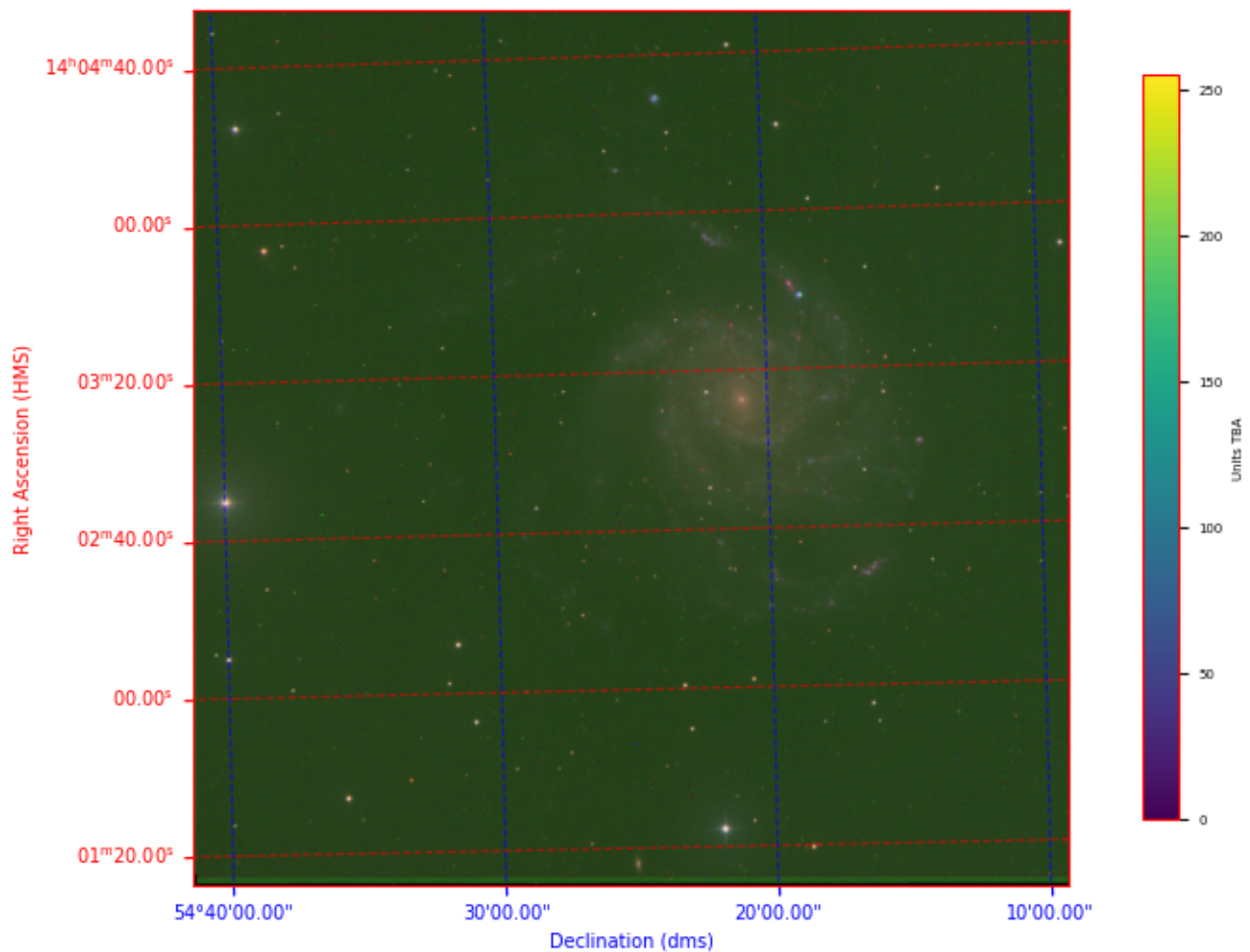
2024-11-26 21:15:12,922 | ApProcess | INFO | Summary color composite plot
generated: composite_summary_plot.png

Wrote plot to composite_summary_plot.png

```



RGB composite image stretch=0.5, Q=8, minimum=None
 Red = Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Green = Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Blue = Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits



```
9]: for key, val in resampled_img_dict.items():
    print(f'Filter: {key:15s} {val}')
```

```
Filter: Blue
Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
Filter: Green
Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
Filter: Lum
Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
Filter: Red
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
```

```
0]: info_table.pprint_all()
```

```
filter_num_navfiles_tot_exptime_m_min_exptime_m_max_exptime_m
resampled_image
-----
Blue          3          15.0          5.0          5.0
Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
Green         3          15.0          5.0          5.0
Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
Lum           3           9.0          3.0          3.0
Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
Red           3          15.0          5.0          5.0
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
```

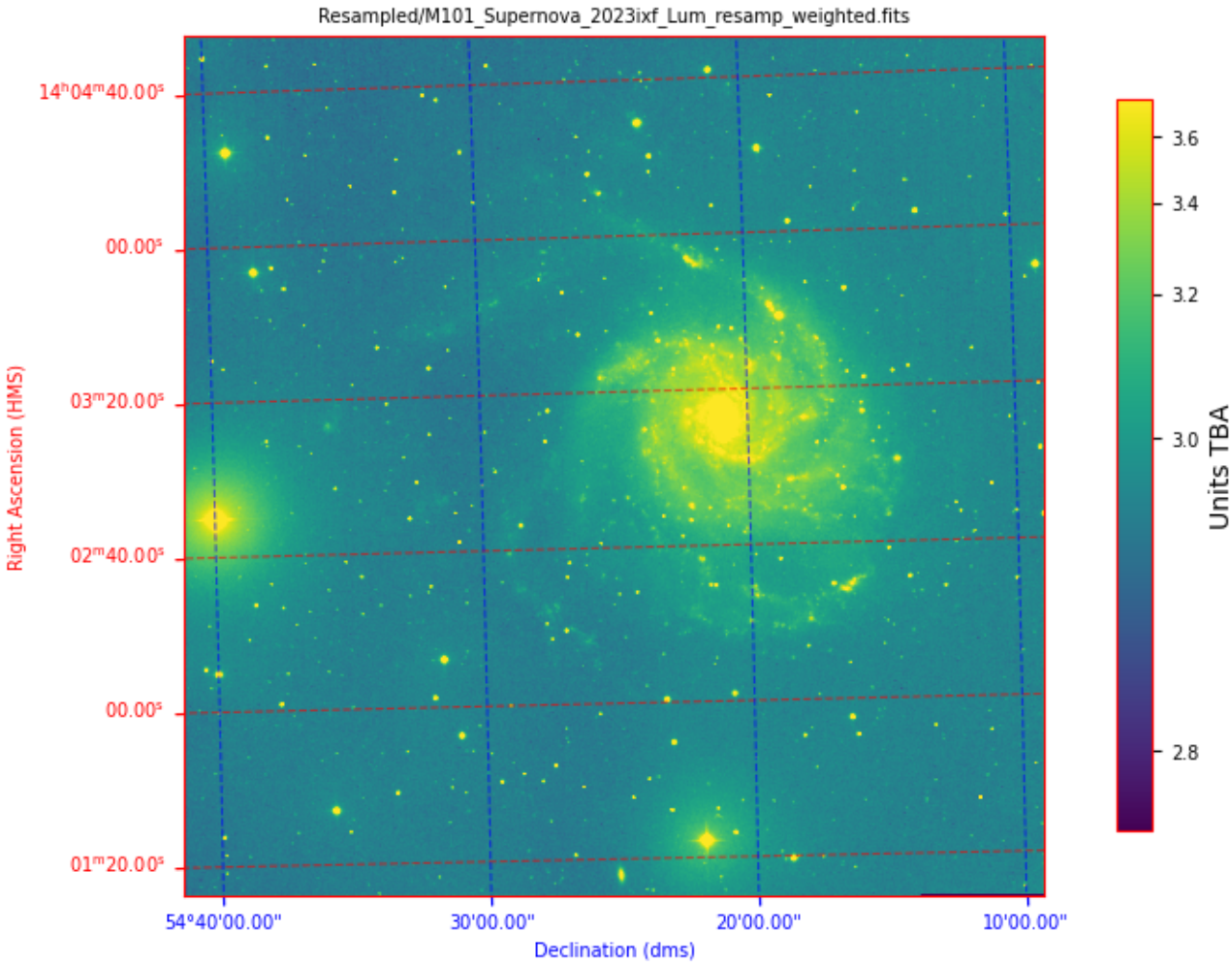
2.5 Visualize the Output

The resampled image dictionary returned by `ApProcess.resample_images_to_match` can be used to plot some or all of the output images.

```
1]: lumf = resampled_img_dict['Lum']
print(f'Final luminance image: {lumf}')
output_plot = 'luminance_plot.png'
ap.util.load_image_and_plot(lumf, 0, output_plot, True, angle_tick_spacing_am=10, swap_radec_axis=True)
```

Final luminance image: Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
Input data 0.5th percentile = 2.7678, 99.5th percentile = 3.7362
Applying asinh stretch between 2.7678 and 3.7362
Using WCS information for axis projection
WCS Keywords

Number of WCS axes: 2
CTYPE : 'RA---TAN' 'DEC--TAN'
CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2 : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2 : -0.000175223532872 -4.64681630389e-06
CDELT : 1.0 1.0
NAXIS : 3056 3056
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Wrote plot to luminance_plot.png



```
2]: bluef = resampled_img_dict['Blue']
greenf = resampled_img_dict['Green']
redf = resampled_img_dict['Red']
valmin = None
valmax = None
```

```

output_plot = None # Name of output PNG file or None
ap.util.load_imlist_and_plot([redf, greenf, bluef], extnum=0, output=output_plot,
                             usewcs=True, vmin=0.6, vmax=1.6, xaxlim=None,
                             yaxlim=None, verbose=True,
                             angle_tick_spacing_am=10, swap_radec_axis=True)

```

The are 3 FITS images in the input imlist

With 3 files the plot layout is 2 rows x 2 columns.

Attempting to plot Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits

Input image data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944

Applying asinh stretch between 0.6000 and 1.6000

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

Attempting to plot Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits

Input image data 0.5th percentile = 0.6154, 99.5th percentile = 0.9078

Applying asinh stretch between 0.6000 and 1.6000

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

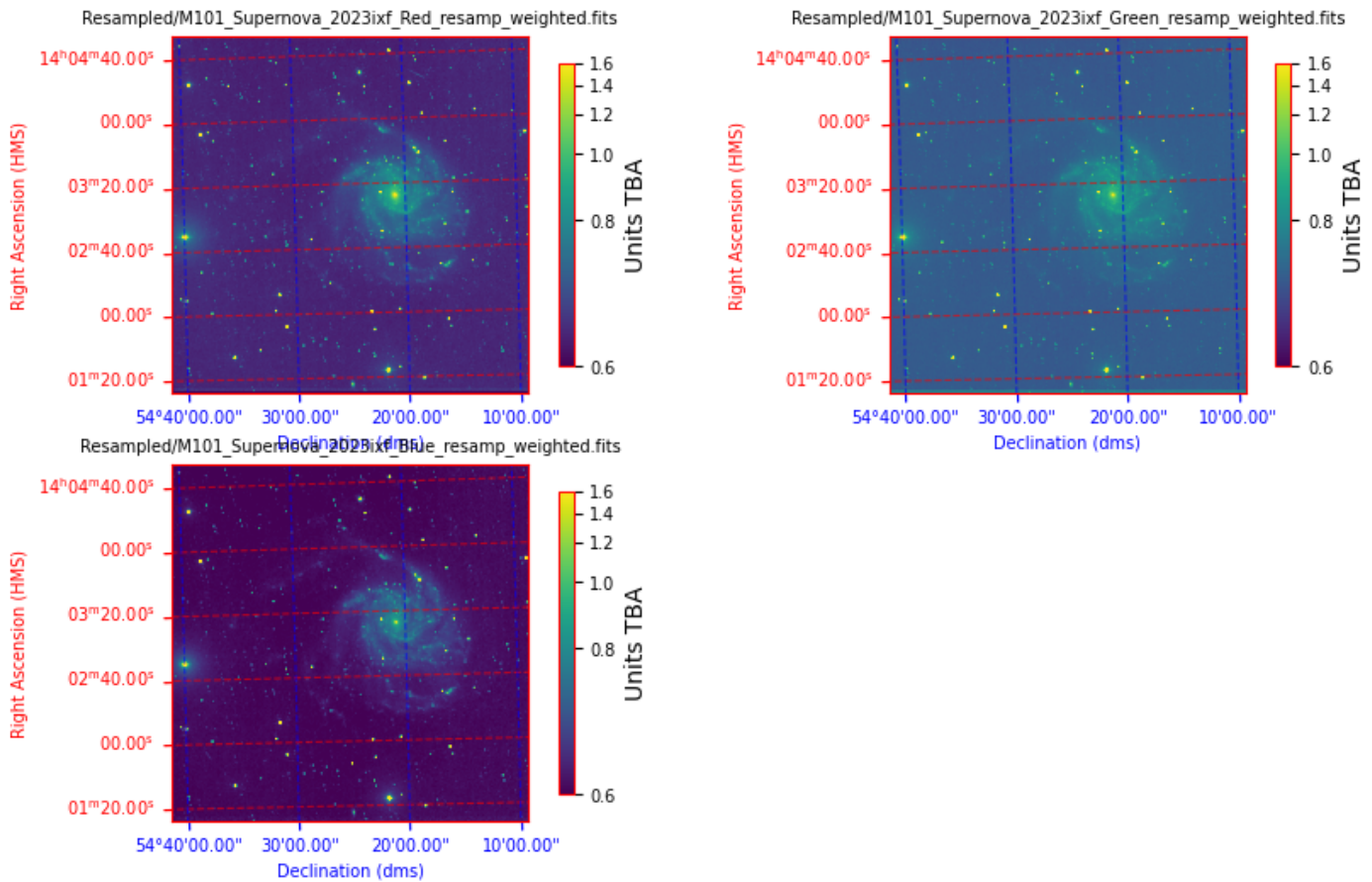
Attempting to plot Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits

Input image data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013

Applying asinh stretch between 0.6000 and 1.6000

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)



In the Blue filter image we can zoom in to look for the bad pixels that made it through iTelescope's calibration process (here in the resampled luminance image)

```

3]: ap.util.load_image_and_plot(bluef, 0, output_plot, True, vmin=0.6, vmax=1.0, xaxlim=[1100, 1500], yaxlim=[350, 750], verbose=True,
    ↪ angle_tick_spacing_am=2, swap_radec_axis=True)

```

Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013

Applying asinh stretch between 0.6000 and 1.0000

Using WCS information for axis projection

WCS Keywords

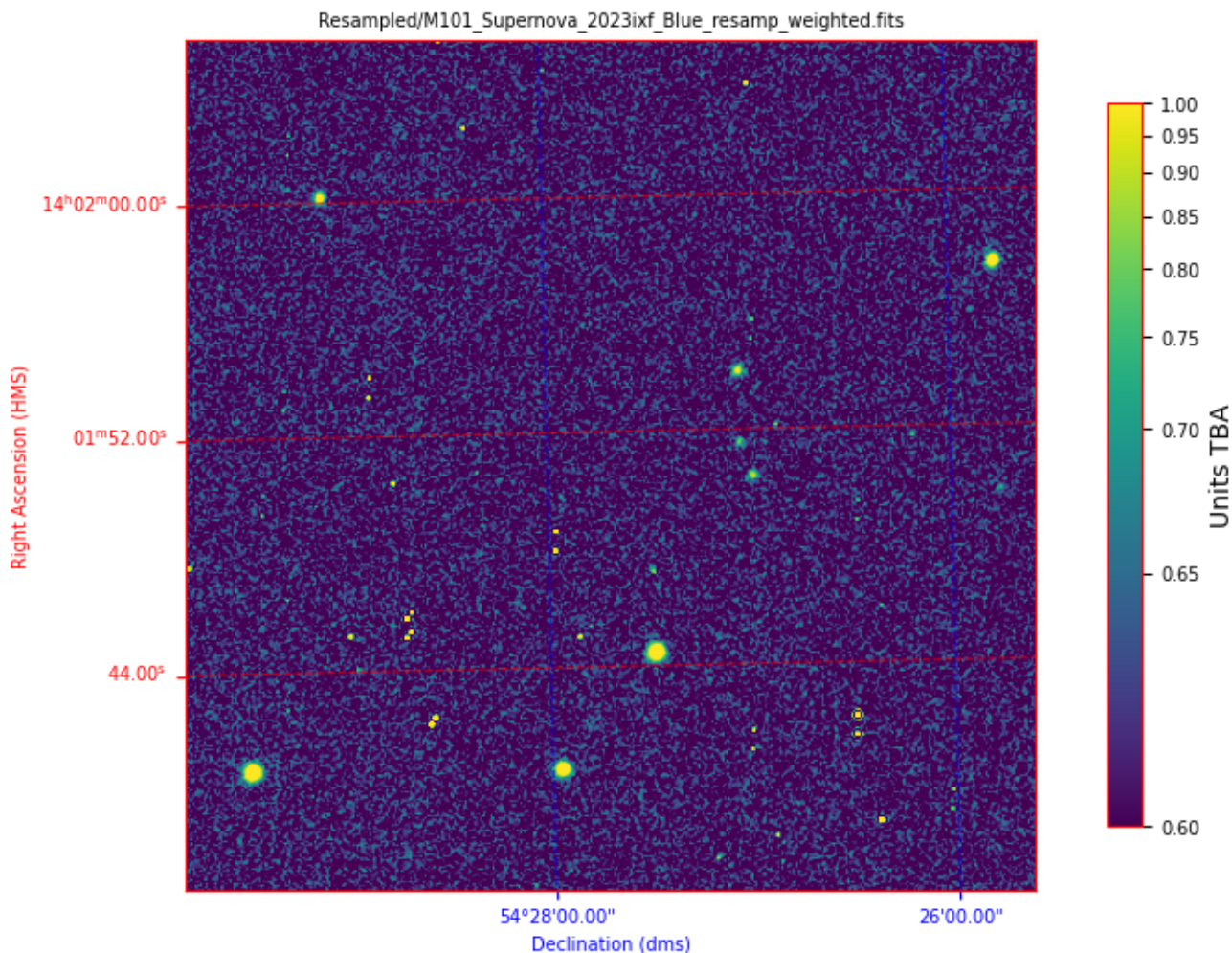
Number of WCS axes: 2

CTYPE : 'RA---TAN' 'DEC--TAN'

```

CRVAL : 210.754804653 54.4174059435
CRPIX : 1528.5 1528.5
PC1_1 PC1_2 : -4.64681630389e-06 0.000175223532872
PC2_1 PC2_2 : -0.000175223532872 -4.64681630389e-06
CDELTA : 1.0 1.0
NAXIS : 3056 3056
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1500]
Setting Y-axis limits to [350, 750]

```



We can use astropy's three-color compositing function

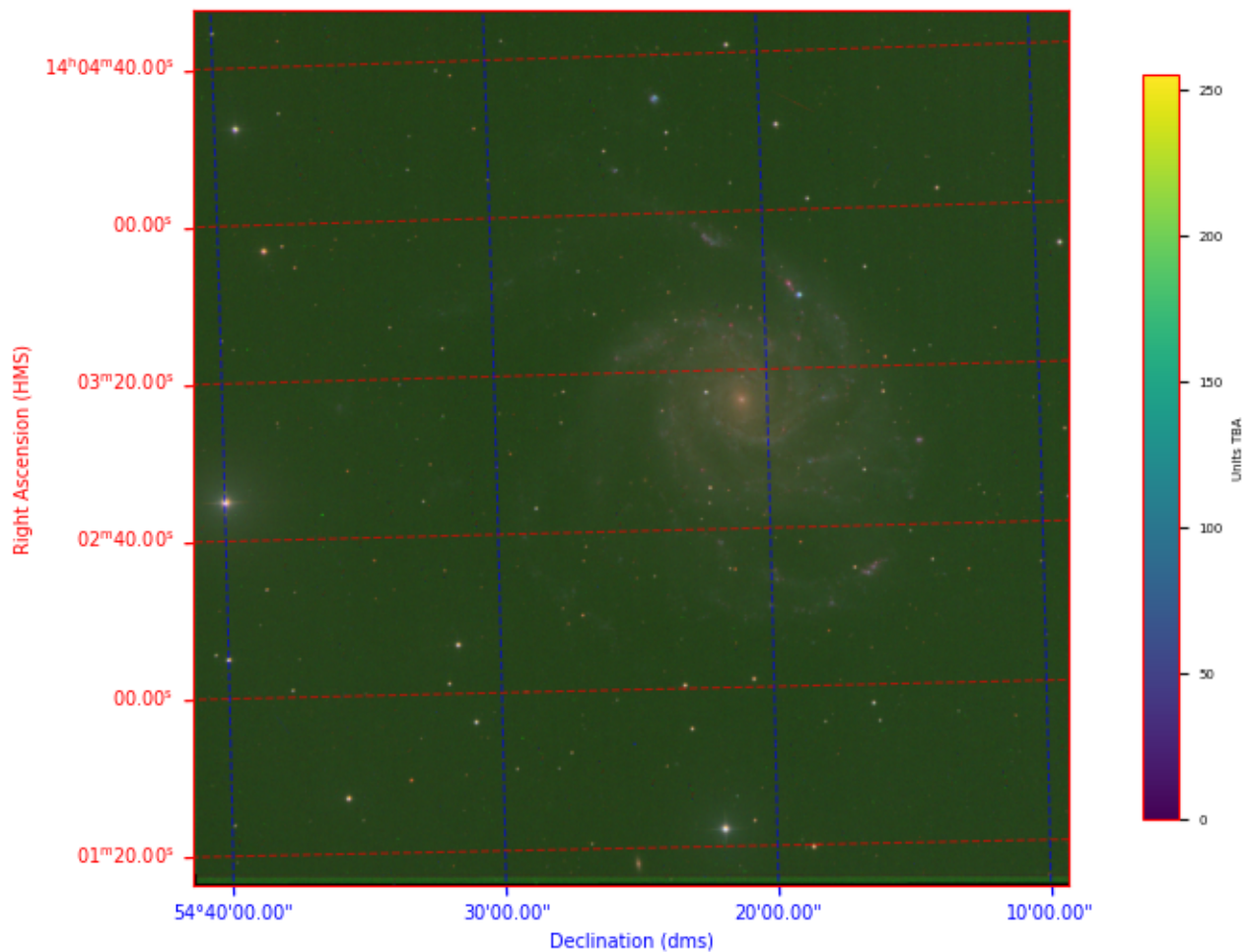
```

4]: qval      = 8
stretch = 0.5
valmin   = None
outf     = 'M101_lupton_RedGreenBlue.png'
rgb = ap.util.plot_lupton_threecolor(redf, greenf, bluef, outf,
                                     extnum=0, usewcs=True, vmin=valmin,
                                     xaxlim=None, yaxlim=None,
                                     qval=qval, stretchval=stretch, verbose=True,
                                     angle_tick_spacing_am=10, swap_radec_axis=True)

```

Input red channel data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944
Using an image minimum of 0.5398805156350136 for all channels.
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Wrote plot to M101_lupton_RedGreenBlue.png

RGB composite image stretch=0.5, Q=8, minimum=None
 Red = Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Green = Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Blue = Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits

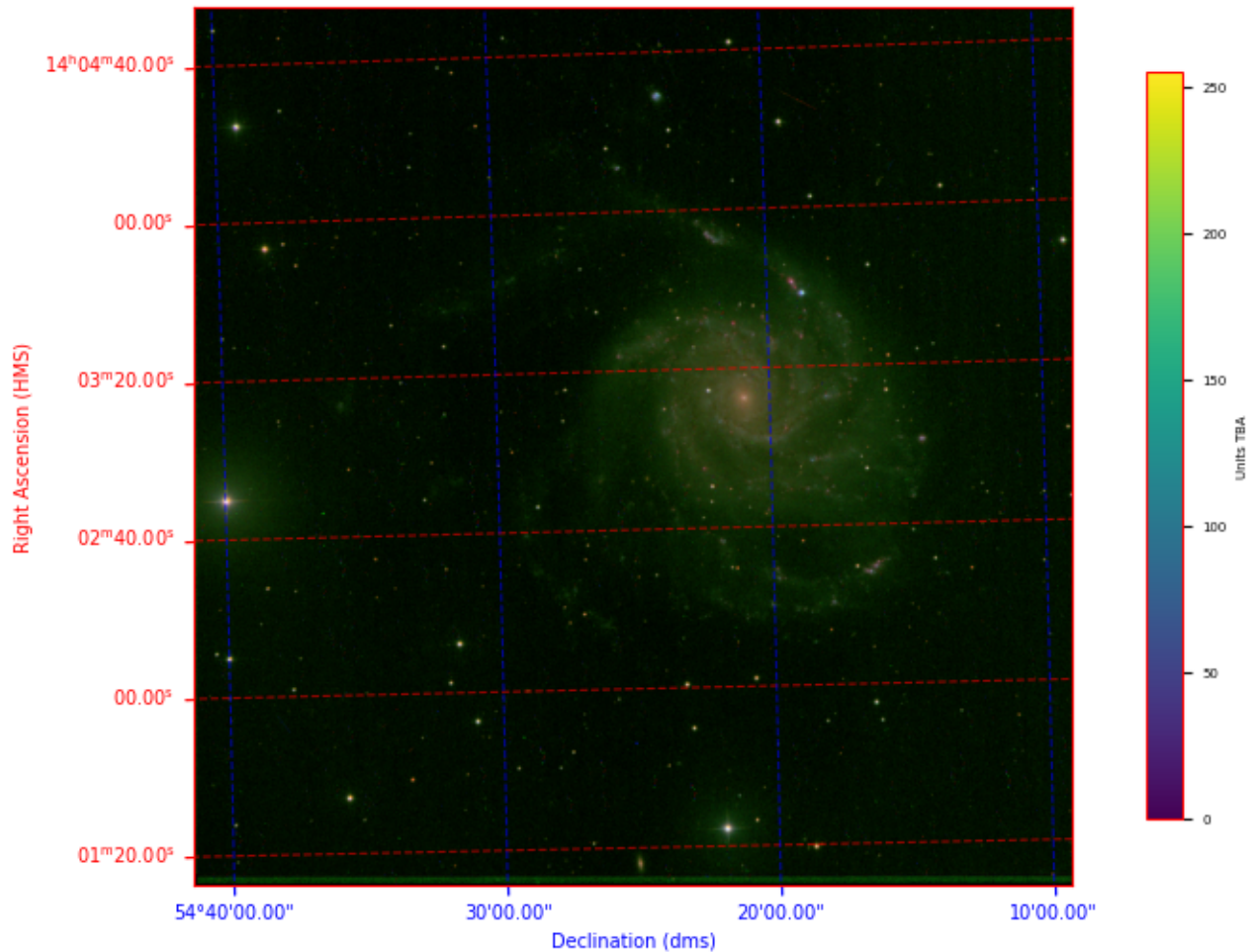


```
5]: # We can adjust the valmin to try to get a dark "black"

qval    = 8
stretch = 0.5
valmin   = 0.65
outf     = 'M101_lupton_RedGreenBlue.png'
rgb = ap.util.plot_lupton_threecolor(redf, greenf, bluef, outf,
                                     extnum=0, usewcs=True, vmin=valmin,
                                     xaxlim=None, yaxlim=None,
                                     qval=qval, stretchval=stretch, verbose=True,
                                     angle_tick_spacing_am=10, swap_radec_axis=True)
```

Input red channel data 0.5th percentile = 0.5399, 99.5th percentile = 0.8944
 Using an image minimum of 0.65 for all channels.
 Default X-axis limits: (-0.5, 3055.5)
 Default Y-axis limits: (-0.5, 3055.5)
 Wrote plot to M101_lupton_RedGreenBlue.png

RGB composite image stretch=0.5, Q=8, minimum=0.65
 Red = Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Green = Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Blue = Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits



Or we can use Astromatic stiff instead to create a TIFF file, and then load and plot that.

Note: this is not wrapped in python yet.

```
5] : stiff_outputf = 'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff'
description = 'M101_Supernova_2023ixf'
user = 'AstroPhotography'
!stiff $redf $greenf $bluef \
  -OUTFILE_NAME $stiff_outputf -GAMMA_TYPE POWER-LAW -GAMMA 2.2 -GAMMA_FAC 1.0 \
  -COLOUR_SAT 1.0 -MAX_TYPE =QUANTILE,QUANTILE,QUANTILE -MAX_LEVEL 0.999,0.999,0.999 -MIN_TYPE =QUANTILE,QUANTILE,QUANTILE \
  -MIN_LEVEL 0.60,0.60,0.60 \
  -BITS_PER_CHANNEL 8 -VERBOSE_TYPE FULL -DESCRIPTION $description -COPYRIGHT $user -WRITE_XML N
```

> WARNING: stiff.conf not found, using internal defaults

----- STIFF 2.7.1 started on 2024-11-26 at 21:15:28 with 16 threads

----- Inputs:

M101_Supernova_2023ixf_Red_resamp_weighted.fits: "no ident" 3056x3056 32 bits (floats)
 Background level: 0.634998 Min level: 0.618508 Max level: 3.07458
 M101_Supernova_2023ixf_Green_resamp_weighted.fits: "no ident" 3056x3056 32 bits (floats)
 Background level: 0.700548 Min level: 0.683503 Max level: 2.88235
 M101_Supernova_2023ixf_Blue_resamp_weighted.fits: "no ident" 3056x3056 32 bits (floats)
 Background level: 0.607307 Min level: 0.590825 Max level: 2.50276

```
----- Output:
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff:
3056x3056      3x8 bits (integers) gamma: x1.00 compression: LZW
```

```
[7]: # where are we?
cwd = pathlib.Path('.')
print(f'Current directory: {cwd.resolve()}')
tiff_file = cwd / 'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff'
if tiff_file.exists():
    fig, (ax1, ax2) = plt.subplots(1,2)
    print(f'Loading {tiff_file.name}')
    tiff_img = plt.imread(tiff_file)
    print(f'Plotting full image {tiff_file}')
    ax1.imshow(tiff_img)

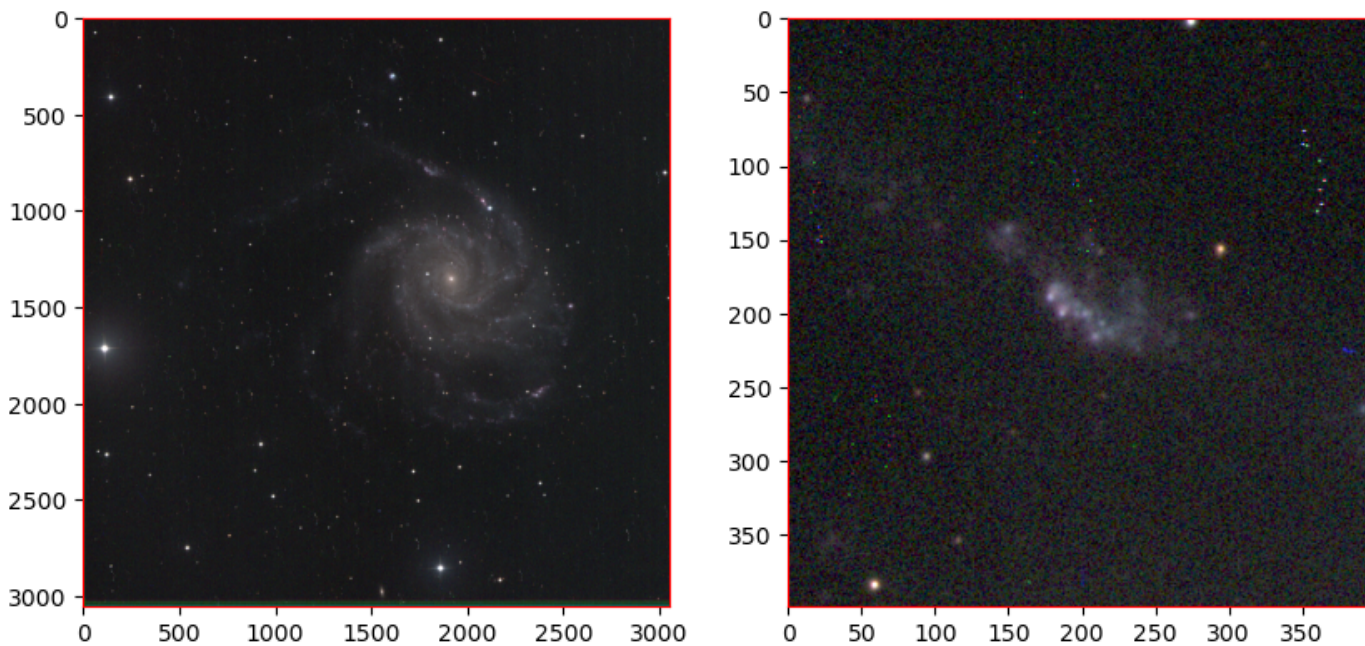
    # Zoom in
    print(f'Plotting zoomed-in part of {tiff_file}')
    print(' Note the bad pixels visible in this zoomed-in view.')
    ax2.imshow(tiff_img[600:1000,1600:2000])
else:
    print(f'File not found: {tiff_file.name}')
    print(' Skipping display of STIFF generated file...')
```

```
Current directory: /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-
Pipeline
Loading M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
Plotting full image
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
```

```
[7]: <matplotlib.image.AxesImage at 0x7f56828d5840>
```

```
Plotting zoomed-in part of
M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
Note the bad pixels visible in this zoomed-in view.
```

```
[7]: <matplotlib.image.AxesImage at 0x7f5682a0bb80>
```



Again, on closer inspection the bad pixels are apparent.

The following lines of code make copies of some of these output files, for later comparison to the same product after attempting fix some of the problems we have highlighted.

```
[8]: orig_bluef_path = pathlib.Path(bluef)
new_name = 'orig_' + orig_bluef_path.name
new_resampled_blue_file = orig_bluef_path.with_name(new_name)
print(f'Copying {bluef} to {str(new_resampled_blue_file)}')
```

```
copyof_original_bluef = shutil.copyfile(bluef, new_resampled_blue_file)
print(f'Created {copyof_original_bluef}')

orig_tiff_file_path = pathlib.Path(tiff_file)
new_name = 'orig_' + orig_tiff_file_path.name
new_tiff_file = orig_tiff_file_path.with_name(new_name)
print(f'Copying {tiff_file} to {str(new_tiff_file)}')
copyof_original_tiff_file = shutil.copyfile(tiff_file, new_tiff_file)
print(f'Created {copyof_original_tiff_file}')
```

```
Copying Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits to
Resampled/orig_M101_Supernova_2023ixf_Blue_resamp_weighted.fits
Created Resampled/orig_M101_Supernova_2023ixf_Blue_resamp_weighted.fits
Copying M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff to
orig_M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
Created
orig_M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
```

```
9]: # Close all the open figures in memory
plt.close('all')
```

2.6 Summary

The main processing pathway used in this section, excluding the plotting functions, is basically

```
loglevel = 'DEBUG'
processor = ap.ApProcess(loglevel)

# Source detection and astrometry
# Set values for: data_dir, include_pattern, exclude_pattern, file suffix = '.fits.gz'
#               quality_summary_file, clean_srclists, clean_navfiles, stop_on_error, quiet
status = processor.navigate_images(data_dir, include_pattern, exclude_pattern, None, None, '.fits.gz',
                                   quality_summary_file, clean_srclists, clean_navfiles, stop_on_error=stop_on_error, quiet=quiet)

# Load and look at the quality summary file
qual_summ_table = Table.read(quality_summary_file, format="ascii.csv")
qual_summ_table['file', 'ncols', 'nrows', 'filter', 'median', 'stddev',
                 'num_detected', 'fwhm_val_pix', 'fwhm_err_pix', 'fwhm_val_arcs', 'fwhm_err_arcs', 'num_data_pts'].pprint_all()

# Get a list of navigated files...
navigated_file_list = processor.get_file_names('navfile', data_dir, 'existing', include_pattern, exclude_pattern, None, None, '.fits.gz', with
print(f'Navigated image files on disk: \n{navigated_file_list}')
first_file = navigated_file_list[0]
w, pix_scales, pix_area, im_scales = ap.util.load_wcs_from_file(first_file)
print(f'\nSummarizing WCS information from {f

# Resample and stack images in all the filters
# Set values for target_wcs_file, resam_prefix, resam_suffix, resampled_dir=resam_dir
resampled_img_dict, info_table = processor.resample_images_to_match(target_wcs_file, resam_prefix, resam_suffix, resampled_dir=resam_dir, fil
```

3 Preprocessing to Fix Bad Pixels and Add Metadata

The previous section demonstrated the use of `ApProcess` to recreate the processing shown in `itelescope_premium-the_hard_way.ipynb`. This section improves on the previous section by applying two pre-processing stages to the calibrated files before navigating them:

1. Add metadata to the iTelescope Premium file headers, both for completeness' sake and to speed up the astrometric solutions.
2. Apply bad pixel, bad column and bad row correction. This will alleviate the issues shown the visualization of the resampled iTelescope premium images shown in Section 2.5.
3. Apply Cosmic Ray rejection. In most cases where bad pixel correction is required it makes sense to perform CR rejection as well after the bad pixel correction.

Note: Currently only a metadata dictionary is supported for adding metadata. In the near term a YAML file will also be an option.

3.1 Metadata Dictionary Format

The python dictionary should consist of `key: (value, comment)` pairs, where `key` and `comment` are strings. The `key` keyword must conform to FITS file conventions, in particular the keyword cannot be longer than 8 letters, cannot start with a numeral, and cannot include whitespace.

In Section 2, running the iTelescope Plan-30 M101 SN 2023ixf data, we saw that the following keywords were not found in the calibrated input image headers:

```
2024-09-20 09:00:50,425 | AstroPhotography.core.ApFindStars | DEBUG | FITS keywords missing from image: ['EXPOSURE', 'DATE-OBS', 'OBJECT', 'O
```

We are going to specify the values for some of these in the metadata dictionary, based on prior knowledge of the telescope/instrument package used for the observations. Others, like the date, time, and airmass, we lack the information necessary to determine.

We could also specify `EXPOSURE` in the metadata dictionary but in practice we know that the header contains one of the commonly used equivalent keywords, so we can tell `ApProcess` to find it and set `EXPOSURE` to match it.

3.2 Bad Pixel Correction

See the `creating_a_master_bad_pixel_file.ipynb` notebook for details on how to create a master bad pixel / bad row / bad column file.

3.3 Preprocessing File Naming Convention

Preprocessing does not modify the original input files. Instead it creates new files with names based on string replacement of the of original inputs files. The new file names are then either explicitly input as an `input_file_list` when running `navigate_images`, or you can modify the `include_pattern`. In both cases the `input_rootname` should be updated.

The diagram below shows a simplified example:

Normal (no-preprocessing) Case

```
data_dir      = r'.'
include_pattern = "Calibrated-*-.fits*"
exclude_pattern = None
processor      = ap.ApProcess(loglevel)
```

```
status = processor.navigate_images(data_dir,
    include_pattern, exclude_pattern, ...)
```

Preprocessing Case

```
data_dir      = r'.'
include_pattern = "Calibrated-*-.fits*"
exclude_pattern = None
processor      = ap.ApProcess(loglevel)
```

```
preprocess_instr = 'Calibrated'
preprocess_outstr = 'recalibrated'
modified_files = processor.preprocess_images(data_dir,
    preprocess_instr, preprocess_outstr,
    include_pattern, exclude_pattern, ...)
```

```
# Then we can either use the modified file list and reset the
# input_rootname, e.g.
```

```
status = processor.navigate_images(data_dir, include_pattern=None,
    input_file_list=modified_files, input_rootname=preprocess_outstr, ...)
```

```
# OR we can modify the input include_pattern, e.g.
```

```
modified_pattern = include_pattern.replace(preprocess_instr, preprocess_outstr)
print(modified_pattern) # produces "recalibrated-*-.fits"
```

```
status = processor.navigate_images(data_dir,
    modified_pattern, exclude_pattern,
    input_file_list=None, input_rootname=preprocess_outstr, ...)
```

3.4 Run the Preprocessing

Note: The following inputs are hard-wired for the iTelescope M101 SN 2023ixf data.

Bad Pixel Files: At the time of writing (2024-10-03) the official T24 calibration files are still December 2021 vintage. The bad pixel file generated from those calibration files is the one pointed to by `old_badpixfile`. Processing using that file generated only partial improvement in the output, so I generated a short series of darks for T24 on 2024-10-01 that were used to generate `new_badpixfile`. This bad pixel file is the one used, and is much more effective at removed bad pixels than the older one.

```
0]: # switch to control running preprocessing and reprocessing "by hand"
do_by_hand_preprocess = True
```

```
1]: # metadata dictionary
meta_dict = {'OBJNAME': ('M101 Supernova 2023ixf', 'Target'),
             'OBJECT': ('M101 Supernova 2023ixf', 'Target'),
             'TELESCOP': ('iTelescope 24', 'Telescope name'),
             'INSTRUME': ('FLI-PL09000', 'Instrument name'),
             'XPIXSZ': (12.0, 'Pixel Width in microns (after binning)'),
             'YPIXSZ': (12.0, 'Pixel Height in microns (after binning)'),
             'EGAIN': (1.53, 'Electronic gain in e-/ADU')}

# Overwrite existing header keywords with the same name
rewrite_existing_kw=False

# Set EXPOSURE if not present and any of the other exposure-like FITS keywords are present
find_exposure_time=True

# Bad pixel file location (e.g. creating_a_master_bad_pixel_file.ipynb)
old_badpixfile='/old_lnx/home/dks/Downloads/iTelescopeScratch/calibration-library/T24/Raw/2021_DEC/
↳Master_Badpix_11_3056x3056_Bin1x1_Temp-25C_ExpTime900s.fit'
new_badpixfile='/old_lnx/home/dks/Downloads/iTelescopeScratch/T24/CALIBRATION/Dark/Bin1/300/
↳master_badpix_T24_20241001_3056x3056_Bin1x1_Temp-25C_ExpTime300s.fits'
badpixfile = new_badpixfile

cr_rejection = True
```

```

# instr is the part of the original calibrated input file name that will be renamed in the modified,
# preprocessed, output files.
preprocess_instr = 'Calibrated'
preprocess_outstr = 'recalibrated'

```

```

2]: del processor
loglevel = 'INFO'
processor = ap.ApProcess(loglevel)

```

```

3]: extnum = 0
if do_by_hand_preprocess:
    resampled_files = processor.preprocess_images(
        data_dir,
        preprocess_instr,
        preprocess_outstr,
        include_pattern,
        exclude_pattern,
        None,
        None,
        ".fits.gz",
        extnum=extnum,
        find_exposure_time=find_exposure_time,
        keyword_dict=meta_dict,
        replace_keywords=rewrite_existing_kw,
        badpixelfile=badpixfile,
        fix_cosmic_rays=cr_rejection,
    )

```

```

2024-11-26 21:15:30,397 | ApProcess | INFO | Attempting to preprocess FITS files
within directory=.
2024-11-26 21:15:30,397 | ApProcess | INFO | Using files that match
include_pattern="Calibrated-*-.fits*"
2024-11-26 21:15:30,397 | ApProcess | INFO | Excluding files that match
exclude_pattern="None"
2024-11-26 21:15:31,878 | ApProcess | INFO | There are 12 input files matching
the parameters given.
2024-11-26 21:15:31,879 | ApProcess | INFO | Loading extension 0 of FITS file /o
ld_lnx/home/dks/Downloads/iTelescopeScratch/T24/CALIBRATION/Dark/Bin1/300/master
_badpix_T24_20241001_3056x3056_Bin1x1_Temp-25C_ExpTime300s.fits
2024-11-26 21:15:32,153 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:15:32,154 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:15:32,154 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:15:34,953 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.791 s, 0.015 ms per bad pixel.
2024-11-26 21:15:37,494 | ApFixCosmicRays | INFO | 9891 pixels in image
identified as affected by cosmic rays.
2024-11-26 21:15:39,198 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:15:39,199 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:15:39,199 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:15:41,981 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.775 s, 0.015 ms per bad pixel.
2024-11-26 21:15:44,490 | ApFixCosmicRays | INFO | 8297 pixels in image
identified as affected by cosmic rays.
2024-11-26 21:15:46,443 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
2024-11-26 21:15:46,443 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:15:46,444 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:15:49,259 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.808 s, 0.015 ms per bad pixel.
2024-11-26 21:15:51,709 | ApFixCosmicRays | INFO | 7111 pixels in image
identified as affected by cosmic rays.
2024-11-26 21:15:53,598 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz

```

2024-11-26 21:15:53,601 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:15:53,601 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:15:56,389 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.779 s, 0.015 ms per bad pixel.

2024-11-26 21:15:58,964 | ApFixCosmicRays | INFO | 78098 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:00,450 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz into recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz

2024-11-26 21:16:00,450 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:16:00,451 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:16:03,244 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.786 s, 0.015 ms per bad pixel.

2024-11-26 21:16:05,887 | ApFixCosmicRays | INFO | 81954 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:07,381 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz into recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz

2024-11-26 21:16:07,382 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:16:07,382 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:16:10,154 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.765 s, 0.015 ms per bad pixel.

2024-11-26 21:16:12,596 | ApFixCosmicRays | INFO | 79343 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:14,068 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz into recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz

2024-11-26 21:16:14,069 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:16:14,069 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:16:16,885 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.809 s, 0.015 ms per bad pixel.

2024-11-26 21:16:19,357 | ApFixCosmicRays | INFO | 53543 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:20,855 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz into recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz

2024-11-26 21:16:20,858 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:16:20,858 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:16:23,677 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.810 s, 0.015 ms per bad pixel.

2024-11-26 21:16:26,184 | ApFixCosmicRays | INFO | 83883 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:27,663 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz into recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz

2024-11-26 21:16:27,663 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:16:27,664 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:16:30,446 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.775 s, 0.015 ms per bad pixel.

2024-11-26 21:16:32,960 | ApFixCosmicRays | INFO | 58394 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:34,442 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz into recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz

2024-11-26 21:16:34,442 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056 rows x 3056 columns, dtype=>f4

2024-11-26 21:16:34,443 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056 rows x 3056 columns, dtype=float32

2024-11-26 21:16:37,259 | ApFixBadPixels | INFO | Processed 185656 pixels in 2.810 s, 0.015 ms per bad pixel.

2024-11-26 21:16:39,775 | ApFixCosmicRays | INFO | 77657 pixels in image identified as affected by cosmic rays.

2024-11-26 21:16:41,253 | ApProcess | INFO | Preprocessing input file Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz into

```

recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
2024-11-26 21:16:41,254 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:16:41,254 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:16:44,063 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.802 s, 0.015 ms per bad pixel.
2024-11-26 21:16:46,583 | ApFixCosmicRays | INFO | 68196 pixels in image
identified as affected by cosmic rays.
2024-11-26 21:16:48,077 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
2024-11-26 21:16:48,077 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:16:48,077 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:16:50,895 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.811 s, 0.015 ms per bad pixel.
2024-11-26 21:16:53,431 | ApFixCosmicRays | INFO | 79765 pixels in image
identified as affected by cosmic rays.
2024-11-26 21:16:54,665 | ApProcess | INFO | Finished preprocessing 12 files in
82.763 seconds.

```

We can inspect the before and after images

```

4]: masterdarkfile = '/old_lnx/home/dks/Downloads/iTelescopeScratch/calibration-library/T24/Raw/2021_DEC/
↳Master_Dark_11_3056x3056_Bin1x1_Temp-25C_ExpTime900s.fit'

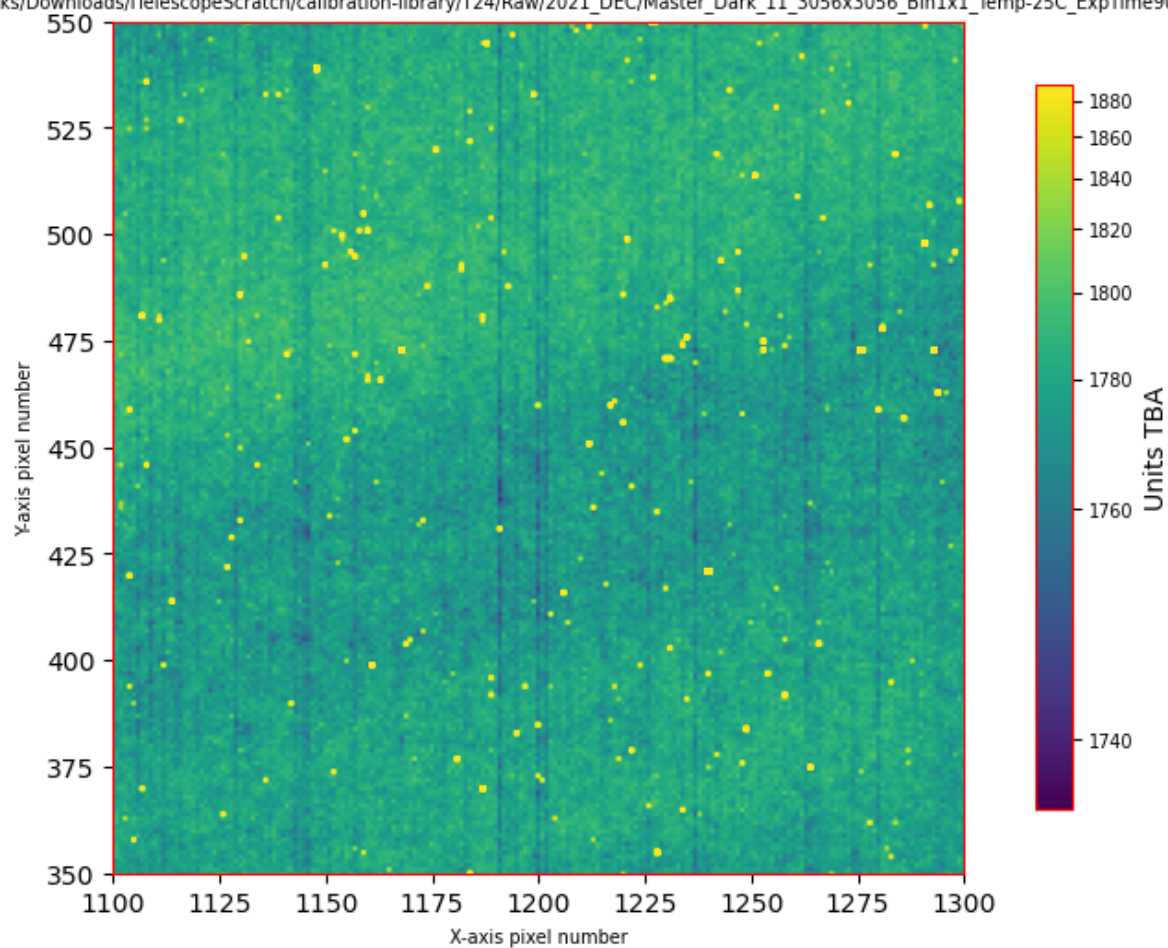
if do_by_hand_preprocess:
    input_files = processor.get_file_names(process_stage, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.
↳gz', no_name_and_dir)
    output_plot = None # Name of output PNG file or None
    ap.util.load_image_and_plot(masterdarkfile, 0, output_plot, False, xaxlim=[1100, 1300], yaxlim=[350, 550], verbose=True)
    ap.util.load_image_and_plot(badpixfile, 0, output_plot, False, xaxlim=[1100, 1300], yaxlim=[350, 550], verbose=True)
    ap.util.load_image_and_plot(input_files[0], 0, output_plot, False, xaxlim=[1100, 1300], yaxlim=[350, 550], verbose=True)
    ap.util.load_image_and_plot(resampled_files[0], 0, output_plot, False, xaxlim=[1100, 1300], yaxlim=[350, 550], verbose=True)
    ap.util.load_image_and_plot(input_files[-3], 0, output_plot, False, xaxlim=[1100, 1300], yaxlim=[350, 550], verbose=True)
    ap.util.load_image_and_plot(resampled_files[-3], 0, output_plot, False, xaxlim=[1100, 1300], yaxlim=[350, 550], verbose=True)

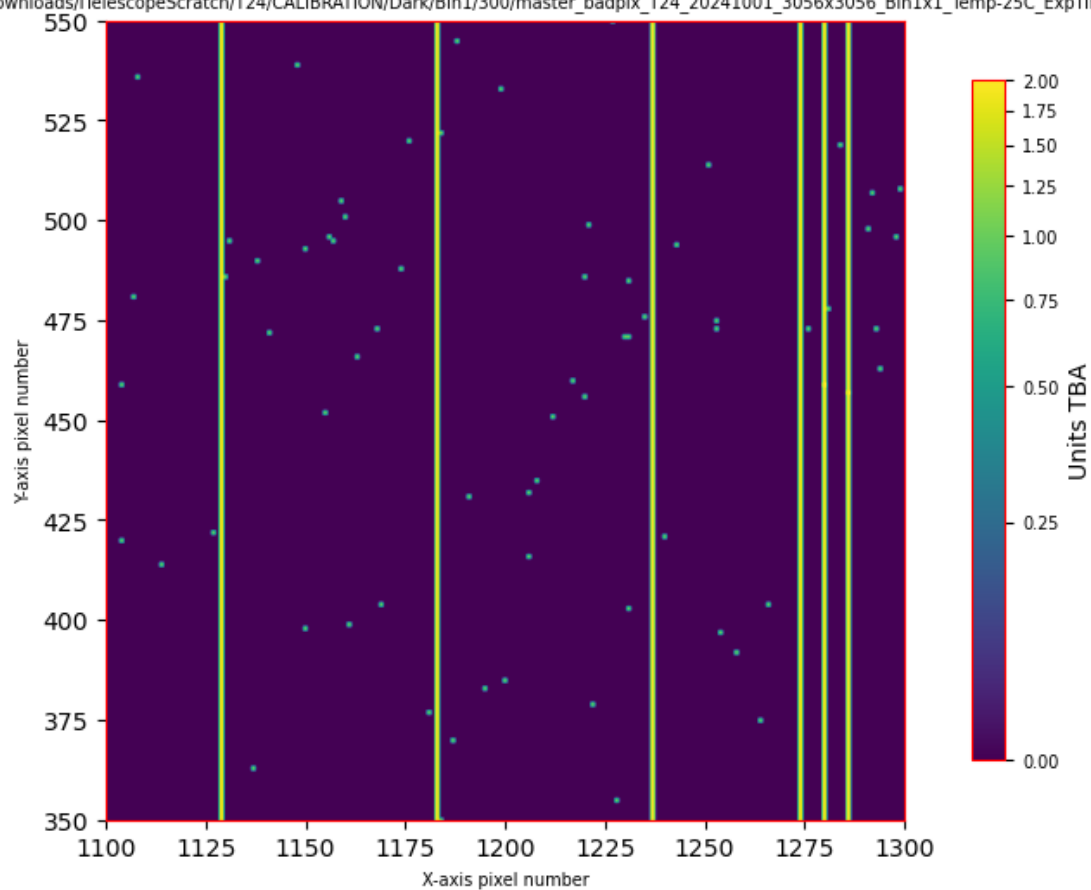
```

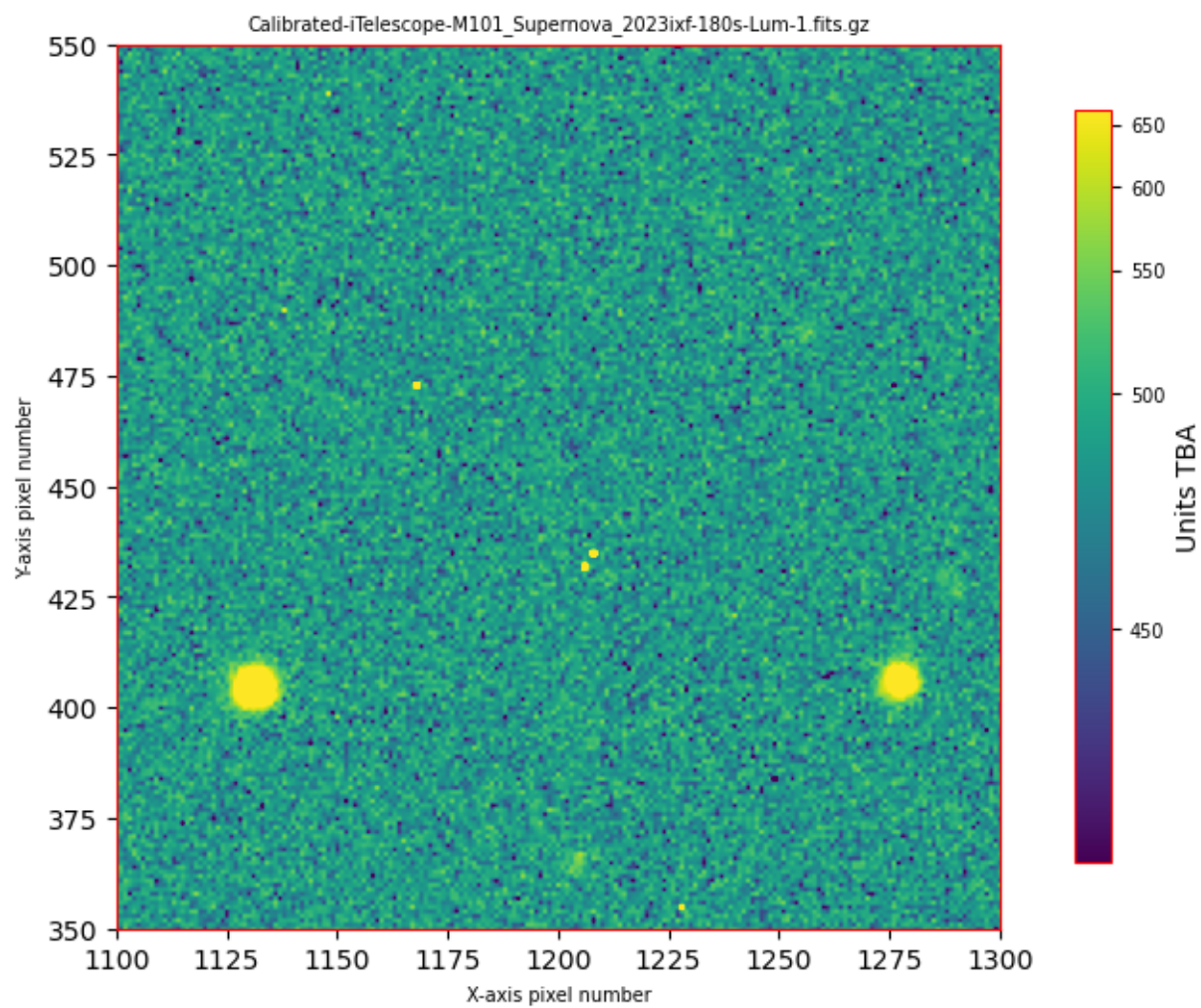
```

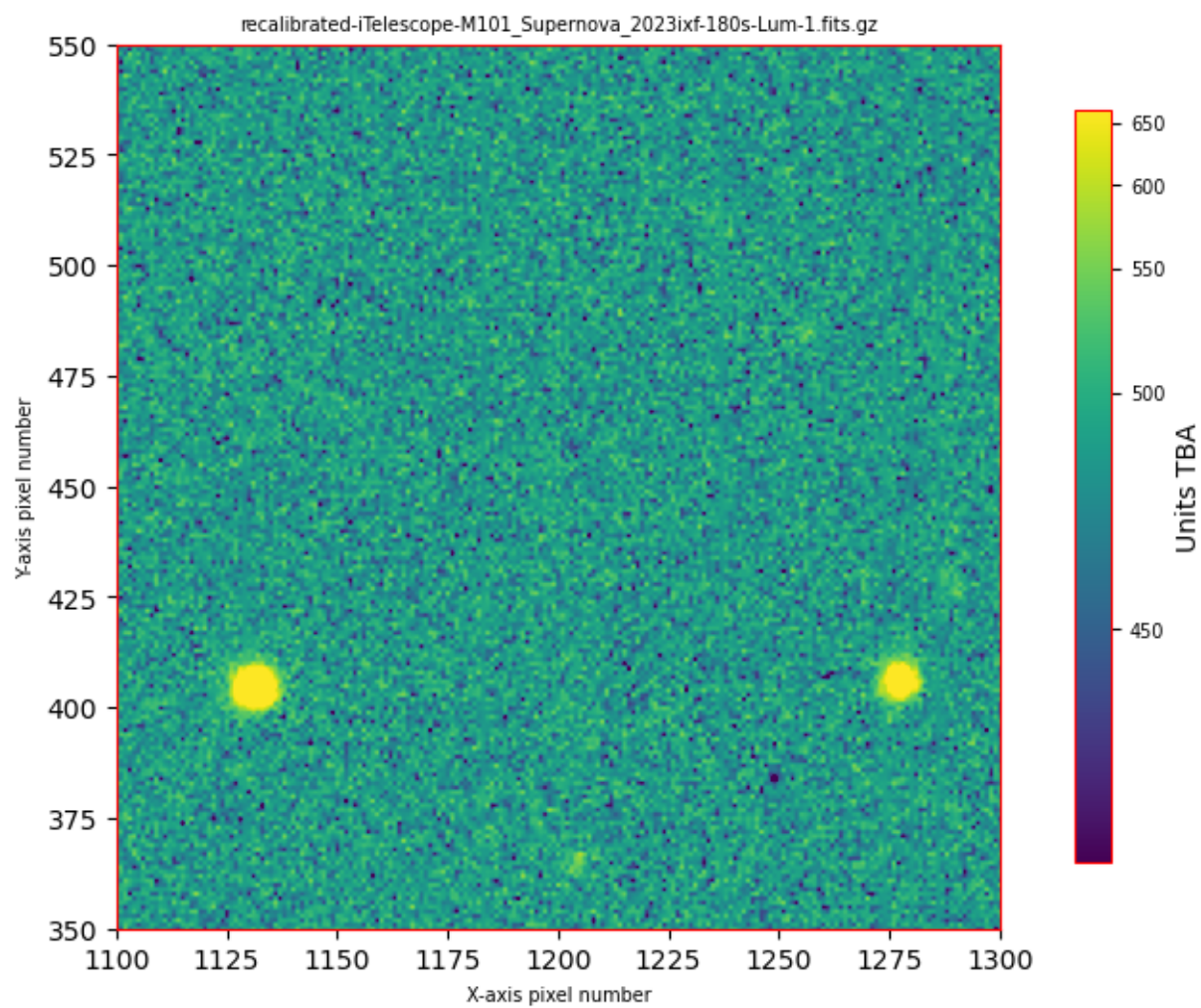
Input data 0.5th percentile = 1735.5000, 99.5th percentile = 1889.5000
Applying asinh stretch between 1735.5000 and 1889.5000
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1300]
Setting Y-axis limits to [350, 550]
Input data 0.5th percentile = 0.0000, 99.5th percentile = 2.0000
Applying asinh stretch between 0.0000 and 2.0000
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1300]
Setting Y-axis limits to [350, 550]
Input data 0.5th percentile = 424.4036, 99.5th percentile = 662.8992
Applying asinh stretch between 424.4036 and 662.8992
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1300]
Setting Y-axis limits to [350, 550]
Input data 0.5th percentile = 424.6816, 99.5th percentile = 661.3711
Applying asinh stretch between 424.6816 and 661.3711
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1300]
Setting Y-axis limits to [350, 550]
Input data 0.5th percentile = 131.3915, 99.5th percentile = 277.7340
Applying asinh stretch between 131.3915 and 277.7340
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1300]
Setting Y-axis limits to [350, 550]
Input data 0.5th percentile = 131.6617, 99.5th percentile = 275.2452
Applying asinh stretch between 131.6617 and 275.2452
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Setting X-axis limits to [1100, 1300]
Setting Y-axis limits to [350, 550]

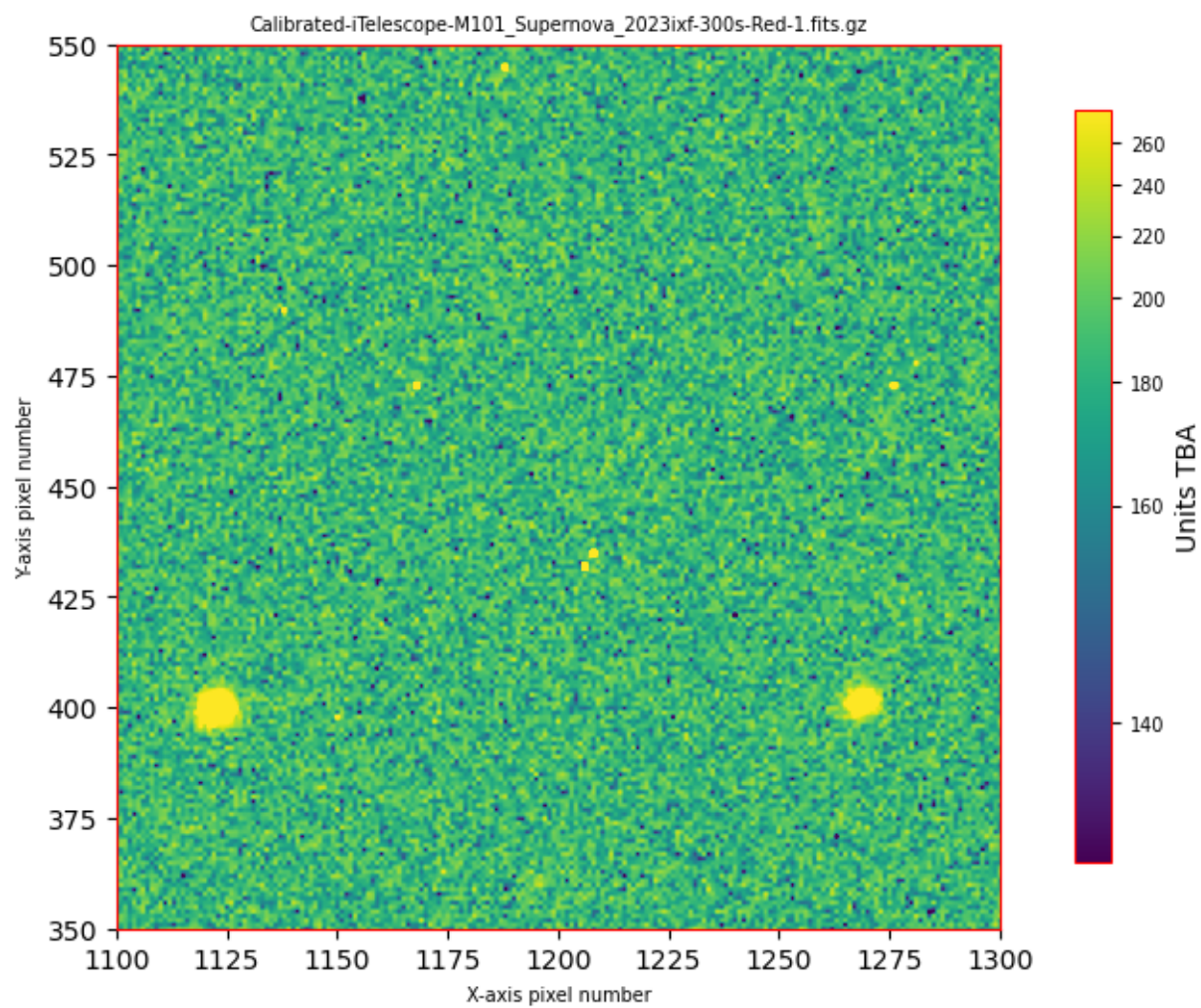
```

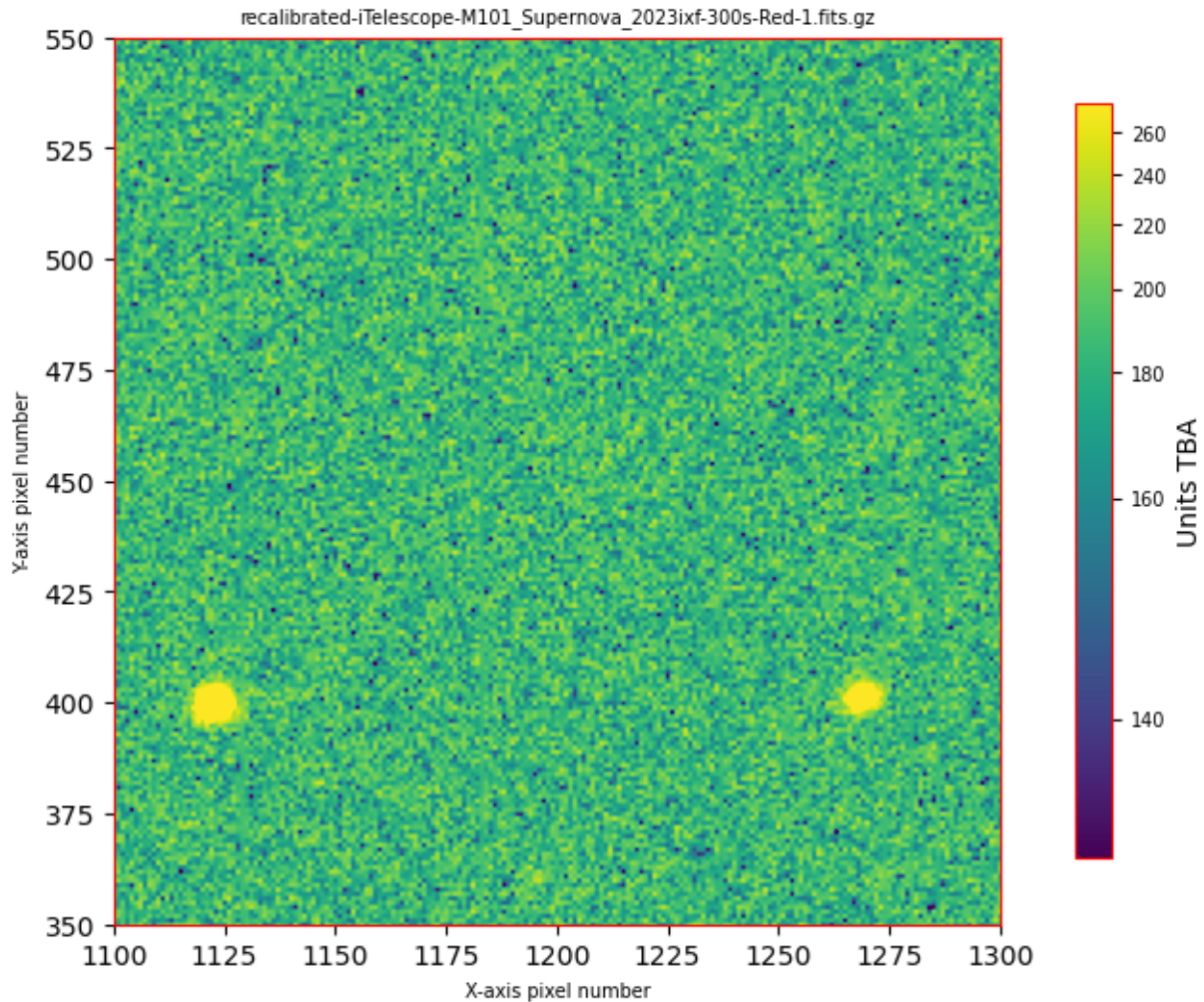












With the more recent bad pixel map the output is very encouraging. In this particular region all the bad pixels were corrected.

Side note: Inspection of the out-of-date master dark file shows the brightest bad pixels in the SN2023ixf data are not active in the 2021-era master dark. Changing the processing parameters used with `ApFindBadPixels` is not going to find those pixels as being bad. Ultimately the problem is that the 2021-era calibration data is not up-to-date. (With the out-of-date bad pixel map the output is disappointing, in that two of the brightest bad pixels are not corrected for in. Only approximately half of the bad pixels were removed.)

3.5 Image Navigation and Resampling on Reprocessed Files

We can proceed with the normal processing using these resampled images.

```
5]: if do_by_hand_preprocess:
    # Need to remove existing quality summary table
    pathlib.Path(quality_summary_file).unlink(True)

6]: if do_by_hand_preprocess:
    # OR we can modify the input include_pattern, e.g.
    print(f'preprocess_instr={preprocess_instr}, prepreprocess_outstr={preprocess_outstr}')
    modified_pattern = include_pattern.replace(preprocess_instr, preprocess_outstr)
    print(f'modified_pattern={modified_pattern}')

preprocess_instr=Calibrated, prepreprocess_outstr=recalibrated
modified_pattern=recalibrated-*-.fits*

7]: if do_by_hand_preprocess:
    # Check that processing will pick up the files we used.
    # Show files names that would be generated {'conceptual', 'processed', 'existing'}
    ap_filestate = 'existing'

    # Note we need to specify that the input_suffix is 'fits.gz' not for the inputs, which
    # are picked up by the include_pattern, but for the output file types.
```

```

process_stage      = 'input'
no_name_and_dir    = False
with_name_and_dir  = True
input_files = processor.get_file_names(process_stage, data_dir, ap_filestate, modified_pattern, exclude_pattern,
↳input_file_list=None, input_rootname=preprocess_outstr,
                                input_suffix='.fits.gz', name_and_dir=no_name_and_dir)

for file in input_files:
    print(f' {file}')

```

```

recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

```

```

8]: if do_by_hand_preprocess:
    # Compare to files we got from preprocess_images()
    for file in resampled_files:
        print(f' {file}')

```

```

recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

```

```

9]: if do_by_hand_preprocess:
    clean_run = True
    if clean_run:
        clean_srclists = True
        clean_navfiles = True
    else:
        clean_srclists = False
        clean_navfiles = False

    stop_on_error = False
    quiet = True

    status = processor.navigate_images(data_dir, modified_pattern, exclude_pattern, None, preprocess_outstr, '.fits.gz',
                                      quality_summary_file, clean_srclists, clean_navfiles, stop_on_error=stop_on_error, quiet=quiet)

```

```

2024-11-26 21:17:00,690 | ApProcess | INFO | Attempting to find stars in FITS
files within directory=.
2024-11-26 21:17:02,239 | ApProcess | INFO | There are 12 input files matching
the parameters given.
2024-11-26 21:17:02,506 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:17:02,507 | ApFindStars | INFO | Loading extension 0 of FITS file
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:17:06,697 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 259 sources.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
    phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:17:07,696 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:17:07,707 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:17:07,947 | ApMeasureStars | INFO | Source average axis ratio =
1.051 +/- 0.083, overall sources are circular.
2024-11-26 21:17:07,949 | ApFindStars | INFO | Median FWHM (over x and y) is
4.33 +/- 0.31 pixels using 50 data points

```

```

2024-11-26 21:17:08,201 | ApFindStars | INFO | Initial source list using
FWHM=4.325 pixels, threshold=7.00 x BG stddev, found 303 sources.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:17:08,253 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:17:08,262 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:17:08,495 | ApMeasureStars | INFO | Source average axis ratio =
1.056 +/- 0.087, overall sources are circular.
2024-11-26 21:17:09,824 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.png
2024-11-26 21:17:09,825 | ApFindStars | INFO | Median FWHM (over x and y) is
4.33 +/- 0.31 pixels using 50 data points
2024-11-26 21:17:10,860 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:17:10,862 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:17:10,862 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:17:10,862 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:17:10,862 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.
2024-11-26 21:17:10,863 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.
2024-11-26 21:17:10,863 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.
2024-11-26 21:17:10,864 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.yaml
2024-11-26 21:17:10,867 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:17:10,868 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:17:10,868 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:17:10,868 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:17:10,869 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astroquery.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astroquery.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astroquery.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:17:10,899 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:17:11,168 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits

Solving...

2024-11-26 21:17:26,079 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:17:26,396 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to ./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:17:26,396 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits with WCS solution.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astroquery.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:17:26,734 | ApProcess | INFO | Processing input file recalibrated-

```

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iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:17:26,735 | ApFindStars | INFO | Loading extension 0 of FITS file
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:17:30,841 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 243 sources.
2024-11-26 21:17:31,762 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:17:31,772 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:17:31,997 | ApMeasureStars | INFO | Source average axis ratio =
1.049 +/- 0.044, overall sources are circular.
2024-11-26 21:17:31,999 | ApFindStars | INFO | Median FWHM (over x and y) is
4.14 +/- 0.19 pixels using 50 data points
2024-11-26 21:17:32,278 | ApFindStars | INFO | Initial source list using
FWHM=4.145 pixels, threshold=7.00 x BG stddev, found 277 sources.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
  phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:17:32,327 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:17:32,337 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:17:32,566 | ApMeasureStars | INFO | Source average axis ratio =
1.046 +/- 0.046, overall sources are circular.
2024-11-26 21:17:33,897 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.png
2024-11-26 21:17:33,900 | ApFindStars | INFO | Median FWHM (over x and y) is
4.14 +/- 0.18 pixels using 49 data points
2024-11-26 21:17:34,772 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:17:34,773 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:17:34,774 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:17:34,774 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:17:34,774 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.
2024-11-26 21:17:34,774 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.
2024-11-26 21:17:34,775 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.
2024-11-26 21:17:34,777 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.yaml
2024-11-26 21:17:34,781 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:17:34,781 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:17:34,782 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:17:34,782 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:17:34,782 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astroquery.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astroquery.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astroquery.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:17:34,809 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:17:35,092 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits

Solving...

2024-11-26 21:17:52,543 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header

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2024-11-26 21:17:52,879 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits

2024-11-26 21:17:52,879 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits with WCS solution.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:17:53,224 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz

2024-11-26 21:17:53,225 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz

2024-11-26 21:17:57,499 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 197 sources.

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10

phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:17:58,382 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 197 sources.

2024-11-26 21:17:58,392 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:17:58,621 | ApMeasureStars | INFO | Source average axis ratio = 1.028 +/- 0.045, overall sources are circular.

2024-11-26 21:17:58,622 | ApFindStars | INFO | Median FWHM (over x and y) is 4.21 +/- 0.16 pixels using 49 data points

2024-11-26 21:17:58,896 | ApFindStars | INFO | Initial source list using FWHM=4.205 pixels, threshold=7.00 x BG stddev, found 234 sources.

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10

phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:17:58,938 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:17:58,947 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:17:59,175 | ApMeasureStars | INFO | Source average axis ratio = 1.019 +/- 0.041, overall sources are circular.

2024-11-26 21:18:00,495 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.png

2024-11-26 21:18:00,497 | ApFindStars | INFO | Median FWHM (over x and y) is 4.18 +/- 0.10 pixels using 47 data points

2024-11-26 21:18:01,524 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:18:01,526 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:18:01,526 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:18:01,526 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:18:01,526 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:18:01,526 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:18:01,527 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:18:01,528 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.yaml

2024-11-26 21:18:01,531 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:18:01,532 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:18:01,533 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:18:01,533 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:18:01,533 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:18:01,560 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz

2024-11-26 21:18:01,859 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/src1ist-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits

Solving...

2024-11-26 21:18:12,133 | AstroPhotography.core.ApAstrometry | INFO | Obtained a WCS header

2024-11-26 21:18:12,489 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits

2024-11-26 21:18:12,489 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/src1ist-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits with WCS soluton.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:18:12,816 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz

2024-11-26 21:18:12,817 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz

2024-11-26 21:18:17,183 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 216 sources.

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10

phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:18:18,149 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:18:18,160 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:18:18,397 | ApMeasureStars | INFO | Source average axis ratio = 1.031 +/- 0.107, overall sources are circular.

2024-11-26 21:18:18,399 | ApFindStars | INFO | Median FWHM (over x and y) is 3.90 +/- 0.28 pixels using 48 data points

2024-11-26 21:18:18,673 | ApFindStars | INFO | Initial source list using FWHM=3.901 pixels, threshold=7.00 x BG stddev, found 234 sources.

2024-11-26 21:18:18,714 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:18:18,724 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:18:18,955 | ApMeasureStars | INFO | Source average axis ratio = 1.045 +/- 0.112, overall sources are circular.

2024-11-26 21:18:20,390 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.png

2024-11-26 21:18:20,392 | ApFindStars | INFO | Median FWHM (over x and y) is 3.95 +/- 0.30 pixels using 50 data points

2024-11-26 21:18:21,322 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:18:21,323 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:18:21,324 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:18:21,324 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:18:21,324 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:18:21,324 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:18:21,325 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:18:21,326 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.yaml

2024-11-26 21:18:21,330 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:18:21,331 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:18:21,331 | ApFindStars | INFO | Approximate image field of view

is 0.750 degrees across.

2024-11-26 21:18:21,331 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:18:21,331 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:18:21,360 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz

2024-11-26 21:18:21,628 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits

Solving...

2024-11-26 21:18:37,469 | AstroPhotography.core.ApAstrometry | INFO | Obtained a WCS header

2024-11-26 21:18:37,790 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits

2024-11-26 21:18:37,791 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits with WCS soluton.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:18:38,103 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz

2024-11-26 21:18:38,104 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz

2024-11-26 21:18:42,567 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 230 sources.

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10

phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:18:43,539 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:18:43,549 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:18:43,802 | ApMeasureStars | INFO | Source average axis ratio = 1.061 +/- 0.093, overall sources are circular.

2024-11-26 21:18:43,803 | ApFindStars | INFO | Median FWHM (over x and y) is 3.47 +/- 0.27 pixels using 50 data points

2024-11-26 21:18:44,014 | ApFindStars | INFO | Initial source list using FWHM=3.472 pixels, threshold=7.00 x BG stddev, found 230 sources.

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10

phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

2024-11-26 21:18:44,054 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:18:44,064 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:18:44,318 | ApMeasureStars | INFO | Source average axis ratio = 1.071 +/- 0.087, overall sources are circular.

2024-11-26 21:18:45,632 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.png

2024-11-26 21:18:45,633 | ApFindStars | INFO | Median FWHM (over x and y) is 3.47 +/- 0.27 pixels using 50 data points

2024-11-26 21:18:46,562 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:18:46,564 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:18:46,564 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:18:46,564 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:18:46,565 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:18:46,565 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:18:46,565 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:18:46,567 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.yaml

2024-11-26 21:18:46,570 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:18:46,570 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:18:46,571 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:18:46,571 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:18:46,571 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:18:46,601 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz

2024-11-26 21:18:46,871 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits

Solving...

2024-11-26 21:18:59,791 | AstroPhotography.core.ApAstrometry | INFO | Obtained a WCS header

2024-11-26 21:19:00,110 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits

2024-11-26 21:19:00,111 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits with WCS solution.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:19:00,424 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz

2024-11-26 21:19:00,425 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz

2024-11-26 21:19:04,582 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 225 sources.

2024-11-26 21:19:05,522 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:19:05,532 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:19:05,815 | ApMeasureStars | INFO | Source average axis ratio = 1.050 +/- 0.122, overall sources are circular.

2024-11-26 21:19:05,817 | ApFindStars | INFO | Median FWHM (over x and y) is 3.64 +/- 0.30 pixels using 48 data points

2024-11-26 21:19:06,092 | ApFindStars | INFO | Initial source list using FWHM=3.644 pixels, threshold=7.00 x BG stddev, found 249 sources.

2024-11-26 21:19:06,135 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:19:06,144 | ApMeasureStars | INFO | Starting to fit

Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:19:06,439 | ApMeasureStars | INFO | Source average axis ratio = 1.105 +/- 0.135, overall sources are circular.

2024-11-26 21:19:07,762 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.png

2024-11-26 21:19:07,764 | ApFindStars | INFO | Median FWHM (over x and y) is 3.60 +/- 0.36 pixels using 48 data points

2024-11-26 21:19:08,676 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:19:08,678 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:19:08,678 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:19:08,678 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:19:08,678 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:19:08,678 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:19:08,679 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:19:08,681 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.yaml

2024-11-26 21:19:08,684 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:19:08,685 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:19:08,685 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:19:08,686 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:19:08,686 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:19:08,715 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz

2024-11-26 21:19:08,985 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits

Solving...

2024-11-26 21:19:25,601 | AstroPhotography.core.ApAstrometry | INFO | Obtained a WCS header

2024-11-26 21:19:25,919 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits

2024-11-26 21:19:25,920 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits with WCS solution.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:19:26,245 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz

2024-11-26 21:19:26,246 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz

2024-11-26 21:19:30,694 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 236 sources.

/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10
 phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])

```

2024-11-26 21:19:31,674 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:19:31,684 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:19:31,918 | ApMeasureStars | INFO | Source average axis ratio =
1.105 +/- 0.096, overall sources are circular.
2024-11-26 21:19:31,920 | ApFindStars | INFO | Median FWHM (over x and y) is
3.67 +/- 0.32 pixels using 49 data points
2024-11-26 21:19:32,194 | ApFindStars | INFO | Initial source list using
FWHM=3.666 pixels, threshold=7.00 x BG stddev, found 280 sources.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527:
RuntimeWarning: invalid value encountered in log10
    phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])
2024-11-26 21:19:32,241 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:19:32,250 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:19:32,480 | ApMeasureStars | INFO | Source average axis ratio =
1.108 +/- 0.107, overall sources are circular.
2024-11-26 21:19:34,021 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Green-1.png
2024-11-26 21:19:34,023 | ApFindStars | INFO | Median FWHM (over x and y) is
3.63 +/- 0.30 pixels using 49 data points
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:323:
RuntimeWarning: More than 20 figures have been opened. Figures created through
the pyplot interface ('matplotlib.pyplot.figure') are retained until explicitly
closed and may consume too much memory. (To control this warning, see the
rcParam `figure.max_open_warning`). Consider using `matplotlib.pyplot.close()`.
    fig, ax = plt.subplots()
2024-11-26 21:19:34,957 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:19:34,958 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:19:34,959 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:19:34,959 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:19:34,959 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.
2024-11-26 21:19:34,959 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.
2024-11-26 21:19:34,959 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.
2024-11-26 21:19:34,961 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Green-1.yaml
2024-11-26 21:19:34,964 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:19:34,965 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:19:34,965 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:19:34,965 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:19:34,966 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits
WARNING: Attribute 'version' of type <class 'dict'> cannot be added to FITS
Header - skipping [astrophy.io.fits.convenience]
WARNING:astroquery:Attribute 'version' of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astrophy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astrophy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:19:34,993 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
2024-11-26 21:19:35,262 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits

Solving...

2024-11-26 21:19:43,656 | AstroPhotography.core.ApAstrometry | INFO | Obtained a

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WCS header

2024-11-26 21:19:43,973 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits

2024-11-26 21:19:43,974 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits with WCS solution.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:19:44,305 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz

2024-11-26 21:19:44,306 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz

2024-11-26 21:19:48,818 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 181 sources.

2024-11-26 21:19:49,716 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 181 sources.

2024-11-26 21:19:49,726 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:19:49,978 | ApMeasureStars | INFO | Source average axis ratio = 1.087 +/- 0.126, overall sources are circular.

2024-11-26 21:19:49,980 | ApFindStars | INFO | Median FWHM (over x and y) is 3.78 +/- 0.29 pixels using 46 data points

2024-11-26 21:19:50,258 | ApFindStars | INFO | Initial source list using FWHM=3.781 pixels, threshold=7.00 x BG stddev, found 210 sources.

2024-11-26 21:19:50,293 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:19:50,303 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:19:50,548 | ApMeasureStars | INFO | Source average axis ratio = 1.088 +/- 0.096, overall sources are circular.

2024-11-26 21:19:51,870 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Green-2.png

2024-11-26 21:19:51,872 | ApFindStars | INFO | Median FWHM (over x and y) is 3.77 +/- 0.27 pixels using 45 data points

2024-11-26 21:19:52,754 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:19:52,755 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:19:52,755 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:19:52,756 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:19:52,756 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.

2024-11-26 21:19:52,756 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.

2024-11-26 21:19:52,756 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:19:52,758 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Green-2.yaml

2024-11-26 21:19:52,761 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:19:52,762 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:19:52,762 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:19:52,763 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:19:52,763 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srcList-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:19:52,791 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
2024-11-26 21:19:53,062 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/src1ist-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits
Solving...

2024-11-26 21:20:09,286 | AstroPhotography.core.ApAstrometry | INFO | Obtained a WCS header
2024-11-26 21:20:09,610 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits
2024-11-26 21:20:09,610 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/src1ist-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits with WCS soluton.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.
2024-11-26 21:20:09,938 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:20:09,939 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:20:14,323 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 264 sources.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10
phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])]
2024-11-26 21:20:15,257 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:20:15,267 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:20:15,556 | ApMeasureStars | INFO | Source average axis ratio = 1.042 +/- 0.178, overall sources are circular.
2024-11-26 21:20:15,557 | ApFindStars | INFO | Median FWHM (over x and y) is 3.65 +/- 0.35 pixels using 50 data points
2024-11-26 21:20:15,831 | ApFindStars | INFO | Initial source list using FWHM=3.651 pixels, threshold=7.00 x BG stddev, found 286 sources.
/home/dks/git/AstroPhotography/AstroPhotography/core/ApFindStars.py:527: RuntimeWarning: invalid value encountered in log10
phot_table["magnitude"] = -2.5 * np.log10(phot_table["adu_per_sec"])]
2024-11-26 21:20:15,879 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:20:15,888 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:20:16,171 | ApMeasureStars | INFO | Source average axis ratio = 1.003 +/- 0.161, overall sources are circular.
2024-11-26 21:20:17,760 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Green-3.png
2024-11-26 21:20:17,762 | ApFindStars | INFO | Median FWHM (over x and y) is 3.66 +/- 0.34 pixels using 50 data points
2024-11-26 21:20:18,642 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:20:18,643 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:20:18,643 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.
2024-11-26 21:20:18,644 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625
2024-11-26 21:20:18,644 | ApFindStars | WARNING | Keyword DATE-OBS not found, not written to quality report.
2024-11-26 21:20:18,644 | ApFindStars | WARNING | Keyword CCD-TEMP not found, not written to quality report.
2024-11-26 21:20:18,644 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.
2024-11-26 21:20:18,645 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Green-3.yaml
2024-11-26 21:20:18,649 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:20:18,649 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:20:18,650 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.
2024-11-26 21:20:18,650 | ApFindStars | INFO | Approximate pixel sizes

(arcseconds) are x=0.625, y=0.625
2024-11-26 21:20:18,650 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srclist-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits
WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]
WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.
2024-11-26 21:20:18,678 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:20:18,943 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from ./SourceLists/srclist-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits
Solving...
2024-11-26 21:20:32,406 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header
2024-11-26 21:20:32,723 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to ./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits
2024-11-26 21:20:32,724 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srclist-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits with WCS soluton.
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]
WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.
2024-11-26 21:20:33,050 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
2024-11-26 21:20:33,051 | ApFindStars | INFO | Loading extension 0 of FITS file
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
2024-11-26 21:20:37,381 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 268 sources.
2024-11-26 21:20:38,328 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:20:38,338 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:20:38,590 | ApMeasureStars | INFO | Source average axis ratio =
1.149 +/- 0.142, overall sources are circular.
2024-11-26 21:20:38,592 | ApFindStars | INFO | Median FWHM (over x and y) is
3.92 +/- 0.43 pixels using 50 data points
2024-11-26 21:20:38,868 | ApFindStars | INFO | Initial source list using
FWHM=3.917 pixels, threshold=7.00 x BG stddev, found 311 sources.
2024-11-26 21:20:38,919 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.
2024-11-26 21:20:38,929 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.
2024-11-26 21:20:39,179 | ApMeasureStars | INFO | Source average axis ratio =
1.167 +/- 0.150, overall sources are circular.
2024-11-26 21:20:40,487 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Red-1.png
2024-11-26 21:20:40,489 | ApFindStars | INFO | Median FWHM (over x and y) is
3.91 +/- 0.43 pixels using 50 data points
2024-11-26 21:20:41,390 | ApFindStars | INFO | Image is 3056 cols x 3056 rows
2024-11-26 21:20:41,391 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg
2024-11-26 21:20:41,392 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.
2024-11-26 21:20:41,392 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625
2024-11-26 21:20:41,392 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.
2024-11-26 21:20:41,392 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.

2024-11-26 21:20:41,392 | ApFindStars | WARNING | Keyword AIRMASS not found, not written to quality report.

2024-11-26 21:20:41,394 | ApFindStars | INFO | Wrote image quality report to ./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Red-1.yaml

2024-11-26 21:20:41,397 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:20:41,398 | ApFindStars | INFO | Approximate coordinates: ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:20:41,398 | ApFindStars | INFO | Approximate image field of view is 0.750 degrees across.

2024-11-26 21:20:41,398 | ApFindStars | INFO | Approximate pixel sizes (arcseconds) are x=0.625, y=0.625

2024-11-26 21:20:41,399 | ApFindStars | INFO | Writing source list to FITS binary table ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:20:41,427 | AstroPhotography.core.ApAstrometry | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz

2024-11-26 21:20:41,696 | AstroPhotography.core.ApAstrometry | INFO | Read 200 source positions from ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits

Solving...

2024-11-26 21:20:54,606 | AstroPhotography.core.ApAstrometry | INFO | Obtained a WCS header

2024-11-26 21:20:54,925 | AstroPhotography.core.ApAstrometry | INFO | Wrote updated file with WCS to ./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits

2024-11-26 21:20:54,926 | AstroPhotography.core.ApAstrometry | INFO | Updating photometry in ./SourceLists/srclist-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits with WCS solution.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is greater than 8 characters or contains characters not allowed by the FITS standard; a HIERARCH card will be created.

2024-11-26 21:20:55,250 | ApProcess | INFO | Processing input file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz

2024-11-26 21:20:55,252 | ApFindStars | INFO | Loading extension 0 of FITS file recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz

2024-11-26 21:20:59,797 | ApFindStars | INFO | Initial source list using FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 221 sources.

2024-11-26 21:21:00,767 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:21:00,778 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:21:01,020 | ApMeasureStars | INFO | Source average axis ratio = 1.125 +/- 0.116, overall sources are circular.

2024-11-26 21:21:01,021 | ApFindStars | INFO | Median FWHM (over x and y) is 3.82 +/- 0.38 pixels using 49 data points

2024-11-26 21:21:01,304 | ApFindStars | INFO | Initial source list using FWHM=3.820 pixels, threshold=7.00 x BG stddev, found 259 sources.

2024-11-26 21:21:01,347 | ApMeasureStars | INFO | Measuring the source extent (Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:21:01,357 | ApMeasureStars | INFO | Starting to fit Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:21:01,596 | ApMeasureStars | INFO | Source average axis ratio = 1.123 +/- 0.117, overall sources are circular.

2024-11-26 21:21:02,905 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Red-2.png

2024-11-26 21:21:02,907 | ApFindStars | INFO | Median FWHM (over x and y) is 3.81 +/- 0.38 pixels using 49 data points

2024-11-26 21:21:03,841 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:21:03,842 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:21:03,843 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.

2024-11-26 21:21:03,843 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625

2024-11-26 21:21:03,843 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.

2024-11-26 21:21:03,843 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.

2024-11-26 21:21:03,843 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.

2024-11-26 21:21:03,845 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Red-2.yaml

2024-11-26 21:21:03,848 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:21:03,849 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:21:03,849 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.

2024-11-26 21:21:03,849 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625

2024-11-26 21:21:03,849 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:21:03,877 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz

2024-11-26 21:21:04,144 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits

Solving...

2024-11-26 21:21:23,077 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header

2024-11-26 21:21:23,403 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to ./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits

2024-11-26 21:21:23,403 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits with WCS solution.

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.

2024-11-26 21:21:23,732 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

2024-11-26 21:21:23,733 | ApFindStars | INFO | Loading extension 0 of FITS file
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

2024-11-26 21:21:28,077 | ApFindStars | INFO | Initial source list using
FWHM=3.000 pixels, threshold=7.00 x BG stddev, found 316 sources.

2024-11-26 21:21:29,013 | ApMeasureStars | INFO | Measuring the source extent
(Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:21:29,023 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:21:29,263 | ApMeasureStars | INFO | Source average axis ratio =
1.026 +/- 0.088, overall sources are circular.

2024-11-26 21:21:29,264 | ApFindStars | INFO | Median FWHM (over x and y) is
3.38 +/- 0.21 pixels using 50 data points

2024-11-26 21:21:29,495 | ApFindStars | INFO | Initial source list using
FWHM=3.375 pixels, threshold=7.00 x BG stddev, found 316 sources.

2024-11-26 21:21:29,547 | ApMeasureStars | INFO | Measuring the source extent

(Gaussian FWHM) from input list of 200 sources.

2024-11-26 21:21:29,557 | ApMeasureStars | INFO | Starting to fit
Const2D+Gaussian2D model to 25 star cutouts.

2024-11-26 21:21:29,798 | ApMeasureStars | INFO | Source average axis ratio =
1.053 +/- 0.081, overall sources are circular.

2024-11-26 21:21:31,489 | ApMeasureStars | INFO | Plotted star FWHM fit cut-outs
to ./SourceLists/fwhmplot-iTelescope-M101_Supernova_2023ixf-300s-Red-3.png

2024-11-26 21:21:31,490 | ApFindStars | INFO | Median FWHM (over x and y) is
3.33 +/- 0.21 pixels using 50 data points

2024-11-26 21:21:32,373 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:21:32,374 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:21:32,375 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.

2024-11-26 21:21:32,375 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625

2024-11-26 21:21:32,375 | ApFindStars | WARNING | Keyword DATE-OBS not found,
not written to quality report.

2024-11-26 21:21:32,375 | ApFindStars | WARNING | Keyword CCD-TEMP not found,
not written to quality report.

2024-11-26 21:21:32,376 | ApFindStars | WARNING | Keyword AIRMASS not found, not
written to quality report.

2024-11-26 21:21:32,377 | ApFindStars | INFO | Wrote image quality report to
./MetaData/qual-iTelescope-M101_Supernova_2023ixf-300s-Red-3.yaml

2024-11-26 21:21:32,381 | ApFindStars | INFO | Image is 3056 cols x 3056 rows

2024-11-26 21:21:32,381 | ApFindStars | INFO | Approximate coordinates:
ra=14.053444 hours, dec=54.349444 deg

2024-11-26 21:21:32,382 | ApFindStars | INFO | Approximate image field of view
is 0.750 degrees across.

2024-11-26 21:21:32,382 | ApFindStars | INFO | Approximate pixel sizes
(arcseconds) are x=0.625, y=0.625

2024-11-26 21:21:32,382 | ApFindStars | INFO | Writing source list to FITS
binary table ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits

WARNING: Attribute `version` of type <class 'dict'> cannot be added to FITS
Header - skipping [astropy.io.fits.convenience]

WARNING:astroquery:Attribute `version` of type <class 'dict'> cannot be added to
FITS Header - skipping

WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.

WARNING: VerifyWarning: Card is too long, comment will be truncated.
[astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Card is too long, comment will be truncated.

2024-11-26 21:21:32,410 | AstroPhotography.core.ApAstrometry | INFO | Loading
extension 0 of FITS file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz

2024-11-26 21:21:32,680 | AstroPhotography.core.ApAstrometry | INFO | Read 200
source positions from ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits

Solving...

2024-11-26 21:21:46,090 | AstroPhotography.core.ApAstrometry | INFO | Obtained a
WCS header

2024-11-26 21:21:46,412 | AstroPhotography.core.ApAstrometry | INFO | Wrote
updated file with WCS to ./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits

2024-11-26 21:21:46,413 | AstroPhotography.core.ApAstrometry | INFO | Updating
photometry in ./SourceLists/srcList-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits with WCS solution.

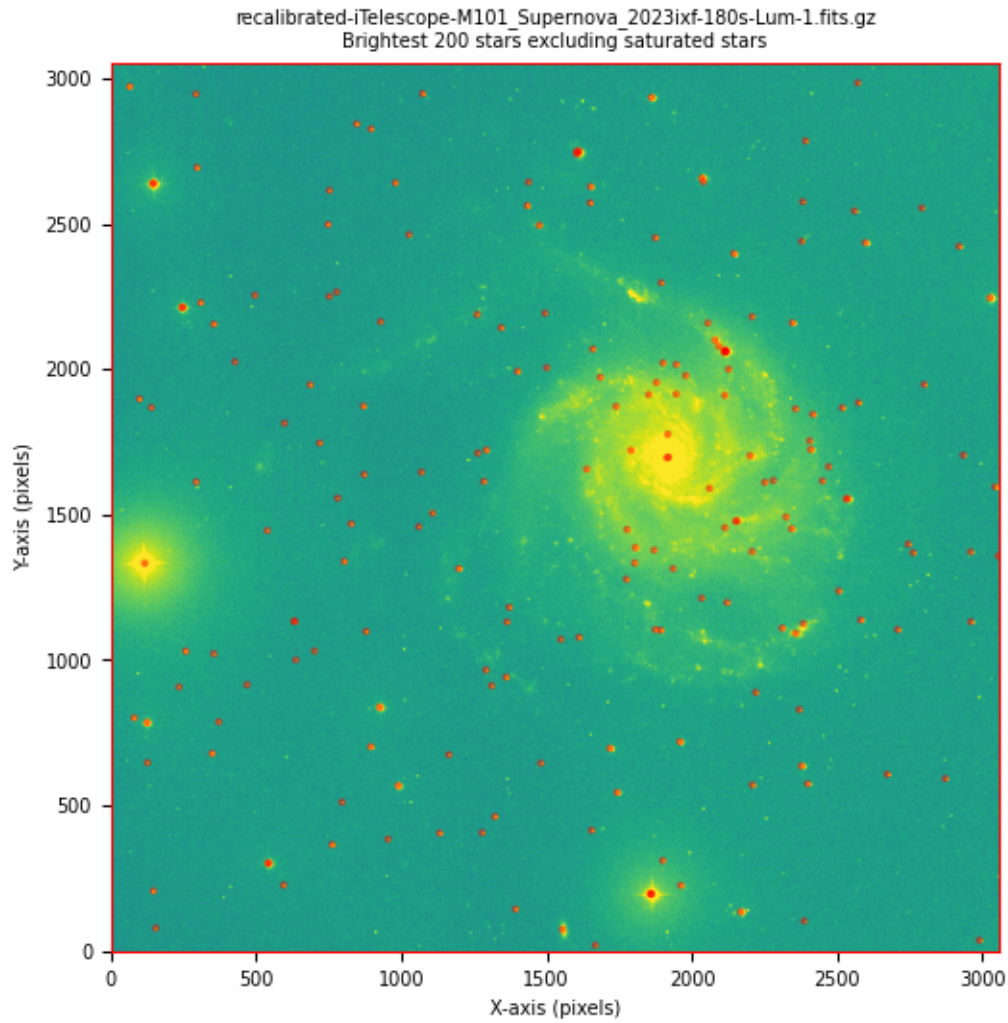
WARNING: VerifyWarning: Keyword name 'aperture_photometry_args' is greater than
8 characters or contains characters not allowed by the FITS standard; a HIERARCH
card will be created. [astropy.io.fits.card]

WARNING:astroquery:VerifyWarning: Keyword name 'aperture_photometry_args' is
greater than 8 characters or contains characters not allowed by the FITS
standard; a HIERARCH card will be created.

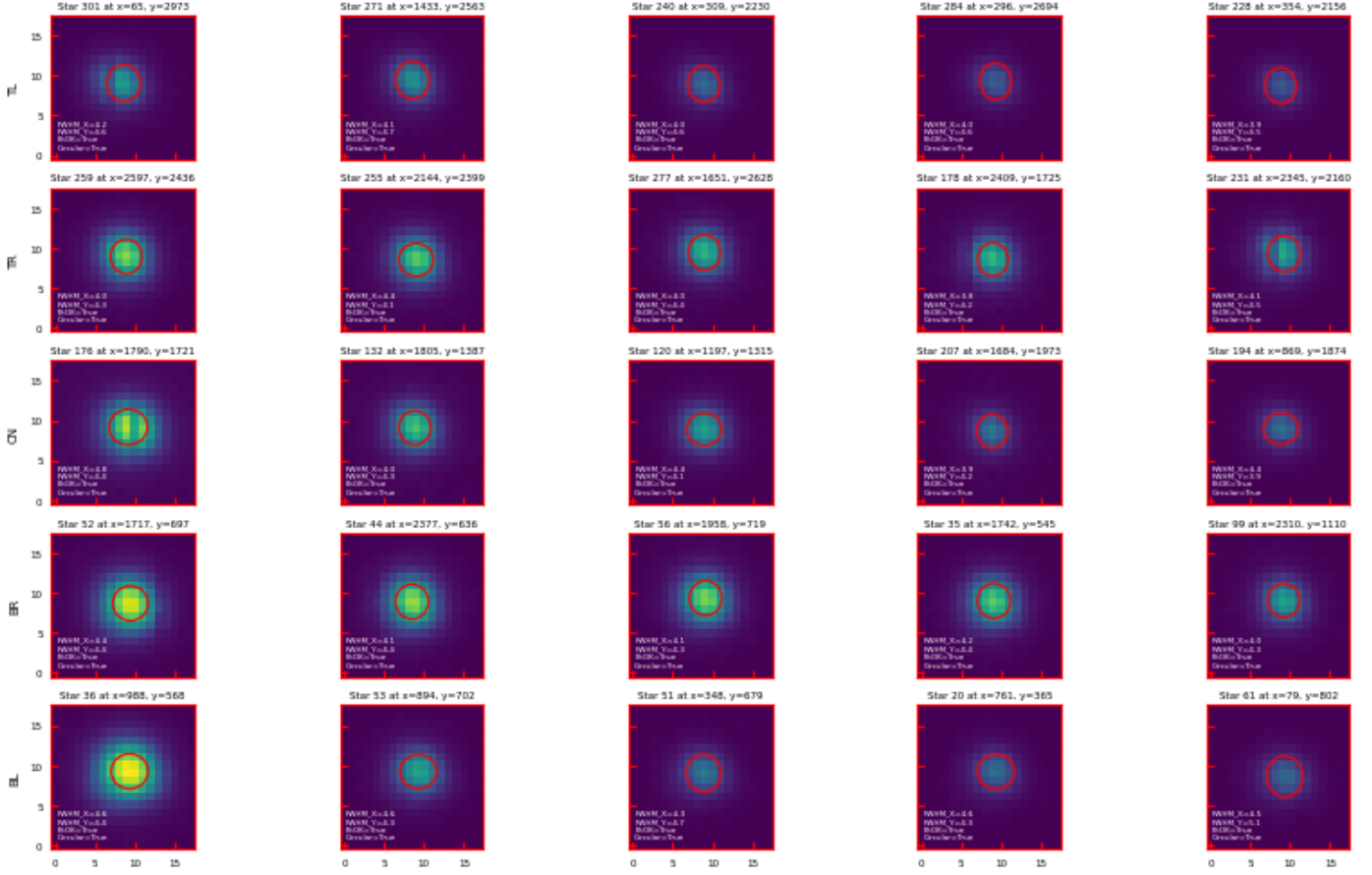
2024-11-26 21:21:46,433 | ApQualitySummarizer | INFO | Searching for file
matching pattern "qual*.yaml" in ./MetaData/

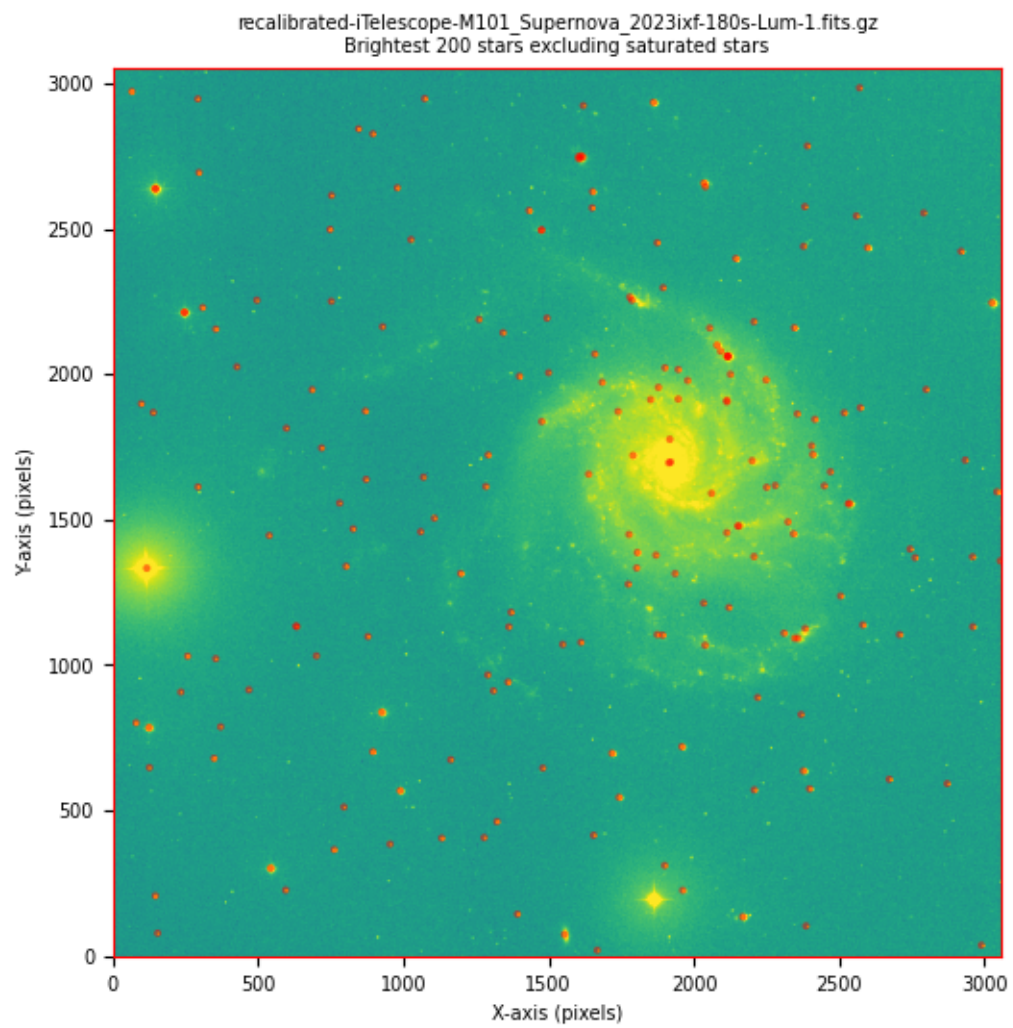
2024-11-26 21:21:46,476 | ApQualitySummarizer | INFO | There are 8 unique
target/telescope/filter groupings: unknown_target:unknown_telescope:Blue,
unknown_target:unknown_telescope:Green, unknown_target:unknown_telescope:Red,
M101_Supernova_2023ixf:iTelescope_24:Green,
M101_Supernova_2023ixf:iTelescope_24:Red, unknown_target:unknown_telescope:Lum,

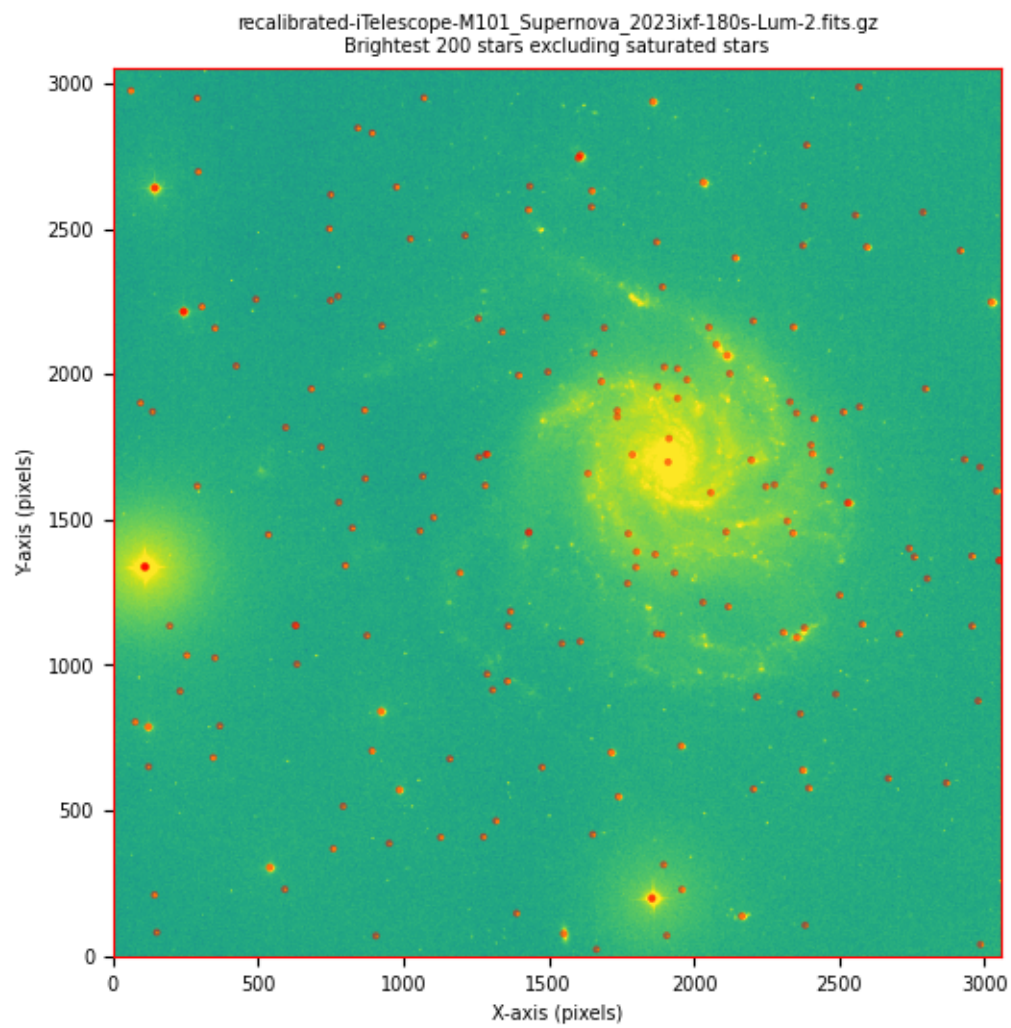
M101_Supernova_2023ixf:iTelescope_24:Lum,
M101_Supernova_2023ixf:iTelescope_24:Blue
2024-11-26 21:21:46,480 | ApQualitySummarizer | INFO | Wrote quality summary to
quality_summary_M101SN.csv
2024-11-26 21:21:46,480 | ApProcess | INFO | Finished processing 12 files in
284.240 seconds.



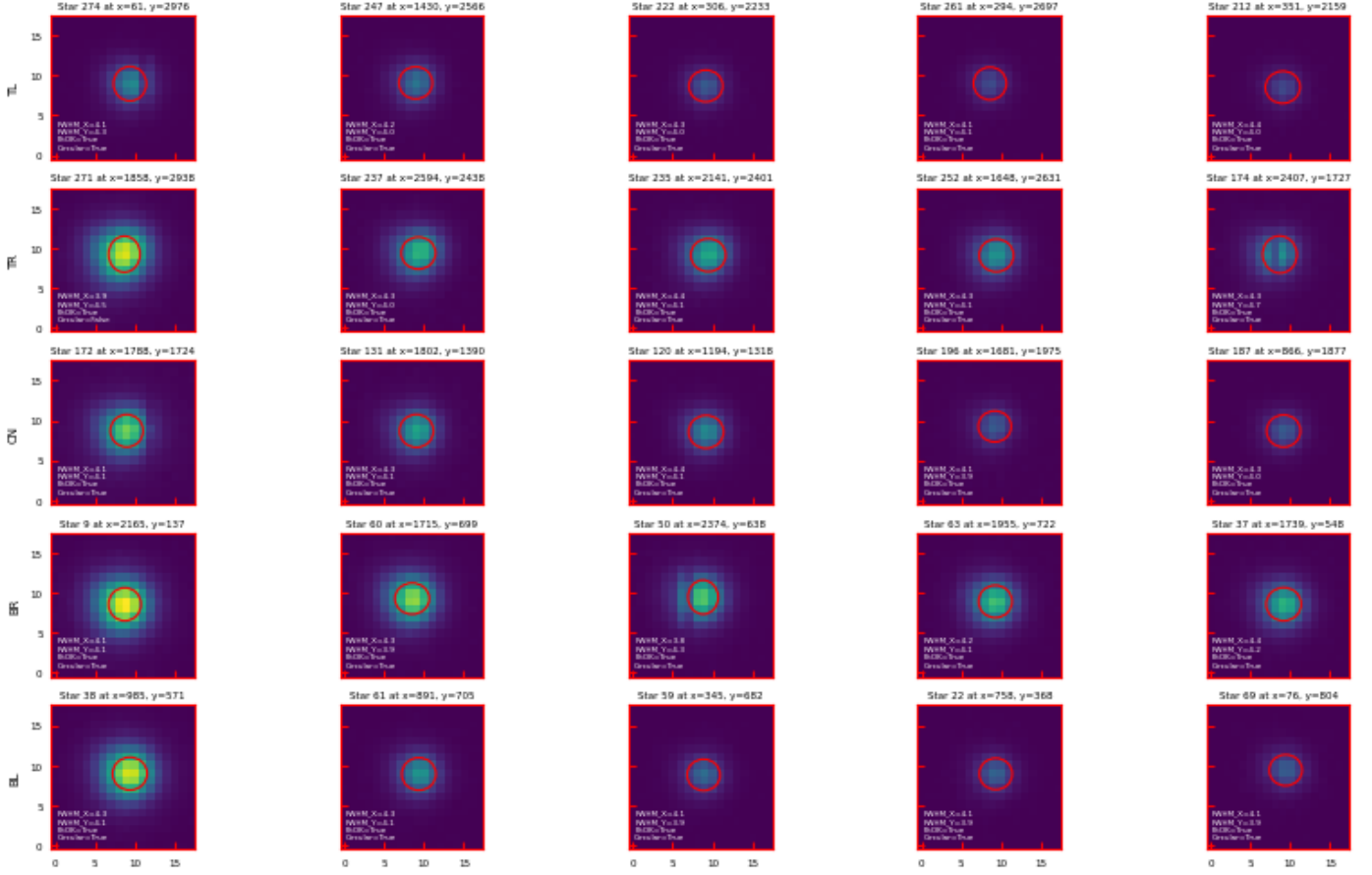
Star FWHM measurements for:
 recalibrated-telescope-M101_Supernova_20231xf-180s-Lum-1.fits.gz
 Median FWHM=4.35 +/- 0.31 (1σ stddev) pixels
 Along X FWHM=4.15 +/- 0.29 pixels, along Y FWHM=4.36 +/- 0.19 pixels

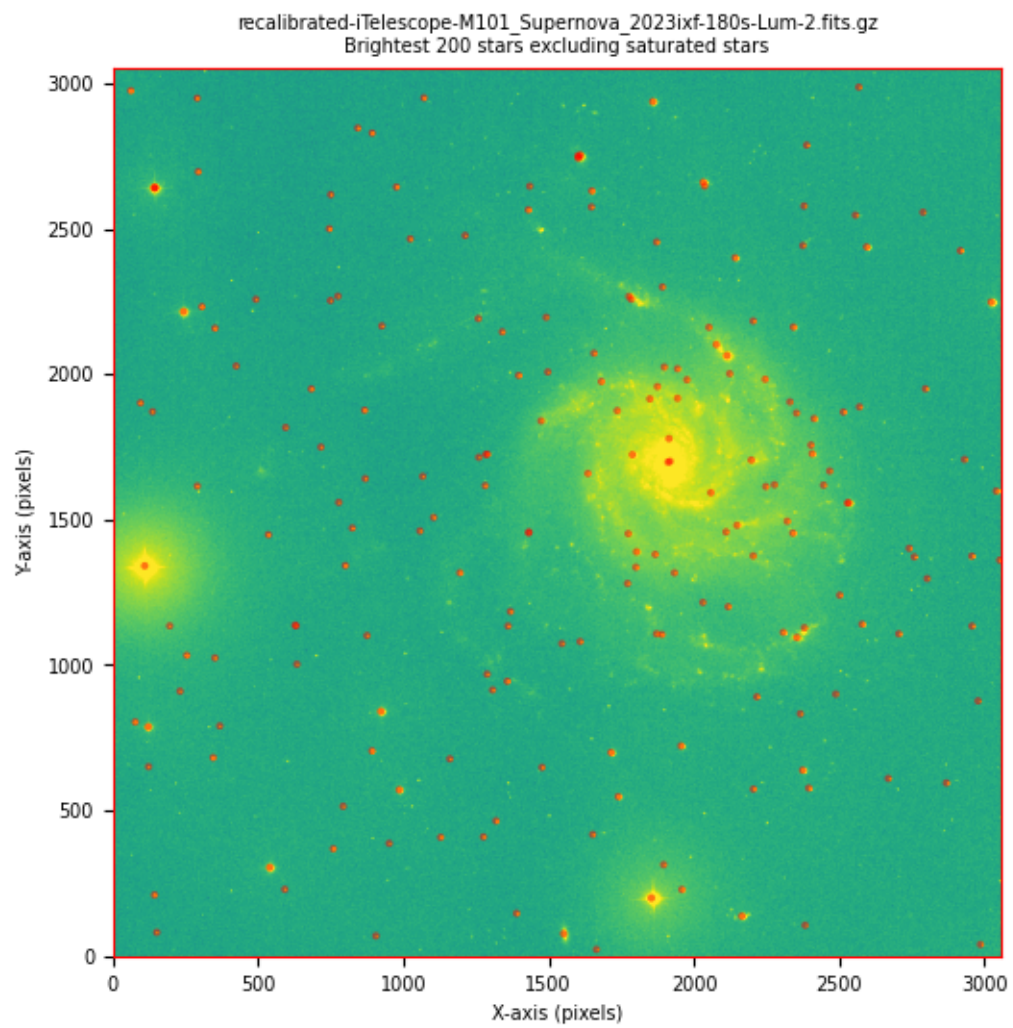


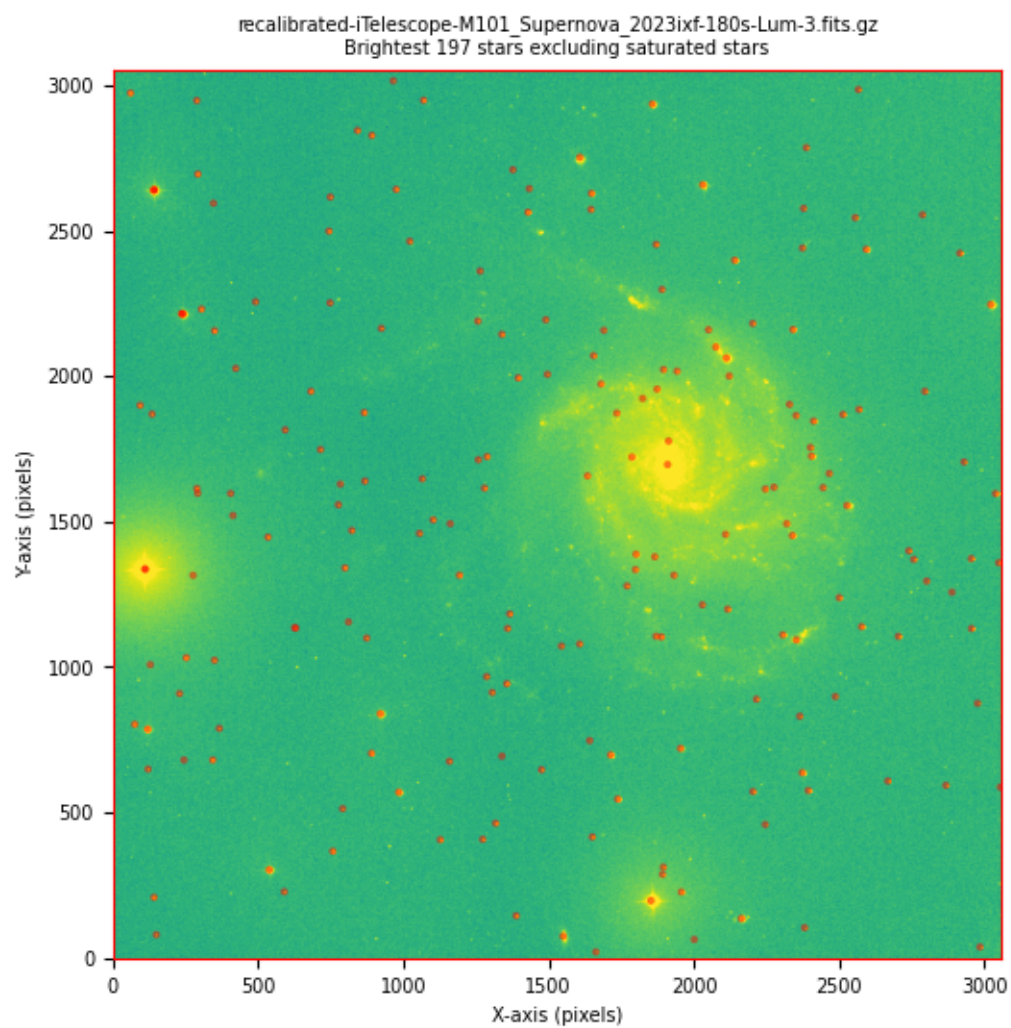




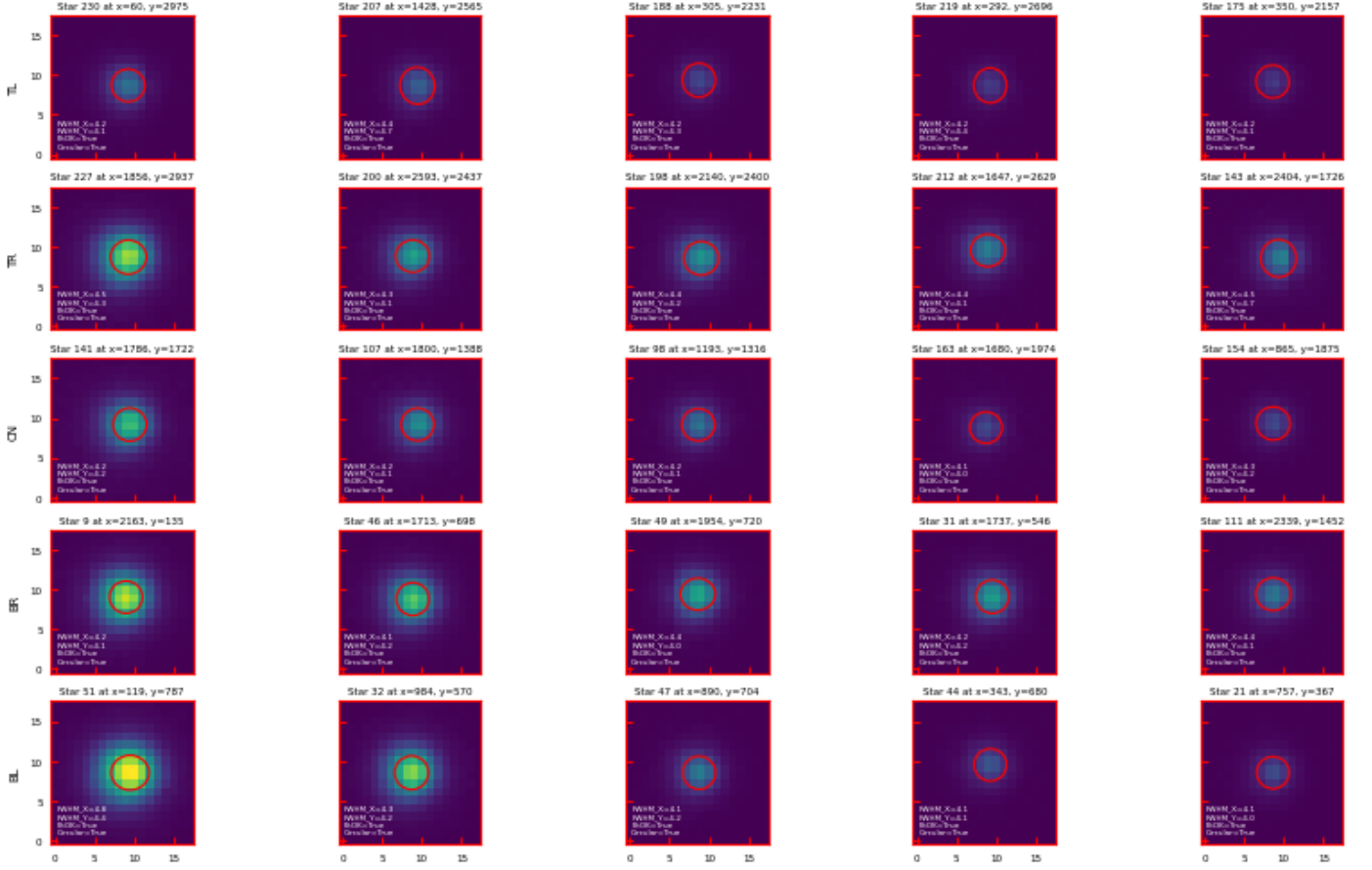
Star FWHM measurements for:
 recalibrated-telescope-M101_Supernova_2023lxr-180s-Lum-2.fits.gz
 Median FWHM=4.14 +/- 0.18 (MAD stddev) pixels
 Along X FWHM=4.26 +/- 0.17 pixels, along Y FWHM=4.07 +/- 0.08 pixels

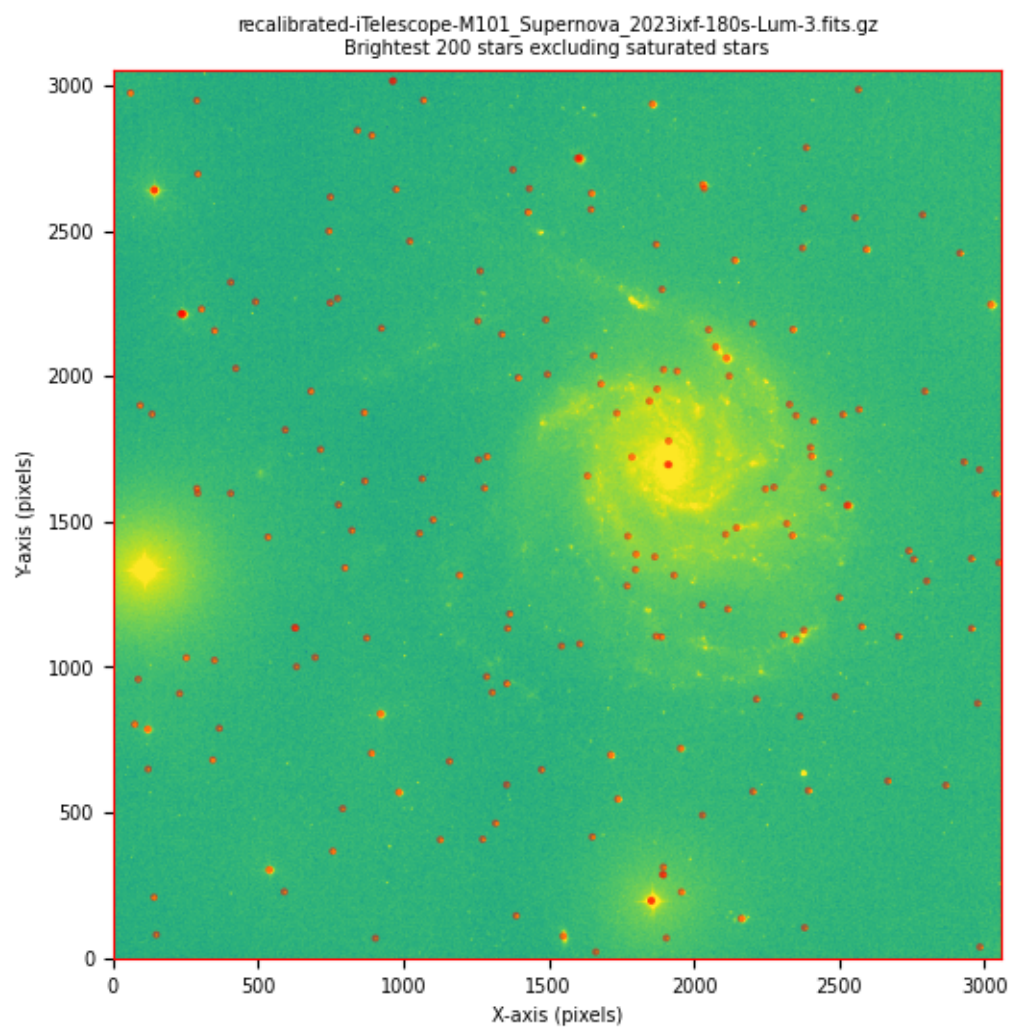


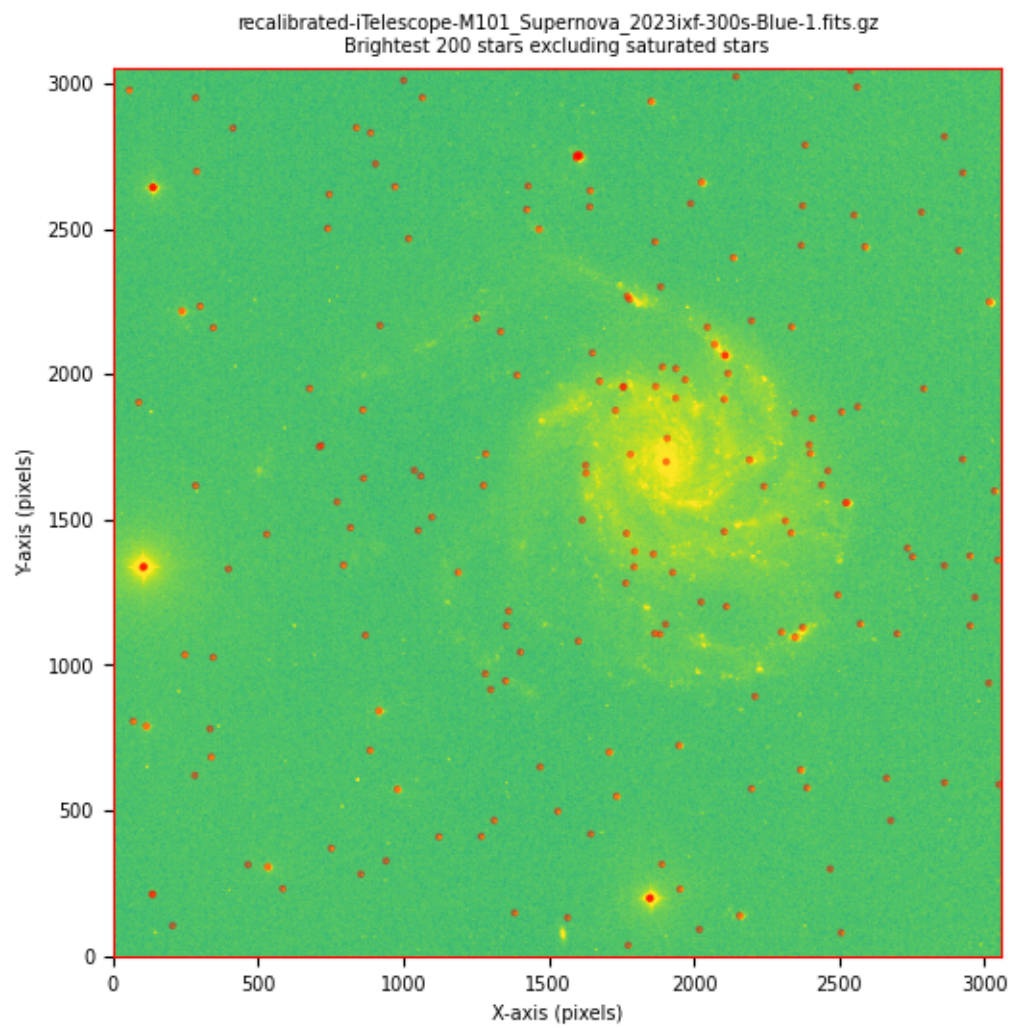




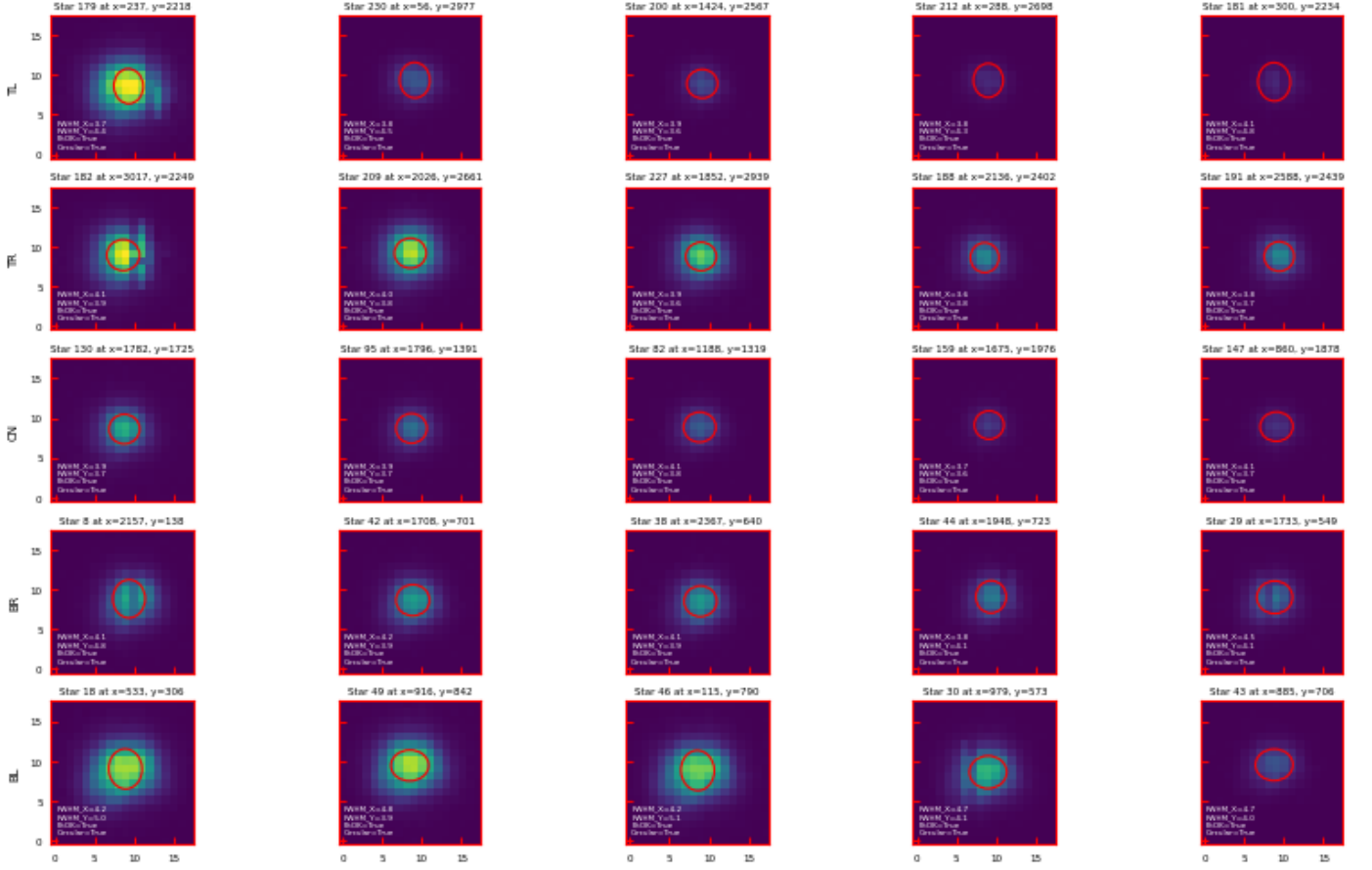
Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_20231x180s-Lum-3.fits.gz
 Median FWHM=4.18 +/- 0.10 (MAD stddev) pixels
 Along X FWHM=4.22 +/- 0.12 pixels, along Y FWHM=4.14 +/- 0.11 pixels

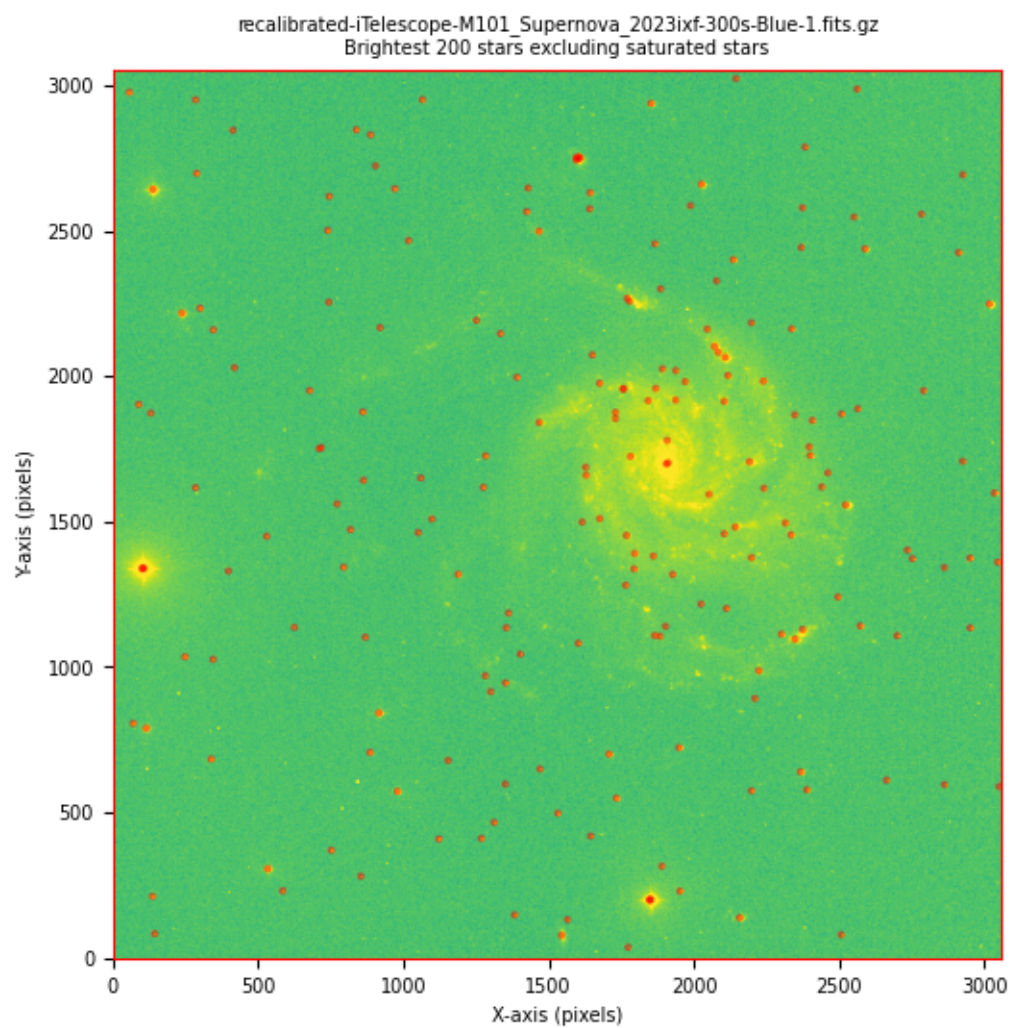


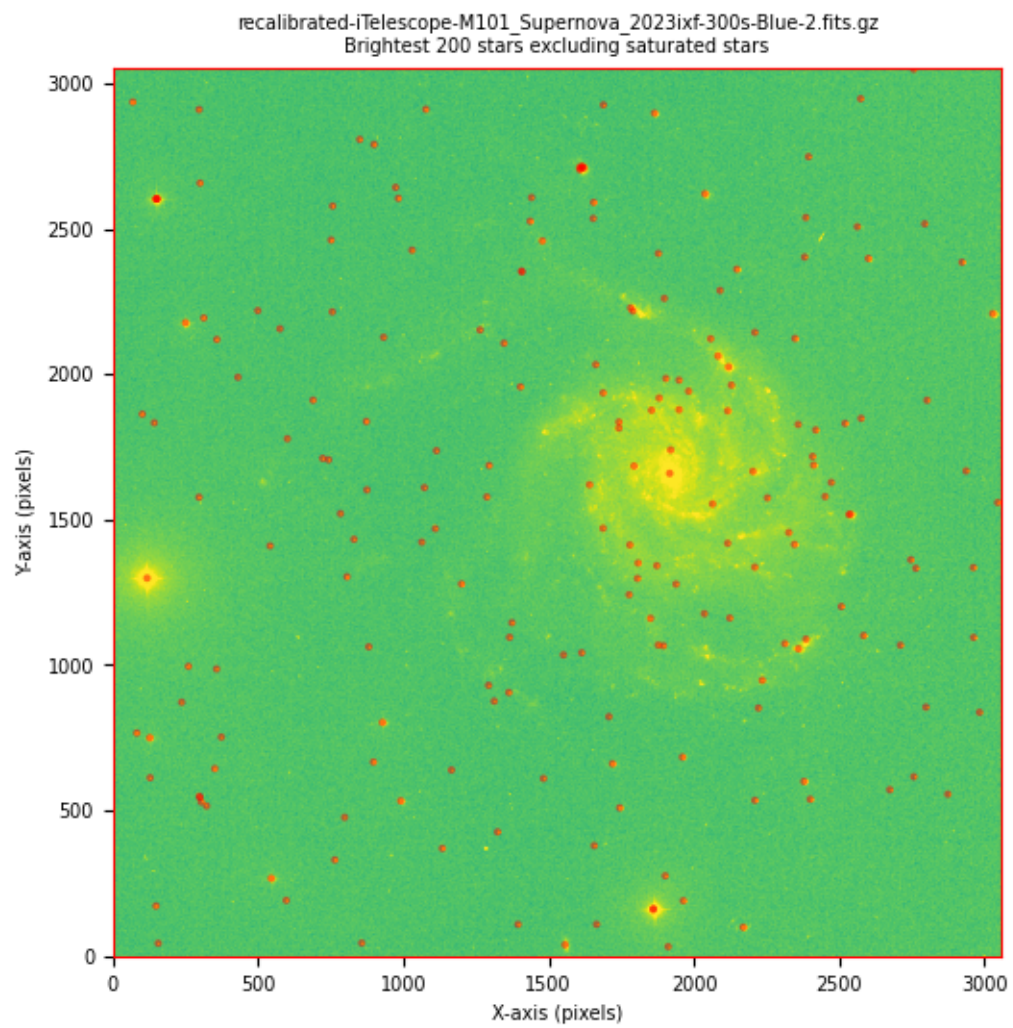




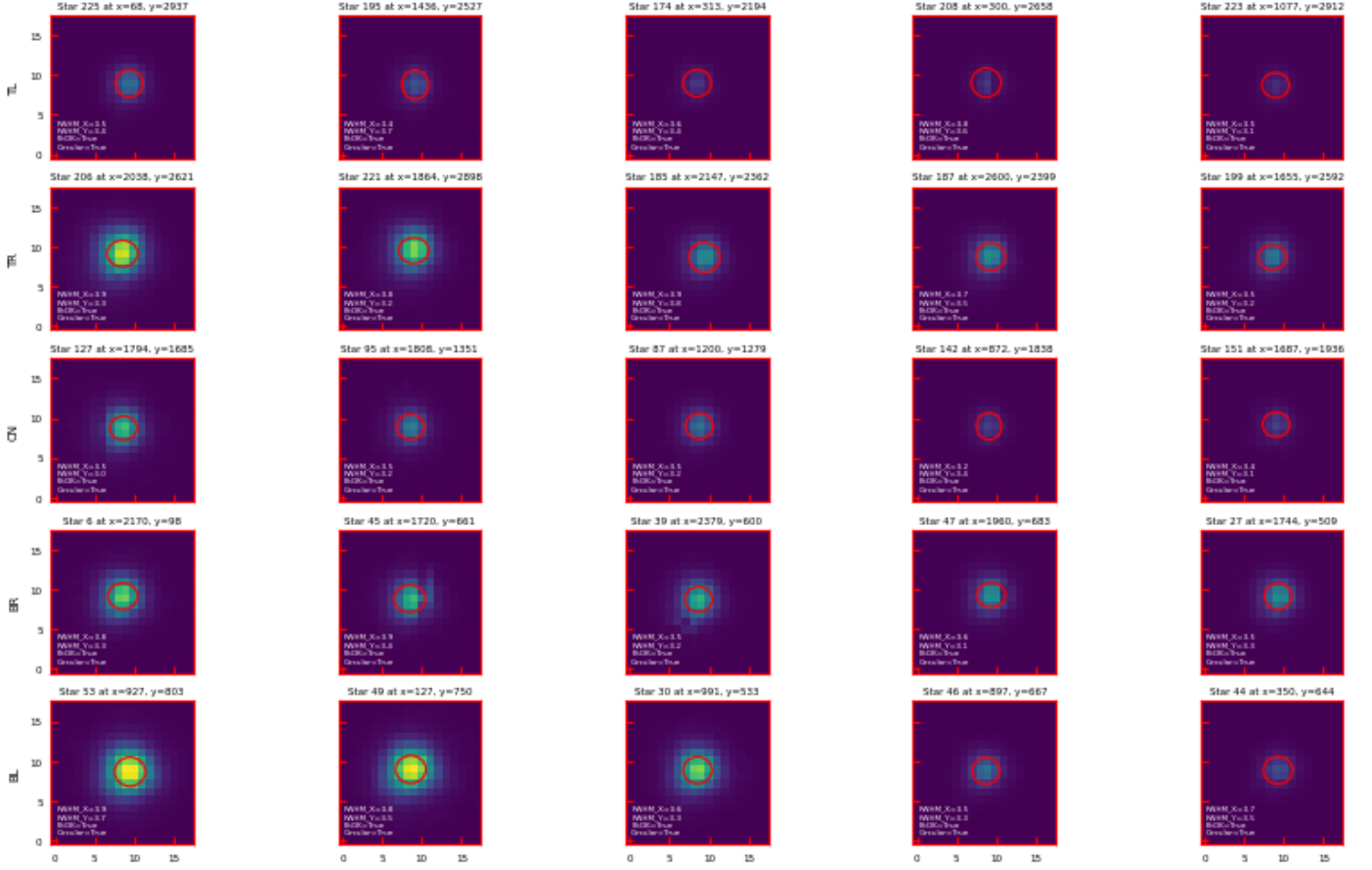
Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_20231x300s-Blue-1.fits.gz
 Median FWHM=3.95 +/- 0.30 (MAD stddev) pixels
 Along X FWHM=4.08 +/- 0.30 pixels, along Y FWHM=3.90 +/- 0.30 pixels

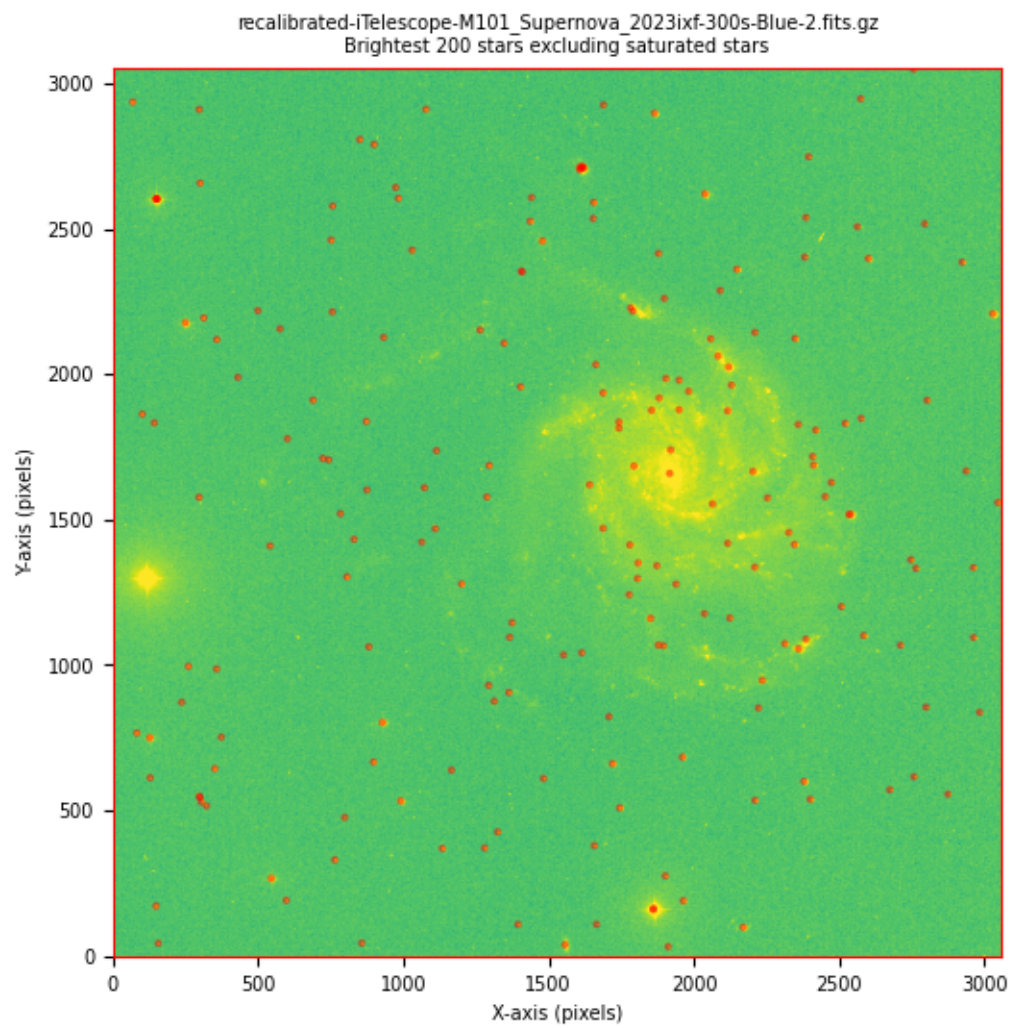


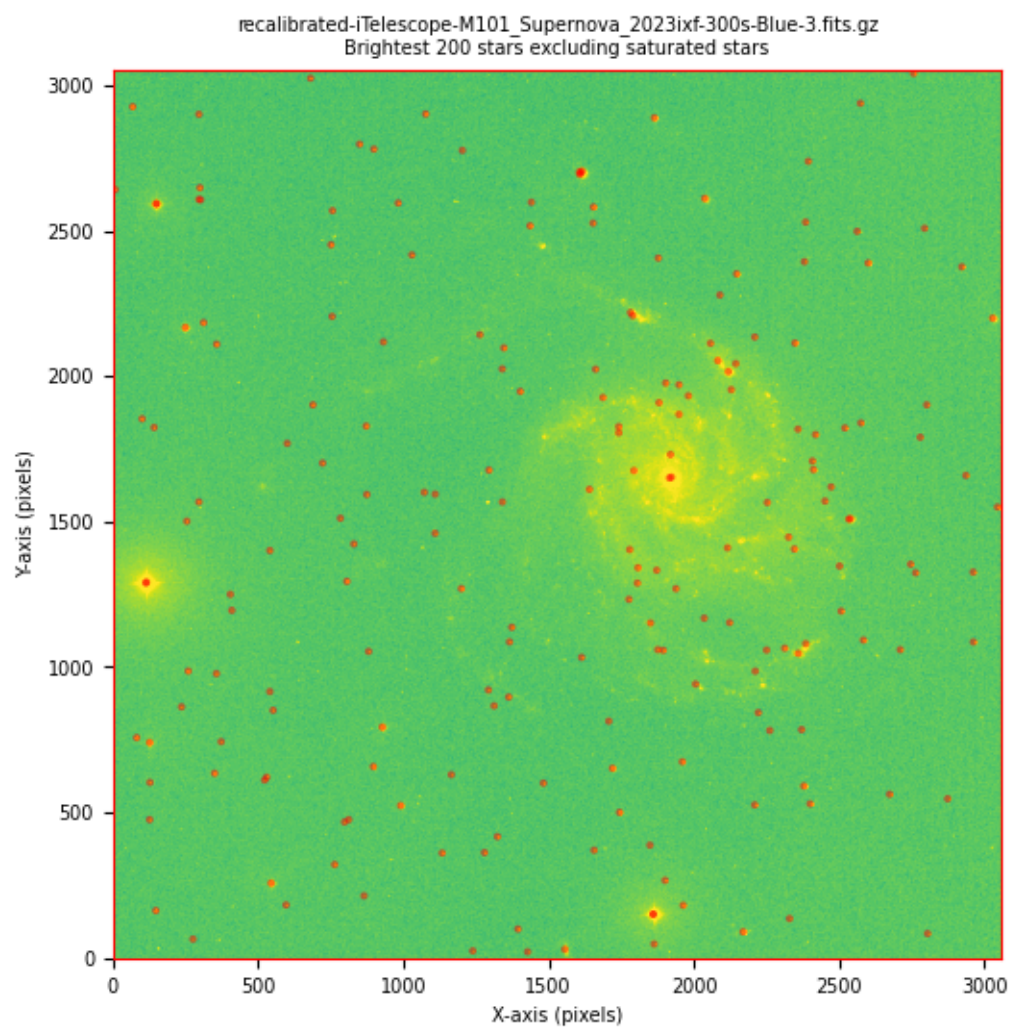




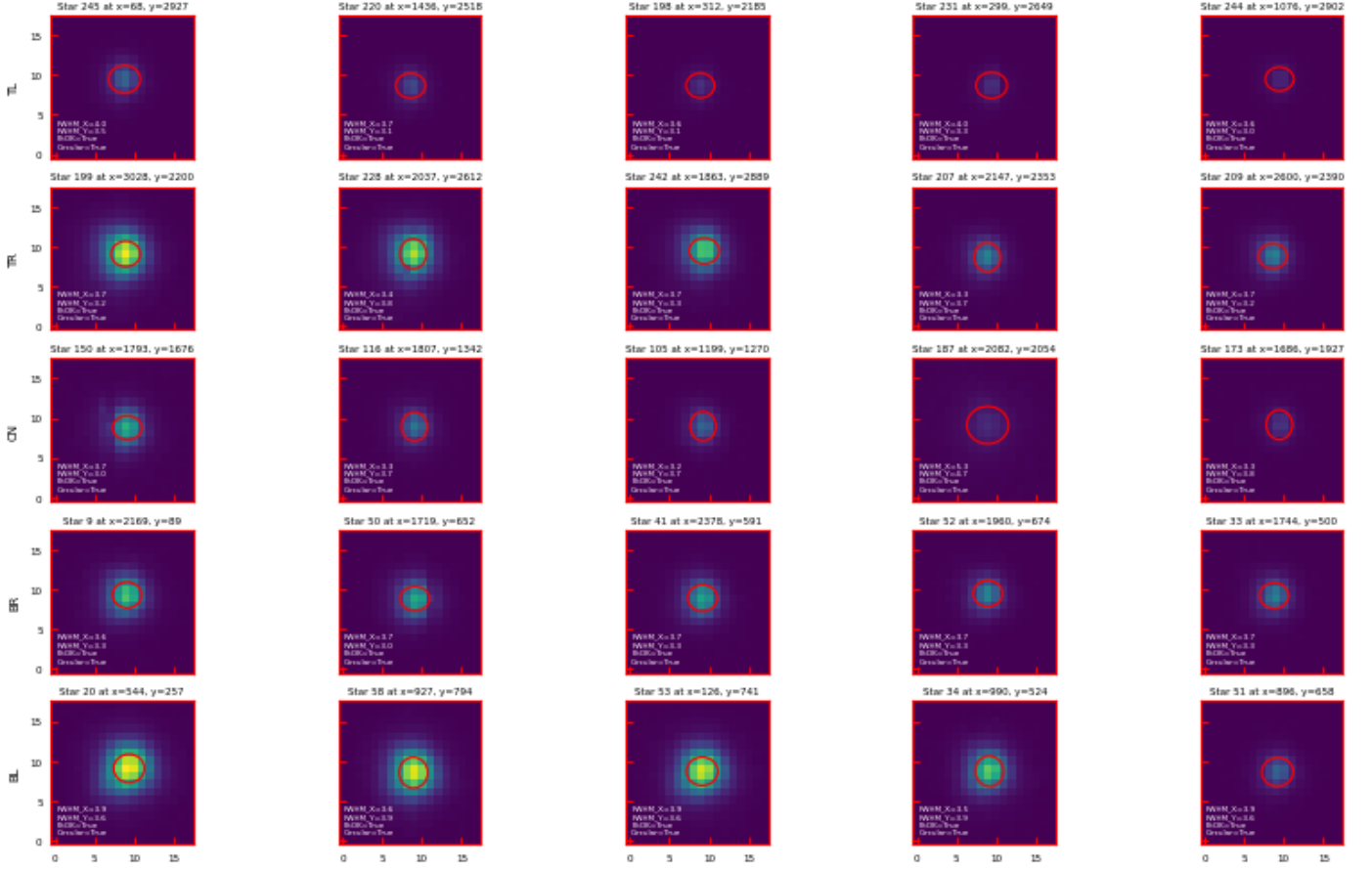
Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_20231x300s-Blue-2.fits.gz
 Median FWHM=3.47 +/- 0.27 (MAD stddev) pixels
 Along X FWHM=3.56 +/- 0.18 pixels, along Y FWHM=3.32 +/- 0.21 pixels

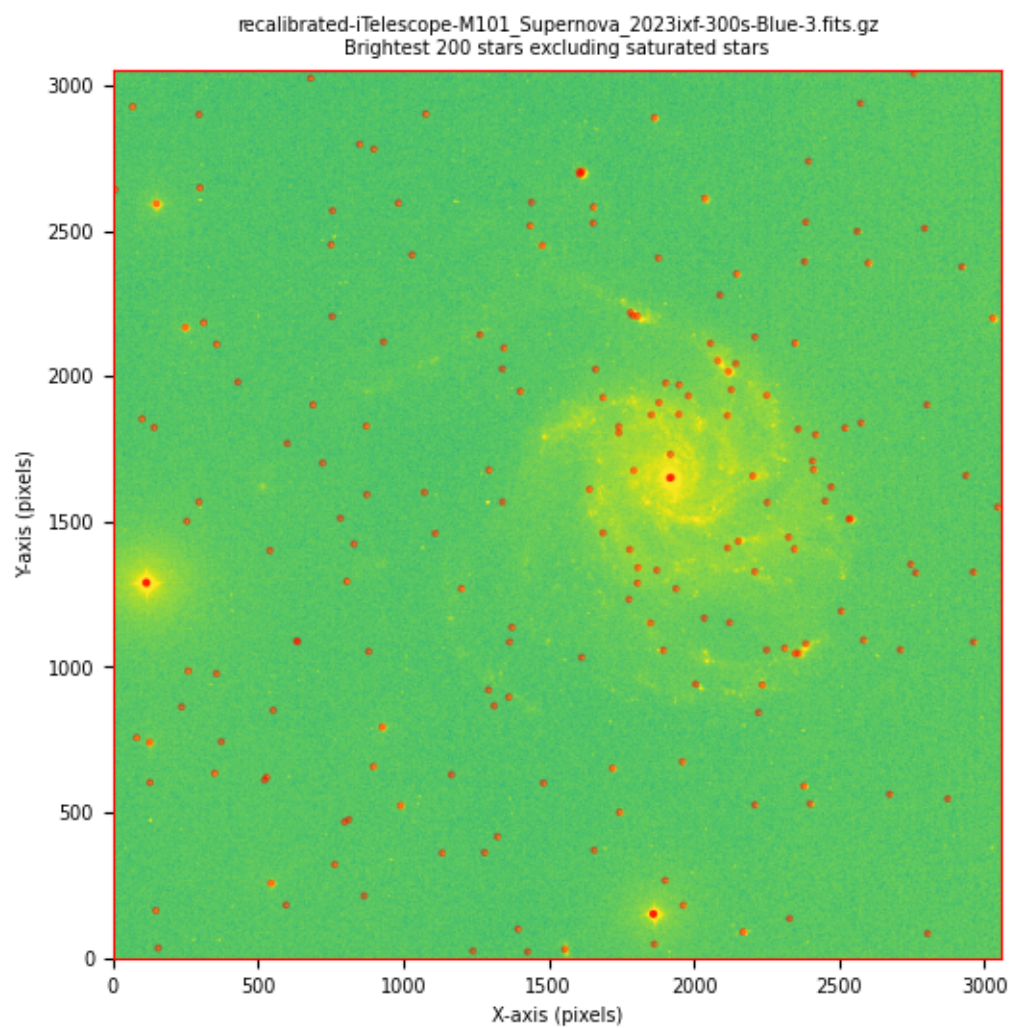


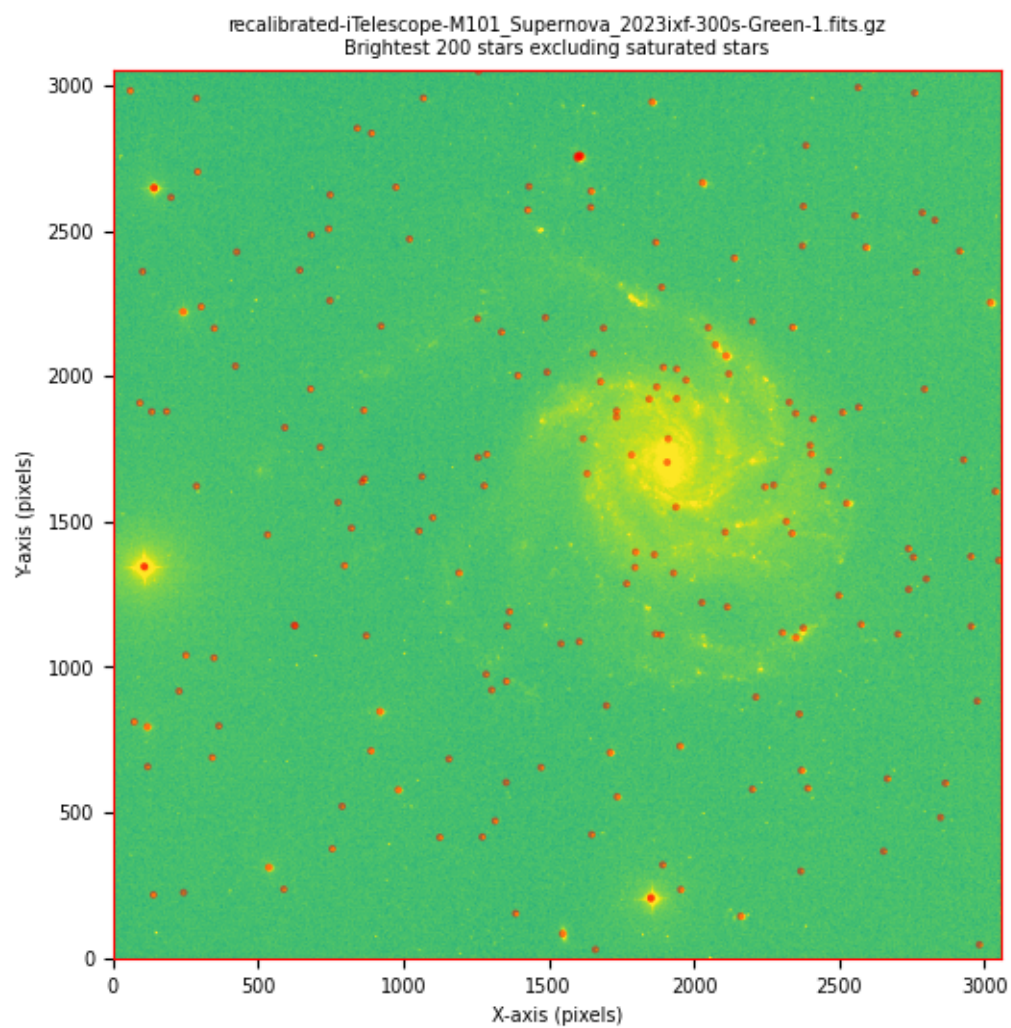




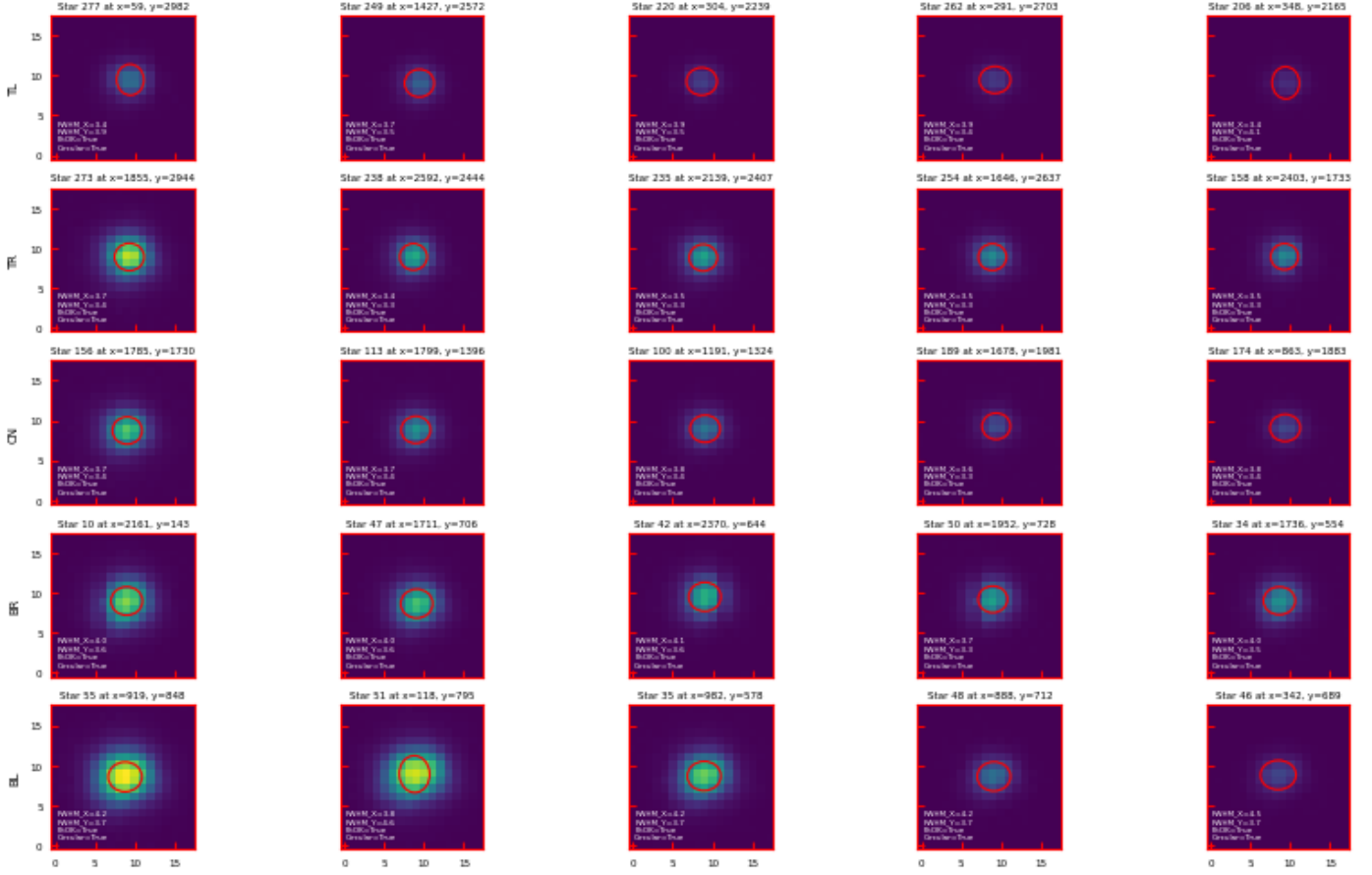
Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_20231xf-300s-Blue-3.fits.gz
 Median FWHM=3.60 +/- 0.36 (MAD stddev) pixels
 Along X FWHM=3.67 +/- 0.14 pixels, along Y FWHM=3.32 +/- 0.39 pixels

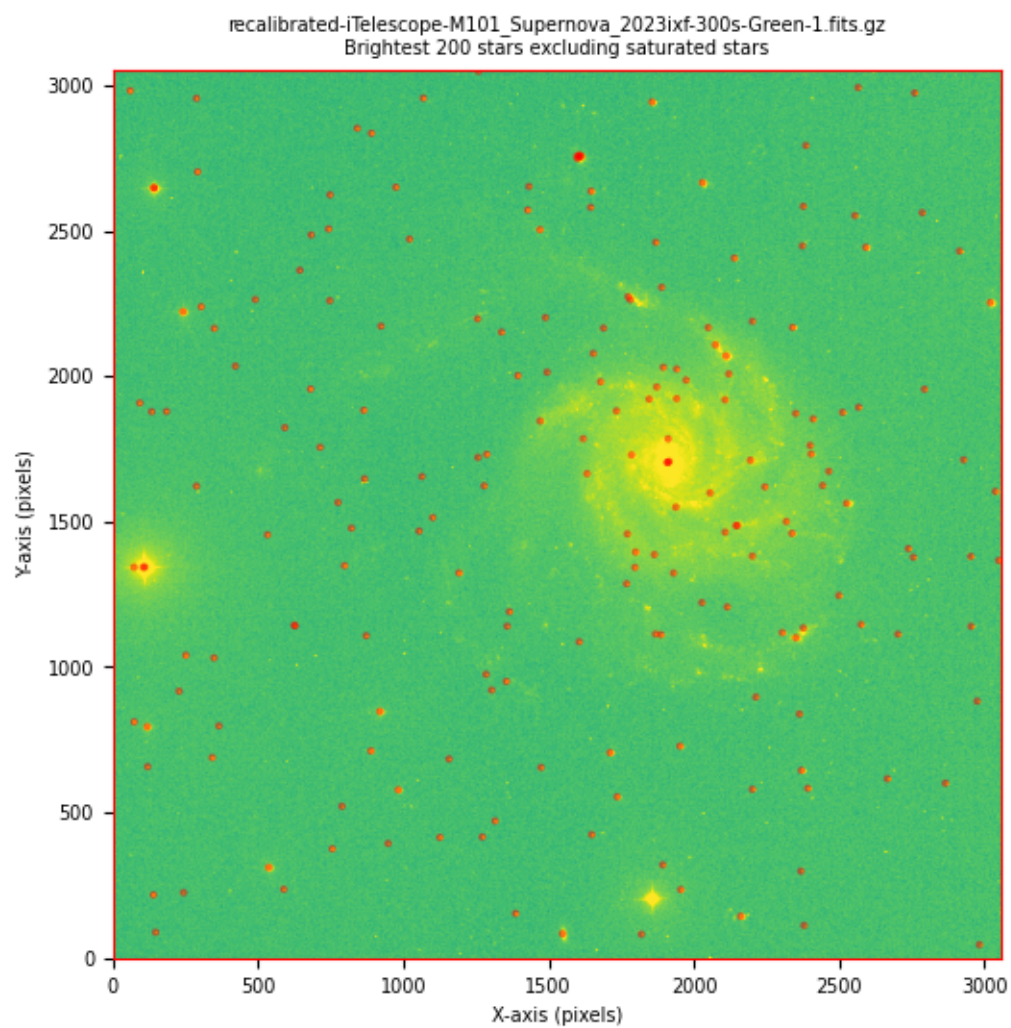


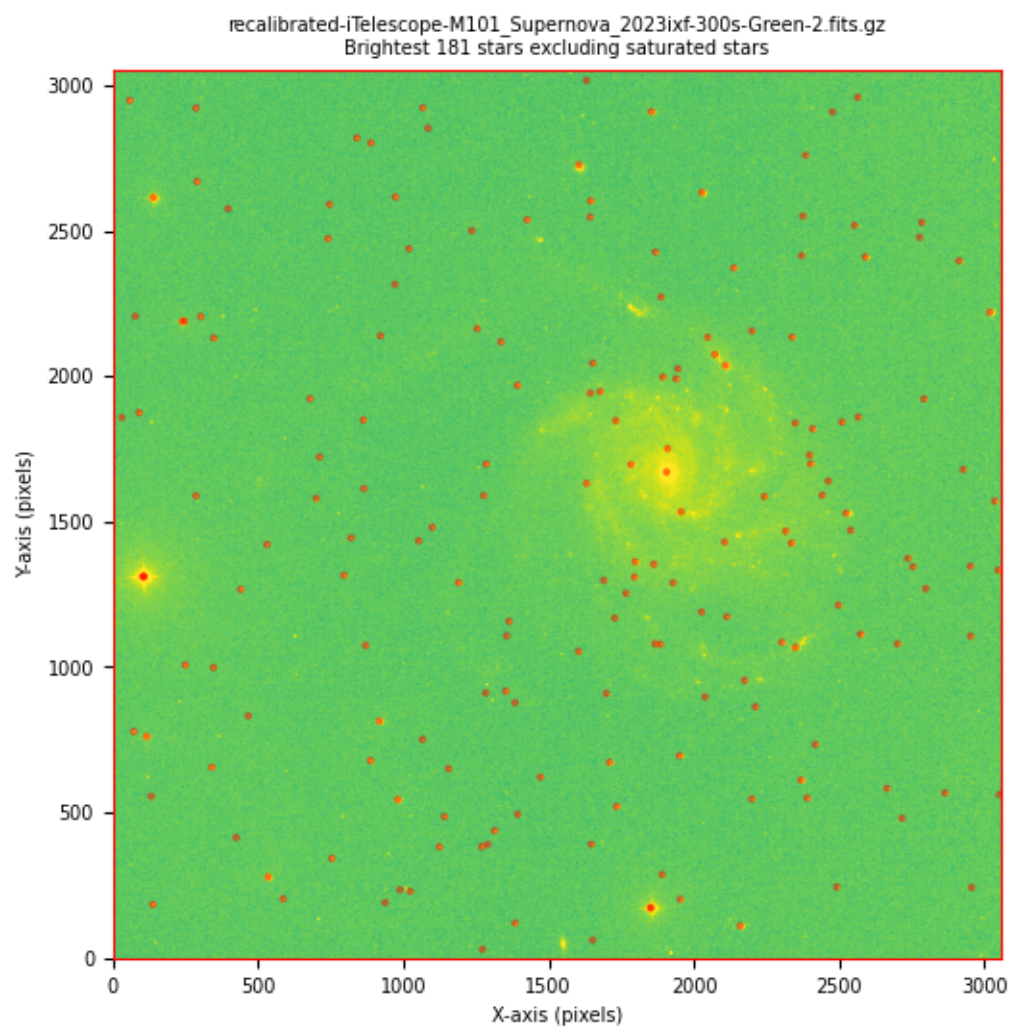




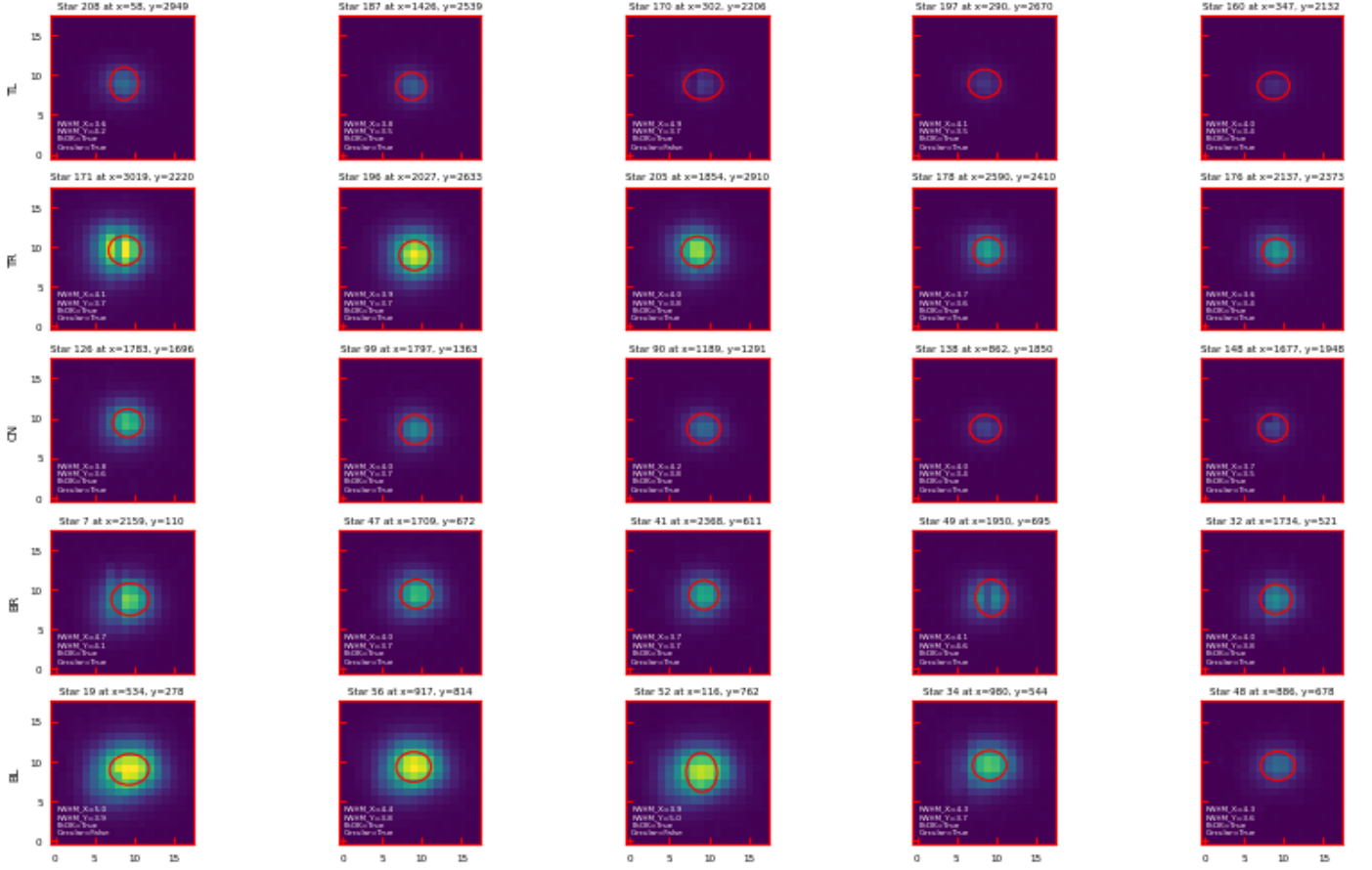
Star FWHM measurements for:
 recalibrated-MTelescope-M101_Supernova_2023xf-300s-Green-1.fits.gz
 Median FWHM=3.63 +/- 0.30 (MAD stddev) pixels
 Along X FWHM=3.80 +/- 0.33 pixels, along Y FWHM=3.43 +/- 0.15 pixels

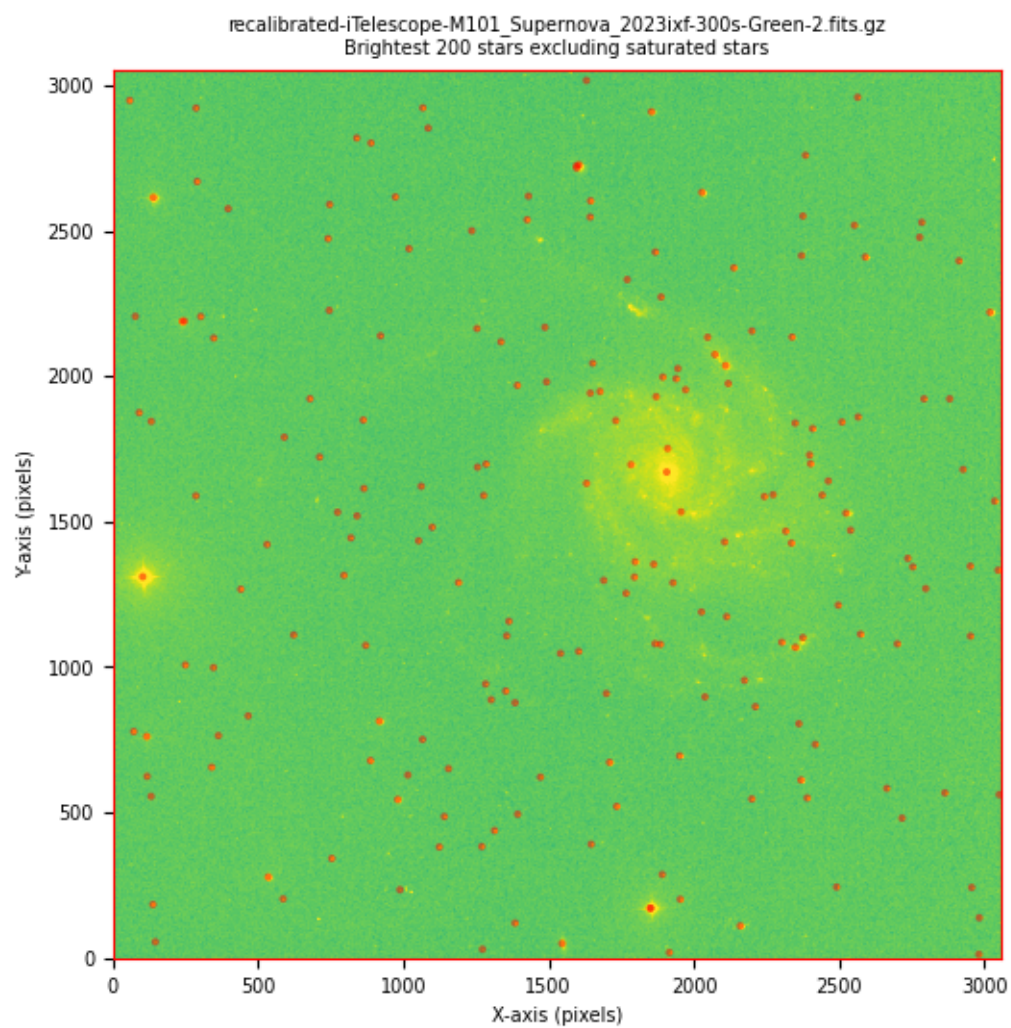


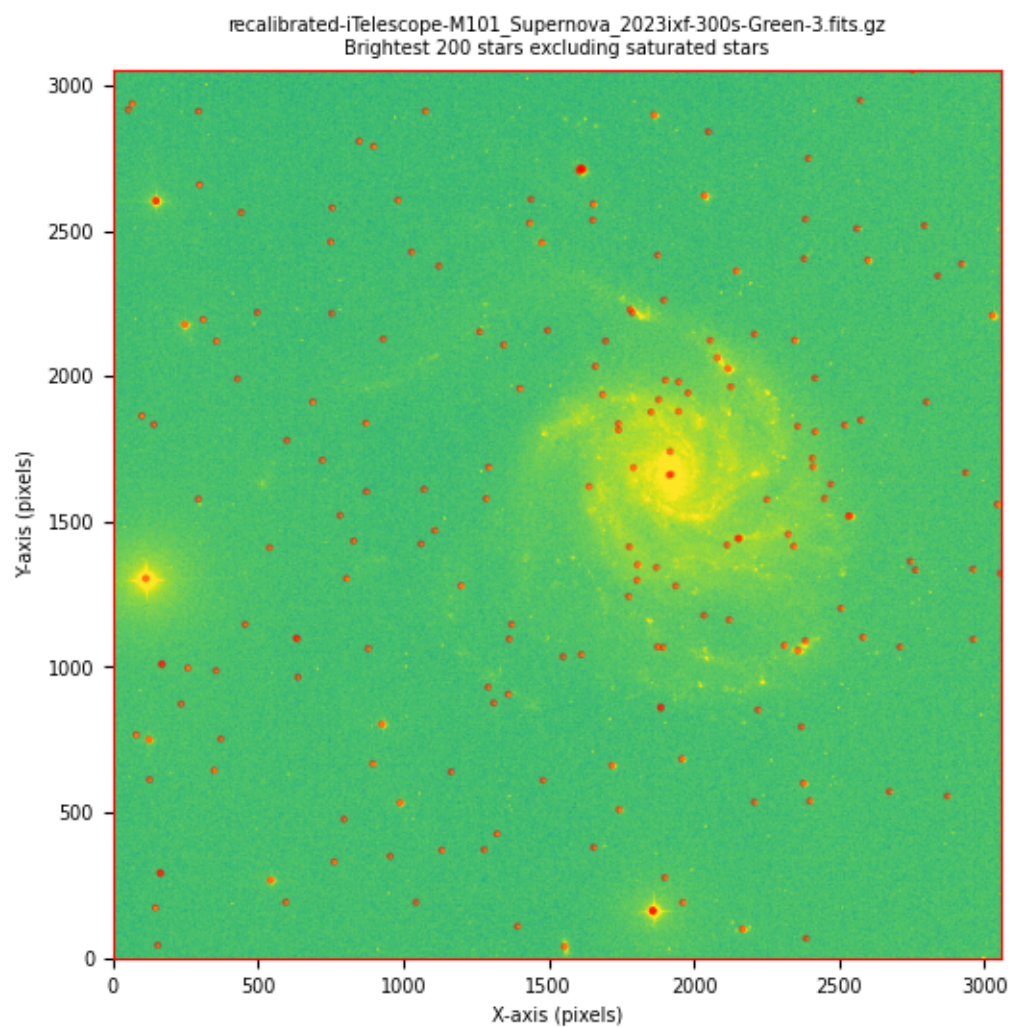




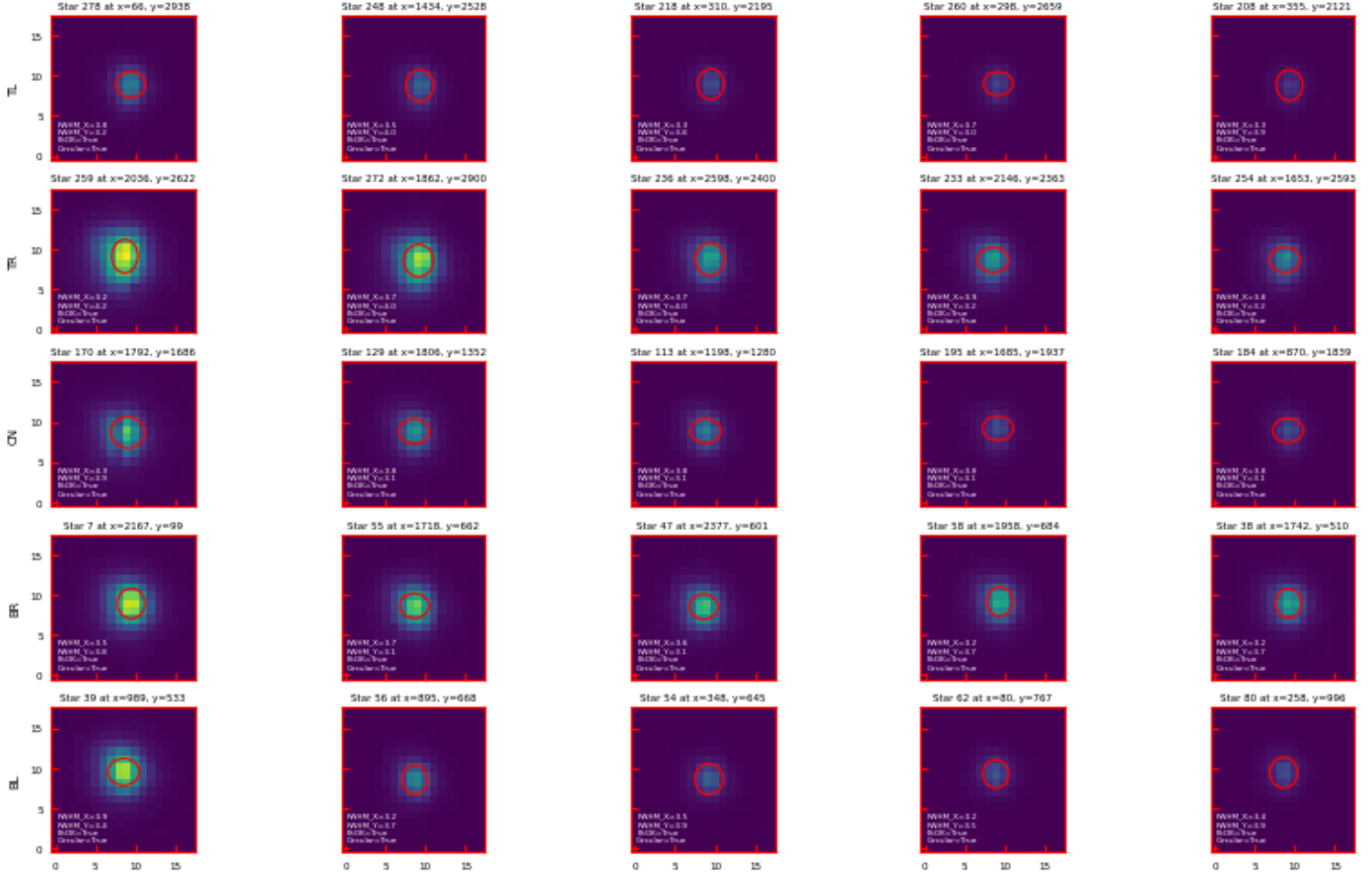
Star FWHM measurements for:
 recalibrated-M101_Supernova_2023xf-300s-Green-2.fits.gz
 Median FWHM=3.77 +/- 0.27 (MAD stddev) pixels
 Along X FWHM=4.02 +/- 0.32 pixels, along Y FWHM=3.76 +/- 0.13 pixels

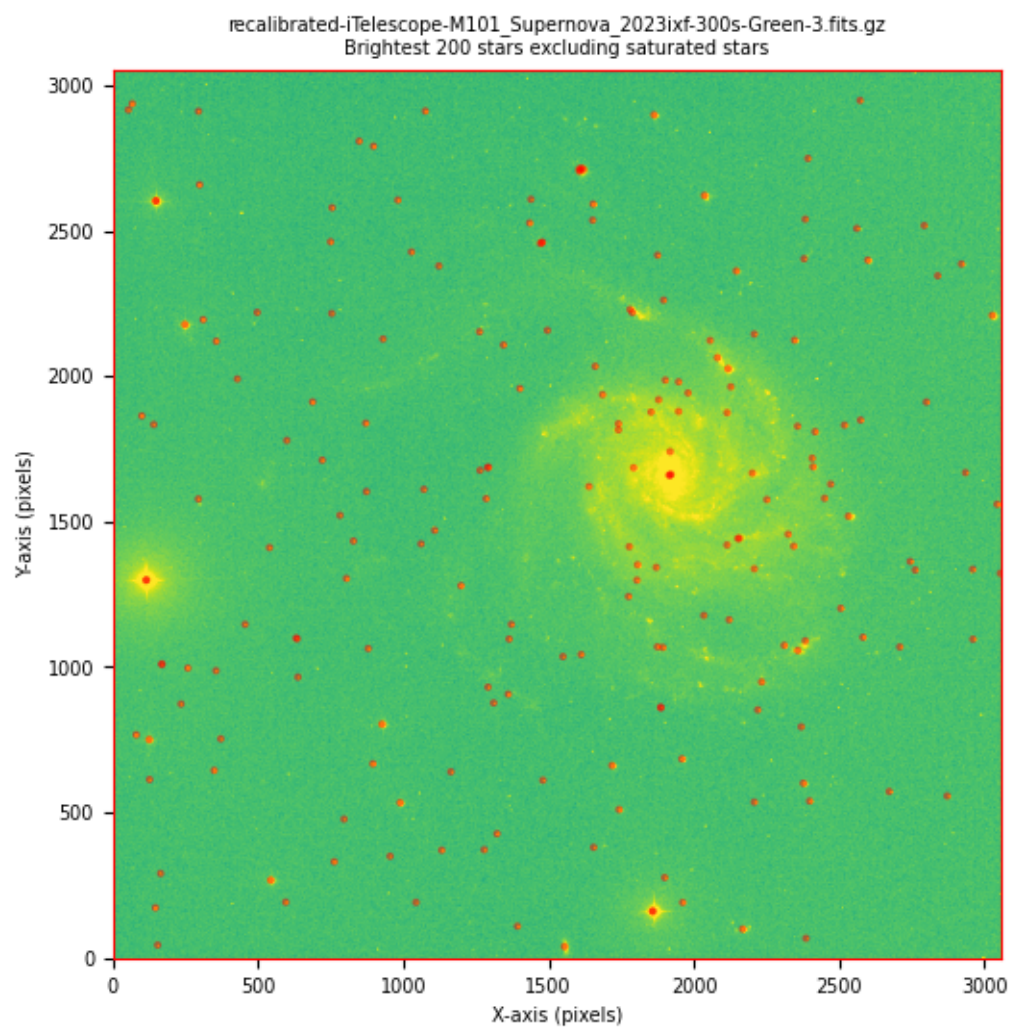


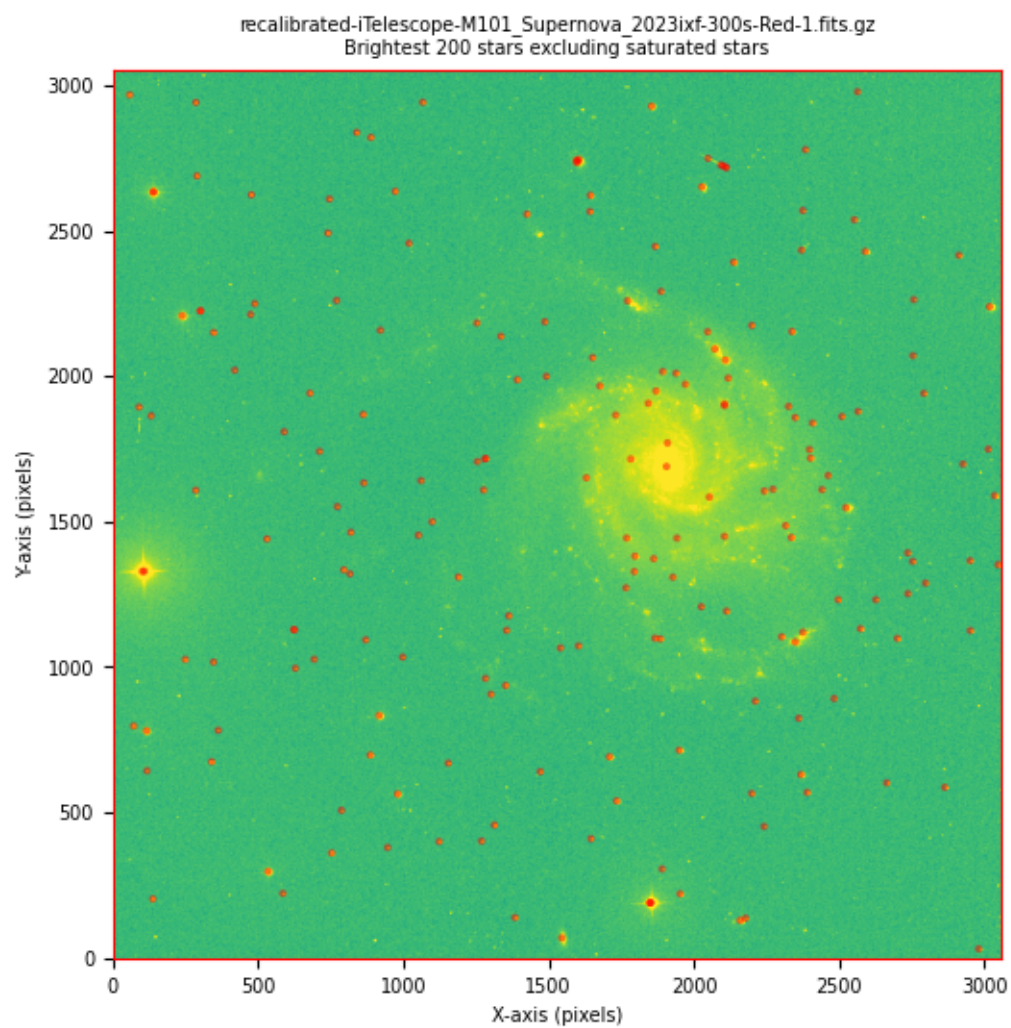




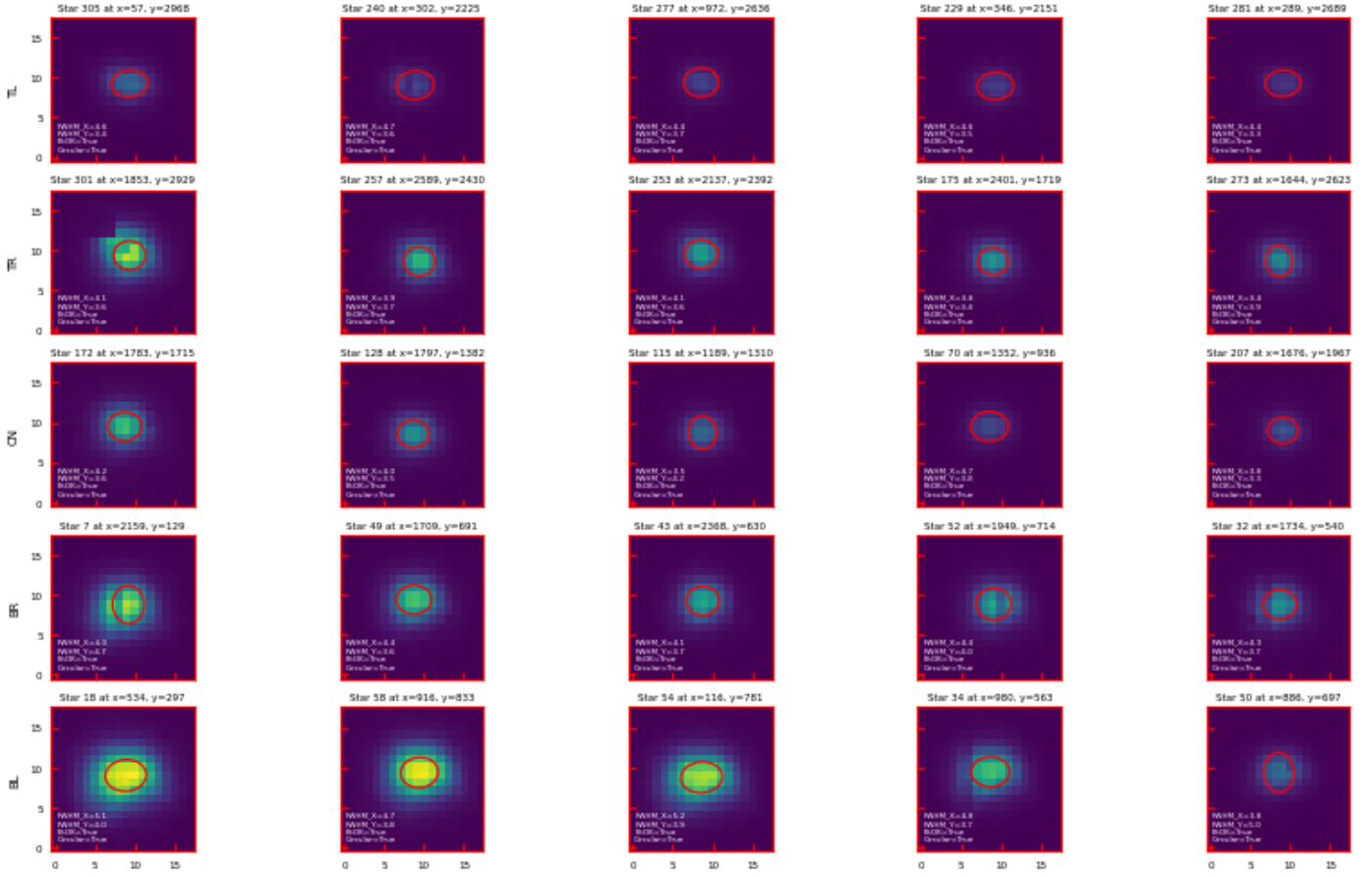
Star FWHM measurements for:
 recalibrated-MTelescope-M101_Supernova_2023xf-300s-Green-3.fits.gz
 Median FWHM=3.66 +/- 0.34 (MAD stddev) pixels
 Along X FWHM=3.65 +/- 0.30 pixels, along Y FWHM=3.67 +/- 0.31 pixels

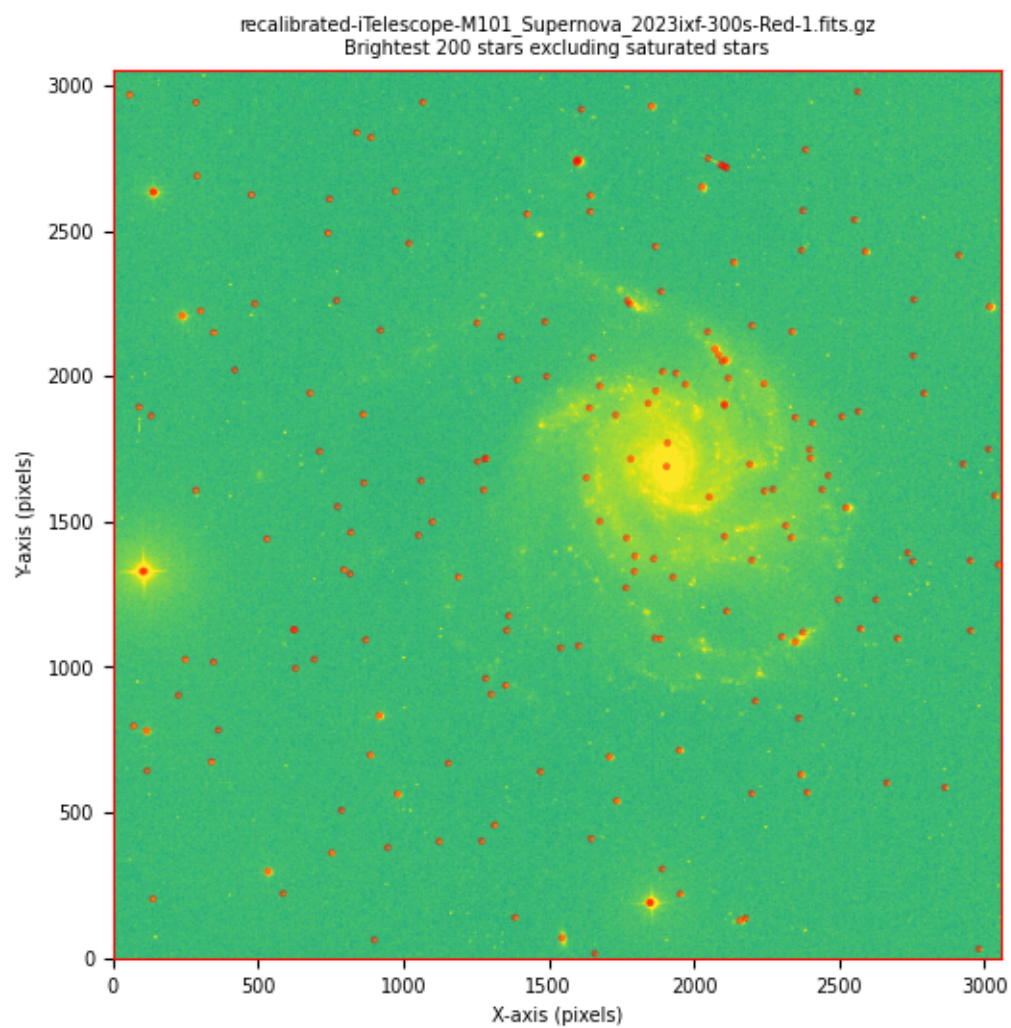


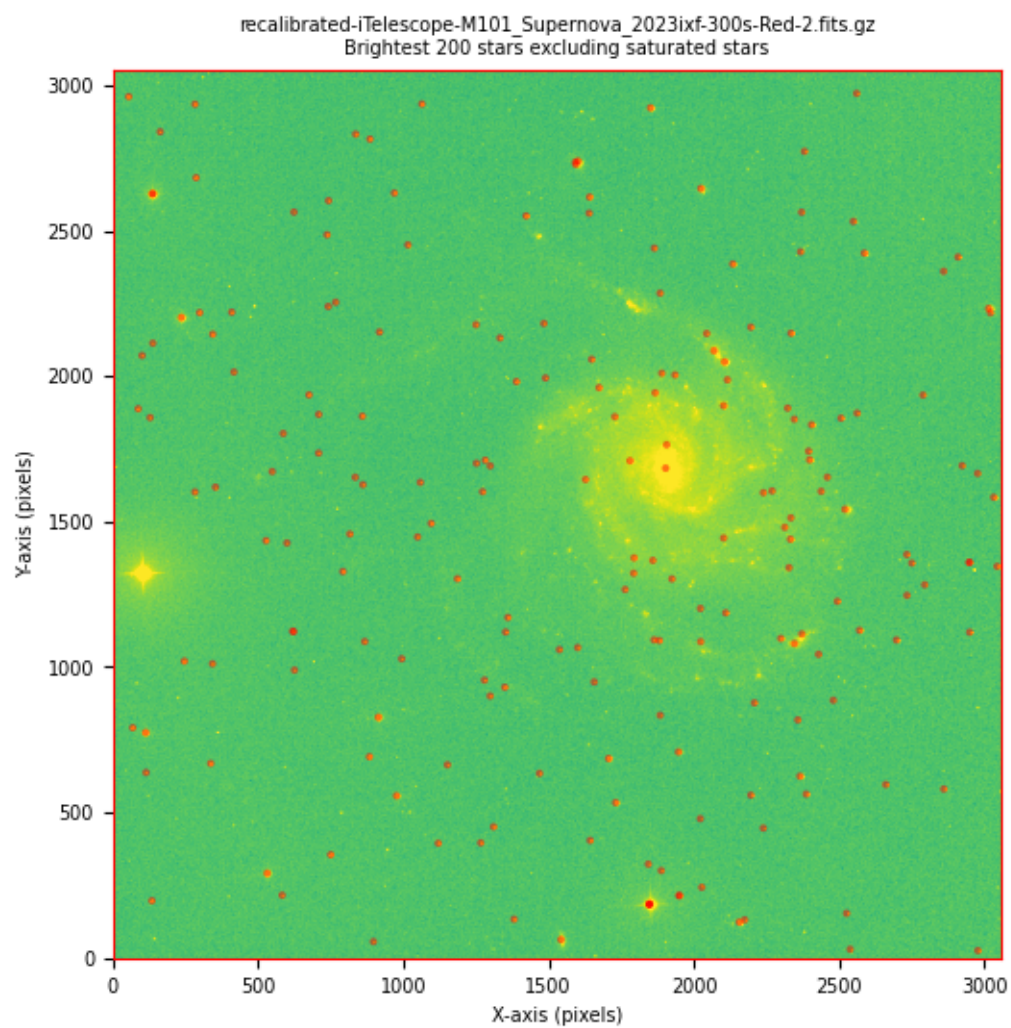




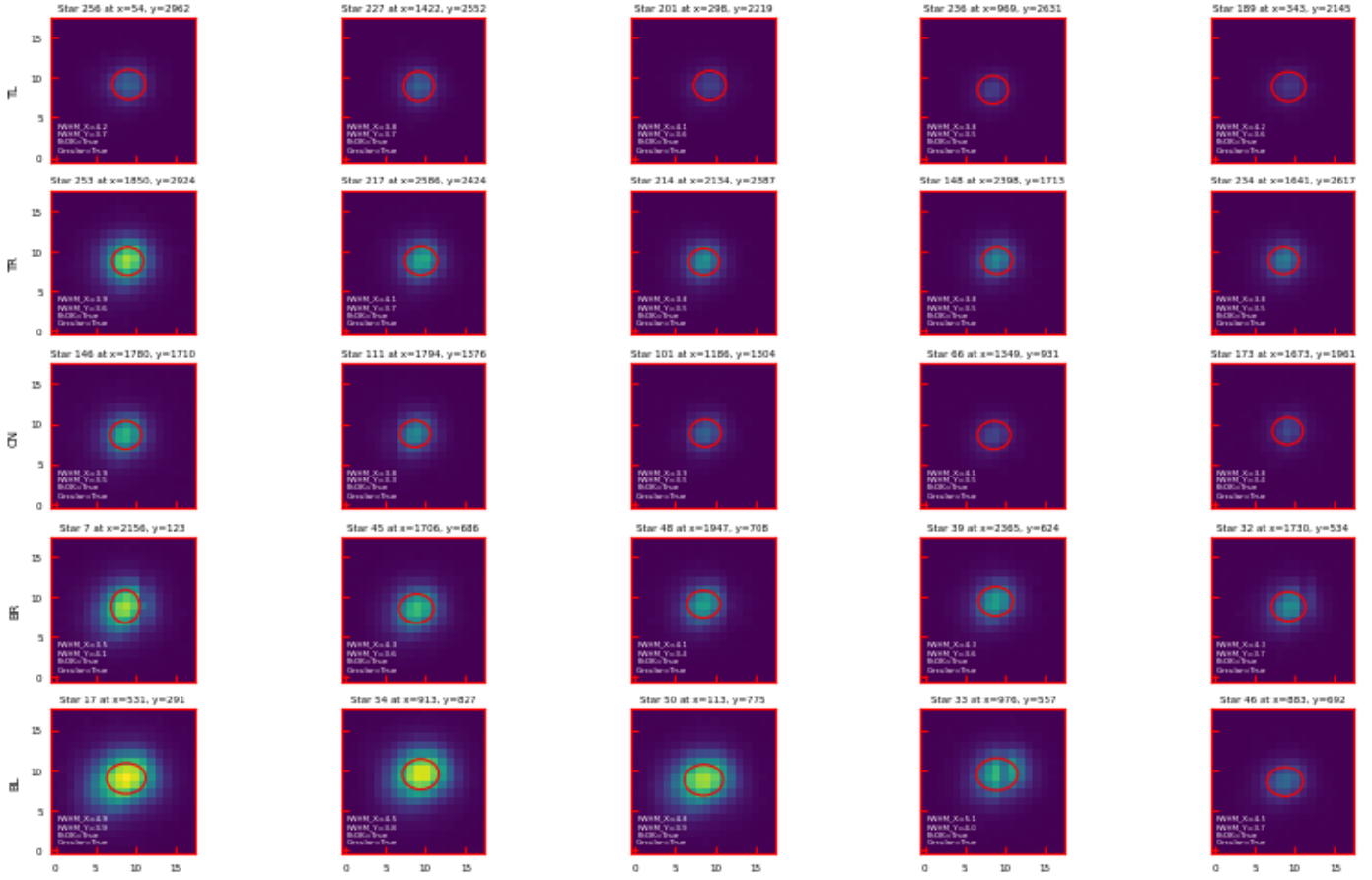
Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
 Median FWHM=3.91 +/- 0.43 (MAD stddev) pixels
 Along X FWHM=4.26 +/- 0.50 pixels, along Y FWHM=3.65 +/- 0.20 pixels

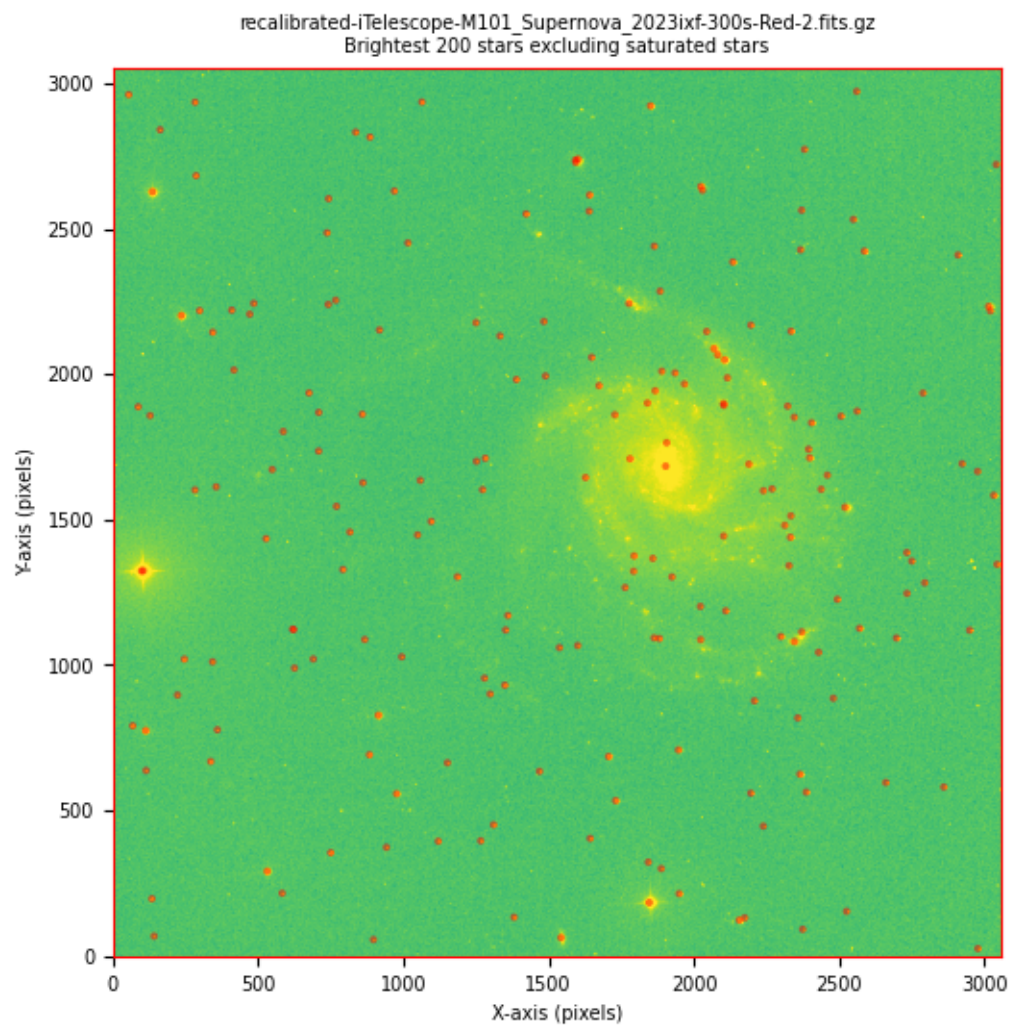


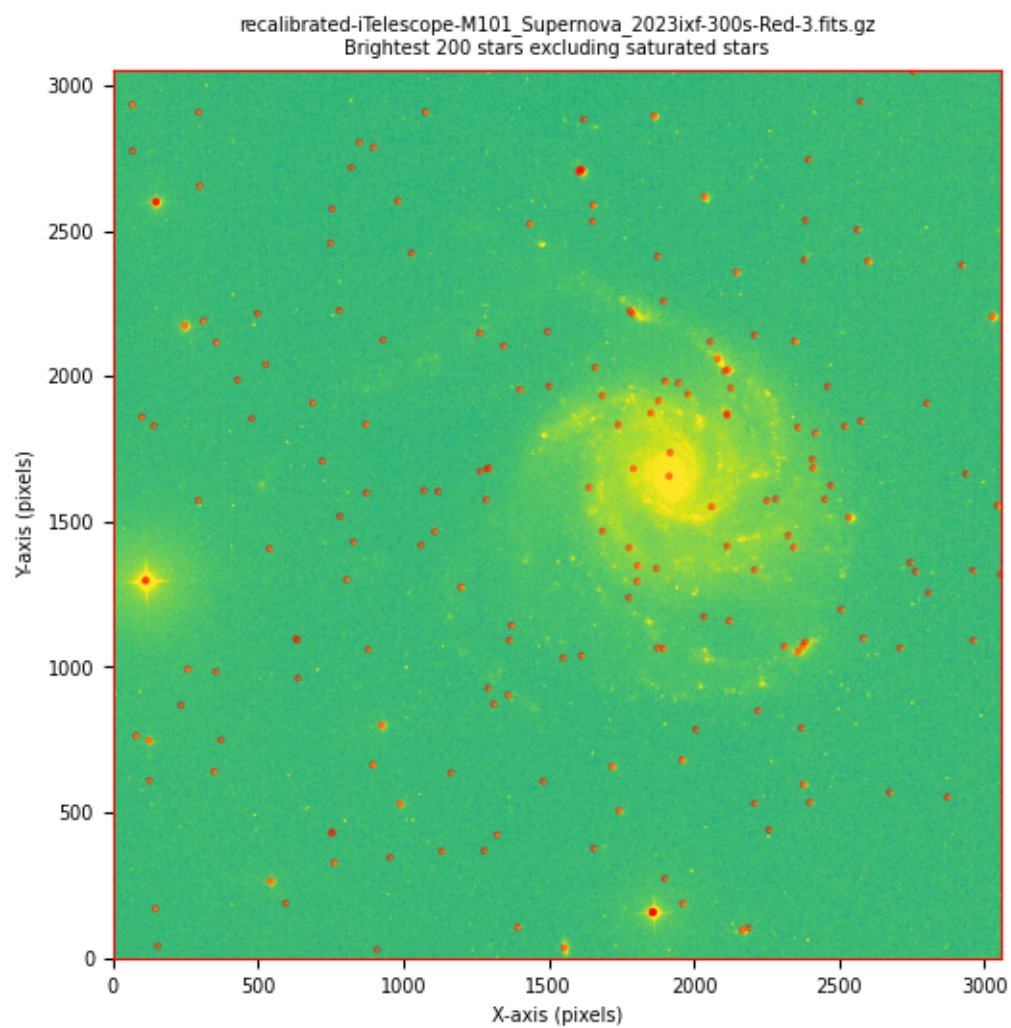




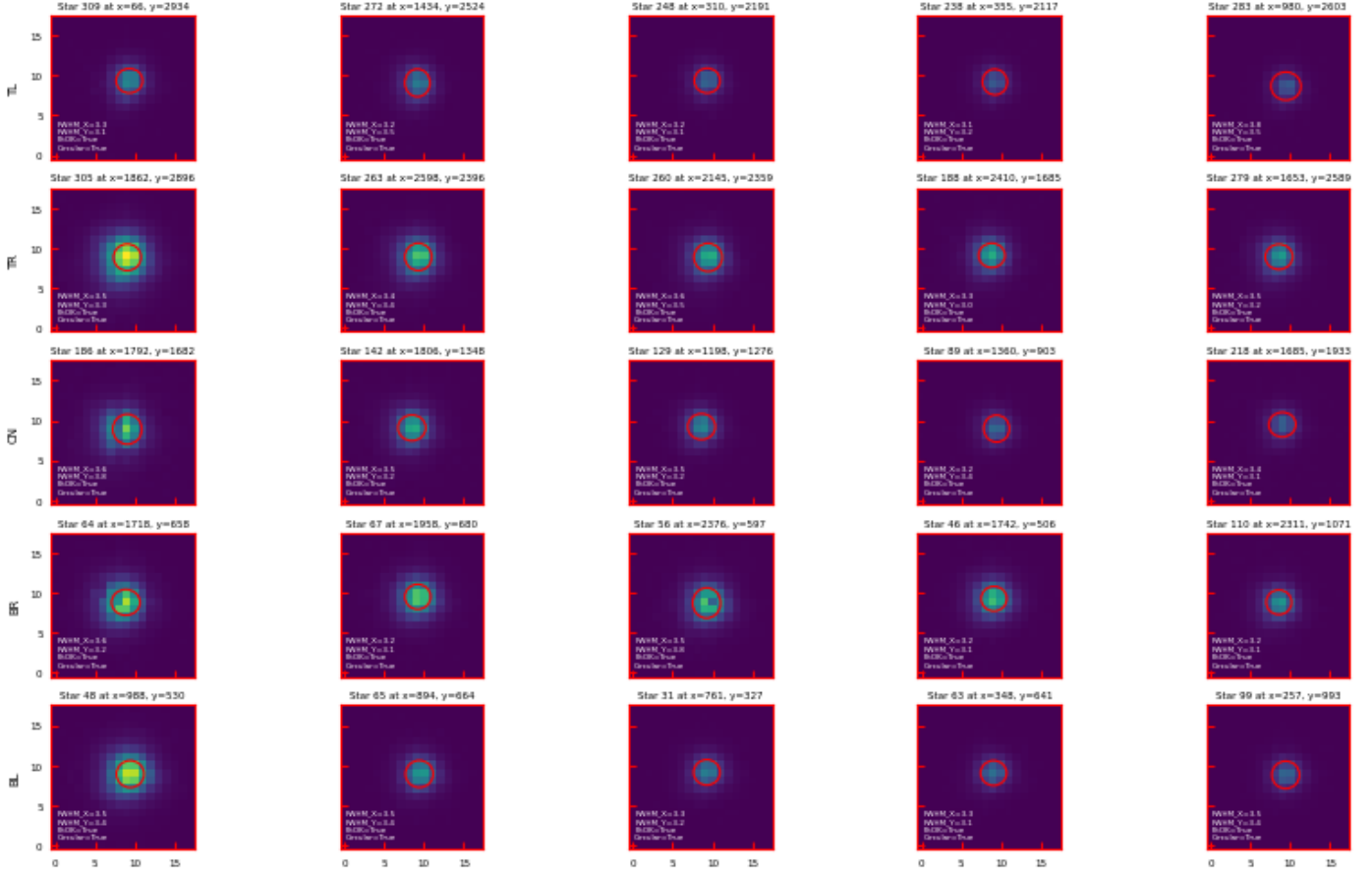
Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
 Median FWHM=3.81 +/- 0.38 (MAD stddev) pixels
 Along X FWHM=4.08 +/- 0.37 pixels, along Y FWHM=3.63 +/- 0.19 pixels

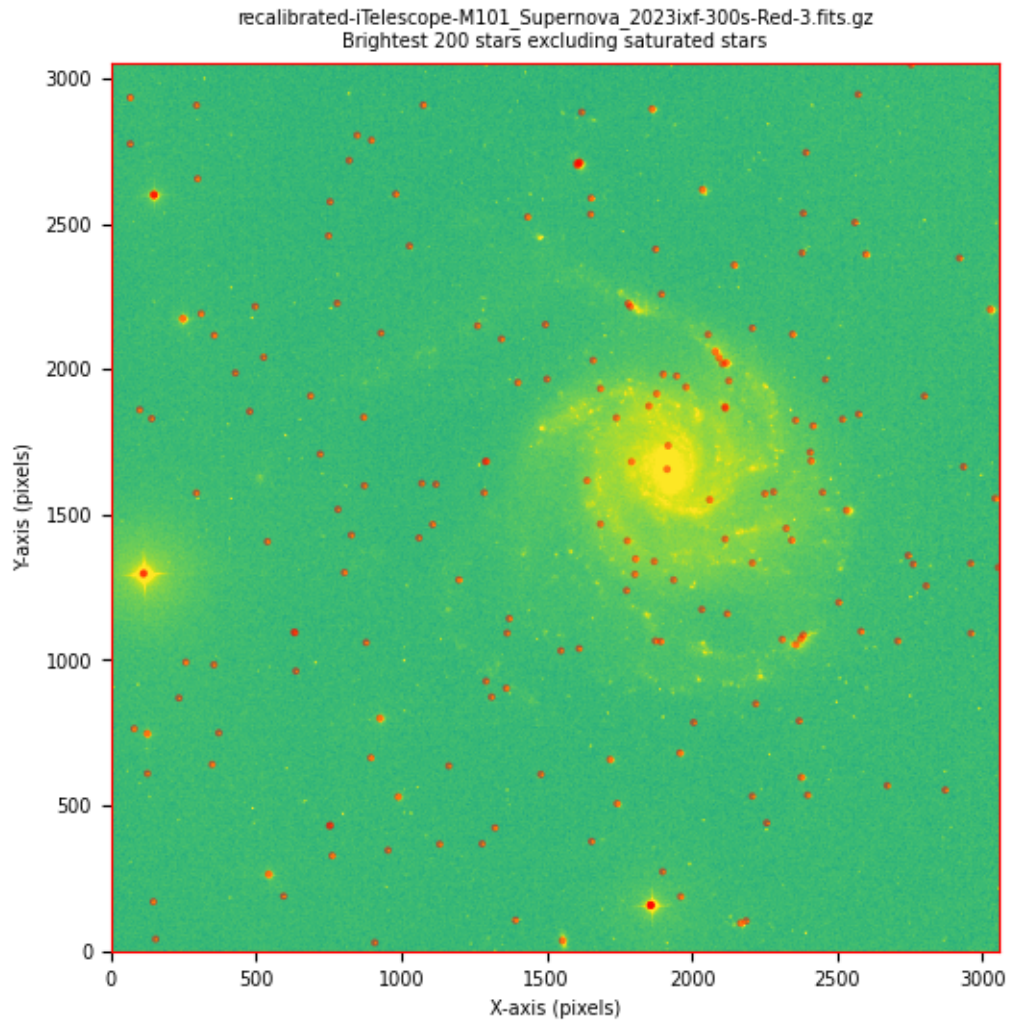






Star FWHM measurements for:
 recalibrated-Telescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
 Median FWHM=3.35 +/- 0.21 (MAD stddev) pixels
 Along X FWHM=3.41 +/- 0.15 pixels, along Y FWHM=3.24 +/- 0.21 pixels





```
0]: if do_by_hand_preprocess:
    # show status table
    status.pprint(max_width=200)
```

```

        filename
find_stars_status find_stars_time astrometry_status astrometry_time
-----
-----
    recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
pass      8.385      pass      15.517
    recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
pass      8.070      pass      18.089
    recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
pass      8.331      pass      10.950
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
pass      8.538      pass      16.451
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
pass      8.490      pass      13.531
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
pass      8.284      pass      17.225
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
pass      8.743      pass      9.001
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
pass      8.481      pass      16.841
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
pass      8.735      pass      14.066
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
pass      8.372      pass      13.519
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
pass      8.621      pass      19.548
    recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
```

pass 8.673 pass 14.023

Performance: Processing all the images sequentially takes about a minute per image, with most of the time taken waiting for [Astrometry.net](#) to provide the WCS solution.

```
1]: if do_by_hand_preprocess:
    # Load and display the quality summary table (NOTE this is a very wide table)
    qual_summ_table = Table.read(quality_summary_file, format="ascii.csv")
    # Show some data from the quality summary data.
    qual_summ_table['file', 'ncols', 'nrows', 'filter', 'median', 'stddev',
                    'num_detected', 'fwhm_val_pix', 'fwhm_err_pix', 'fwhm_val_arcs', 'fwhm_err_arcs', 'num_data_pts'].pprint_all()
```

	file	ncols	nrows			
filter	median	stddev	num_detected	fwhm_val_pix	fwhm_err_pix	fwhm_val_arcs
fwhm_err_arcs	num_data_pts					

	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz	3056	3056			
Blue	179.874023 21.801167	278	3.465217	0.284413	-999.0	
-999.0	50					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz	3056	3056			
Blue	183.536133 21.837498	266	3.625976	0.287341	-999.0	
-999.0	48					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz	3056	3056			
Blue	184.711258 21.853487	243	3.883582	0.247384	-999.0	
-999.0	48					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz	3056	3056			
Green	233.455261 22.624113	286	3.671051	0.350572	-999.0	
-999.0	49					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz	3056	3056			
Green	223.168976 22.500845	299	3.641141	0.365279	-999.0	
-999.0	50					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz	3056	3056			
Green	176.803589 21.651184	222	3.77143	0.249502	-999.0	
-999.0	48					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz	3056	3056			
Red	187.759628 21.953953	327	3.863207	0.549708	-999.0	
-999.0	50					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz	3056	3056			
Red	202.138245 22.116404	266	3.795447	0.351523	-999.0	
-999.0	50					
	Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz	3056	3056			
Red	182.942108 21.920551	349	3.324884	0.218721	-999.0	
-999.0	49					
	recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz	3056	3056			
Green	176.420837 21.084484	210	3.770194	0.272545	2.355351	
0.170267	45					
	recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz	3056	3056			
Green	222.861832 22.006407	286	3.659576	0.33953	2.286245	
0.212114	50					
	recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz	3056	3056			
Green	233.173233 22.156723	280	3.628631	0.299203	2.266913	
0.186921	49					
	recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz	3056	3056			
Red	187.362381 21.391273	311	3.910228	0.428611	2.442834	
0.267766	50					
	recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz	3056	3056			
Red	201.800049 21.592646	259	3.809383	0.37515	2.379833	
0.234367	49					
	recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz	3056	3056			
Red	182.529343 21.347113	316	3.330542	0.214698	2.080687	
0.134128	50					
	Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz	3056	3056			
Lum	535.460938 26.379311	287	4.139703	0.168107	-999.0	
-999.0	50					
	Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz	3056	3056			
Lum	488.849457 25.656437	306	4.295939	0.2659	-999.0	
-999.0	49					
	Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz	3056	3056			
Lum	573.592041 26.749031	240	4.180753	0.101949	-999.0	
-999.0	48					
	recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz	3056	3056			
Lum	535.367676 26.065695	277	4.136064	0.175152	2.583921	
0.109423	49					
	recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz	3056	3056			

Lum 573.510498	26.436098	234	4.181885	0.103848	2.612546
0.064877	47				
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz					3056 3056
Lum 488.744263	25.344429	303	4.325183	0.307606	2.702069
0.192171	50				
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz					3056 3056
Blue 179.470123	21.228891	230	3.468269	0.267846	2.166729
0.167331	50				
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz					3056 3056
Blue 183.147446	21.275547	249	3.603248	0.361569	2.251055
0.225883	48				
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz					3056 3056
Blue 184.327057	21.292282	234	3.946385	0.295515	2.465422
0.184617	50				

```
2] if do_by_hand_preprocess:
    target_wcs_file = './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits'
    resam_prefix = 'M101_Supernova_2023ixf_'
    resam_suffix = '_resamp_weighted.fits'
    resam_dir = 'Resampled'
    resampled_summary_plot = 'resampled_summary_plot.png'
    composite_summary_plot = 'composite_summary_plot.png'
    resampled_img_dict, info_table = processor.resample_images_to_match(target_wcs_file, resam_prefix, resam_suffix,
↪resampled_dir=resam_dir, filter_list=None,
                                                                    resampled_summary_plot=resampled_summary_plot,
↪composite_summary_plot=composite_summary_plot)
```

```
2024-11-26 21:22:03,696 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:22:03,708 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:22:03,709 | ApProcess | INFO | Found the following 3 files for
filter Blue: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:22:03,710 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:22:03,714 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:22:03,716 | ApProcess | INFO | Found the following 3 files for
filter Green: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:22:03,716 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:22:03,720 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:22:03,722 | ApProcess | INFO | Found the following 3 files for
filter Lum: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:22:03,722 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:22:03,726 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:22:03,728 | ApProcess | INFO | Found the following 3 files for
filter Red: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:22:03,728 | ApProcess | INFO | There are 4 unique filters among 12
files: ['Blue', 'Green', 'Lum', 'Red']
2024-11-26 21:22:03,731 | ApProcess | INFO | Target WCS file is 32.14 x 32.14
arcmin (wxh) with 0.63 x 0.63 arcsec pixels at an angle of 91.52 deg from North.
2024-11-26 21:22:03,731 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:22:03,736 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:22:03,736 | ApProcess | INFO | Processing 3 images for filter Blue
2024-11-26 21:22:03,750 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:22:03,752 | ApProcess | INFO | Filter Blue with 3 images totalling
900.0 seconds.
2024-11-26 21:22:08,392 | ApProcess | INFO | Success: swarp process took 4.586
```

```

seconds, return code 0
2024-11-26 21:22:08,464 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
2024-11-26 21:22:08,464 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:22:08,469 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:22:08,470 | ApProcess | INFO | Processing 3 images for filter
Green
2024-11-26 21:22:08,484 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits
2024-11-26 21:22:08,486 | ApProcess | INFO | Filter Green with 3 images
totalling 900.0 seconds.
2024-11-26 21:22:13,103 | ApProcess | INFO | Success: swarp process took 4.565
seconds, return code 0
2024-11-26 21:22:13,130 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
2024-11-26 21:22:13,130 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:22:13,135 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:22:13,135 | ApProcess | INFO | Processing 3 images for filter Lum
2024-11-26 21:22:13,149 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:22:13,151 | ApProcess | INFO | Filter Lum with 3 images totalling
540.0 seconds.
2024-11-26 21:22:17,835 | ApProcess | INFO | Success: swarp process took 4.630
seconds, return code 0
2024-11-26 21:22:17,862 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
2024-11-26 21:22:17,862 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:22:17,868 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:22:17,869 | ApProcess | INFO | Processing 3 images for filter Red
2024-11-26 21:22:17,883 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits
2024-11-26 21:22:17,885 | ApProcess | INFO | Filter Red with 3 images totalling
900.0 seconds.
2024-11-26 21:22:22,560 | ApProcess | INFO | Success: swarp process took 4.617
seconds, return code 0
2024-11-26 21:22:22,587 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
2024-11-26 21:22:22,587 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:22:22,593 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:22:22,594 | ApProcess | INFO | Found the following 3 files for
filter Blue: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:22:22,594 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:22:22,599 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:22:22,601 | ApProcess | INFO | Found the following 3 files for
filter Green: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:22:22,601 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:22:22,606 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:22:22,607 | ApProcess | INFO | Found the following 3 files for
filter Lum: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:22:22,608 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:22:22,613 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:22:22,614 | ApProcess | INFO | Found the following 3 files for
filter Red: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits']

```

2024-11-26 21:22:22,615 | ApProcess | INFO | Finished. Generated 4 resampled images in 18.940 seconds: {'Blue': 'Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits', 'Green': 'Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits', 'Lum': 'Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits', 'Red': 'Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits'}

The are 4 FITS images in the input imlist

With 4 files the plot layout is 2 rows x 2 columns.

Attempting to plot Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits

Input image data 0.5th percentile = 0.5265, 99.5th percentile = 0.7889

Applying asinh stretch between 0.5265 and 0.7889

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

Attempting to plot Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits

Input image data 0.5th percentile = 0.6168, 99.5th percentile = 0.8938

Applying asinh stretch between 0.6168 and 0.8938

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

Attempting to plot Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits

Input image data 0.5th percentile = 2.7715, 99.5th percentile = 3.7196

Applying asinh stretch between 2.7715 and 3.7196

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

Attempting to plot Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits

Input image data 0.5th percentile = 0.5434, 99.5th percentile = 0.8794

Applying asinh stretch between 0.5434 and 0.8794

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

2024-11-26 21:22:24,575 | ApProcess | INFO | Resampled image summary plot generated: resampled_summary_plot.png

Wrote plot to resampled_summary_plot.png

There are enough files to generate a RGB threecolor composite.

NB: insufficient files to generate a SHO threecolor composite.

Input red channel data 0.5th percentile = 0.5434, 99.5th percentile = 0.8794

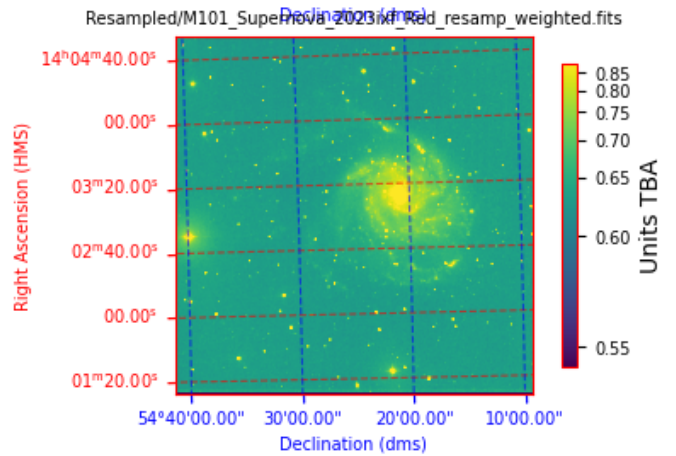
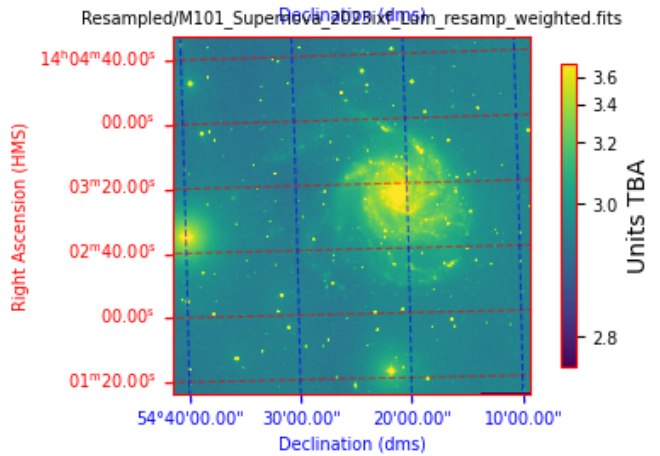
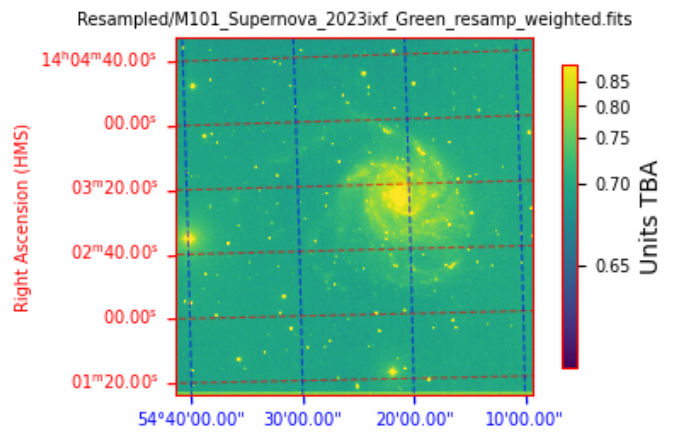
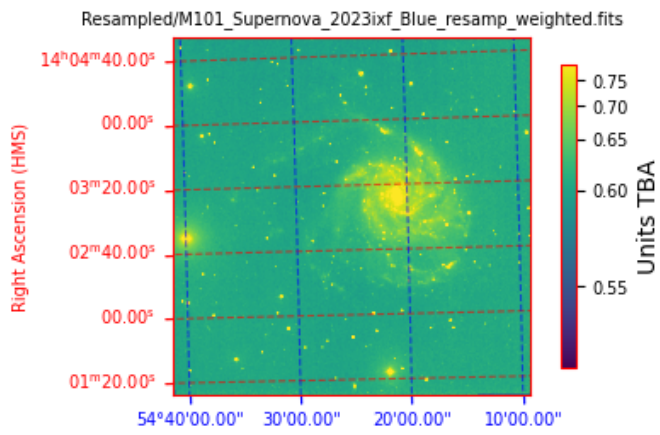
Using an image minimum of 0.5433955729007721 for all channels.

Default X-axis limits: (-0.5, 3055.5)

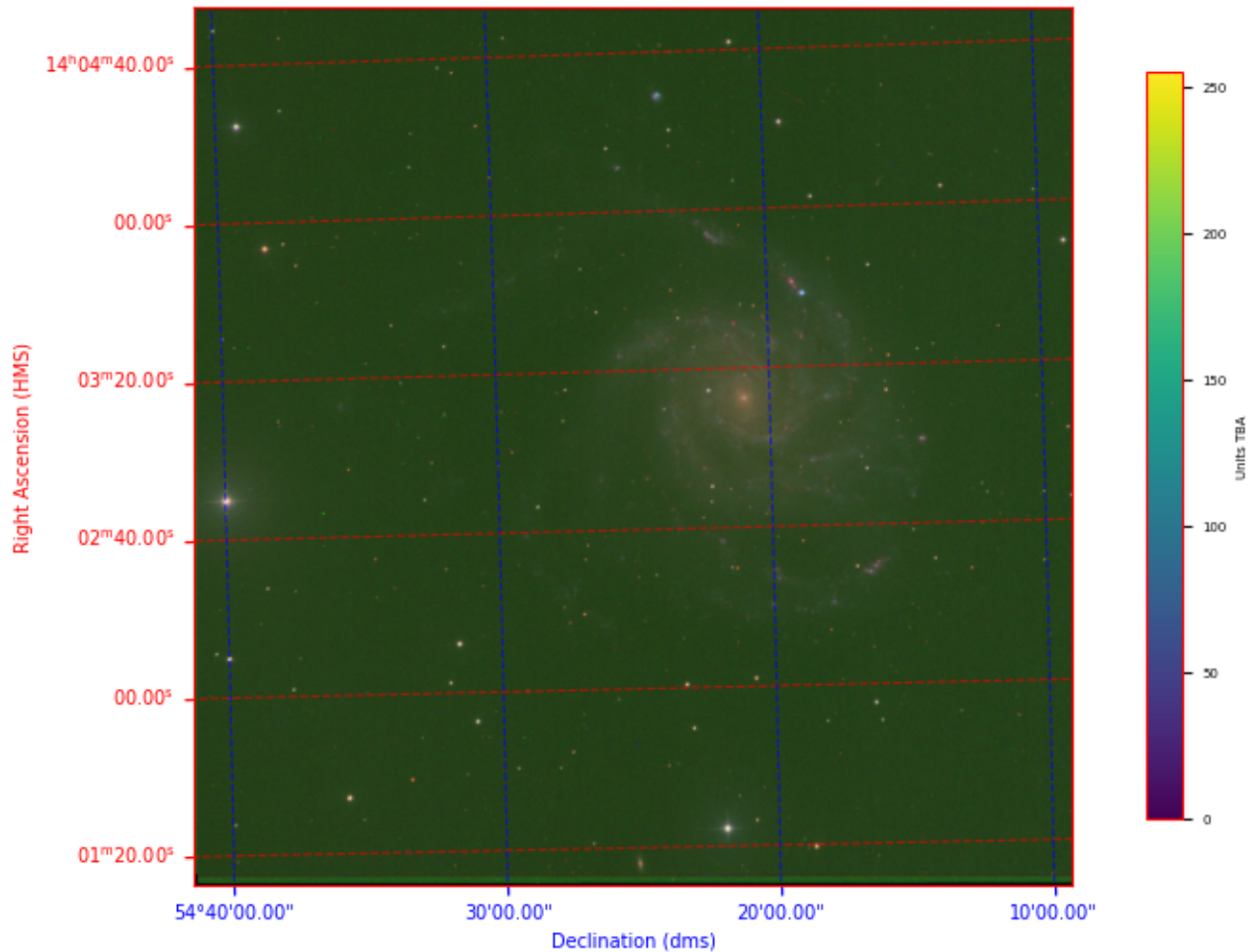
Default Y-axis limits: (-0.5, 3055.5)

2024-11-26 21:22:25,960 | ApProcess | INFO | Summary color composite plot generated: composite_summary_plot.png

Wrote plot to composite_summary_plot.png



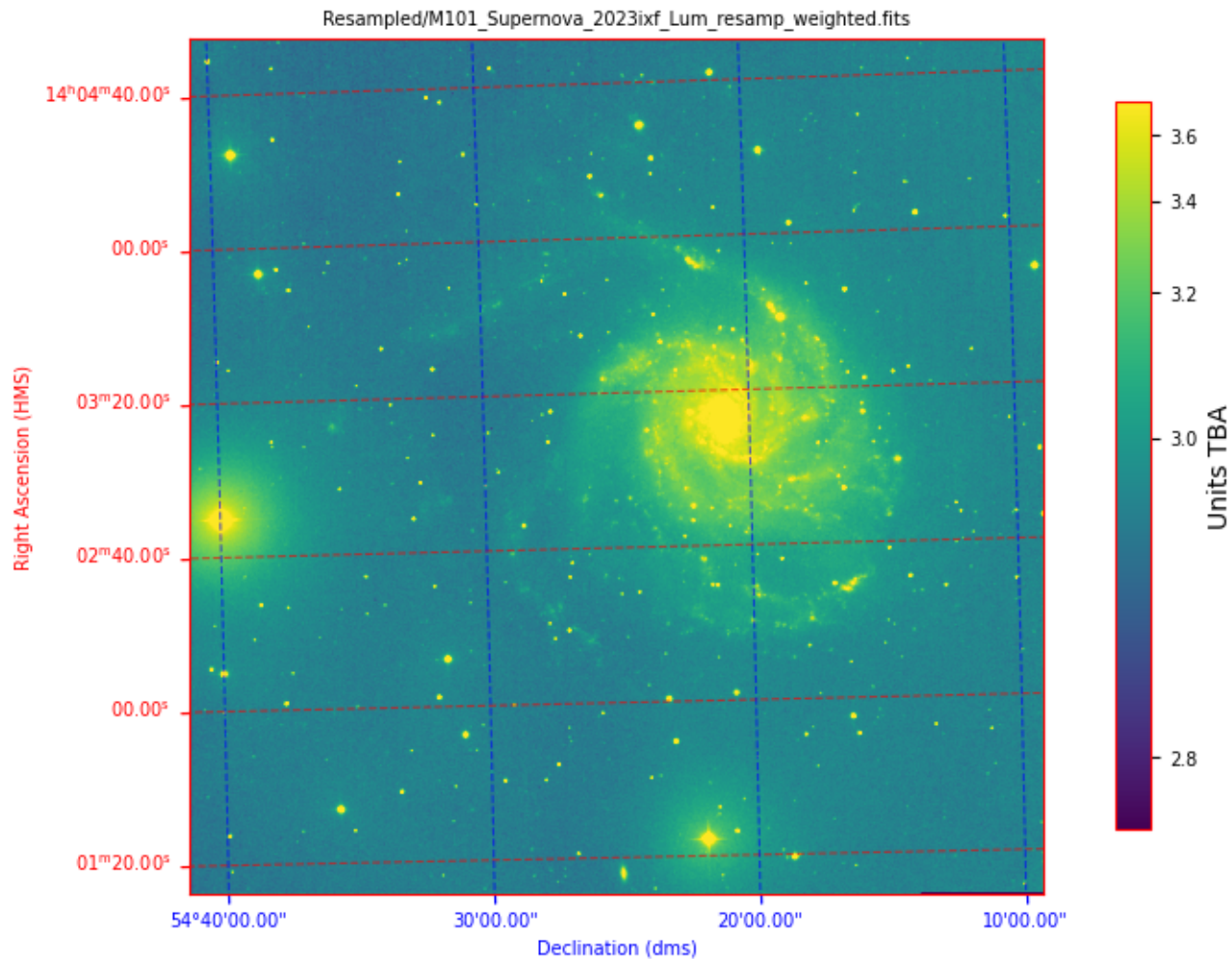
RGB composite image stretch=0.5, Q=8, minimum=None
 Red = Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Green = Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Blue = Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits



```
3]: if do_by_hand_preprocess:
    lumf = resampled_img_dict['Lum']
    print(f'Final luminance image: {lumf}')
    output_plot = 'luminance_plot.png'
    ap.util.load_image_and_plot(lumf, 0, output_plot, True, angle_tick_spacing_am=10, swap_radec_axis=True)
```

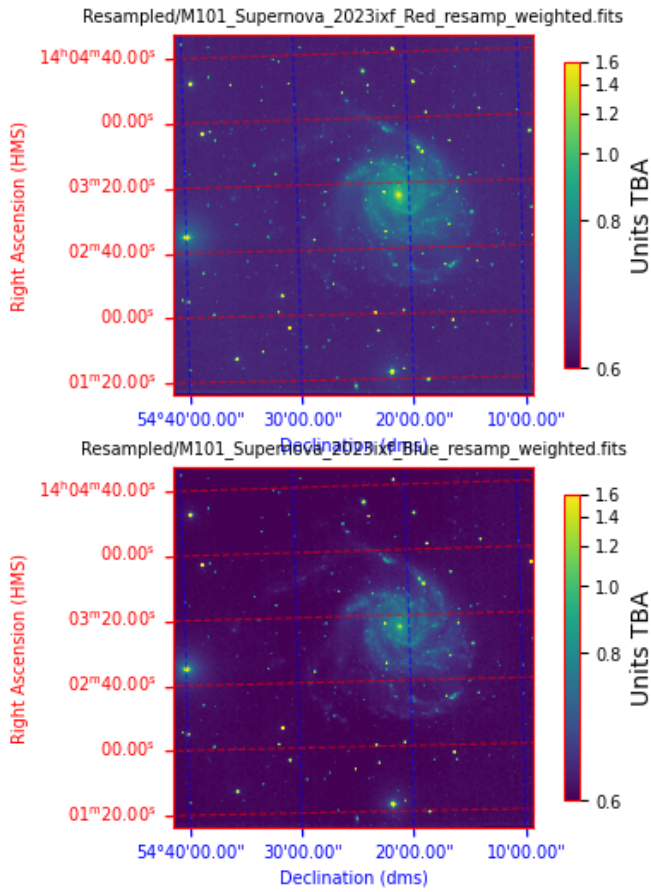
Final luminance image: Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
 Input data 0.5th percentile = 2.7715, 99.5th percentile = 3.7196
 Applying asinh stretch between 2.7715 and 3.7196
 Using WCS information for axis projection
 WCS Keywords

Number of WCS axes: 2
 CTYPE : 'RA---TAN' 'DEC--TAN'
 CRVAL : 210.754804653 54.4174059435
 CRPIX : 1528.5 1528.5
 PC1_1 PC1_2 : -4.64681630389e-06 0.000175223532872
 PC2_1 PC2_2 : -0.000175223532872 -4.64681630389e-06
 CDELT : 1.0 1.0
 NAXIS : 3056 3056
 Default X-axis limits: (-0.5, 3055.5)
 Default Y-axis limits: (-0.5, 3055.5)
 Wrote plot to luminance_plot.png



```
4]: if do_by_hand_preprocess:
    bluef = resampled_img_dict['Blue']
    greenf = resampled_img_dict['Green']
    redf = resampled_img_dict['Red']
    valmin = None
    valmax = None
    output_plot = None # Name of output PNG file or None
    ap.util.load_imlist_and_plot([redf, greenf, bluef], extnum=0, output=output_plot,
                                usewcs=True, vmin=0.6, vmax=1.6, xaxlim=None,
                                yaxlim=None, verbose=True,
                                angle_tick_spacing_am=10, swap_radec_axis=True)
```

The are 3 FITS images in the input imlist
 With 3 files the plot layout is 2 rows x 2 columns.
 Attempting to plot Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Input image data 0.5th percentile = 0.5434, 99.5th percentile = 0.8794
 Applying asinh stretch between 0.6000 and 1.6000
 Default X-axis limits: (-0.5, 3055.5)
 Default Y-axis limits: (-0.5, 3055.5)
 Attempting to plot Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Input image data 0.5th percentile = 0.6168, 99.5th percentile = 0.8938
 Applying asinh stretch between 0.6000 and 1.6000
 Default X-axis limits: (-0.5, 3055.5)
 Default Y-axis limits: (-0.5, 3055.5)
 Attempting to plot Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
 Input image data 0.5th percentile = 0.5265, 99.5th percentile = 0.7889
 Applying asinh stretch between 0.6000 and 1.6000
 Default X-axis limits: (-0.5, 3055.5)
 Default Y-axis limits: (-0.5, 3055.5)



```
5]: if do_by_hand_preprocess:
    ap.util.load_image_and_plot(bluef, 0, output_plot, True, vmin=0.6, vmax=1.0, xaxlim=[1100, 1500], yaxlim=[350, 750], verbose=True,
    ↪angle_tick_spacing_am=2, swap_radec_axis=True)
```

Input data 0.5th percentile = 0.5265, 99.5th percentile = 0.7889

Applying asinh stretch between 0.6000 and 1.0000

Using WCS information for axis projection

WCS Keywords

Number of WCS axes: 2

CTYPE : 'RA---TAN' 'DEC--TAN'

CRVAL : 210.754804653 54.4174059435

CRPIX : 1528.5 1528.5

PC1_1 PC1_2 : -4.64681630389e-06 0.000175223532872

PC2_1 PC2_2 : -0.000175223532872 -4.64681630389e-06

CDELT : 1.0 1.0

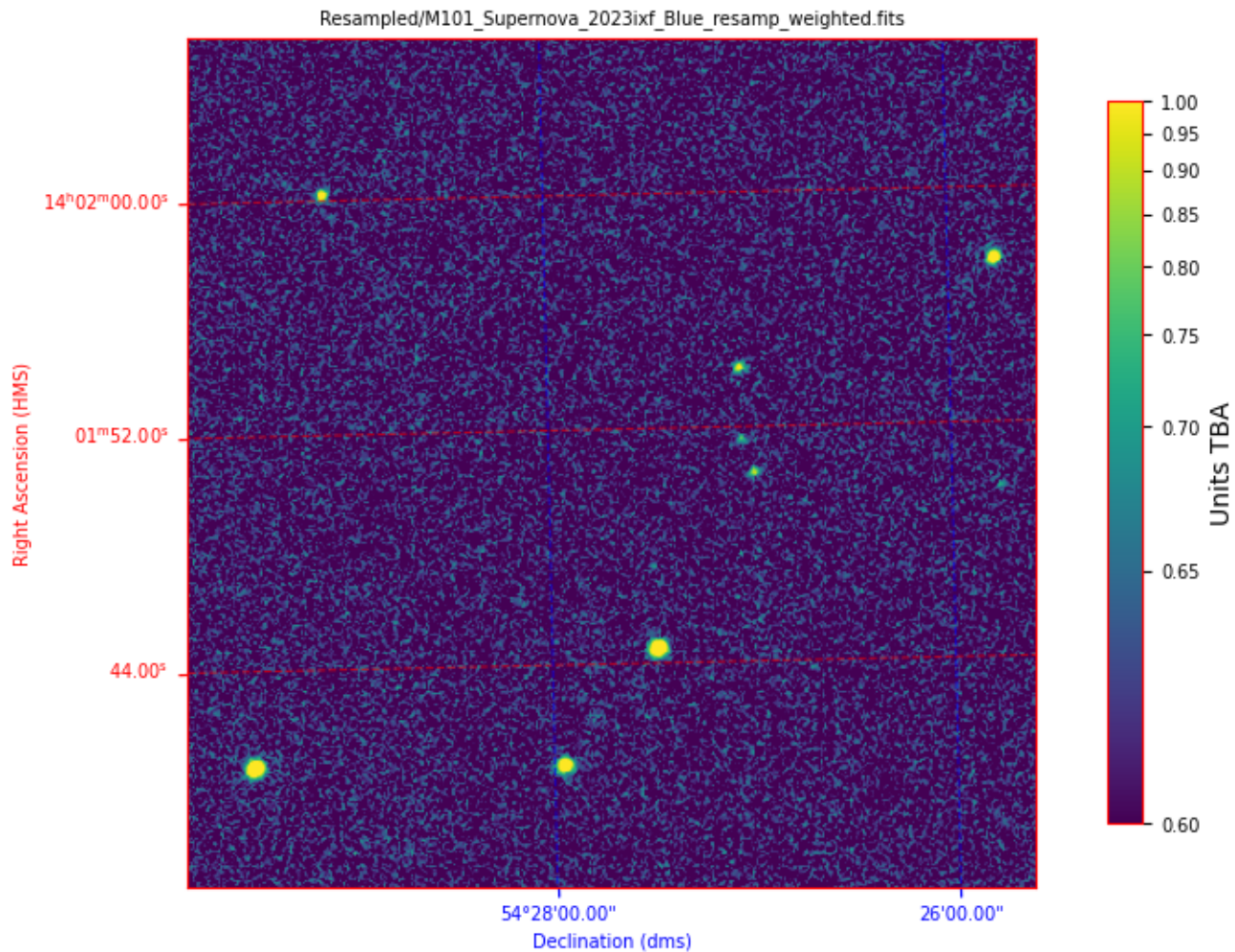
NAXIS : 3056 3056

Default X-axis limits: (-0.5, 3055.5)

Default Y-axis limits: (-0.5, 3055.5)

Setting X-axis limits to [1100, 1500]

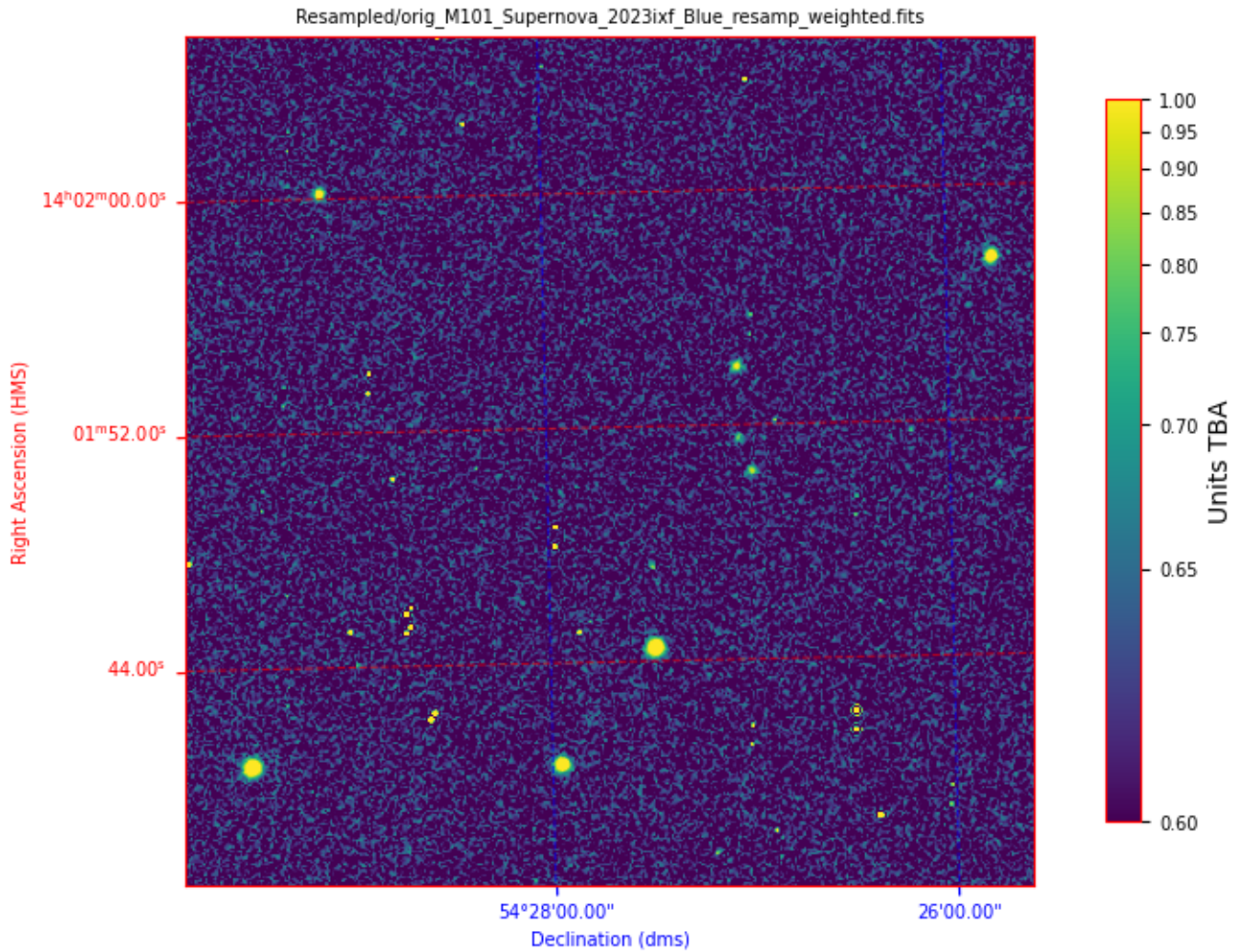
Setting Y-axis limits to [350, 750]



```
6]: if do_by_hand_preprocess:
    ap.util.load_image_and_plot(copyof_original_bluef, 0, output_plot, True, vmin=0.6, vmax=1.0, xaxlim=[1100, 1500], yaxlim=[350, 750], verbose=True, angle_tick_spacing_am=2, swap_radec_axis=True)
```

Input data 0.5th percentile = 0.5249, 99.5th percentile = 0.8013
 Applying asinh stretch between 0.6000 and 1.0000
 Using WCS information for axis projection
 WCS Keywords

Number of WCS axes: 2
 CTYPE : 'RA---TAN' 'DEC--TAN'
 CRVAL : 210.754804653 54.4174059435
 CRPIX : 1528.5 1528.5
 PC1_1 PC1_2 : -4.64681630389e-06 0.000175223532872
 PC2_1 PC2_2 : -0.000175223532872 -4.64681630389e-06
 CDELT : 1.0 1.0
 NAXIS : 3056 3056
 Default X-axis limits: (-0.5, 3055.5)
 Default Y-axis limits: (-0.5, 3055.5)
 Setting X-axis limits to [1100, 1500]
 Setting Y-axis limits to [350, 750]

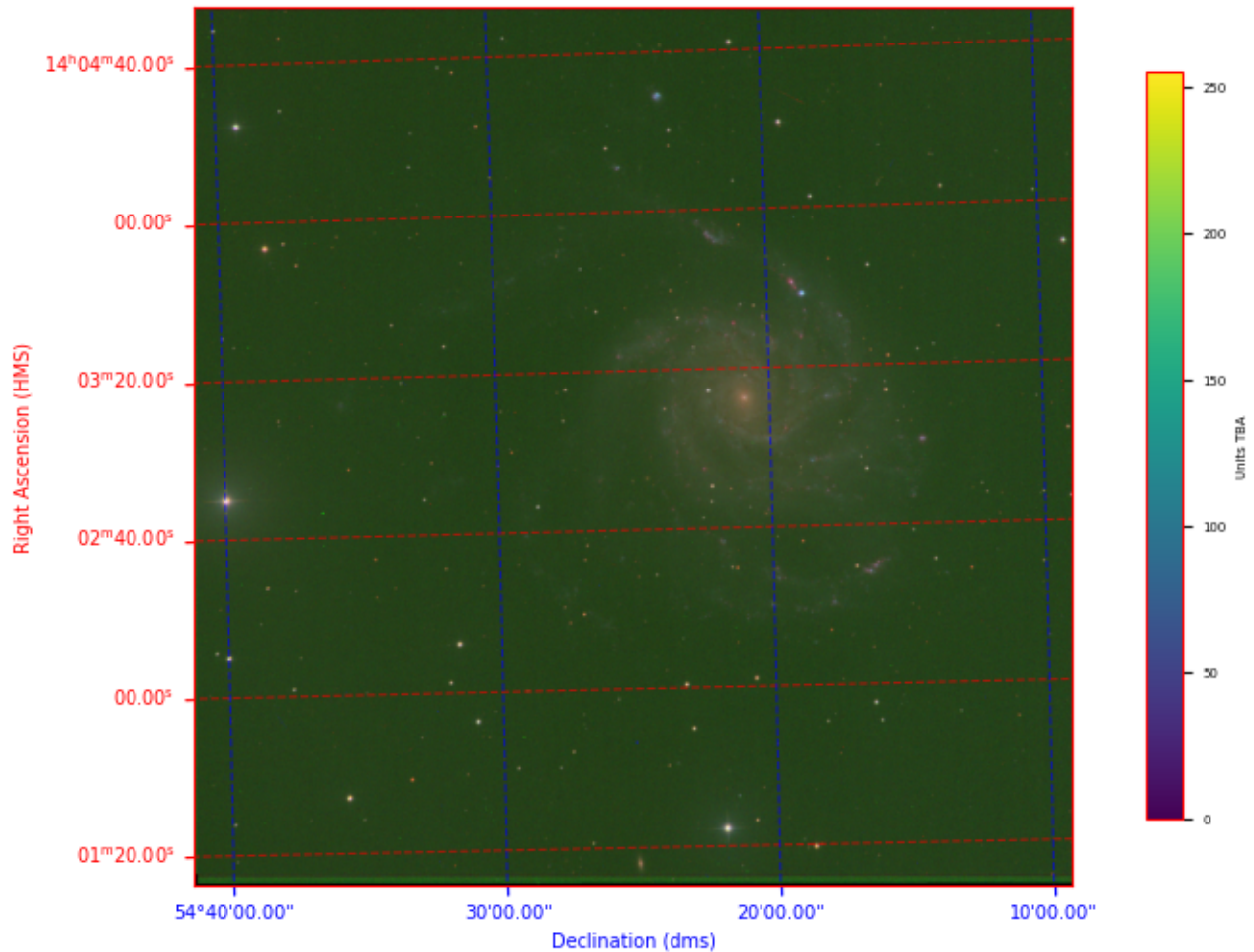


So, approximately 95% or more of the bad pixels were removed by the preprocessing with the recent dark-file bad pixel map. (Only 50% with the out-of-date bad pixel map.)

```
7]: if do_by_hand_preprocess:
    qval = 8
    stretch = 0.5
    valmin = None
    outf = 'M101_lupton_RedGreenBlue.png'
    rgb = ap.util.plot_lupton_threecolor(redf, greenf, bluef, outf,
                                         extnum=0, usewcs=True, vmin=valmin,
                                         xaxlim=None, yaxlim=None,
                                         qval=qval, stretchval=stretch, verbose=True,
                                         angle_tick_spacing_am=10, swap_radec_axis=True)
```

Input red channel data 0.5th percentile = 0.5434, 99.5th percentile = 0.8794
Using an image minimum of 0.5433955729007721 for all channels.
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Wrote plot to M101_lupton_RedGreenBlue.png

RGB composite image stretch=0.5, Q=8, minimum=None
 Red = Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Green = Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Blue = Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits



```
3]: if do_by_hand_preprocess:
    stiff_outputf = 'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff'
    description = 'M101_Supernova_2023ixf'
    user = 'AstroPhotography'
    !stiff $redf $greenf $bluef \
      -OUTFILE_NAME $stiff_outputf -GAMMA_TYPE POWER-LAW -GAMMA 2.2 -GAMMA_FAC 1.0 \
      -COLOUR_SAT 1.0 -MAX_TYPE =QUANTILE,QUANTILE,QUANTILE -MAX_LEVEL 0.999,0.999,0.999 -MIN_TYPE =QUANTILE,QUANTILE,QUANTILE \
      -MIN_LEVEL 0.60,0.60,0.60 \
      -BITS_PER_CHANNEL 8 -VERBOSE_TYPE FULL -DESCRIPTION $description -COPYRIGHT $user -WRITE_XML N
```

> WARNING: stiff.conf not found, using internal defaults

----- STIFF 2.7.1 started on 2024-11-26 at 21:22:36 with 16 threads

----- Inputs:

M101_Supernova_2023ixf_Red_resamp_weighted.fits: "M101 Supernova 2023i"
 3056x3056 32 bits (floats)
 Background level: 0.633776 Min level: 0.617579 Max level: 2.52959
 M101_Supernova_2023ixf_Green_resamp_weighted.fits: "M101 Supernova 2023i"
 3056x3056 32 bits (floats)
 Background level: 0.698947 Min level: 0.682215 Max level: 2.26172
 M101_Supernova_2023ixf_Blue_resamp_weighted.fits: "M101 Supernova 2023i"
 3056x3056 32 bits (floats)
 Background level: 0.606025 Min level: 0.589985 Max level: 1.95145

----- Output:

M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff:

3056x3056 3x8 bits (integers) gamma: x1.00 compression: LZW

```
9]: if do_by_hand_preprocess:
    # where are we?
    cwd = pathlib.Path('.')
    print(f'Current directory: {cwd.resolve()}')
    tiff_file = cwd / 'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff'
    if tiff_file.exists():
        fig, (ax1, ax2) = plt.subplots(1,2)
        print(f'Loading {tiff_file.name}')
        tiff_img = plt.imread(tiff_file)
        print(f'Plotting full image {tiff_file}')
        ax1.imshow(tiff_img)

        # Zoom in
        print(f'Plotting zoomed-in part of {tiff_file}')
        print(' Note the bad pixels visible in this zoomed-in view.')
        ax2.imshow(tiff_img[600:1000,1600:2000])
    else:
        print(f'File not found: {tiff_file.name}')
        print(' Skipping display of STIFF generated file...')
```

Current directory: /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline

Loading M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff

Plotting full image

M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff

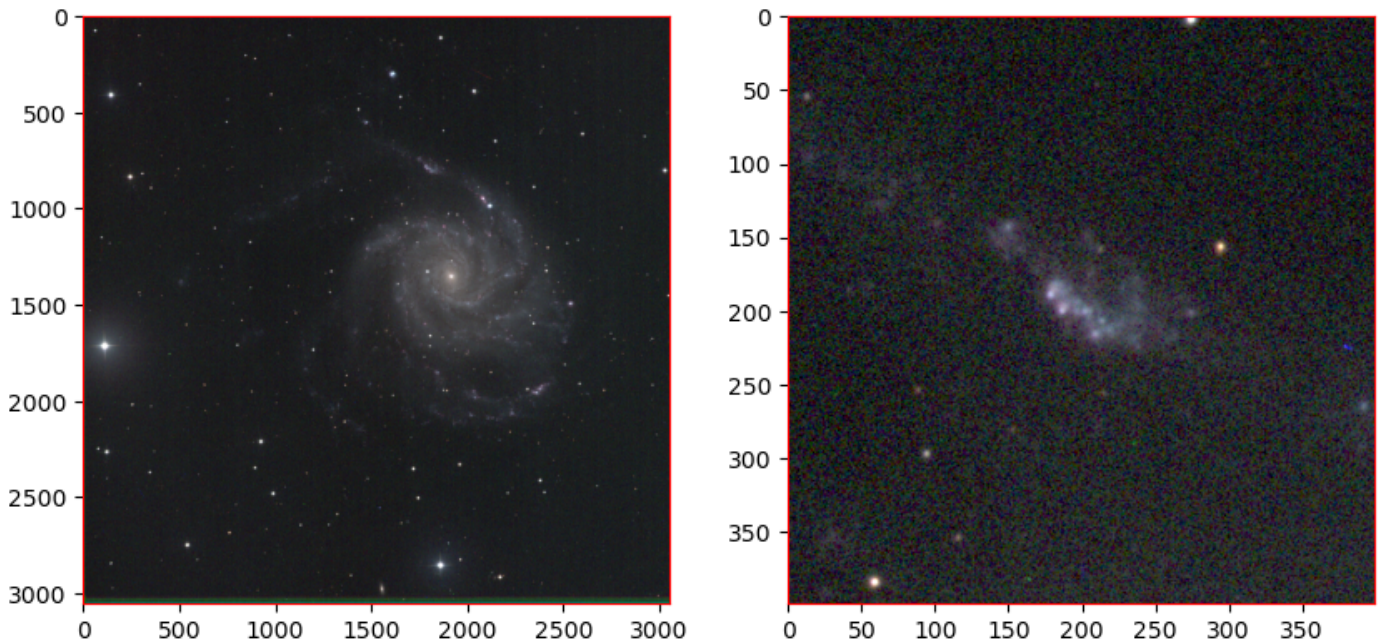
9]: <matplotlib.image.AxesImage at 0x7f5683d44eb0>

Plotting zoomed-in part of

M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff

Note the bad pixels visible in this zoomed-in view.

9]: <matplotlib.image.AxesImage at 0x7f5683d74070>



Compare the old and new TIFF files... The new ones are on the top row and the old ones are on the bottom row.

```
0]: if do_by_hand_preprocess:
    fig, (ax1, ax2) = plt.subplots(1,2)
    print(f'Loading {tiff_file.name}')
    tiff_img = plt.imread(tiff_file)
    print(f'Plotting full image {tiff_file}')
    ax1.imshow(tiff_img)
```

```

ax1.set_title(tiff_file, fontsize=7)

# Zoom in
print(f'Plotting zoomed-in part of {tiff_file}')
print(' Note the bad pixels visible in this zoomed-in view.')
ax2.imshow(tiff_img[600:1000,1600:2000])
ax2.set_title('recalibrated image zoom-in', fontsize=7)

fig, (ax1, ax2) = plt.subplots(1,2)
print(f'Loading {copyof_original_tiff_file}')
tiff_img = plt.imread(copyof_original_tiff_file)
print(f'Plotting full image {tiff_file}')
ax1.imshow(tiff_img)
ax1.set_title(tiff_file, fontsize=7)

# Zoom in
print(f'Plotting zoomed-in part of {copyof_original_tiff_file}')
print(' Note the bad pixels visible in this zoomed-in view.')
ax2.imshow(tiff_img[600:1000,1600:2000])
ax2.set_title('original image zoom-in', fontsize=7)

```

Loading M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
 Plotting full image
 M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff

0]: <matplotlib.image.AxesImage at 0x7f5683f848b0>

0]: Text(0.5, 1.0,
 'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff')

Plotting zoomed-in part of
 M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
 Note the bad pixels visible in this zoomed-in view.

0]: <matplotlib.image.AxesImage at 0x7f5683f05420>

0]: Text(0.5, 1.0, 'recalibrated image zoom-in')

Loading
 orig_M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
 Plotting full image
 M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff

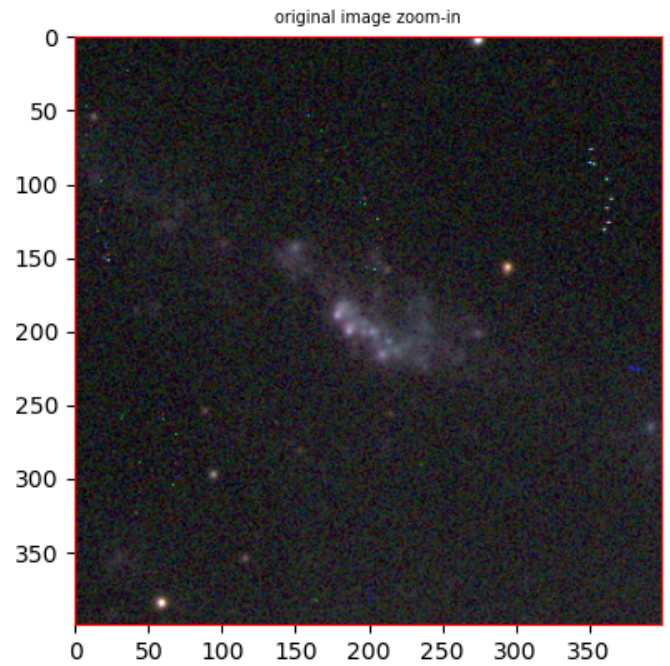
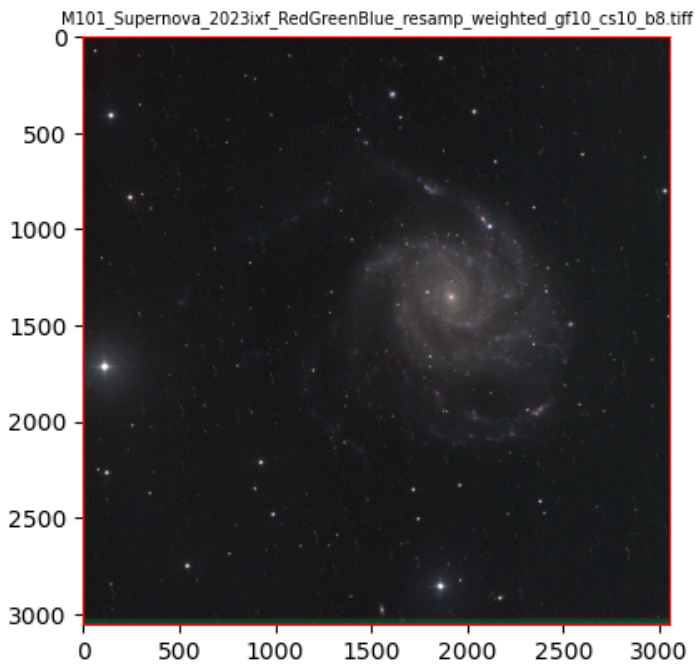
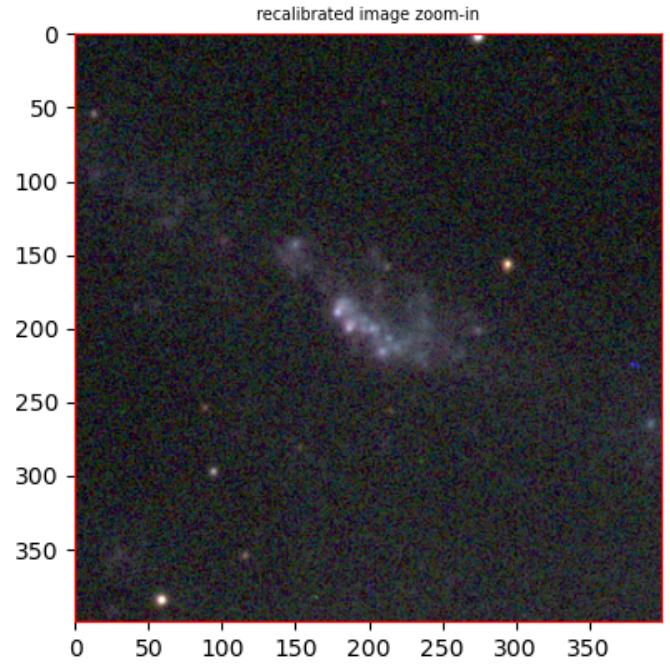
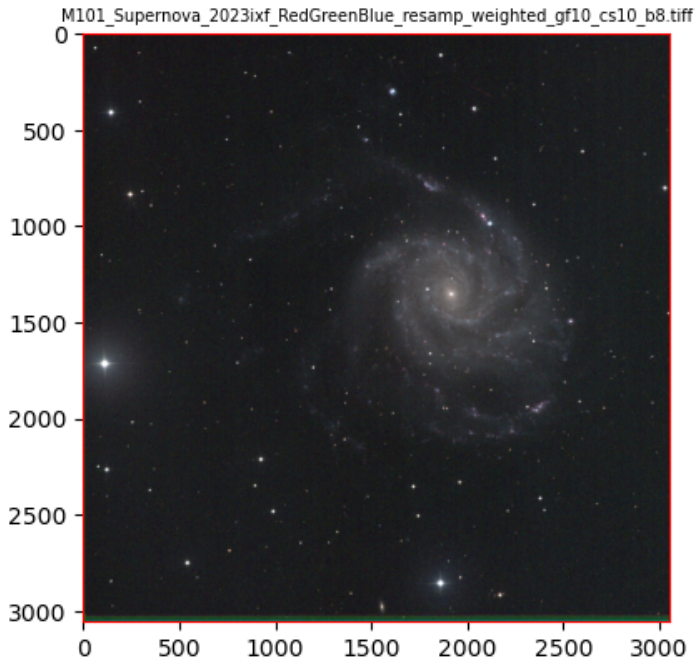
0]: <matplotlib.image.AxesImage at 0x7f5683fc82b0>

0]: Text(0.5, 1.0,
 'M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff')

Plotting zoomed-in part of
 orig_M101_Supernova_2023ixf_RedGreenBlue_resamp_weighted_gf10_cs10_b8.tiff
 Note the bad pixels visible in this zoomed-in view.

0]: <matplotlib.image.AxesImage at 0x7f5683720dc0>

0]: Text(0.5, 1.0, 'original image zoom-in')



A significant improvement in this zoomed-in region, with only a hint of some cosmic rays or possibly bad pixels in the green and blue channels.

```
1]: # Close all the open figures in memory
plt.close('all')
```

4 Batch Processing with a Single Function Call

To further simplify batch processing `ApProcess` provides a function that wraps all the the steps together. So far the processing stages, including preprocessing. **Warning:** This does have an unwieldy number of function parameters.

All the necessary function parameters have been collected and listed below.

```
2]: del processor
loglevel = 'INFO'
```

```
processor = ap.ApProcess(loglevel)
```

Standard input file patterns for iTelescope premium datasets...

```
3]: # Parameters
data_dir = r'./'

# Make sure to check that your pattern works on the command line using `ls`
default_include_pattern = "Calibrated-*-.fits*"
default_exclude_pattern = None
```

Code for selecting the specific directory we will process

```
4]: default_dir = '/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline'
print(f'Default directory for input files: {default_dir}')

print('Enter the path to the directory containing the premium image set (or return for the default): ')
wdir = input('Image set path:').strip() or default_dir
try:
    os.chdir(wdir.strip())
    print('Switched directory to ' + os.getcwd())
except:
    print(f'Error, os.chdir threw an exception changing to {wdir}')
    print('Check that the path you supplied is a valid filesystem path.')
    raise

print(f'Default include file pattern for input files: {default_include_pattern}')
print(f'Default exclude file pattern for input files: {default_exclude_pattern}')

msg = 'Enter new include input file pattern (or return to accept default pattern)'
include_pattern = input(msg).strip() or default_include_pattern
print(f'Using "{include_pattern}" as the input include file pattern.')

msg = 'Enter new exclude input file pattern (or return to accept default pattern)'
exclude_pattern = input(msg).strip() or default_exclude_pattern
print(f'Using "{exclude_pattern}" as the input exclude file pattern.')
```

```
Default directory for input files:
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
Enter the path to the directory containing the premium image set (or return for
the default):
Image set path:
Switched directory to
/old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
Default include file pattern for input files: Calibrated-*-.fits*
Default exclude file pattern for input files: None

Enter new include input file pattern (or return to accept default pattern)
Using "Calibrated-*-.fits*" as the input include file pattern.

Enter new exclude input file pattern (or return to accept default pattern)
Using "None" as the input exclude file pattern.
```

Clean up any existing source-detection related quality summary file. Both the master summary and the individual YaML files for each image...

```
5]: if 'M101' in wdir:
    quality_summary_file = 'quality_summary_M101SN.csv'
    print(f'M101 observations detected, adopted default quality summary file name: {quality_summary_file}')
else:
    default_qual_sum = 'quality_summary.csv'
    quality_summary_file = input('Name for final image quality summary CSV (enter for default): ').strip() or default_qual_sum
    print(f'Adopted quality summary file name: {quality_summary_file}')

# Need to remove existing quality summary table, and remove existing per-file quality files.
print('Deleting any existing quality summary file...')
pathlib.Path(quality_summary_file).unlink(True)

# Determine the metadata (quality file) directory name
metadata_dir = processor.get_directories('qualfile', data_dir, False)
print(f'YaML quality metadata files are in {metadata_dir}')
```

```
# We know the
print(f'Deleting any existing quality YaML files in {metadata_dir}')
for file_path in pathlib.Path(metadata_dir).glob("qual*.yaml"):
    print(f'Deleting existing quality file {str(file_path)}')
    file_path.unlink(True)
```

M101 observations detected, adopted default quality summary file name:

```
quality_summary_M101SN.csv
Deleting any existing quality summary file...
YaML quality metadata files are in ./MetaData/
Deleting any existing quality YaML files in ./MetaData/
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Blue-2.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Green-1.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Red-1.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Red-2.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-180s-Lum-2.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-180s-Lum-1.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Green-3.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Green-2.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Blue-3.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Red-3.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-300s-Blue-1.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.yaml
Deleting existing quality file
MetaData/qual-M101_Supernova_2023ixf-180s-Lum-3.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.yaml
Deleting existing quality file MetaData/qual-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.yaml
```

The following settings are for preprocessing. Remember, preprocessing is only advised if you already know that the existing calibration is lacking.

```
5]: # metadata dictionary in dict[str, tuple[Any, str]] form
meta_dict = {
    "OBJNAME": ("M101 Supernova 2023ixf", "Target"),
    "OBJECT": ("M101 Supernova 2023ixf", "Target"),
    "TELESCOP": ("iTelescope 24", "Telescope name"),
    "INSTRUME": ("FLI-PL09000", "Instrument name"),
    "XPIXSZ": (12.0, "Pixel Width in microns (after binning)"),
    "YPIXSZ": (12.0, "Pixel Height in microns (after binning)"),
    "EGAIN": (1.53, "Electronic gain in e-/ADU"),
}

# Note that dict[str, list[Any, str]] will fail to process
```

```

fail_meta_dict = {
    "EGAIN": [1.53, "Electronic gain in e-/ADU"],
    "INSTRUME": ["FLI-PL09000", "Instrument name"],
    "OBJECT": ["M101 Supernova 2023ixf", "Target"],
    "OBJNAME": ["M101 Supernova 2023ixf", "Target"],
    "TELESCOP": ["iTelescope 24", "Telescope name"],
    "XPIXSZ": [12.0, "Pixel Width in microns (after binning)"],
    "YPIXSZ": [12.0, "Pixel Height in microns (after binning)"],
}

# Overwrite existing header keywords with the same name
rewrite_existing_kw = False

# Set EXPOSURE if not present and any of the other exposure-like FITS keywords are present
find_exposure_time = True

# Bad pixel file location (e.g. creating_a_master_bad_pixel_file.ipynb)
old_badpixfile = "/old_lnx/home/dks/Downloads/iTelescopeScratch/calibration-library/T24/Raw/2021_DEC/
↳Master_Badpix_11_3056x3056_Bin1x1_Temp-25C_ExpTime900s.fit"
new_badpixfile = "/old_lnx/home/dks/Downloads/iTelescopeScratch/T24/CALIBRATION/Dark/Bin1/300/
↳master_badpix_T24_20241001_3056x3056_Bin1x1_Temp-25C_ExpTime300s.fits"
badpixfile = new_badpixfile

# instr is the part of the original calibrated input file name that will be renamed in the modified,
# preprocessed, output files.
preprocess_instr = "Calibrated"
preprocess_outstr = "recalibrated"

```

Additional settings for image navigation

```

7]: inp_flist = None
inp_rootname = None
clean_run = False ##True
if clean_run:
    clean_srclists = True
    clean_navfiles = True
else:
    clean_srclists = False
    clean_navfiles = False

stop_on_error = False
quiet = True

# accept the defaults on the other navigation parameters....

```

Resampling and image stacking settings.

```

8]: target_wcs_file = None
# target_wcs_file = './NavigatedImages/navigated-M101_Supernova_2023ixf-180s-Lum-1.fits'
resam_prefix = "M101_Supernova_2023ixf-"
resam_suffix = "_resamp_weighted.fits"
resam_dir = "Resampled"
filter_list = None
resampled_summary_plot = 'resampled_summary_plot.png'
composite_summary_plot = 'composite_summary_plot.png'

```

It is often a good idea to investigate some of the out file names in cases where the input files are preprocessed or have a file prefix other than `cal` or `calibrated` or `Calibrated`. Failure to do so may result in hard to diagnose errors later in processing.

```

9]: # Full set of ftypes
ftypes = ['srclist', 'regfile', 'plotfile', 'qualfile', 'fwhmplot', 'navfile']

# Shorter subset of the files than must be generated for WCS generated and resampling/stacking to work
#ftypes = ['srclist', 'navfile']

show_files = False
if show_files:
    for ftype in ftypes:
        print(80*'=')
        # Get directory info, relative path or absolute path
        rel_filedir = processor.get_directories(ftype, data_dir, False)
        abs_filedir = processor.get_directories(ftype, data_dir, True)
        print(f'File type {ftype} output directory names:')

```

```

print(f' Relative path={rel_filedir}\n Absolute path={abs_filedir}')
print(80*'-')

# File names, without directory path
print(f'File type {ftype} output file names (no directory info):')
ofiles = processor.get_file_names(ftype, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.gz',
↳no_name_and_dir)
print(f' There are {len(ofiles)} files in the file list.')
for file in ofiles:
    print(f' {file}')
print(80*'-')

# File names **with** directory path
print(f'File type {ftype} output file names (including directory):')
ofiles = processor.get_file_names(ftype, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.gz',
↳with_name_and_dir)
print(f' There are {len(ofiles)} files in the file list.')
for file in ofiles:
    print(f' {file}')

# Files that this instance has already processed (without directory)
print(f'File type {ftype} output file names (no directory info), files that have been processed by this instance:')
ofiles = processor.get_file_names(ftype, data_dir, 'processed', include_pattern, exclude_pattern, None, None, '.fits.gz',
↳no_name_and_dir)
print(f' There are {len(ofiles)} files in the file list.')
for file in ofiles:
    print(f' {file}')
print(80*'-')

# File that exist on disk, even if they haven't been processed by this instance
print(f'File type {ftype} output file names (no directory info), files that exist on disk:')
ofiles = processor.get_file_names(ftype, data_dir, 'existing', include_pattern, exclude_pattern, None, None, '.fits.gz',
↳no_name_and_dir)
print(f' There are {len(ofiles)} files in the file list.')
for file in ofiles:
    print(f' {file}')

else:
    print(f'File name display disabled. To enable, set show_files=True')

```

File name display disabled. To enable, set show_files=True

Finally, run process all

```

0]: nav_stat_tbl, stacked_imlist, stacked_im_tbl, preproc_flist = processor.process_all(
    data_dir,
    include_pattern,
    exclude_pattern,
    input_file_list=inp_flist,
    input_rootname=inp_rootname,
    input_suffix=".fits.gz",
    final_quality_file=quality_summary_file,
    clean_star_detection=clean_srclists,
    clean_astrometry=clean_navfiles,
    stop_on_error=stop_on_error,
    quiet=quiet,
    target_wcs_file=target_wcs_file,
    resampled_file_prefix=resam_prefix,
    resampled_file_suffix=resam_suffix,
    resampled_dir=resam_dir,
    filter_list=None,
    resampled_summary_plot=resampled_summary_plot,
    composite_summary_plot=composite_summary_plot,
    preprocess_replace=preprocess_instr,
    preprocess_with=preprocess_outstr,
    find_exposure_time=True,
    keyword_dict=meta_dict,
    replace_keywords=rewrite_existing_kw,
    badpixelfile=badpixfile,
)

```

2024-11-26 21:22:51,514 | ApProcess | INFO | Attempting to preprocess FITS files
within directory=./
2024-11-26 21:22:51,514 | ApProcess | INFO | Using files that match
include_pattern="Calibrated-*?.fits*"

2024-11-26 21:22:51,514 | ApProcess | INFO | Excluding files that match
exclude_pattern="None"

2024-11-26 21:22:53,072 | ApProcess | INFO | There are 12 input files matching
the parameters given.

2024-11-26 21:22:53,073 | ApProcess | INFO | Loading extension 0 of FITS file /o
ld_lnx/home/dks/Downloads/iTelescopeScratch/T24/CALIBRATION/Dark/Bin1/300/master
_badpix_T24_20241001_3056x3056_Bin1x1_Temp-25C_ExpTime300s.fits

2024-11-26 21:22:53,372 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz

2024-11-26 21:22:53,373 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:22:53,373 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:22:56,150 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.768 s, 0.015 ms per bad pixel.

2024-11-26 21:22:57,848 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz

2024-11-26 21:22:57,849 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:22:57,849 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:23:00,657 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.799 s, 0.015 ms per bad pixel.

2024-11-26 21:23:02,607 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz

2024-11-26 21:23:02,608 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:23:02,608 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:23:05,405 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.790 s, 0.015 ms per bad pixel.

2024-11-26 21:23:07,307 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz

2024-11-26 21:23:07,307 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:23:07,308 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:23:10,088 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.774 s, 0.015 ms per bad pixel.

2024-11-26 21:23:11,617 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz

2024-11-26 21:23:11,622 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:23:11,622 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:23:14,404 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.772 s, 0.015 ms per bad pixel.

2024-11-26 21:23:15,950 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz

2024-11-26 21:23:15,951 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:23:15,951 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:23:18,741 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.782 s, 0.015 ms per bad pixel.

2024-11-26 21:23:20,256 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz

2024-11-26 21:23:20,257 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:23:20,257 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32

2024-11-26 21:23:23,060 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.794 s, 0.015 ms per bad pixel.

2024-11-26 21:23:24,576 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz

2024-11-26 21:23:24,576 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4

2024-11-26 21:23:24,577 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056

```

rows x 3056 columns, dtype=float32
2024-11-26 21:23:27,360 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.777 s, 0.015 ms per bad pixel.
2024-11-26 21:23:28,878 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:23:28,879 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:23:28,879 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:23:31,686 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.800 s, 0.015 ms per bad pixel.
2024-11-26 21:23:33,197 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
2024-11-26 21:23:33,202 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:23:33,202 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:23:35,978 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.767 s, 0.015 ms per bad pixel.
2024-11-26 21:23:37,510 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
2024-11-26 21:23:37,510 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:23:37,511 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:23:40,283 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.764 s, 0.015 ms per bad pixel.
2024-11-26 21:23:41,803 | ApProcess | INFO | Preprocessing input file
Calibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz into
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
2024-11-26 21:23:41,804 | ApFixBadPixels | INFO | fix_bad_pixels: data has 3056
rows x 3056 columns, dtype=>f4
2024-11-26 21:23:41,805 | ApFixBadPixels | INFO | fix_bad_pixels: mask has 3056
rows x 3056 columns, dtype=float32
2024-11-26 21:23:44,602 | ApFixBadPixels | INFO | Processed 185656 pixels in
2.789 s, 0.015 ms per bad pixel.
2024-11-26 21:23:45,874 | ApProcess | INFO | Finished preprocessing 12 files in
52.769 seconds.
2024-11-26 21:23:45,880 | ApProcess | INFO | Navigating images using modified
inputs: include_pattern=None exclude_pattern=None input_rootname=recalibrated
input_file_list=['recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz', 'recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz']
2024-11-26 21:23:45,880 | ApProcess | INFO | Attempting to find stars in FITS
files within directory=./
2024-11-26 21:23:47,462 | ApProcess | INFO | There are 12 input files matching
the parameters given.
2024-11-26 21:23:47,730 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
2024-11-26 21:23:48,024 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
2024-11-26 21:23:48,311 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
2024-11-26 21:23:48,589 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
2024-11-26 21:23:48,853 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
2024-11-26 21:23:49,132 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
2024-11-26 21:23:49,392 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
2024-11-26 21:23:49,675 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz

```

```

2024-11-26 21:23:49,939 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
2024-11-26 21:23:50,219 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
2024-11-26 21:23:50,482 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
2024-11-26 21:23:50,759 | ApProcess | INFO | Processing input file recalibrated-
iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
2024-11-26 21:23:50,760 | ApQualitySummarizer | INFO | Searching for file
matching pattern "qual*.yaml" in ./MetaData/
2024-11-26 21:23:50,760 | ApQualitySummarizer | WARNING | Found zero quality
files under ./MetaData/. Nothing to do.
2024-11-26 21:23:50,761 | ApProcess | INFO | Finished processing 12 files in
3.297 seconds.
2024-11-26 21:23:50,761 | ApProcess | INFO | Adopting
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits as
the target WCS file for resample_images_to_match.
2024-11-26 21:23:50,780 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:23:50,785 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:23:50,786 | ApProcess | INFO | Found the following 3 files for
filter Blue: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:23:50,786 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:23:50,791 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:23:50,792 | ApProcess | INFO | Found the following 3 files for
filter Green: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:23:50,792 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:23:50,797 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:23:50,798 | ApProcess | INFO | Found the following 3 files for
filter Lum: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:23:50,798 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:23:50,804 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:23:50,805 | ApProcess | INFO | Found the following 3 files for
filter Red: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:23:50,806 | ApProcess | INFO | There are 4 unique filters among 12
files: ['Blue', 'Green', 'Lum', 'Red']
2024-11-26 21:23:50,809 | ApProcess | INFO | Target WCS file is 32.14 x 32.14
arcmin (wxh) with 0.63 x 0.63 arcsec pixels at an angle of 91.52 deg from North.
2024-11-26 21:23:50,810 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:23:50,814 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:23:50,814 | ApProcess | INFO | Processing 3 images for filter Blue
2024-11-26 21:23:50,829 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits
2024-11-26 21:23:50,830 | ApProcess | INFO | Filter Blue with 3 images totalling
900.0 seconds.
2024-11-26 21:23:55,535 | ApProcess | INFO | Success: swarp process took 4.644
seconds, return code 0
2024-11-26 21:23:55,596 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
2024-11-26 21:23:55,597 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:23:55,603 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:23:55,603 | ApProcess | INFO | Processing 3 images for filter
Green
2024-11-26 21:23:55,616 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits

```

```

2024-11-26 21:23:55,618 | ApProcess | INFO | Filter Green with 3 images
totalling 900.0 seconds.
2024-11-26 21:24:00,337 | ApProcess | INFO | Success: swarp process took 4.665
seconds, return code 0
2024-11-26 21:24:00,363 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
2024-11-26 21:24:00,363 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:24:00,368 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:24:00,369 | ApProcess | INFO | Processing 3 images for filter Lum
2024-11-26 21:24:00,378 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits
2024-11-26 21:24:00,380 | ApProcess | INFO | Filter Lum with 3 images totalling
540.0 seconds.
2024-11-26 21:24:05,189 | ApProcess | INFO | Success: swarp process took 4.756
seconds, return code 0
2024-11-26 21:24:05,215 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
2024-11-26 21:24:05,215 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:24:05,220 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:24:05,220 | ApProcess | INFO | Processing 3 images for filter Red
2024-11-26 21:24:05,233 | ApProcess | INFO | Reading FITS header keywords from
./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits
2024-11-26 21:24:05,235 | ApProcess | INFO | Filter Red with 3 images totalling
900.0 seconds.
2024-11-26 21:24:09,901 | ApProcess | INFO | Success: swarp process took 4.612
seconds, return code 0
2024-11-26 21:24:09,928 | ApProcess | INFO | Modified FITS header keywords in
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
2024-11-26 21:24:09,928 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Blue
2024-11-26 21:24:09,933 | ApProcess | INFO | Found 3 files with FILTER=Blue
2024-11-26 21:24:09,935 | ApProcess | INFO | Found the following 3 files for
filter Blue: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits']
2024-11-26 21:24:09,935 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Green
2024-11-26 21:24:09,940 | ApProcess | INFO | Found 3 files with FILTER=Green
2024-11-26 21:24:09,942 | ApProcess | INFO | Found the following 3 files for
filter Green: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits',
'./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits']
2024-11-26 21:24:09,942 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Lum
2024-11-26 21:24:09,947 | ApProcess | INFO | Found 3 files with FILTER=Lum
2024-11-26 21:24:09,949 | ApProcess | INFO | Found the following 3 files for
filter Lum: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits']
2024-11-26 21:24:09,949 | ApProcess | INFO | Searching 12 files for headers in
extension=0 that have FILTER=Red
2024-11-26 21:24:09,954 | ApProcess | INFO | Found 3 files with FILTER=Red
2024-11-26 21:24:09,956 | ApProcess | INFO | Found the following 3 files for
filter Red: ['./NavigatedImages/navigated-
iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits',
'./NavigatedImages/navigated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits']
2024-11-26 21:24:09,956 | ApProcess | INFO | Finished. Generated 4 resampled
images in 19.195 seconds: {'Blue':
'Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits', 'Green':
'Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits', 'Lum':
'Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits', 'Red':
'Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits'}

```

The are 4 FITS images in the input imlist

With 4 files the plot layout is 2 rows x 2 columns.

Attempting to plot Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits

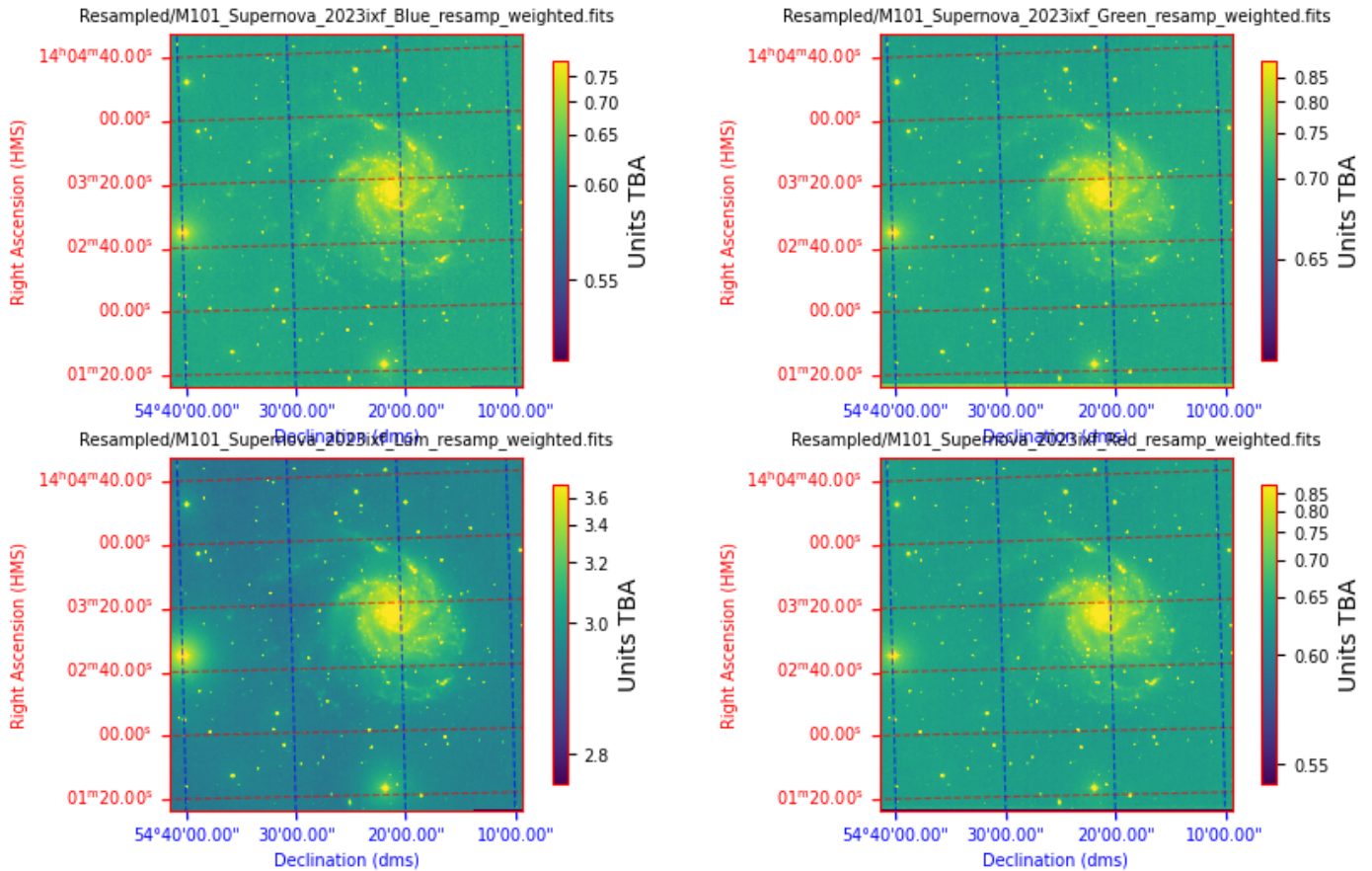
Input image data 0.5th percentile = 0.5265, 99.5th percentile = 0.7889
Applying asinh stretch between 0.5265 and 0.7889
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Attempting to plot Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
Input image data 0.5th percentile = 0.6168, 99.5th percentile = 0.8938
Applying asinh stretch between 0.6168 and 0.8938
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Attempting to plot Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
Input image data 0.5th percentile = 2.7715, 99.5th percentile = 3.7197
Applying asinh stretch between 2.7715 and 3.7197
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)
Attempting to plot Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
Input image data 0.5th percentile = 0.5434, 99.5th percentile = 0.8794
Applying asinh stretch between 0.5434 and 0.8794
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)

2024-11-26 21:24:11,899 | ApProcess | INFO | Resampled image summary plot
generated: resampled_summary_plot.png

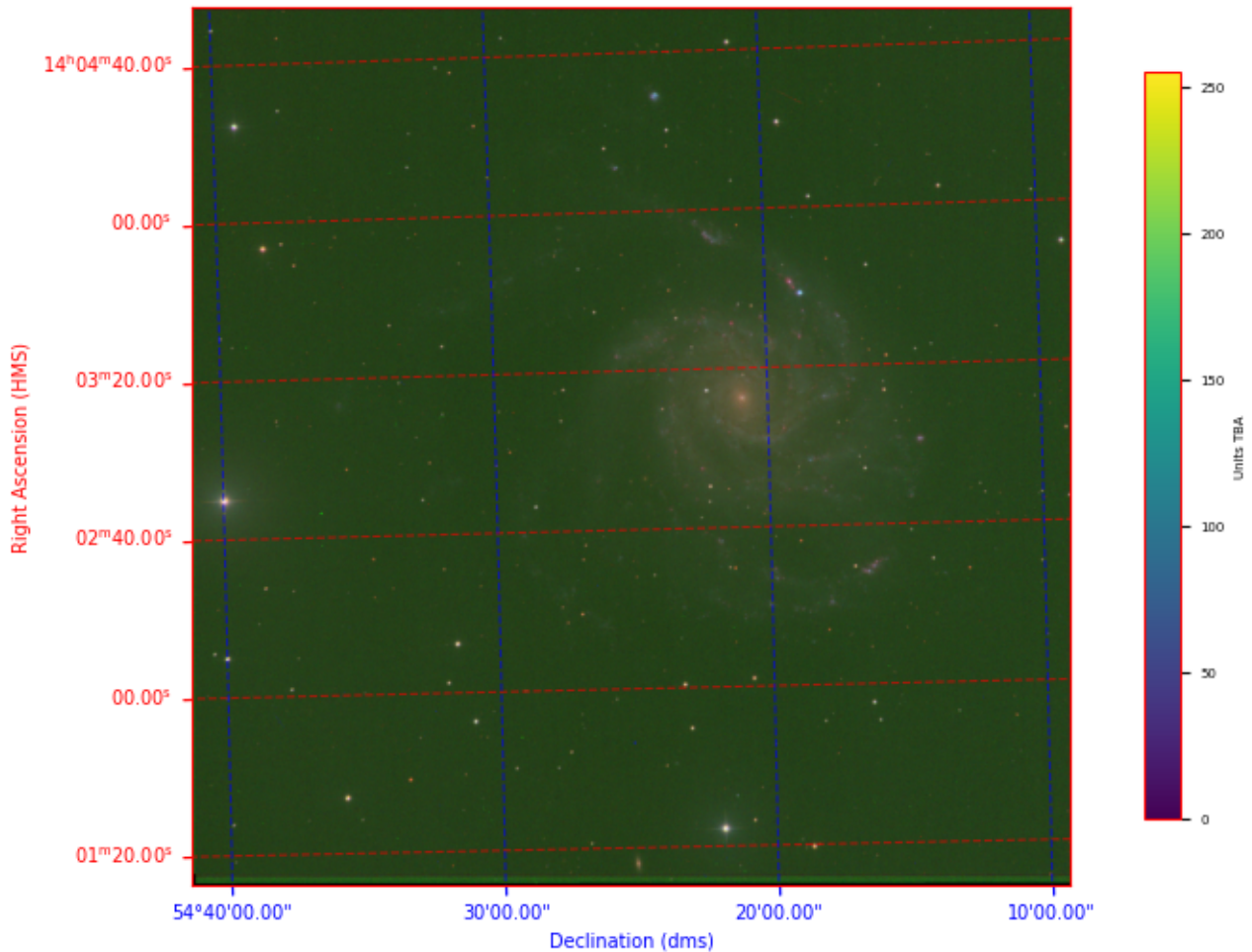
Wrote plot to resampled_summary_plot.png
There are enough files to generate a RGB threecolor composite.
NB: insufficient files to generate a SHO threecolor composite.
Input red channel data 0.5th percentile = 0.5434, 99.5th percentile = 0.8794
Using an image minimum of 0.543389481306076 for all channels.
Default X-axis limits: (-0.5, 3055.5)
Default Y-axis limits: (-0.5, 3055.5)

2024-11-26 21:24:13,304 | ApProcess | INFO | Summary color composite plot
generated: composite_summary_plot.png
2024-11-26 21:24:13,304 | ApProcess | INFO | Finished. Processing took 81.791
seconds.

Wrote plot to composite_summary_plot.png



RGB composite image stretch=0.5, Q=8, minimum=None
 Red = Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
 Green = Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
 Blue = Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits



```
1]: # Output files generated by preprocessing
if preproc_flist is not None:
    print('--- Files generated by preprocessing: ---')
    for im in preproc_flist:
        print(f' {im}')
else:
    print('--- No preprocessing applied. ---')

# show status table
print('--- Navigation status table ---')
nav_stat_tbl.pprint(max_width=200)

# # Load and display the quality summary table (NOTE this is a very wide table)
print('--- Source detection image quality summary table ---')
try:
    qual_summ_table = Table.read(quality_summary_file, format="ascii.csv")
    qual_summ_table['file', 'ncols', 'nrows', 'filter', 'median', 'stddev',
    ↪ 'num_detected', 'fwhm_val_pix', 'fwhm_err_pix', 'fwhm_val_arcs', 'fwhm_err_arcs', 'num_data_pts'].pprint_all()
except:
    print('Quality summary table was not generated')
```

```
--- Files generated by preprocessing: ---
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
```

```

recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
--- Navigation status table ---
      filename
find_stars_status find_stars_time astrometry_status astrometry_time
-----
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
skipped_exists      0.000      skipped_exists      0.000
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
skipped_exists      0.000      skipped_exists      0.000
--- Source detection image quality summary table ---
Quality summary table was not generated

```

```

2]: # Stacked images etc
print('--- List of stacked/resampled images: ---')
for im in stacked_imlist.items():
    print(f' {im}')
print('--- Resampled Image Info Table ---')
stacked_im_tbl.pprint_all()

--- List of stacked/resampled images: ---
('Blue', 'Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits')
('Green', 'Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits')
('Lum', 'Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits')
('Red', 'Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits')
--- Resampled Image Info Table ---
filter num_navfiles tot_exptime_m min_exptime_m max_exptime_m
resampled_image
-----
Blue      3      15.0      5.0      5.0
Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
Green     3      15.0      5.0      5.0
Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
Lum       3       9.0      3.0      3.0
Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
Red       3      15.0      5.0      5.0
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits

```

```

3]: # An alternate method of getting the file names at this stage is to use get_directories()
# and/or get_file_names() on the preprocessed and resampled files with the ap_filestate='processed'
ap_filestate = 'processed'
name_and_dir = True
for ftype in ['preproc_out', 'resampled']:
    print(80*'-')
    tmp_dir = processor.get_directories(ftype, data_dir, False, resampled_dir=resam_dir)
    print(f'For {ftype} the directory is {tmp_dir}')
    tmp_flist = processor.get_file_names(ftype, data_dir, ap_filestate, include_pattern, exclude_pattern, None, None, '.fits.gz',
    ↪False)
    print(f' There are {len(tmp_flist)} files in the file list.')

```

```
for file in tmp_flist:
    print(f' {file}')
```

```
-----
For preproc_out the directory is ./
There are 12 files in the file list.
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-180s-Lum-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Blue-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Green-3.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-1.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-2.fits.gz
recalibrated-iTelescope-M101_Supernova_2023ixf-300s-Red-3.fits.gz
-----
```

```
For resampled the directory is Resampled
There are 4 files in the file list.
Resampled/M101_Supernova_2023ixf_Blue_resamp_weighted.fits
Resampled/M101_Supernova_2023ixf_Green_resamp_weighted.fits
Resampled/M101_Supernova_2023ixf_Lum_resamp_weighted.fits
Resampled/M101_Supernova_2023ixf_Red_resamp_weighted.fits
```

Given the processed files and quick-look summary images the user can now start working on making display-quality images.

4.0.1 Running from the command line

TBA

```
4]: # experiment with the meta_dict
for key, val in meta_dict.items():
    print(key, val)
```

```
OBJNAME ('M101 Supernova 2023ixf', 'Target')
OBJECT ('M101 Supernova 2023ixf', 'Target')
TELESCOP ('iTelescope 24', 'Telescope name')
INSTRUME ('FLI-PL09000', 'Instrument name')
XPIXSZ (12.0, 'Pixel Width in microns (after binning)')
YPIXSZ (12.0, 'Pixel Height in microns (after binning)')
EGAIN (1.53, 'Electronic gain in e-/ADU')
```

```
5]: import yaml

fname = 'm101sn_metadict.yml'
with open(fname, 'w') as f:
    yaml.safe_dump(meta_dict, f,
        default_flow_style=False,
        explicit_start=True,
        explicit_end=True)
    print(f'Wrote example to {fname}')

!cat $fname

with open(fname, 'r') as f2:
    new_data = yaml.safe_load(f2)
    print('Reloaded data:')
    print(new_data)
```

```
Wrote example to m101sn_metadict.yml
---
EGAIN:
- 1.53
- Electronic gain in e-/ADU
INSTRUME:
- FLI-PL09000
- Instrument name
OBJECT: &id001
- M101 Supernova 2023ixf
- Target
OBJNAME: *id001
TELESCOP:
- iTelescope 24
- Telescope name
```

```
XPIXSZ:
- 12.0
- Pixel Width in microns (after binning)
YPIXSZ:
- 12.0
- Pixel Height in microns (after binning)
...
```

```
Reloaded data:
{'EGAIN': [1.53, 'Electronic gain in e-/ADU'], 'INSTRUME': ['FLI-PL09000',
'Instrument name'], 'OBJECT': ['M101 Supernova 2023ixf', 'Target'], 'OBJNAME':
['M101 Supernova 2023ixf', 'Target'], 'TELESCOP': ['iTelescope 24', 'Telescope
name'], 'XPIXSZ': [12.0, 'Pixel Width in microns (after binning)'], 'YPIXSZ':
[12.0, 'Pixel Height in microns (after binning)']}
```

Note that the tuples in the original dict[str, tuple[Any, str]] have been replaced with lists, e.g. dict[str, list[Any, str]]. Does this matter for our purposes?

Unfortunately yes, it does matter, as the list form will fail. So we need to process the output from a yamll safe_load.

Although we didn't need it in this notebook this functionality is required for the command line script to function. The util.read_yaml_into_metadatadict function performs this task.

```
6]: new_dict = ap.util.read_yaml_into_metadatadict(fname)
for key,val in new_dict.items():
    print(key, val)
```

```
EGAIN (1.53, 'Electronic gain in e-/ADU')
INSTRUME ('FLI-PL09000', 'Instrument name')
OBJECT ('M101 Supernova 2023ixf', 'Target')
OBJNAME ('M101 Supernova 2023ixf', 'Target')
TELESCOP ('iTelescope 24', 'Telescope name')
XPIXSZ (12.0, 'Pixel Width in microns (after binning)')
YPIXSZ (12.0, 'Pixel Height in microns (after binning)')
```

5 Batch Processing from the Command Line

Notebooks are fine for development and demonstration, but aren't great for repeated use. The ap_process.py script provides a command line wrapper for ApProcess.process_all, which the following section demonstrates for documentation purposes. (You normally would not create a notebook to run a command line script.)

Note that:

- the following example does not represent a clean run on new data. It reruns the commands we have already used in directories containing existing output files.
- not all of the command line options are used.
- it does not handle the case where an input file list or an filter list is used.

```
7]: # Off by default
run_script=True
script_path='~/git/AstroPhotography/AstroPhotography/scripts/ap_process.py'
add_help=False

extra_parameters=[]
if run_script:
    # binary T/F parameters are handled differently on the command line
    if add_help:
        extra_parameters.append( '--help' )
    if clean_srclists:
        extra_parameters.append( '--clean_star_detection' )
    if clean_navfiles:
        extra_parameters.append( '--clean_astrometry' )
    if stop_on_error:
        extra_parameters.append( '--stop_on_error' )
    if quiet:
        extra_parameters.append( '--quiet' )
    if find_exposure_time:
        extra_parameters.append( '--find_exposure_time' )
    if rewrite_existing_kw:
        extra_parameters.append( '--replace_keywords' )

    # Generate a string that we will run using the ! command
    cmd_str = (f"python3 {script_path} {data_dir} --loglevel=DEBUG"
               f" --include_pattern=\"{include_pattern}\""
               f" --exclude_pattern=\"{exclude_pattern}\""
               f" --input_suffix=\".fits.gz\""
               f" --final_quality_file={quality_summary_file}"
               f" --target_wcs_file={target_wcs_file}"
               f" --resampled_file_prefix={resam_prefix}")
```

```

f" --resampled_file_suffix={resam_suffix}"
f" --resampled_dir={resam_dir}"
f" --resampled_summary_plot={resampled_summary_plot}"
f" --composite_summary_plot={composite_summary_plot}"
f" --preprocess_replace={preprocess_instr}"
f" --preprocess_with={preprocess_outstr}"
f" --metadata_yaml={fname}"
f" --badpixelfile={badpixfile}"
f" {' '.join(extra_parameters)}")

print('To run the ap_process enter the following commands in a terminal:\n\n'
      '# Make sure to activate the appropriate python virtual environment or conda environment.\n'
      f'cd {wdir}\n'
      f'{cmd_str}')

```

To run the ap_process enter the following commands in a terminal:

```

# Make sure to activate the appropriate python virtual environment or conda
environment.
cd /old_lnx/home/dks/Downloads/iTelescopeScratch/Plan-20-M101SN-Pipeline
python3 ~/git/AstroPhotography/AstroPhotography/scripts/ap_process.py ./
--loglevel=DEBUG --include_pattern="Calibrated-*?.fits*"
--exclude_pattern="None" --input_suffix=".fits.gz"
--final_quality_file=quality_summary_M101SN.csv --target_wcs_file=None
--resampled_file_prefix=M101_Supernova_2023ixf_
--resampled_file_suffix=_resamp_weighted.fits --resampled_dir=Resampled
--resampled_summary_plot=resampled_summary_plot.png
--composite_summary_plot=composite_summary_plot.png
--preprocess_replace=Calibrated --preprocess_with=recalibrated
--metadata_yaml=m101sn_metadict.yml --badpixelfile=/old_lnx/home/dks/Downloads/i
TelescopeScratch/T24/CALIBRATION/Dark/Bin1/300/master_badpix_T24_20241001_3056x3
056_Bin1x1_Temp-25C_ExpTime300s.fits --quiet --find_exposure_time

```

6 Versions and Changes

Version	Date	Description
0.5.2-alpha	2024-04-26	issue-002: created script skeleton.
0.6.0	2024-11-26	Final version worked under issue-002

```

3]: # Close all the open figures in memory
plt.close('all')

```